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Agriculture And Veterinary
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**UNIVERZITET U NOVOM SADU
POLJOPRIVREDNI FAKULTET**

**ZBORNIK RADOVA
SA 48. SMOTRE NAUČNIH RADOVA STUDENATA POLJOPRIVREDE I
VETERINARSKE MEDICINE SA MEĐUNARODNIM UČEŠĆEM**

**PROCEEDINGS
OF THE 48th CONFERENCE FOR STUDENTS OF AGRICULTURE AND
VETERINARY MEDICINE
WITH INTERNATIONAL PARTICIPATION**

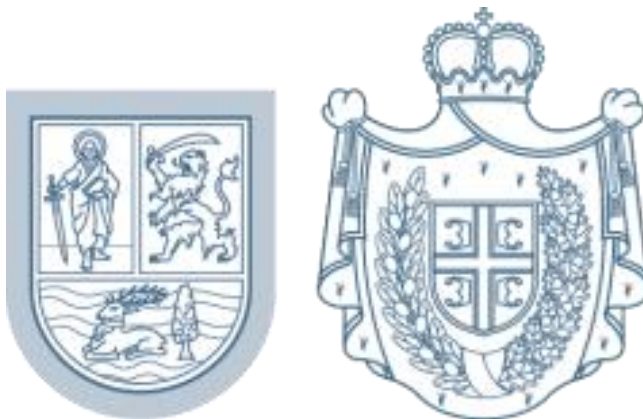
15 November 2024

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**REPUBLIKA SRBIJA
AUTONOMNA POKRAJINA VOJVODINA
POKRAJINSKI SEKRETARIJAT ZA POLJOPRIVREDU, VODOPRIVREDU I
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**REPUBLIC OF SERBIA
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**48th CONFERENCE FOR STUDENTS OF AGRICULTURE AND VETERINARY
MEDICINE WITH INTERNATIONAL PARTICIPATION**

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Council of the University of Novi Sad, Faculty of Agriculture, at its fourth meeting on 18th of February 1981 made a decision that the Conference for students of agriculture and veterinary medicine with international participation, held each year at the Faculty of Agriculture will be held in the memory of academician ***Dr Petar Drezgić***, professor, and the prizes awarded at the Conference will bear his name.

PREFACE

The first Conference for students of agriculture and veterinary medicine with international participation was held in Novi Sad 48 years ago. To the present date, many papers from national and international authors were presented (Hungary – Gödölö and Debrecen, Germany – Kassel, Slovakia – Nitra, Poland – Wroclaw and Warsaw, Romania – Timisoara, Cluj, Bucharest and Sibiu, Bulgaria – Plovdiv, Russia – Moscow and Kemerovo, Macedonia – Skopje and Bitola, Bosnia and Herzegovina – Sarajevo, Banja Luka, Croatia – Zagreb). Since 2009, papers presented at the Conference are published in a special publication named “PROCEEDINGS OF THE 48th CONFERENCE FOR STUDENTS OF AGRICULTURE AND VETERINARY MEDICINE WITH INTERNATIONAL PARTICIPATION” where all published papers have all elements of the original scientific paper. We encourage students and mentors who are interested in benefiting more from this Conference to publish their papers in the journal Contemporary Agriculture published by the University of Novi Sad, Faculty of Agriculture.

As the host and organizer of the Conference University of Novi Sad, Faculty of Agriculture has a special honor and a pleasure to wish a warm welcome to all participants, successful work and amusing socializing in Novi Sad.

Vice-dean for science and international cooperation
Dr Mila Grahovac, full professor

President of the organizing committee
Dr Marta Loc, assistant professor

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GREENHOUSE GAS EMISSIONS AND MICROCLIMATE PARAMETERS ON DAIRY FARMS

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This research was conducted to determine greenhouse gas emissions and microclimate parameters on dairy farm in Slovenia. A total of 48 measurements from different locations, 40 inside and 8 outside the farm. The results of the measurements were influenced by many factors, such as the size of the herd, the method of keeping, the ventilation system, the place from which the measurements were taken, as well as the air flow during the measurements. Based on the results, we can conclude that the GHG as well as the microclimate parameters according to the norms are within the permitted values.

Key words: greenhouse gas, microclimate parameters, dairy farm

INTRODUCTION

Livestock is considered as one of the primary sources of anthropogenic greenhouse gas (GHG) emission, and dairy farms are major contributors in this regard (Rotz, 2018). Within the farm, important emissions include enteric CH₄ from the animals (Verge et al., 2007), CH₄ and N₂O from manure in housing facilities during long-term storage and during field application, and N₂O from nitrification and denitrification processes in the soil used to produce feed crops and pasture. Dairy farming is responsible for approximately 20% of the greenhouse gas (GHG) emissions caused by all global livestock (Gerber et al., 2013). These GHG emissions exacerbate the occurrence of extreme weather events, such as severe heat and drought (IPCC et al., 2022; Donnelly et al., 2024).

Dairy production contributes to GHG emissions along the chain through several processes, which include feed production, enteric fermentation, manure management, and energy use (Van Middelaar et al., 2014). Major GHGs, i.e., carbon dioxide (CO₂), methane (CH₄) and nitrous oxide (N₂O), are released during these processes, and GHG emission intensity (EI), which is defined as emissions per unit of milk produced, is usually used to express the climate impact of dairy production. The emission of carbon dioxide (CO₂) from land use depends on the demand for food grains, grazing land, and energy usage for farm operations. In addition, the decomposition of lime applied to crop and pasture land also contributes to the emission of CO₂ (Herron et al., 2022). Recently, compost-bedded pack (CBP) barn systems have received increasing attention. This system consists of an open resting area (between 20 and 30 m² per cow) where cows lie over their own manure, which is composted daily “in situ” by the tillage of a rotary harrow or cultivator (Black et al., 2013). An alternative stocking rate density may require less space (minimum of 15 m²) when feed alleys are daily scraped and the resultant slurry is removed and stored in a pile (Klaas et al., 2010). Because of the low cost and positive effects on animal welfare, health, and milk quality, this system has become an alternative to the loose-housing systems based on cubicles (Biasato et al., 2019). Nevertheless, its environmental impact on contaminant gas emissions, such as CH₄ and NH₃ is quite unknown yet, and the fact of disrupting the manure surface by tillage may support a rise in such emissions. Dairy cattle barns are generally open, and naturally ventilated and the gas emission rate is dependent on several factors, such as thermal buoyancy forces, temperature, air humidity, and air pressure on the openings of the building (Leso et al., 2020; Poteko et al., 2019). Thus, choosing the right procedure to determine gas emissions in these systems is vital to obtain reliable information. Enteric emissions are normally the largest source of GHG on a dairy farm. On well-managed confinement farms, they contribute about 45% of the total GHG emission of the full farm system, and on more-extensive grazing farms the proportion may be a little greater (Rotz and Thoma, 2017). Models using each level of detail have been used to represent enteric GHG production, which is primarily CH₄ but may include minor amounts of N₂O (Hamilton et al., 2009).

Carbon Dioxide (CO₂) is the most prevalent GHG, primarily released from the burning of fossil fuels and land-use changes. Approximately 75% of global CO₂ emissions come from industrial activities and transportation. CO₂ is also produced as a by-product of the cow's breathing process. Reducing CO₂ emissions requires a transition to renewable energy sources, improving energy efficiency, and adopting low-carbon technologies (IPCC, 2014).

Methane (CH₄) is the second most significant GHG, with emissions stemming from agriculture, particularly livestock production, as well as landfills and natural gas. Methane is most often produced during digestion in cows, especially through enteric fermentation. This gas has the potential to contribute to global warming much more than carbon dioxide on an annual basis (Gerber et al., 2013). Strategies to reduce methane emissions include optimizing livestock diets, using additives that decrease methane production during digestion, and improving manure management. The use of anaerobic digesters can also significantly reduce methane emissions from manure (A.N. Hristov et. al., 2022)

Nitrous Oxide (N₂O) is primarily released from agricultural activities, especially through the use of synthetic fertilizers. N₂O has the potential to be 298 times stronger than CO₂ in terms of heat retention (Smith et al., 2008). Reducing N₂O emissions can be achieved through precise fertilizer application, the use of cover crops, and the implementation of sustainable practices in agriculture, such as crop rotation and agroecological methods.

The aim of this research was to determine the emission of greenhouse gases on dairy cows.

MATERIAL AND METHODS

Measurements were done August 2024 on dairy farm Černivec in Srednje Jarše Slovenia. The farm owns 195 dairy cows, which are free-range. At least 24 measurements inside the barn, 50 % at a height of 1,5 m and 50 % at a height of 0,2 m and 8 measurements outside the barn at a height of 1,5 m. Each measurement took 5 minutes on the same spot. The 40 measurements were taken inside the farm in different locations, milking parlor, feeding area, maternity ward, hospital, rest area, as well as 8 measurements outside the farm. Use of GT 5000 Terra – Splashproof multigas FTIR analyser and Multifunctional TESTO 435 for micro-climate parameters (temperature, humidity).

We measured: H₂O, CO₂, NH₃, CH₄, N₂O, Temperature (°C), Humidity (%), Air flow rate (m/s). The daily temperature humidity index values (THI) were calculated using the equation by Kibler (1964): $THI = 1,8 Ta - (1 - RH) (Ta - 14,3) + 32$ where: THI – temperature humidity index, Ta – Temperature detected in stable, RH – relative humidity.

Statistical analysis: General variability of observed parameters was calculated using the descriptive statistical methods within the Statistica software package version 14.0 (TIBCO Software Inc., Palo Alto, CA, USA).

RESULTS AND DISCUSSION

In Table 1, we can see the statistical processing of the average measurement values taken from a height of 1.5m and a height of 0.2m, as well as the total average values of both heights. Gas measurement results are expressed in the ppm unit. These results are consistent with N.M. Ngwabie et al., (2009) who used a similar method of measuring greenhouse gases in dairy farms using a photoacoustic multi-gas analyzer 1412 and a multiplexer 1309. In their study, gas concentrations measured were as follows: 0.16 to 0.75 ppm N₂O, 1.70 to 17.93 ppm NH₃, 9 to 283 ppm CH₄, and 644 to 3530 ppm CO₂. EFSA (2009) concludes that cows are negatively affected by the concentration of gases in barns if it is over 3000 ppm carbon dioxide, 10 ppm ammonia and 0.5 ppm hydrogen sulphide.

Table 1. Results of gases emissions measured inside (at different heights) and outside

Different heights, (m)	Gases	N	Mean	Std. Deviation	Minimum	Maximum	Range	CV%
1.5	H ₂ O	20	2.18	0.22	1.69	2.44	0.75	9.97
	CO ₂	20	638.44	78.88	499.72	794.6	294.88	12.35
	CH ₄	20	19.56	5.81	8.16	29.18	21.02	29.70
	N ₂ O	20	0.17	0.02	0.13	0.22	0.09	13.40
	NH ₃	20	2.39	0.99	0.65	3.77	3.12	41.50
0.2	H ₂ O	20	2.18	0.19	1.75	2.46	0.71	8.71
	CO ₂	20	622.94	87.68	474.68	771.82	297.14	14.07
	CH ₄	20	19.84	7.87	7.47	35.1	27.63	39.67
	N ₂ O	20	0.17	0.02	0.13	0.21	0.08	11.11
	NH ₃	20	2.37	0.98	1.07	4.11	3.04	41.49
Total inside	H ₂ O	40	2.18	0.20	1.69	2.46	0.77	9.24
	CO ₂	40	630.69	82.69	474.68	794.6	319.92	13.11
	CH ₄	40	19.70	6.83	7.47	35.1	27.63	34.67
	N ₂ O	40	0.17	0.02	0.13	0.22	0.09	12.15
	NH ₃	40	2.38	0.97	0.65	4.11	3.46	40.96
Total outside	H ₂ O	8	2.25	0.07	2.17	2.35	.18	3.01
	CO ₂	8	438.55	30.07	418.92	509.95	91.03	6.86
	CH ₄	8	4.13	2.80	2.31	10.79	8.48	67.69
	N ₂ O	8	0.17	0.00	.17	.17	0.00	0.00
	NH ₃	8	0.58	0.46	.31	1.67	1.36	79.70

In Table 2, we can see the statistical processing of the average measurement values taken from a height of 1.5 m and a height of 0.2 m, as well as the total average values of both heights. The microclimatic parameters processed include: Air (m/s), RH (%), T (°C), THI (%). The accuracy of the measurements was influenced by numerous factors, because the barn, in addition to artificial ventilation, also has natural ventilation in the form of large openings on the side walls. Many authors have defined optimal temperature values in facilities for dairy cows. So, for example, according to Hristov (2002) and Ensminger (1977), the optimal temperature values in facilities for dairy cows are 10-15°C. According to Hristov (2002) and Ensminger (1977) the comfort zone is 5-21°C, according to Sambraus (1997) 0-20°C. Acceptable relative air humidity in dairy barns is 50-75%, according to Čobić and Antov (1996) and Hristov (2002), while Sambraus et al. (2002) believe that in the biozone of dairy cows, it is necessary to ensure a relative humidity of 60-80%.

Table 2. Results of microclimate parameters and THI values measured at different heights

Different heights, (m)	Microclimate parameters	N	Mean	Std. Deviation	Minimum	Maximum	Range	CV%
1.5	AIR	20	0.39	0.20	0.15	0.84	0.69	49.96
	RH	20	63.09	5.12	53.8	69.1	15.3	8.11
	Tc	20	26.05	1.57	23.7	28.5	4.8	6.01
	THI	20	74.49	1.81	71.68	77.18	5.5	2.44
0.2	AIR	20	0.60	0.50	.13	1.93	1.80	83.52
	RH	20	62.63	5.30	50.80	68.20	17.40	8.46
	Tc	20	26.30	1.66	23.90	28.60	4.70	6.32
	THI	20	74.77	1.82	71.86	77.30	5.44	2.43
Total inside	AIR	40	0.49	0.39	.13	1.93	1.80	78.38
	RH	40	62.86	5.15	50.80	69.10	18.30	8.19
	Tc	40	26.17	1.60	23.70	28.60	4.90	6.11
	THI	40	74.63	1.80	71.68	77.30	5.62	2.41
Total outside	AIR	8	1.10	0.90	.09	2.36	2.27	81.64
	RH	8	34.93	2.56	30.90	38.40	7.50	7.32
	Tc	8	35.81	0.91	34.20	36.90	2.70	2.53

In Table 3, the first column represents the temperature of the object, the first row shows the percentage of air humidity, the values within the table itself represent the calculated value of THI, which is obtained by a specific formula.

Interpretation of THI values:

- ❖  Optimal, without heat stress.
- ❖  Warning; possible slight increase in stress.
- ❖  Moderate stress; intervention is required.
- ❖  Heavy stress; livestock health may be at risk.
- ❖  Extreme stress; serious consequences for health and productivity.

Temperatura		vlažnost vazduha [%]																			
°F	°C	0	5	10	15	20	25	30	35	40	45	50	55	60	65	70	75	80	85	90	
72	22.0	64	65	65	65	66	66	67	67	67	67	68	68	69	70	70	70	70	71	71	
73	23.0	65	65	66	66	66	67	67	68	68	68	69	69	70	70	71	71	71	72	72	
74	23.5	65	66	66	66	67	67	68	68	69	69	70	70	70	71	71	72	72	73	73	
75	24.0	66	66	67	67	68	68	68	69	69	70	70	71	71	72	72	73	73	74	74	
76	24.5	66	67	67	67	68	69	69	70	70	71	71	72	72	73	73	74	74	75	75	
77	25.0	67	67	68	68	69	69	70	70	71	71	72	72	73	73	74	74	75	75	76	
78	25.5	67	68	68	68	69	70	70	71	71	72	73	73	74	74	75	75	76	76	77	
79	26.0	67	68	69	69	70	70	71	71	72	73	73	74	74	75	76	76	77	77	78	
80	26.5	68	69	69	69	70	71	72	72	73	74	74	75	75	76	76	77	78	78	79	
81	27.0	68	69	70	70	71	72	72	73	73	75	75	75	76	77	77	78	78	79	80	
82	28.0	69	69	70	70	71	72	73	73	74	75	75	76	77	77	78	79	79	80	81	
83	28.5	69	70	71	71	72	73	73	74	75	76	76	77	78	78	79	80	80	81	82	
84	29.0	70	70	71	72	73	73	74	75	75	77	77	78	78	79	80	80	81	82	83	
85	29.5	70	71	72	72	73	74	75	75	76	78	78	79	80	81	81	82	83	84	84	
86	30.0	71	71	72	73	74	74	75	76	77	78	78	79	80	81	81	82	83	84	85	
87	30.5	71	72	73	73	74	75	76	77	77	79	79	80	81	81	82	83	84	85	85	
88	31.0	72	72	73	74	75	76	76	77	78	80	80	81	81	82	83	84	85	86	86	
89	31.5	72	73	74	75	75	76	77	78	79	80	80	81	82	83	84	85	86	86	87	
90	32.0	72	73	74	75	76	77	78	79	79	81	81	82	83	84	85	86	86	87	88	
91	33.0	73	74	75	76	76	77	78	79	80	82	82	83	84	85	86	86	87	88	89	
92	33.5	73	74	75	76	77	78	79	80	81	82	83	84	85	85	86	87	88	89	90	
93	34.0	74	75	76	77	78	79	80	80	81	83	83	85	85	86	87	88	89	90	91	
94	34.5	74	75	76	77	78	79	80	81	82	84	84	86	86	87	88	89	90	91	92	
95	35.0	75	76	77	78	79	80	81	82	83	85	85	86	87	88	89	90	91	92	93	
96	35.5	75	76	77	78	79	80	81	82	83	85	86	87	88	89	90	91	92	93	94	
97	36.0	76	77	78	79	80	81	82	83	84	86	86	87	88	89	91	92	93	94	95	
98	36.5	76	77	78	80	80	82	83	83	85	87	87	88	89	90	91	92	93	94	95	
99	37.0	76	78	79	80	81	82	83	84	85	87	88	89	90	91	92	93	94	95	96	
100	38.0	77	78	79	81	82	83	84	85	86	88	88	90	91	92	93	94	95	96	98	
101	38.5	77	79	80	81	82	83	84	86	87	89	89	90	92	93	94	95	96	98	99	
102	39.0	78	79	80	82	83	84	85	86	87	89	90	91	92	94	95	96	97	98	100	
103	39.5	78	79	81	82	83	84	86	87	88	90	91	92	93	94	96	97	98	99	101	
104	40.0	79	80	81	83	84	85	86	88	89	91	91	92	94	95	96	98	99	100	101	
105	40.5	79	80	82	83	84	86	87	88	89	91	92	93	95	96	97	99	100	101	102	
106	41.0	80	81	82	84	85	87	88	89	90	91	93	94	95	97	98	99	101	102	103	
107	41.5	80	81	83	84	85	87	88	89	91	92	94	95	96	98	99	100	102	103	104	

Table 3: Temperature range according to temperature-humidity index (THI) (modified according to Lallemand Animal Nutrition, 2015)

CONCLUSION

Higher values were determined at a height of 0.2m, while lower values were determined at a height of 1.5m. Based on the results of greenhouse gas measurements and the obtained values, it can be concluded that the measurement height affects the measurement results. Also, a farm that was built in compliance with all measures and regulations, has good ventilation and enough space for the conditional throat, does not emit enormous amounts of gases that can have a harmful effect on the environment.

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CONSTRUCTION OF PHYLOGENETIC TREE FROM LIPIZZAN HORSE MITOCHONDRIAL DNA DATA

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The Lipizzan horse breed comprises 62 mare family lines, and mitochondrial DNA (mtDNA) plays a vital role in tracing maternal lineage and preserving these historic family lines. In this study, 13 individuals were sequenced from five different maternal lines: Thais, Drava, Wera, Batosta, and Zora. The primary aim is to construct phylogenetic trees using Neighbor-Joining and Median-Joining methods, thereby analyzing the genetic diversity and evolutionary relationships among these mare family lines.

Key words: mtDNA, Mare family lines, Horses, Neighbor-Joining method, Median-Joining method

INTRODUCTION

The mitochondrial genome (mtDNA) comprises a 16.6 kb circle of double-stranded DNA, located in multiple copies within the mitochondria of each cell (Schaefer et al., 2001). The sequence of the horse (*Equus caballus*) mitochondrial DNA was sequenced by Xu and Árnason (1994), NCBI Reference Sequence: NC_001640.1. The prevailing view is that animal mtDNA is maternally transmitted, non-recombining and with elevated mutation rate compared to nuclear DNA (Ladoukakis and Zouros 2017). The gene content and gene order are strongly conserved and include 22 tRNA genes, 2 rRNA genes and 13 protein coding regions, coding different subunits of the mitochondrial respiratory chain. The control region is in the majority of mtDNA molecules in the mitochondrion characterized by the typical structure, called displacement loop (D-loop). Due to the high mutation rate in the D-loop region and strictly maternal mode of inheritance, became sequencing of the control region a widely used tool for tracing back the mtDNA sequence diversity among founder mares in pedigreed populations (Kavar et al., 1999; Bowling et al., 2000; Hill et al., 2002; Kavar et al., 2002). The Lipizzan horse breed was established in Lipica in 1580. Lipizzan population was formed from nine stallions and 24 mares, imported from Spain, and from a number of mares belonging to the local breed of white Karst horses. Later imports of Andalusians, Barbs, Neapolitans and Arab horses contributed significantly to the breeding of the Lipizzan horse (Clutton-Brock, 1999). Lipizzan stallion lines were developed from very outstanding males which succeeded to form their own dynasties during more than a century. There are eight Lipizzan stallion lines: six belong to the classical Lipizzan stallion lines (Pluto, Neapolitano, Favory, Conversano, Maestoso and Siglavy), whereas two are considered non-classical (Tulipan and Incitato). Similar to the formation of stallion lines, the establishment of mare family lines was inevitable for the development of the Lipizzan breed. Within the Lipizzan breed, 62 mares established dominant families which are recognized mare family lines today (Dovč et al., 2006).

Phylogeny is the study of the evolutionary histories of living organisms and represents the evolutionary divergences by finite directed (weighted) graphs, or directed (weighted) trees (Peng, 2007). Any biological information that can be used to infer the evolutionary relationship among the taxa is known as a phylogenetic information marker. It can be anything like DNA, RNA, protein, RFLP, AFLP, ISSR, allozymes, and conserved intronic positions, etc. Identification of conserved genetic loci (coding- or non-coding) is the first step in analyzing the phylogenetic relationship (Reddy, 2011). There are two types of phylogenetic methods, character-based methods, which are based on discrete characters from molecular sequences from individual taxa, and those that are based on the distance, the degrees of differences between pairs of sequences. In distance-based methods there are two different algorithms, the cluster-based (UPMGA - Unweighted Pair Group Method using Arithmetic average, and Neighbor Joining) and the optimality based (Fitch-Margoliash and Minimum Evolution) (Peng, 2007). We distinguish two types of phylogenetic trees, rooted and unrooted trees. A rooted tree has a node identified as the root from which ultimately all other nodes descend, hence a rooted tree has direction. This direction corresponds to evolutionary time; the closer a node is to the root of the tree, the older it

is in time. Unrooted trees lack a root and hence do not specify evolutionary relationships in quite the same way, and they do not allow us to talk of ancestors and descendants (Page and Holmes, 1998). In this paper, focus will be on constructing phylogenetic trees based on mitochondrial DNA sequence data from mares of the Lipizzan horse breed. The paper includes two types of trees using different algorithms, the Neighbor-Joining method and the Median-Joining method. The Neighbor-Joining method constructs bifurcating phylogenetic trees, offering a single hierarchical solution that minimizes branch lengths but cannot handle complex evolutionary events like recombination. In contrast, the Median-Joining method builds networks that can include loops and multiple evolutionary paths, making it more suitable for analyzing intraspecific data with recombination or genetic variability (Bandelt et al., 1999; Saitou and Nei, 1987). The main aim of this study is to present phylogenetic trees based on sequencing results.

MATERIAL AND METHOD

DNA isolation. Samples of hairs were collected from horses in Serbia. DNeasy Blood & Tissue Kits is used for DNA extraction, and this was done by the protocol issued by a producer. The verification of the quantity and the quality of the extracted DNA samples was done by NanoPhotometer® N60 and also using gel electrophoresis. This study included 13 horses and mares from different mare family lines including Thais (Rebecca-Thais) (n=2), Drava-Dubovina (Djebrin) (n=2), Wera (Theodorosta) (n=4), Batosta (Africa) (n=3) and Zora (n=2) which currently is not recognized as purebred Lipizzan mare family line by LIF (Lipizzan International Federation - <https://www.lipizzan-online.com>).

Amplification and mitochondrial DNA sequencing. Amplification of fragments of the control region (D-loop) of mtDNA was done using primers Forward primer 5'-CAC CCA AAG CTG AAA TTC TAC-3' and Reverse primer 5'-ATA ACA CCT TAT GGT TGC TG-3' (Hristov et al., 2016). The length of the amplified fragments is 665bp, which was visualized on a 1% agarose gel electrophoresis. The PCR reaction conditions are shown in Table 1.

Table 1: Polymerase chain reaction program for amplification of control region of mtDNA

Phase		Temperature	Time
Initial denaturation		95 °C	3 min
35 cycles	Denaturation	95 °C	30 s
	Annealing	46 °C	30 s
	Extension	72 °C	40 s
Final extension		72 °C	5 min
Hold		4 °C	∞

The amplified fragments were sequenced by Sanger sequencing with the Reverse primer 5'-ATA ACA CCT TAT GGT TGC TG-3' (Hristov et al., 2016) using the Applied Biosystems 3500 Genetic Analyzer. The obtained sequencing results were then processed using bioinformatics software and online tools.

Bioinformatics analysis. After obtaining the sequencing results, chromatograms were primarily manually reviewed and processed using the MEGA 11 software (Tamura et al., 2021). The mitochondrial DNA sequences, specifically the control region (D-loop), from 13 Lipizzan horses belonging to four standard mare family lines and the local family line Zora were aligned using the same software. Additionally, all obtained sequences were aligned with sequences (385bp) from mare family lines downloaded from GenBank (Benson et al., 2013). Multiple alignment of all obtained and downloaded sequences was performed with the reference mtDNA sequence downloaded from GenBank (X79547) using the MEGA 11 software. The sequences (385bp) representing 45 different mare family lines downloaded from GenBank, along with their accession numbers, are shown in Table 2.

Table 2: Mare family lines along with their accession numbers (GenBank).

Haplotype	Accession Number	Haplotype	Accession Number	Haplotype	Accession Number	Haplotype	Accession Number
Allegra	AY057413.1	Gaetana	AY057433.1	MH2L	KP731356.1	Thais	AF168698.1
A	AY057420.1	Gratiosa	AF168696.1	MH3L	KP731357.1	Trompeta	AF168693.1
Batosta	AF168691.1	G	AY057426.1	MH4L	KP731358.1	T	AY057412.1
Betalka	AF168689.1	H	AY057419.1	MH5L	KP731359.1	U	AY057410.1
B	AY057421.1	I	AY057427.1	N	AY057434.1	V	AY057411.1
Boka	DQ233731.1	J	AY057428.1	O	AY057415.1	Wera	AF168697.1
Capriola	AF168690.1	K	AY057432.1	P	AY057414.1	X	AY057418.1
C	AY057422.1	L	AY057430.1	Q	AY057417.1	Z	AY057429.1
Dubovina	AF168695.1	Monteaura	AY057416.1	R	AY057409.1	Y	DQ233732.1
D	AY057423.1	M	AY057431.1	Slavina	AF168692.1		
E	AY057424.1	MH1aL	KP731354.1	Strana	AF168694.1		
F	AY057425.1	MH1bL	KP731355.1	S	AY057408.1		

After obtaining the multiple alignment, the results were exported in NEXUS format and then converted to PHY file, suitable for importing into PopART 1.7 (Leigh and Bryant, 2015). In this program, a Median-Joining network was constructed from the aligned sequences. The MEGA 11 (Tamura et al., 2021) program was used to build the second type of tree, the Neighbor-Joining tree, from which the tree was exported in text format. Online tool iTOL v6. (Interactive Tree of Life - <https://itol.embl.de/>) was used for the aesthetic correction of this tree. To have a better insight into the distances between the mentioned mare family lines, a heat map with numerical values was constructed based on the alignment. The heat map was constructed in the R software.

RESULTS AND DISCUSSION

The main goal of this study is to use data obtained from sequencing to construct phylogenetic trees of two different types: Neighbor-Joining and Median-Joining. After obtaining the results and analyzing 13 sequences of mitochondrial DNA from three classical mare family lines of Lipizzan horses (Drava - Dubovina (n=2), Wera (n=4), Batosta (n=3)), one Croatian mare family line (Thais (n=2)) and the Zora family line from Serbia (n=2) which currently not recognized as purebred Lipizzan, the aforementioned phylogenetic trees were formed. As a result of aligning the sequences and those downloaded from GenBank, all sequences corresponded to a specific mare family line. Additionally, the Zora family line formed a cluster along with several classical mare family line haplotypes of Lipizzan horses (Monteaura, Betalka, Wera, Allegra). Based on the calculated genetic distances, a heat map was created showing the numerical values.

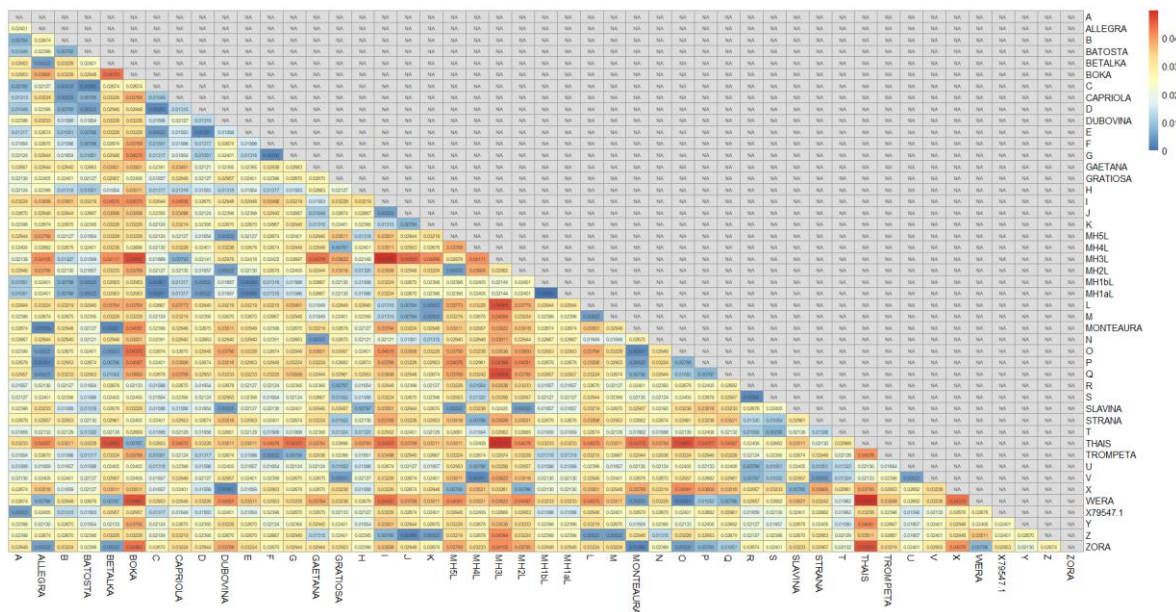


Fig 1. The heat map generated in R software based on the calculated distances

Two types of phylogenetic trees were formed; the first is the Median-Joining network, that can include loops and multiple evolutionary paths, which displays the topological position of haplotypes as well as the differences in nucleotides between haplotypes represented by dashes.

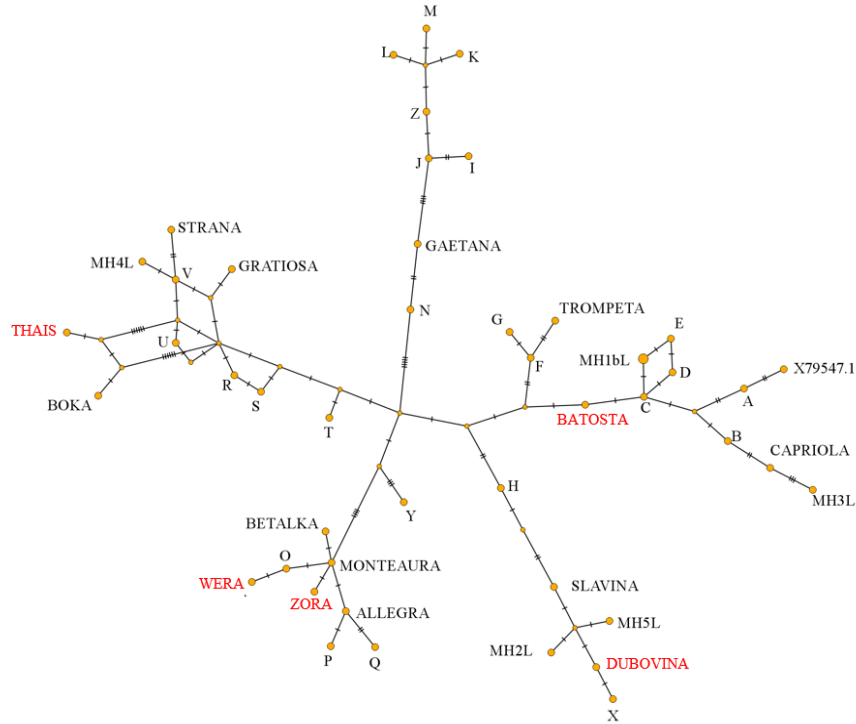


Fig 2. Median-Joining network constructed in PopART 1.7 (Leigh and Bryant, 2015)

Additionally, the second type of tree, the Neighbor-Joining tree, was constructed and displayed with color-coded clusters, that is, groups of haplotypes. In the display, Figure 3., the clusters are highlighted in colors, which is significant for visualization and easier recognition of clusters during the construction and presentation of the phylogenetic trees. Also, on this tree, individuals of the Zora family line formed a cluster along with other classical mare family lines (Monteaura, Betalka, Wera, Allegra), highlighted in gray blue.

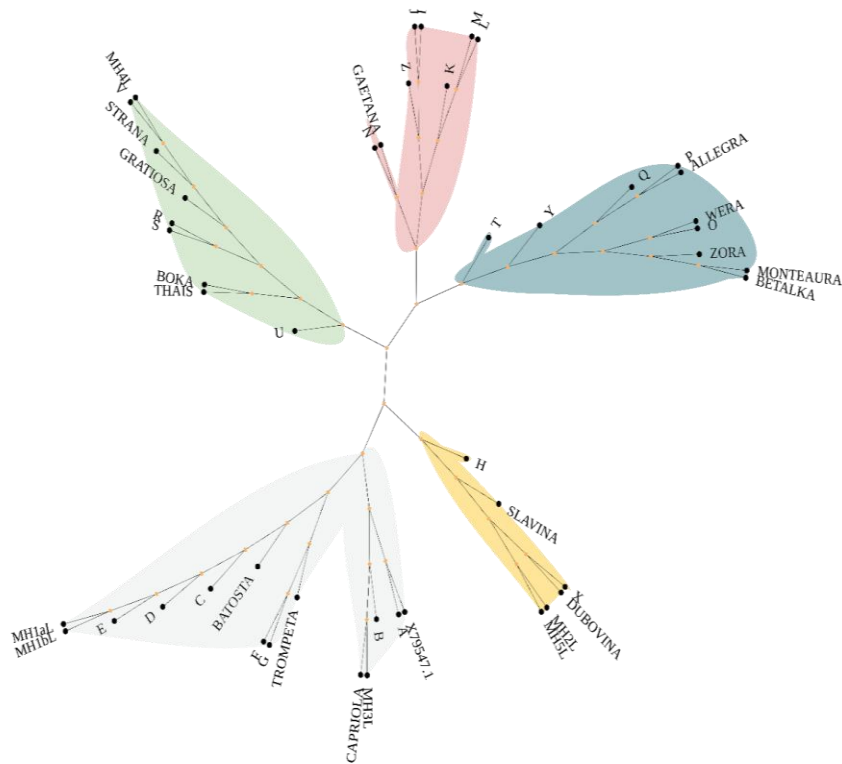


Fig 3. Neighbor-Joining tree constructed in MEGA 11 (Tamura et al., 2021) and aesthetically refined in iTOL v6.

As for the mares of the Zora family line, to our knowledge, there is one study (Zorc et al., 2023) conducted on the characterization of this mare family line. In the study by Zorc et al. (2023), individuals from the Zora family line formed a compact cluster with other classical mare family lines (Monteaura, Betalka, Wera, Allegra). According to this study, as well as many other studies related to phylogenetic analyses (Kavar et al., 2002; Dovč et al., 2006; Čačić et al., 2005; Cardinali et al., 2016), phylogenetic trees are often found as a means of visualizing the distances between individuals, which indicates that phylogenetic trees are an extremely important representation for this type of research.

CONCLUSION

In conclusion, this study analyzed mitochondrial DNA sequences from Lipizzan horses to construct two types of phylogenetic trees: Neighbor-Joining and Median-Joining. Phylogenetic trees are essential for visualizing genetic relationships and tracing evolutionary lineages. They also play a key role in verifying pedigrees, ensuring the accuracy of lineage tracking and the conservation of rare genetic lines. The use of mitochondrial DNA, specifically the D-loop region, helps trace maternal lineages, which is crucial for maintaining genetic diversity. The Zora family line, though not officially recognized, formed a genetic cluster with classical Lipizzan mare lines, indicating its close genetic relationship to the breed. This finding suggests that Zora could play a role in preserving genetic diversity within the Lipizzan population.

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METHODS FOR EVALUATING THE LEG QUALITY OF BROILER CHICKENS

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In this study, the quality of broiler legs was analyzed using three methods: assessment of valgus-varus deformities, foot-pad dermatitis, and gait score. Based on these methods, it was determined that the severity of lesions increases with the age of the chickens, with a significant impact from bedding and environmental conditions, while the gait score is useful for analyzing other factors such as diet. The research results indicate that the age of chickens and bedding materials significantly affect leg quality, enabling an objective assessment of leg quality.

Key words: leg quality, broilers, footpad dermatitis, gait score, varus-valgus

INTRODUCTION

The assessment of leg quality in chickens plays a key role in maintaining animal welfare and production efficiency in poultry farming. For live chickens, various assessment methods are used, such as varus-valgus deformities, which measure the degree of leg bone curvature; foot pad dermatitis, which evaluates damage to the footpads caused by contact with the bedding; and gait score, which determines the quality of walking and mobility in chickens (Wilcox et al., 2023). These methods allow for quick and non-invasive data collection on leg condition during the growth of chickens.

After slaughter, opportunities for more detailed analysis expand, with methods such as X-ray imaging providing insights into the internal bone structure and potential damage that were not visible on live chickens. X-ray images enable precise diagnosis of bone irregularities and detection of microscopic changes that could indicate early stages of developmental issues. In addition to X-rays, other laboratory techniques are employed, offering a comprehensive overview of leg conditions and enabling a better understanding of the impact of rearing factors on leg quality in chickens (Chen et al., 2012). In this paper, the focus will be on methods performed on live chickens.

Valgus-varus deformity (VVD) is one of the leg disorders that affects the health and welfare of broiler chickens and can lead to economic losses due to mortality or reduced growth rates in affected chickens (Shim et al., 2012). Valgus-varus deformity refers to the angular deviation in the joint between the tibia and the tarsal segment of the foot, resulting in the tarsal part of the foot deviating outward (valgus) or inward (varus) (Van Den Brand et al., 2022). The frequency of VVD varies widely, with potential causes for this variance associated with factors such as growth rate, genetics, body weight, housing system, age of the chickens, bedding quality, and diet composition. However, another factor that may contribute to the variation in VVD prevalence rates is the method used to determine VVD. Leterrier and Nys (1992) held live broilers by the wings and classified VVD into four categories based on the angulation of the joint between the tibia and the tarsal segment of the leg through visual inspection. The angulation was classified as (1) normal (angle between the tibia and tarsal segment of the foot $<10^\circ$); (2) mild (angle between 10 and 25°); (3) moderate (angle between 25 and 45°); and (4) severe (angle $>45^\circ$). During VVD assessment, the legs of the broilers were not fixed; instead, the angle between the tibia and the third toe was measured using a protractor.

Gait score (GS) is a standardized method for the subjective assessment of lameness and walking ability in broiler chickens, and their impact on chicken welfare. A six-point gait scoring scale, from 0 to 5, was developed by Kestin et al. (1992) to evaluate the movement ability of broiler chickens, with higher scores indicating increasing walking difficulties. Chickens with a GS greater than 3 are considered severe cases, as this is positively correlated with their movement speed and acceleration. It has been found that an increase in this GS correlates with pain, which further affects their health and welfare (Hothersall et al., 2016; Kittelsen et al., 2017; Granquist et al., 2019).

Foot-pad dermatitis (FPD) constitutes a significant challenge in broiler production, impacting both animal welfare and economic viability within the poultry industry. FPD is a type of contact dermatitis that is presented as inflammatory lesions on the plantar surfaces of the feet and toes. This condition, caused by a combination of factors such as wet bedding, poor ventilation, and high stocking density, can lead to painful lesions and tissue necrosis, significantly impairing poultry welfare and performance (Mayne, 2005; Shepherd and Fairchild, 2010). Wet and contaminated litter allows bacteria to penetrate the skin, further worsening the condition, as foot lesions can serve as entry points for bacterial infections, including *Staphylococcus aureus*, commonly found as a secondary agent in foot ulcers in poultry (Olsen et al., 2018). Foot-pad dermatitis impacts both health and the

performance of broiler chickens (Bjedov et al., 2013; De Jong et al., 2014). Alabi et al. (2024) demonstrated that birds with severe foot lesions had lower daily weight gain, which may be associated with pain and discomfort, leading them to avoid movement and reduce feed intake.

MATERIAL AND METHODS

The incidence and severity of FPD were measured at 4 and 6 weeks of age by the method described by Eichner (2007): no lesions (0), lesions cover less than 25% of the footpad (1), lesions covering between 25% and 50% of the footpad (2) and lesions covering more than 50% of the footpad (3).

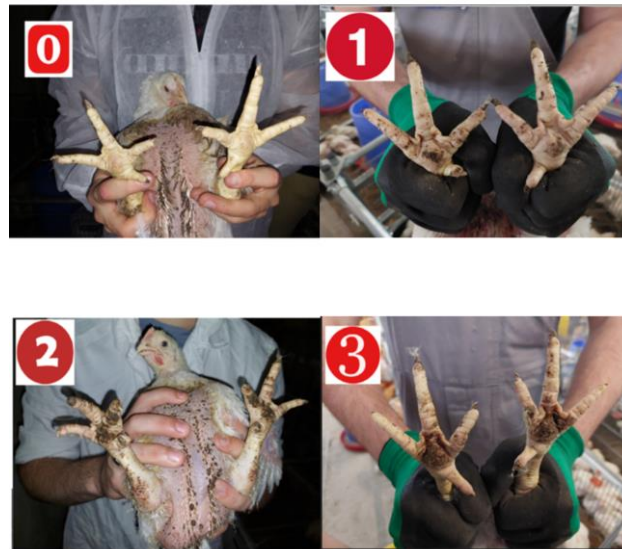


Fig. 1 Footpad dermatitis in broilers, scores from 0 to 3.

The gait score assessment was conducted by randomly selecting five chickens from each box containing 40 chickens, resulting in a sample size of 20 chickens per treatment, a total of 40 chickens. Gait score evaluation was performed according to the method by Kestin (1992), which enables precise differentiation between various levels of gait in chickens. Each chicken was individually allowed to move freely through a corridor, where we observed its gait through subjective assessment and assigned a score ranging from 0 to 5. A score of 0 represents normal gait with no abnormalities, score 1 represents gait with mild abnormalities, score 2 represents gait with noticeable undefined abnormalities, score 3 with an obvious gait abnormality, score 4 with a significant gait abnormality, while a score of 5 indicates a complete inability to walk or stand.



Fig. 2 Assessment of gait score in broilers at 4 weeks of age.

Varus-valgus deformity was assessed by measuring leg angles in live chickens. The chickens were gently restrained by the wings and carefully turned backward to extend their legs. The leg angle was measured with a

digital protractor at four and six weeks of age. During angle determination, one side of the protractor was aligned parallel along the tibiotarsus, while the other side was positioned parallel along the tarsometatarsus.



Fig. 3 Assessment of leg angulation in broilers

RESULTS AND DISCUSSION

Foot pad dermatitis assessment was conducted to compare the effects of two different bedding materials at the 4 and 6 weeks of age. The treatment involved different types of bedding: straw (treatment 1) and peat (treatment 2).

Table 1. Effect of different bedding material and age on incidence and severity of FPD

Score	4 weeks of age				Score	6 weeks of age			
	Treatment 1		Treatment 2			Treatment 1		Treatment 2	
	No.	%	No.	%		No.	%	No.	%
0	8	8.4	86	91.5	0	0	0.0	59	62.8
1	65	68.4	8	8.5	1	10	11.1	32	34.0
2	22	23.2	0	0.0	2	24	26.7	3	3.2
3	0	0.0	0	0.0	3	56	62.2	0	0.0
Sum	95	100.0	94	100.0	Sum	90	100.0	94	100.0
Mean	1.15**		0.09		Mean	2.51**		0.40	

** statistically significant differences ($P < 0.01$)

Based on the results in Table 1, it can be observed that the average score for foot-pad dermatitis (FPD) increases over the period from 4 to 6 weeks. By analyzing these results, we can observe that the age of the chickens and the different types of bedding significantly influence the occurrence and severity of FPD. The use of standardized assessment methods allowed for objective monitoring of footpad dermatitis development on chicken feet and the differences between groups subjected to different conditions. Throughout the experiment, visual assessment of lesions was employed, ranking FPD by severity, enabling quantitative comparison between groups. Results after the fourth week clearly demonstrate that the assessment method allows for early detection of FPD even in its initial stages. Applying this assessment method is essential for similar research, as it provides a standardized approach for precise analysis of FPD under various conditions. These results highlight the importance of timely and continuous FPD assessment to allow for swift interventions and improvements in poultry welfare. Due to its simplicity, the FPD assessment method is feasible even in larger commercial broiler farms with minimal investment.

Table 2. Effect of different bedding material and age on gait score

4 weeks of age					6 weeks of age				
Score	Treatment 1		Treatment 2		Score	Treatment 1		Treatment 2	
	No.	%	No.	%		No.	%	No.	%
0	16	80.0	18	85.7	0	0	0.0	1	5.0
1	4	20.0	0	0.0	1	0	0.0	8	40.0
2	0	0.0	2	9.5	2	12	60.0	8	40.0
3	0	0.0	1	4.8	3	7	35.0	1	5.0
4	0	0.0	0	0.0	4	1	5.0	2	10.0
Sum	20	100.0	21	100.0	Sum	20	100.0	20	100.0
Mean	0.2		0.33		Mean	2.45*		1.75*	

* statistically significant differences (P<0.05)

The gait score (GS) assessment method was crucial for monitoring changes in chicken gait and detecting potential movement issues from the fourth to the sixth week, as well as for identifying any differences between the treatments: straw (treatment 1) and peat (treatment 2). Results in Table 2. show that the difference was much smaller at 4 weeks of age but increased in chickens aged 6 weeks. The standardized and quantitative approach in GS evaluation allowed for precise tracking of differences in motor abilities between experimental groups over time, providing reliable data on potential mobility issues. Gait score assessment in chickens was conducted according to clearly defined criteria for gait evaluation, including aspects such as stability, coordination, and movement symmetry. The GS method proved reliable for distinguishing mild lameness or unsteady gait, which is essential for early identification of risk factors that may impact chicken leg health. Based on the results, we can conclude that motor abilities in chickens change with age and are also influenced by the different treatments, such as bedding material.

Regarding leg assessment by monitoring valgus-varus deformity, the examination was conducted as part of an experiment in which we observed the effect of added phytase enzyme on chicken leg quality. Treatment 1 did not have added phytase in the feed, while treatment 2 added phytase in the feed. The assessment was also carried out at the ages of 4 and 6 weeks of age, and the results are presented in Table 3.

Table 3. Effect of different feeding programs on varus-valgus deformity

4 weeks of age					6 weeks of age				
Score	Treatment 1		Treatment 2		Score	Treatment 1		Treatment 2	
	No.	%	No.	%		No.	%	No.	%
< 10	14	70	15	75	< 10	8	40	15	75
10 - 25	6	30	5	25	10 - 25	12	60	5	25
Sum	20	100	20	100	Sum	20	100	20	100
Mean	8.13		7.05		Mean	9.91*		7.74*	

* statistically significant differences (P<0.05)

This method is based on standardized criteria for assessing the angle and symmetry of leg, allowing for a quantitative analysis of leg deviation, which is crucial for the early detection and monitoring of varus and valgus deviations. The precisely defined criteria within the method enabled differentiation between normal limb positioning and mild to moderate deviations, which is essential for detecting early changes that may indicate potential joint or skeletal development issues. The method is sensitive enough to detect small changes in limb positioning, making it crucial for monitoring differences between control and experimental groups throughout the growth period. Van den Brand et al. (2022) state that by fixating the legs of live animals, highly reliable data on leg angles can be obtained, with minimal differences compared comparable with the angle in dissected legs.

CONCLUSION

The analysis of footpad dermatitis shows that this parameter is significantly influenced by the age of the chickens, type of bedding, and environmental factors within the broiler house. Since the presence of FPD is one of the key indicators in assessing poultry welfare, it is crucial for producers to regularly conduct its evaluation

on farms and for slaughterhouse workers to assess it on the slaughter line. Regarding gait score, this parameter can be used to evaluate bedding, and different feeding programs, though its assessment is somewhat more complex, harder to implement, and requires a certain level of evaluator experience. Varus-valgus can be used to assess nutrition (especially mineral deficiencies) and test different hybrids, as leg quality is an inheritable parameter that may be compromised by growth traits selection. Based on the results of this study, we can conclude that methods for assessing leg quality in broiler chickens can clearly indicate differences in rearing conditions, as well as differences in genotype, age, nutrition, and other factors.

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THE IMPORTANCE OF THE PHYSICAL STRUCTURE OF RUMINANT RATIONS

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Based on the analysis of the results, the study concludes that while silage particle size does not directly influence milk fat content, the ratio of concentrates to bulky feeds plays a key role. Farms with a higher proportion of concentrates and lower hay content tend to have lower milk fat, likely due to the starch-rich nature of concentrates. Conversely, farms with more fibrous feeds, especially hay, show higher milk fat levels. To improve milk fat content, it is recommended to reduce concentrates and increase bulky feeds in the diet.

Key words: physical structure, milk fat, penn state separator, silage, haylage.

INTRODUCTION

Before the automation and modernization of machinery, ruminant diets were simple. Animals grazed in the pastures and were fed grains that the household or farm had available at the time. As machinery developed, farmers gained new opportunities for food production. In addition to simplifying meal preparation, the range of available feed options expanded. With the growth of industry, the number of by-products that could be used as animal feed also increased, providing excellent options for nutrition. However, with every new era or technological advancement comes a set of challenges. Modernization of technology allowed for the production of more feed, but these same feeds introduced new problems. In the case of cows, which are ruminants by nature, concentrated feeds and feeds with smaller particle sizes, such as silage and chopped haylage, were introduced, replacing the coarser hay they had been consuming. To meet the nutritional needs of high-producing cows in lactation, especially in terms of energy and specific nutrients, it is essential to formulate diets that contain a high proportion of concentrates, along with quality fibrous feeds (Stojanović et al., 2009). The increase in production that resulted from the introduction of concentrated feeds into the cows' diet was not inevitable, and even with this increase, farmers still faced problems with their animals. An excess of concentrated feeds led to various health issues, including acidosis and displaced abomasum. These two conditions, of course, often led to secondary diseases if not treated in time. Problems with hoof health could arise, reproductive issues might occur, and most significantly, these health problems would have a direct impact on milk production, which would drastically decrease. In addition to reduced production, the quality of the milk would also suffer. The fat content of the milk would decrease, and consequently, the overall quality of the milk produced would be lower. As Blaxter (1944) noted, it is essential to balance fiber intake with concentrates to avoid detrimental effects on cow health and milk production. In their study, Krause et al. (2002) found that excessive inclusion of coarse fibrous feeds in the diet can reduce digestibility and milk yield. Furthermore, research by Couvreur et al. (2007) confirmed that silage and forage particle size influence milk fat composition, with smaller particles resulting in lower milk fat content. In addition, Murphy et al. (2000) found that the fermentation process in the rumen and the balance between fiber and concentrate in the diet were key factors influencing milk fat content in lactating cows. They showed that cows fed diets with higher proportions of fibrous feeds had better fermentation in the rumen, which positively influenced milk fat content. O'Mara et al. (1998) observed similar effects, noting that replacing grass silage with maize silage in the diet of dairy cows led to an increase in milk yield but did not significantly impact milk composition. On the other hand, Dewhurst (2013) highlighted the importance of comparing different silage types, such as grass, legume, and maize silages, as they differ in their effects on milk production. Grass silage, for instance, was found to support better milk production when combined with legume silage, while maize silage contributed to higher energy intake. In summary, while the introduction of more advanced feeding strategies has improved milk production potential, it has also brought with it a range of health challenges that must be managed to maintain both the quantity and quality of milk production.

Rumen acidosis or acid indigestion is the most common form of digestive disorder in cattle. It occurs due to an excessive amount of easily digestible carbohydrates in the diet, such as starch and sugars, as well as an imbalanced ratio between fibrous and concentrated feeds. The high content of easily fermentable carbohydrates leads to a significant drop in pH and acidification of the rumen. In such conditions, the number of beneficial bacteria responsible for cellulose digestion decreases. Additionally, rumen contractions and rumination may

cease, leading to a reduction in saliva production, which normally helps maintain the pH of the rumen. As a result, lactic acid is absorbed from the rumen into the bloodstream, leading to metabolic acidosis. Displaced abomasum (DA) is a condition in which the abomasum (the true stomach) shifts from its normal position, moving to the left or right side of the abdomen or more dorsally. This condition most commonly occurs after calving, though it can also happen before calving, and is more frequent in cows aged 3-8 years compared to heifers. When we talk about fibrous feeds, they should form the foundation of a ruminant's diet because they stimulate rumination. However, from a production standpoint, fibrous feeds alone will not yield significant results. Therefore, they should always be combined with concentrated feeds. The appropriate particle size of fibrous feeds in the total diet (the size of feed particles) is an important parameter when formulating rations and should be considered equally alongside its chemical composition and nutritional value (Stojanović et al., 2009). Today, we have a wide variety of fibrous feeds that can be used in milk production, including silage, haylage, and hay. The most commonly used in our region is corn whole-plant silage, while alfalfa haylage is also used, though to a lesser extent and mostly on larger farms. Additionally, a part of the ruminant's diet includes alfalfa hay, which has replaced meadow hay due to a shortage of meadow areas.

An experiment conducted by Wing and Wilcox (1963) showed that there were no significant differences in milk yield or milk composition among cows that were fed differently. The cows were divided into three groups, each of which had been in lactation for 200 days. All three groups received the same fibrous feeds. The first group received concentrated feed individually according to the Morrison standard, the second group received the concentrated feed individually, and the third group received the concentrated feed mixed with the fibrous feed, given together in a group. The scientific studies conducted by Grant et al. (1990) explored how the particle size of corn silage and alfalfa haylage affects milk production, milk composition, and the rumination time in cows. In both experiments, the cows were divided into two groups and fed diets containing different amounts of NDF (Neutral Detergent Fiber). In one group, the NDF content was 29%, while in the other group, it was 32%. The cows also received silage and haylage with different particle sizes. The results of their experiments indicated that particle size did not have a significant impact on milk production, but there was a dramatic difference in milk fat content. The milk fat dropped from 3.8% to 3%. Additionally, cows fed the more finely chopped diet had a reduced rumination time due to the finer nature of the feed. The study conducted by Kononoff et al. (2003), which investigates how the particle size of corn silage affects rumination, cow activity during the day, and ruminal fermentation, shows that there are no significant differences between the groups fed diets with different silage particle sizes. The only noticeable difference was related to the duration of rumination in cows, with a clear distinction observed between cows consuming finer versus coarser particles.

The goal of the study is to determine the quality of particle size in silage and the relationship between silage particle size and milk fat content.

MATERIAL AND METHODS

The research was conducted on 8 dairy farms located in the northern part of Vojvodina, within the Subotica district. Information about the farms can be found in Table 1.

Table 1. Information about the farm

N°	Year	N° IHH	System	MA	MF	MP
1.	35	11	Fs.	23	4.128	3.313
2.	30	6	Fs.	15	4.266	3.364
3.	10	21	Fs.	10	3.991	3.241
4.	40	11	Fs.	25	3.550	3.388
5.	35	58	Fs.	24	4.056	3.305
6.	5	21	Fs.	26	4.203	3.258
7.	45	180	Fs.	26	3.486	3.356
8.	20	145	Fs.	31	3.252	3.431
Average	27.5	56.6	100% fs.	22.5	3.866	3.332

Table legend: Year - number of years engaged in milk production, N° IHH– number of individuals in the herd, system – housing system (fs. – free stall), MA – milk average, MF – milk fat, MP – milk protein

From Table 1, we get information about the herd size on the farms, which varies (ranging from 6 to 180 animals). The farms where the research was conducted are family owned farms, which have been engaged in milk production for many years. The average milk yield on the farms is solid, with generally small differences between the farms. All the farms use a free stall housing system.

The sieving of fibrous feeds was performed using the older model of the Penn State Particle Separator. The Penn State Particle Separator consists of 3 sieves of different diameters. The bottom of the first and middle sieves is made of plastic, while the bottom of the lower sieve is made of wire mesh, and the box representing the bottom is also made of plastic. The top or first sieve has a diameter of 19 mm, the middle sieve has a diameter of

8 mm, and the lower or last sieve has a diameter of 1.18 mm. Samples were taken from whole-plant corn silage and finely chopped alfalfa haylage. The samples were directly taken from the silage bunkers. To ensure the accuracy and reliability of the samples, 5 samples were taken from different parts of the silage bunkers. The amount of each sample collected was approximately 1.4 liters. The final result was the average of all 5 samples. The sieving procedure follows the guidelines set by Penn State. The Penn State Particle Separator must be placed on a flat surface. Sieving is done immediately after the sample is collected. One motion consists of a horizontal movement of the sieve back and forth. It is necessary to perform 5 forward-backward movements of equal strength and speed, each covering about 20 cm. After 5 movements, the sieves are rotated 90 degrees, and the process is repeated. The sieving method is shown in the image below.

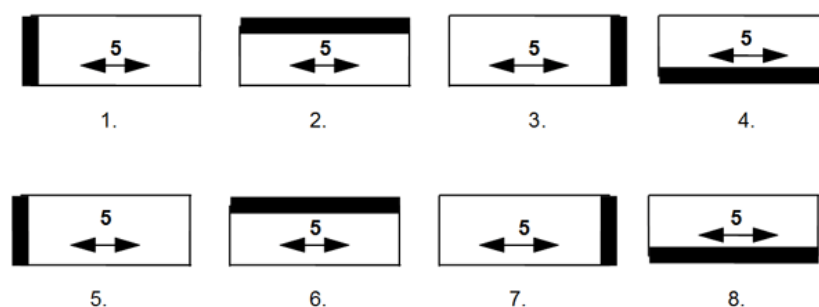


Figure 1: The way of sifting along Penn State Particle Separator

The sieved particles were weighed separately from each sieve using a scale. The results were then recorded, and the obtained masses were summed to determine the total sample mass. After the sampling was completed, the results were processed using Excel. Since 5 samples were taken on each farm, their masses were summed and divided by 5 to obtain the average. The obtained results were then compared with the standards set by Penn State. The standards are shown in Table 2.

Table 2. Standards prescribed by Penn State Particle Separator

Sieve diameter size	Silage from the whole corn plant	Alfalfa hay
19 mm	3-8%	10-20%
8 mm	45-65%	45-75%
1.18 mm	30-40%	20-30%
<1.18 mm	<5%	<5%

In addition to sieving using the Penn State Particle Separator, a survey was conducted with the producers, where we gathered information about their production processes, the feed production methods used on their farms, as well as the diseases that occur in their operations. The survey was in written form, with descriptive-type questions to gather as much information as possible.

RESULTS AND DISCUSSION

All eight of the surveyed farms prepare and use whole-plant corn silage, so eight samples were analyzed. Table 3 shows the proportions of specific particle sizes, with the standards listed in the last column. The values that deviate from the standards are highlighted in bold.

Table 3. Proportion of fractions of silage with specific particle sizes

Farm	1.	2.	3.	4.	5.	6.	7.	8.	Norm
>19 mm	14.73	18.50	10.01	6.83	7.41	10.46	7.80	6.90	3-8 %
8-19 mm	54.38	56.52	56.23	53.21	62.87	63.39	64.63	77.22	45-65 %
1.18-8 mm	30.47	23.36	32.29	37.80	27.89	24.03	25.16	14.45	30-40 %
<1.18 mm	0.42	1.62	1.47	2.16	1.82	2.12	2.40	1.43	<5 %

As shown in Table 3, the results indicate significant variations in compliance with the standards set by Penn State regarding the particle size of silage on the farms. Only one farm (12.5%) fully meets all the requirements related to particle size according to the Penn State standards, which suggests that most farms do not fully follow the recommended guidelines for proper silage preparation. These standards are crucial, as the correct particle size in silage can affect digestion efficiency, cows' eating behavior, and, of course, the overall milk quality, including the composition of milk fat. Interestingly, four samples (50%) show deviations on only one sieve,

which suggests that most farms face minor issues in achieving the optimal particle size but still exhibit some level of non-compliance with the ideal parameters. This could be the result of various factors, such as variations in silage preparation techniques, the types of plant materials used, or the equipment utilized on the farms. However, three samples (37.5%) come from farms that do not meet the standards for two particle fractions, indicating more significant problems in silage preparation. This may be a sign that these farms are experiencing issues with grinding equipment, storage, or other aspects of the production process that result in improper distribution of particle sizes in the silage. The deviations that mostly occur on the first and last sieves also present an interesting pattern. On the first sieve, which retains the largest particles, there may be non-compliance in the grinding process, meaning the silage is not finely enough chopped to allow for efficient digestion in cows. On the last sieve, which retains the smallest particles, there may be issues with over-chopping or incorrect determination of the optimal size, which can affect rumen fermentation rate and reduce the quality of nutrients. In both cases, this can have long-term consequences for the health and productivity of the cows.

In addition to whole-plant corn silage, two out of the eight farms also produce chopped alfalfa haylage. Table 4 shows the proportions of specific particle sizes, with the standards listed in the last column. The values that deviate from the standards are highlighted in bold.

Table 4. Proportion of fractions of haylage with specific particle sizes

Farm	7.	8.	Norm
>19 mm	40.15	46.15	10-20 %
8-19 mm	39.26	35.46	45-75 %
1.18-8 mm	18.67	16.27	20-30 %
<1.18 mm	1.92	2.12	<5 %

As can be seen from Table 4, neither of the two farms meets the standards prescribed by Penn State. On both farms, the only fractions that comply with the prescribed standards are those that passed through the last sieve (<1.18 mm). This indicates that while some fine particles are within the required size range, the larger fractions on both farms do not meet the recommended specifications. The failure to meet the standards for the larger fractions may suggest problems with the equipment used for chopping or with the processing technique. Such deviations could have significant implications for the quality of silage, as the size distribution of the particles directly affects the digestibility and fermentation processes in the rumen.

On all the farms, a survey was conducted with the producers, from whom we gathered information about the amount of feed in the rations. Information about the proportion of feed in the rations is provided in Table 5.

Table 5. Feeds and their quantities used in the cows' rations

Nutritious	1.	2.	3.	4.	5.	6.	7.	8.	Average
Concentrate	3.5	5	6	8.5	8	4	7	9	100%
Silage	10	10	6	20	18	7	22	20	100%
Alfalfa haylage	/	/	/	2	/	/	2	2	37.5%
Rye haylage	/	/	/	2	5	/	3	/	37.5%
Sugar beet pulp	/	/	/	3	/	/	/	/	12.5%
Alfalfa hay	3	10	6	5	5	9	2	1.7	100%
Meadow hay	4	/	4	/	/	/	/	/	25%
Brewers' grain	/	/	/	2	/	/	4	/	25%
Do they all get the same?	YES	YES	NO	NO	NO	YES	NO	NO	YES (37.5%) NO (62.5%)

From Table 5, we can conclude that the three main feeds used on all the farms are concentrate, whole-plant corn silage, and alfalfa hay. These feeds form the foundation of the cows' diet, providing essential nutrients for milk production and overall animal health. Interestingly, only one farm includes sugar beet pulp in the cows' rations, which is often added for energy. Other feeds used on the farms include alfalfa and rye haylage (on three farms), which is a good source of fermentable fiber, and meadow hay (used on two farms), which provides roughage and essential minerals. Additionally, brewers' grains are used on two farms, contributing protein and energy, especially for lactating cows. The diversity in feed types across farms suggests a tailored approach to meet the specific nutritional needs of the cows, although the most common ingredients are still focused on ensuring a balanced diet rich in fiber and energy for optimal milk production.

CONCLUSION

Based on the analysis of the results, several important conclusions can be drawn regarding the factors influencing milk fat content in cows' diets. While silage particle size could appear to be a potential factor, the data indicates that it is not causally related to low milk fat content. Instead, the key factors affecting milk fat

levels are the ratio of concentrates to bulky feeds (such as hay) in the diet. On farms where the diet includes a higher proportion of concentrates and a lower amount of hay, milk fat content is lower. This suggests that a high proportion of concentrates, which are often rich in starch, can reduce the fat content in milk. On the other hand, farms that use a higher proportion of fibrous feeds, especially hay, tend to have higher milk fat content. Bulky feeds contain more fiber, which can positively affect digestion and metabolic processes, leading to a higher fat percentage in milk.

For producers who want to increase milk fat content, it is recommended to reduce the amount of concentrates in the diet and increase the proportion of bulky feeds. This could positively influence the quality of milk fat, thereby improving the efficiency of milk production with higher fat content.

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INFLUENCE OF BREED ON MILK COAGULATION PROPERTIES

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Milk coagulation is a critical process in cheese making, as it directly affects the texture and quality of the cheese. This study aimed to determine the influence of breed on milk coagulation properties. The present research included 138 individual milk samples from 3 breeds. The research showed that there are no statistically significant differences among the breeds. However, the results indicate that the coagulum firmness of the Brown Swiss breed is better than that of the Holstein Friesian and Simmental breeds. Still, to confirm these differences more samples must be analyzed.

Key words: milk, coagulation, breed

INTRODUCTION

In Serbia, around 1.4 billion liters of cow's milk are produced annually. A significant amount of milk is used for cheese production. According to the Statistical Office of the Republic of Serbia (2023) data, approximately 70 different types of cheese are produced in the dairy industries in Serbia each year, totaling around 38,000 tons. Cheese yield (% CY), expressed as the percentage ratio between the cheese produced and the milk processed, is of global economic importance, while daily cheese yield (dCY), expressed in kilograms of cheese produced daily per cow, is the final direct or indirect production target of many dairy farmers (Stocco et al., 2017). Milk coagulation property (MCP) is an important characteristic and a basic requirement of milk for cheese production (Summer et al., 2002). They can be measured with direct rheological methods measuring differences in viscosity, such as computerized renneting meters, or indirect optical methods using different spectroscopic approaches (Bittante et al., 2012). The key milk properties for cheesemaking are good reactivity with rennet, a high degree of curd firming capacity, and good syneresis ability and whey expulsion (De Marchi et al., 2007). Traditionally, MCP are expressed with the following parameters: rennet coagulation time (RCT), time to curd firmness of 20 mm (k20) and curd firmness at 30 min after enzyme addition (a30) (Bittante et al., 2012). These parameters describe the milk coagulation process: the enzymatic or primary phase after the chymosin effect, the nonenzymatic or secondary phase after the casein aggregation, and the last phase after the gel structure formation (De Marchi et al., 2007). In addition to genetic factors, several other elements influence MCP, including casein content (Van Hooydonk and Walstra, 1987; Caron et al., 1997), variants of κ -casein (Mariani et al., 1976; Okigbo et al., 1985; Davoli et al., 1990; Ikonen et al., 1997; Tyrisev  et al., 2003, 2004), β -LG genotypes (K ubarsepp et al., 2005), titratable acidity (Summer et al., 2002), pH levels (Lindstr m et al., 1984; Ikonen et al., 2004), and calcium content (Tervala et al., 1985; K ubarsepp et al., 2005). Milk proteins play critical roles in milk coagulation, although the detailed reason remains yet unknown (Zhang et al., 2023). The total concentration of milk protein in bovine milk in total population in Vojvodina given by Main breeding organization (2024) was about 3.43% for Brown Swiss (BS), 3.27% for Holstein- Friesian (HF) breed and 3.31% for Simmental (SIM) breed in 2023. The milk protein in bovine milk can be mainly classified as caseins (~80%), whey proteins (~16%), and milk fat globule membrane proteins (1–4%) (D'Alessandro et al., 2011). The important roles of caseins in milk coagulation have been widely studied. For example, lower relative kappa casein (κ -CN) content in poor coagulation and non-coagulation milk has been related to the negative correlation between κ -CN content and casein micelle size (Frederiksen et al., 2011). Generally, milk coagulation starts with the hydrolysis of κ -CN by chymosin, the key enzyme in rennet (Amalfitano et al., 2019). When the hydrolyzation degree of κ -CN reaches ~80%, the formed paracasein micelles start to aggregate in the presence of ionic calcium to form a coagulum, in which the fat globules are trapped (Guinee, 2003). Subsequently, the increasing number of secondary interactions within the curd result in its contraction and partial expulsion of whey over time (Zhang et al. 2023). Optimal values for RCT for milk intended for cheese production typically range between 10-18 minutes, as shorter coagulation times enable a faster and more efficient cheese-making process. Ikonen et al. (2004) found longer coagulation times (15-25 minutes) in HF cows compared to other

breeds, due to the genetic characteristics of this breed. The coagulation time for BS cows usually ranges between 10-18 minutes, making them more suitable for cheese production. De Marchi et al. (2008) indicated that SIM has better coagulation characteristics (12-20 minutes) than HF but slightly worse than BS. On the other side, optimal values for k20 in commercial cheese production range between 1.5-3 minutes, indicating a rapid formation and firming of the curd. These values facilitate efficient cheese production by enabling quick firming of the milk mass. Wedholm et al. (2006) concluded that Holstein has values around 2-4 minutes for k20, which is higher compared to other breeds. The optimal k20 values are similar for BS (1.5-3 minutes) and for the SIM breed (1.8-3 minutes) (De Marchi et al., 2008). The value that De Marchi et al. (2007) cite as optimal for curd firmness (a30) after 30 minutes is usually between 25-35 mm, as these values indicate a firm curd suitable for further processing in the cheese industry. The a30 values for Holstein are typically between 15-25 mm, which indicates a softer curd and may affect cheese quality (Wedholm et al., 2006). In contrast, values ranging from 20-35 mm in the BS breed indicate a firmer curd (De Marchi et al., 2008), as do values between 20-30 mm for the SIM, making them suitable for the cheese industry. Different cow breeds have proven suitable for the production of various dairy products due to variations in the chemical composition of their milk, which directly affects processing properties and the quality of the final product. Holstein milk is characterized by a lower fat and protein content compared to other breeds (Ikonen et al., 2004), but it has a higher overall milk yield. Because of these characteristics, Holstein milk is most suitable for producing liquid dairy products, such as pasteurized milk and yogurt (De Marchi et al., 2008), where high yield and stable production play an important role. Lower values for curd firmness (a30) and a slower coagulation process (RCT) often reduce efficiency in the production of hard cheeses, although the milk can be used for softer cheeses or fermented products where curd firmness and coagulation time have less impact on quality (Wedholm et al., 2006). Brown Swiss milk is rich in proteins and has a high casein content, which contributes to better coagulation characteristics. BS milk has optimal values for RCT and k20, as well as curd firmness (a30) that falls within a range suitable for cheese production (De Marchi et al., 2008). This breed has proven to be ideal for the production of hard and semi-hard cheeses, where stronger coagulation characteristics of the milk are required. The high protein content further contributes to a greater cheese yield, making BS milk valuable for cheese production. Simmental milk balances between Holstein and Brown Swiss in terms of fat and protein content, as well as coagulation properties. Simmental milk exhibits sufficiently high protein and fat content, making it suitable for products such as butter and cream, where a high fat content is desirable (De Marchi et al., 2008).

The aim of this study was to determine the influence of breed on milk coagulation properties.

MATERIALS AND METHODS

The study included 3 dairy farms located in Slovenia with a total of 64 Holstein cows (29 from Bovec and 35 from Cerklje na Gorenjskem), 39 Brown Swiss cows from Bovec and 35 Simmental cows from Ljubljana. Individual milk samples were collected between July and August. Individual milk samples were analysed using Milkoscan 6000 FT, according to the standard method ISO 9622/IDF 141 (ISO/IDF, 2013) and for somatic cell count (SCC) using Fossomatic 5000 according to ISO 13366-2/IDF 148-2 (ISO/IDF, 2006). Analysis of milk quality traits were performed in the accredited milk laboratory of the Institute of Dairy Science and Probiotics at Biotechnical faculty in Slovenia. Milk coagulation properties were evaluated using a computerized renneting meter (CRM) (Polo Trade, Monselice, Italy). The analysis involved measuring the rennet coagulation time (RCT), which is the interval from the addition of rennet to the start of milk gelation; the curd-firming rate (k20), defined as the time from the beginning of gelation until the curve's width reaches 20 mm; and curd firmness at 30 minutes (a30), represented by the width of the curve at a specific time after rennet was added. First, 10 mL aliquots of milk were heated to 35 °C in an aluminum block and then 200 µl of enzyme solution (NATUREN® Premium 225 (Chr. Hansen, Denmark)), diluted to 1.2% (w/v) in distilled water to achieve a final concentration of 0.0516 international milk clotting units (IMCU) per ml of milk, was added. The samples were mixed for 30 seconds and subjected to 30-minute analysis. MCP were analysed using a PROC GLM procedure in SAS version 9.4 (SAS Institute Inc., Cary, NC). Correlations were calculated using PROC CORR.

RESULTS AND DISCUSSION

The average results for milk fat, protein, lactose, solids- non- fat (SNF), urea, freezing point and somatic cell count (SCC) are presented in Table 1.

Table 1. Average milk composition of different breeds

Breed	Milk fat (%)	Protein (%)	Lactose (%)	SNF (%)	Milk urea (mg/ml)	SCC (*1000/ml)
HF	3.83 ^a	3.29 ^b	4.46 ^a	8.48 ^a	27.58 ^a	445.38 ^a
SIM	4.15 ^a	3.44 ^{ab}	4.18 ^a	8.29 ^a	12.91 ^b	206.54 ^a
BS	3.95 ^a	3.65 ^a	4.43 ^a	8.8 ^a	33.78 ^c	492.9 ^a
Mean	3.98	3.46	4.36	8.52	24.76	381.6

- SNF – solid non-fat; SCC - somatic cell count
- Values in the same column with the same lower- case letters do not differ statistically significantly ($p>0.05$)
- Between arithmetic means in the same column, with different labels (a, b, c, d), there is a statistically significant difference ($p<0.05$)

The observed differences in milk components such as fat, lactose, solids-not-fat (SNF) and somatic cell count (SCC) are not statistically significant, in contrast to milk urea and proteins which are statistically significant. Mean values, determined in this study, for milk fat (3.98%) was lower and protein contents (3.46%) was approximately the same as average values for total amount of cows milk collected in Slovenia from January to May, 2023 (milk fat 4.25%, protein 3.45%) given by the Statistical Office of the Republic of Slovenia (SURs). Furthermore, the protein content varied among breeds, with Brown Swiss cows exhibiting the highest protein percentage (3.65%). This finding aligns with previous studies indicating that Brown Swiss cows are often selected for higher protein yields, making their milk more suitable for dairy products requiring substantial protein content (De Marchi et al., 2008).

Although differences in MCP parameters were observed among breeds, these differences were not statistically significant ($p>0.05$). This could be due to the limited sample size or other variables influencing MCP, such as feeding conditions or individual variations among cows. The average results for RCT, k20 and a30 are presented in Table 2.

Table 2. Coagulation parameters of different breeds

Parameter	HF	SIM	BS	Mean
RCT (min)	14.97 ^a	16.10 ^a	13.48 ^a	14.85
k20 (min)	2.54 ^a	2.97 ^a	2.80 ^a	2.77
a30 (mm)	34.67 ^a	37.51 ^a	40.67 ^a	37.62

- Between arithmetic means in the same row, with the same labels (a) there is no statistically significant difference ($p>0.05$)

The rennet coagulation time (RCT) varied, with BS cows showing the shortest average time (13.48 minutes), while Simmental cows had the longest (16.10 minutes). This difference suggests that the coagulation properties of milk can vary significantly by breed, which may influence the cheese-making process. Notably, Holstein cows exhibited the fastest curd-firming rate (k20 at 2.54 minutes) and the highest curd firmness at 30 minutes exhibited BS breed (a30 at 40.67 mm), indicating their milk may be particularly suited for cheese production. Mean values, determined in this study, for RCT (HF: 14.97; SIM: 16.10; BS: 13.48) were lower than reported by De Marchi et al. (2007) (HF: 18.0; SIM: 16.2; BS: 16.1).

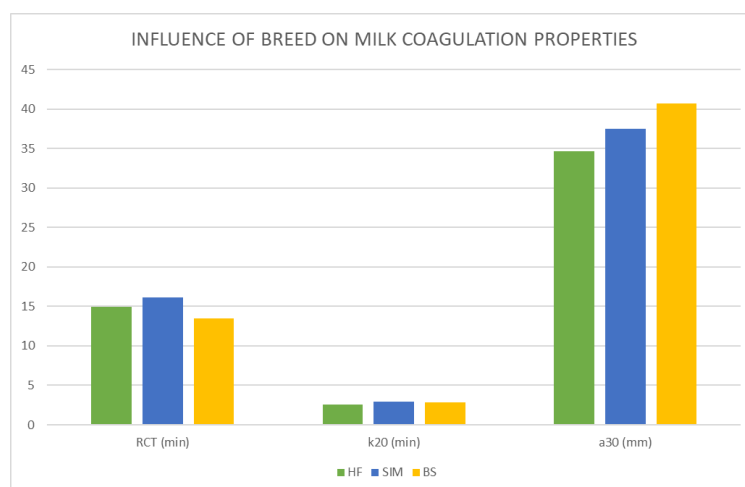


Fig.1 Influence of breed on milk coagulation properties

Table 3 shows the correlations between the average values of coagulation parameters and milk ingredients.

Table 3. Correlations coagulation parameters and milk ingredients.

	Proteins	Milk fat	Lactose	SNF	Urea	SCC
RCT	-0.022545	-0.001648	-0.066091	-0.052132	-0.2667*	0.057329
k20	0.048044	0.1752*	0.107272	0.085379	-0.061079	-0.002852
r30	0.009485	0.105123	0.052101	0.035537	0.067992	-0.084712

- SNF - solid non- fat; SCC - somatic cell count,

* Significant difference $p < 0.05$

Based on the results shown in Table 3, RCT showed a significant negative correlation with urea ($r = -0.2667$), suggesting that higher urea levels might be associated with shorter coagulation times. Conversely, k20 demonstrated a significant positive correlation with milk fat ($r = 0.1752$), indicating that higher fat content could enhance curd-firming rates. These correlations suggest that the milk's compositional attributes and coagulation properties are interrelated, highlighting the complexity of milk quality assessment.

These results may have practical implications for herd selection depending on the final product, such as cheese. Breeds like Brown Swiss, which demonstrate better MCP, could be more beneficial for producing cheese with specific characteristics.

CONCLUSION

From the results of this study, it can be concluded that no statistically significant differences were found between the breeds, observing the parameters of milk quality and coagulation properties. However, the results indicate that the coagulum firmness of the Brown Swiss breed is better than that of the Holstein Friesian and Simmental breeds, but more samples need to be analyzed to confirm these differences. The variations in milk composition and coagulation parameters among the three breeds highlight the importance of breed selection based on specific dairy production goals. Breeders and dairy farmers may consider these differences when formulating feeding strategies and selecting breeds that align with their production objectives, whether that be higher fat, protein, or overall milk yield.

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PARASITOLOGICAL STUDY OF THE RED FOX (*VULPES VULPES*)

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The red fox (*Vulpes vulpes*), a widespread predator in Poland, shows significant adaptability across various environments, including forests, fields, and urban areas. This adaptability has driven a recent population increase, with red foxes frequently found near human residences, often attracted by easily accessible food sources such as garbage bins. Additionally, some individuals feed foxes directly, which has led to a growing presence of these animals in close proximity to humans. Given this increased population and close interaction with human environments, it is crucial to monitor the potential health risks posed by zoonotic parasites transmitted by foxes. This study aims to assess the prevalence of endoparasites in red foxes within the Lublin Province of Poland. The study examined nine red foxes (two females and seven males) weighing between 6.3 and 8.4 kg, sourced from road accidents and hunting activities in the Lublin region. To mitigate epidemiological risks, the foxes were frozen for two weeks at -80°C prior to necropsy. During the examination, thorough inspections were conducted on the thoracic and abdominal cavities, focusing specifically on the lungs, heart, kidneys, liver, and gallbladder. Mucosal scrapings were collected from the gastrointestinal tract, and the contents of the stomach, small intestine, and large intestine were analyzed using flotation and decantation techniques. The anal glands of the foxes were also examined. Results indicated that all examined foxes (100%) were infected with at least one parasite. In the lungs, *Crenosoma vulpis* and *Capillaria aerophila* were each found in 22.2% of individuals. In the small intestine, the parasites detected included *Mesocostoides lineatus* (55.6%), *Taenia pisiformis* (33.3%), *Toxocara canis* (44.4%), *Echinococcus multilocularis* (11.1%), and *Alaria alata* (33.3%). In the large intestine, eggs of *Trichuris vulpis* (11.1%) and *Uncinaria stenocephala* (22.2%) were found. Additionally, fecal analysis revealed eggs of *Capillaria aerophila*, *Toxocara canis*, and *Uncinaria stenocephala*. In the anal glands, 22.2% of the foxes showed the presence of *Aonchoteca paranalisis*, and *Capillaria plica* was detected in the urinary bladder of 55.6% of individuals. The most frequently detected parasites were *Capillaria aerophila* (55.6%), *Mesocostoides lineatus* (55.6%), and *Capillaria plica* (55.6%). This study highlights the diverse range of endoparasites present in red foxes within the Lublin Province, with a particular emphasis on the detection of *Echinococcus multilocularis*, a parasite that poses a significant zoonotic risk. The presence of this parasite indicates a considerable potential health risk for humans, especially as the red fox population continues to grow across Poland. Systematic monitoring of pathogens within fox populations is necessary, alongside public education initiatives to raise awareness of the risks associated with human-fox cohabitation in shared environments.

Key words: Red fox (*Vulpes vulpes*), endoparasites, zoonotic risk, *Echinococcus multilocularis*

INTRODUCTION

The red fox (*Vulpes vulpes*), a common predator in Poland from the canid family, is widely distributed across the country. While its primary diet consists of small rodents, it also preys on hares and birds, and its proximity to humans is often reinforced by occasional feeding, which encourages foxes to approach human settlements. The red fox is highly adaptable, thriving in a range of environments from forests and fields to urban areas. Its ability to utilize alternative food sources, such as food scraps, pet food, and animal carcasses, enables it to survive and reproduce even under challenging environmental conditions. (Górski et al. 2004) In recent years, there has been a noticeable increase in fox populations in Poland, especially in urban areas. The population has nearly quadrupled—from 56,000 in 1990 to over 210,000 today—partly due to the anti-rabies vaccination program implemented across the country (Świątalska, 2018). In the Lublin region (Figure 1.), foxes are frequently spotted in city streets and near garbage bins, searching for food and occasionally interacting with domestic animals like dogs and cats. This migration toward human settlements raises the risk of contact between humans and foxes, which are hosts to numerous endoparasites, including zoonotic species that can cause diseases in humans. One significant zoonotic threat is *Echinococcus multilocularis*, which causes alveolar echinococcosis—a dangerous parasitic disease where larvae form cysts in the liver, damaging internal organs.

Due to the difficulty of treating this condition, which often requires surgery, it poses serious public health concern. (Karamon et al., 2018) Therefore, research on the parasitic fauna of foxes is crucial for assessing epidemiological risks and developing protective measures for the health of humans and domestic animals. The aim of our study was to determine the prevalence of endoparasite infections in red foxes in the Lublin region, with a particular focus on species that are directly hazardous to human health.

MATERIAL AND METHODS

In 2023-2024, necropsies were performed on 9 foxes from the Lublin region of Poland (Figure 1). These animals were victims of traffic accidents or were acquired by hunters in the area. Among the examined individuals were 2 females and 7 males, with body weights ranging from 6.3 to 8.4 kg. The fox carcasses were frozen for two weeks at -80°C prior to dissection to eliminate potential epidemiological threats, including the transmission of infectious diseases that could pose risks to humans or other animals. During the parasitological examination, the thoracic and abdominal cavities of the foxes were carefully examined, with particular attention given to the internal organs such as the lungs, heart, kidneys, liver, and gallbladder. The dissection focused on detecting any pathological changes in these organs that could suggest parasitic infections. What's more, the perianal glands were excised, then dissected and examined under a magnifying glass to identify adult specimens, while the washings from these glands allowed for the detection of eggs of perianal parasites. Additionally, a thorough analysis of the digestive tract was conducted, including the stomach contents, small intestine, large intestine, and cecum. Each sample was first examined macroscopically for adult parasites, followed by the application of various parasitological methods—such as decantation, flotation, and scraping—to identify and study eggs or worms of parasites in the animals' digestive systems. The decantation method, as outlined by Żarnowski and Josztowa, was used to separate parasite eggs from other components of the fecal matter. A 4-gram sample of feces was placed in a beaker with a strainer submerged in water. After homogenizing the sample, the heaviest elements, such as parasite eggs, settled at the bottom of the beaker. After 10 minutes, the liquid above the sediment was poured off, and this process was repeated three times. The collected sediment was transferred to a watch glass or Petri dish for examination and identification of parasites under a binocular magnifying glass. Concurrently, flotation was performed, a method that relies on differences in density between parasite eggs and other fecal components. The 2-gram fecal sample was mixed with an appropriate flotation solution, such as concentrated NaCl, and passed through a strainer into an Erlenmeyer flask. After transferring to the flask, a cover slip was placed on the surface of the solution, and the sample was left to sit for 15-20 minutes to allow the parasite eggs to float to the top. The cover slip was then carefully transferred to a microscope slide for examination under the microscope. This method allowed for the identification of coccidia oocysts, nematode eggs, and certain tapeworm eggs. Additionally, to investigate other potential sources of parasites, rinsings were collected from the urinary bladder, gallbladder, and trachea, and these were also subjected to microscopic examination. Scraping of the mucous membrane from the small and large intestines provided additional material for analysis, aiding in the detection of potential cysts or larvae of parasites residing in these parts of the digestive tract. Parasites were identified based on morphological features such as dimensions, shape and structural features, using the keys described by Khalil (Khalil et al., 1994) and Anderson (Anderson et al., 2009).



Fig. 1 Location of the study area

RESULTS AND DISCUSSION

Parasitological studies conducted on 9 foxes from the Lublin region revealed the presence of parasites from two main groups: flatworms (*Platyhelminthes*) and roundworms (*Nematoda*). These parasites were found in various parts of the animals' bodies, including the respiratory system, digestive tract, perianal glands, and urinary bladder. Parasites were detected in all of the examined foxes, representing 100% of the population studied.

Table 1. Number of adult parasite specimens found in the examined foxes.

Parasite	Number of occurrences
<i>Crenosoma vulpis</i>	3
<i>Capillaria aerophila</i>	5
<i>Mesocostoides lineatus</i>	6
<i>Toxocara canis</i>	5
<i>Taenia pisiformis</i>	3
<i>Alaria alata</i>	3
<i>Trichuris vulpis</i>	1
<i>Uncinaria stenocephala</i>	2
<i>Echinococcus multilocularis</i>	1
<i>Aonchoteca paranalisis</i>	2
<i>Capillaria plica</i>	6

In the respiratory system, particularly in the lungs, *Crenosoma vulpis* (22.2%) and *Capillaria aerophila* (22.2%) were identified. *Crenosoma vulpis* is a parasite whose presence in the lungs can lead to chronic respiratory issues, particularly with prolonged infections. *Capillaria aerophila*, also found in the lungs, can cause inflammation and other respiratory disorders, leading to reduced gas exchange efficiency. Parasites were also found in various parts of the digestive tract. In the small intestine, *Mesocostoides lineatus* (55.6%), a tapeworm, was the most commonly identified parasite. The adult form of this parasite can cause damage to the intestinal wall and digestive disturbances. *Taenia pisiformis* (33.3%) was present both as adult individuals and eggs, and its presence can lead to severe intestinal infections. *Toxocara canis* (44.4%) and *Alaria alata* (33.3%) were detected in both adult and egg forms. Their presence may lead to general infection symptoms, including diarrhea and abdominal pain. In one case, the presence of adult *Echinococcus multilocularis* (Figure 2) was also identified, which poses a significant zoonotic threat. *Echinococcus multilocularis* is responsible for alveococcosis, a potentially fatal condition for humans. Although only one of the 9 foxes tested positive, the detection of *Echinococcus multilocularis* in 11.1% of the samples is significant, as it poses a potential risk to both human and animal health, particularly in rural areas where contact with wildlife is frequent. In the large intestine, eggs of *Trichuris vulpis* (11.1%) and *Uncinaria stenocephala* (22.2%) were found. Both *Trichuris*

vulpis and *Uncinaria stenocephala* are hematophagous parasites that feed on blood, which can lead to serious health consequences in infected animals, such as anemia and damage to the intestinal tissues.

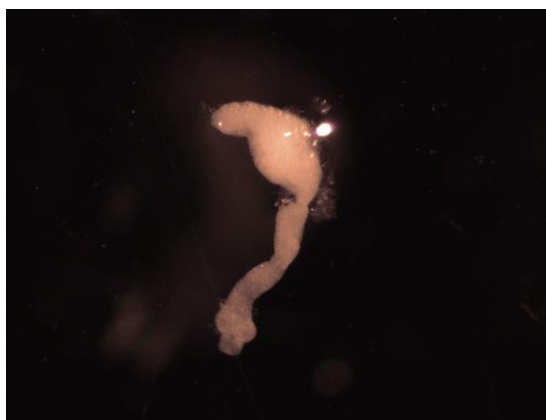


Fig. 2 *Echinococcus multilocularis*

Additionally, in the perianal glands of 22.2% of the examined foxes, eggs of *Aonchoteca paranalis* (Figure 3) were detected, and in the urinary bladder, *Capillaria plica* was found in 55.6% of the animals. *Capillaria plica* infects the urinary system, causing inflammation of the urinary bladder and other urinary issues. Its presence can lead to damage to urinary structures, resulting in problems with urination and general disturbances in the urinary system. Eggs of *Capillaria aerophila*, *Toxocara canis*, and *Uncinaria stenocephala* were detected in the feces of all 9 examined foxes, confirming the patent invasion of these parasites. These parasites can cause health issues related to intestinal infections, abdominal pain, and other symptoms of chronic parasitic infections. In total, the most commonly detected parasites in the examined foxes were *Capillaria aerophila*, *Mesocestoides lineatus*, and *Capillaria plica*, which were identified in 55.6% of the animals.



Fig. 3 *Aonchoteca paranalis*

CONCLUSION

In total, the most commonly detected parasites in the examined foxes were *Capillaria aerophila*, *Mesocestoides lineatus*, and *Capillaria plica*, identified in 55.6% of the animals. Although *Echinococcus multilocularis* was detected in only one of the nine foxes (11.1%), its presence represents a significant zoonotic threat. Alveococcosis caused by this parasite is extremely dangerous, particularly for humans, as it can lead to liver proliferation through cytoplasmic extensions and metastasis to other organs, often with fatal outcomes (Pilarczyk et al., 2019). Therefore, despite its low occurrence in the sample, the presence of *Echinococcus multilocularis* in this group of animals underscores the importance of monitoring and research to prevent its spread within wildlife populations and to protect human and domestic animal health. All the detected parasites pose a serious health risk to both wild animals and humans. Therefore, further studies are necessary to gain a more thorough understanding of the risks associated with these parasites and to develop effective methods for preventing their spread. Public education on parasitology and zoonotic threats is also crucial to raise awareness about the dangers of contact with wildlife and their parasites. It is important to note that the population of the red fox in Poland, particularly in the Lublin Voivodeship, is growing, with an increasing number of foxes inhabiting urban areas or living in close proximity to human settlements. The rising number of these animals in areas where humans have more frequent contact with wildlife increases the risk of spreading zoonotic diseases and parasites.

Therefore, monitoring fox populations and educating the public about the risks associated with these animals and their parasites is essential.

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CONSUMER BEHAVIOR IN GAME MEAT CONSUMPTION IN CROATIA

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Given the growing awareness of health and environmental issues, game meat offers high nutritional value, yet its consumption remains low. A survey conducted with 603 respondents revealed that consumers have positive attitudes toward game meat but face barriers such as limited availability and seasonal variations. Subjective norms, such as the influence of family and friends, also play a significant role. The results suggest the need for better consumer education about the benefits of game meat to encourage higher consumption.

Keywords: game meat, consumer behavior, consumption

INTRODUCTION

Consumer eating habits have significantly changed, and health is recognized as a very important motivator in food choices (Sorić et al., 2021). Although there is a push to increase the consumption of plant-based foods, consumers (especially in Eastern and Central Europe, where meat is frequently consumed) find it difficult to change their eating habits. The transition toward reduced consumption of animal-based products and an increase in plant-based food consumption is slow (Denver et al., 2023). Additionally, the main barriers to reducing meat consumption include a lack of knowledge about the link between animal production and climate change, as well as social influences (Whittall et al., 2023; Mylan, 2018; Macdiarmid et al., 2016; O'Keefe et al., 2016).

Game meat is considered a prestigious food, and the game meat market still represents a niche market (Bodnar & Szel, 2014). Furthermore, game meat is an excellent source of protein, with a favorable n-6/n-3 fatty acid ratio. The amounts of essential amino acids, vitamins (B1, B2, B6, E) (Soriano et al., 2021; Marescotti et al., 2019), and micro and macroelements (phosphorus, magnesium, iron, manganese, and zinc) (Karwowska et al., 2021; Marescotti et al., 2019) are significantly higher in game meat compared to other animal meats.

Despite these advantages, the consumption of game meat in Europe remains much lower compared to other types of meat. According to data from the Croatian Bureau of Statistics (CBS, 2019), the consumption of meat in the *other* category (including horsemeat and game) per household member in 2019 was 0.2 kg, which represents a decline from 2003 when it was 0.55 kg (Tolušić et al., 2006). The consumption of game meat in European countries ranges from 0.2 to 1.1 kg per capita annually (Czarniecka-Skubina et al., 2022).

Food scandals and various diseases also affect consumer behavior in food choices. Therefore, the aim of this paper is to investigate Croatian consumers' behavior regarding game meat consumption and determine whether attitudes toward game meat will influence the intention to consume it.

MATERIALS AND METHODS

In this research, an online survey was conducted to analyze consumer behavior related to game meat. The survey covered a convenience sample of 603 respondents in Croatia, conducted between January 23 and February 17, 2024. The survey was created using Google Forms, and the link was distributed via social networks (Facebook and LinkedIn), as well as through email to selected participants.

The questionnaire included the following categories of questions: sociodemographic characteristics (sex, age, education, place of residence, household monthly income, employment status, eating habits). Attitudes toward game meat were measured using 9 statements adopted from Corradini et al. (2022). To measure subjective norms, 4 statements were used, and to measure the intention to consume game meat, 3 statements were adopted from Tomić et al. (2016). The statements were adapted to the research topic. Perceived behavioral control (PBC) was measured using 4 statements adopted from Seo et al. (2011). To measure barriers to game meat consumption, 3 statements were used, taken from Corradini et al. (2022). Subjective knowledge was measured using 3 statements, adapted from Pieniak et al. (2010). Attitudes toward hunting were measured with 4 statements, formulated based on Ljung et al. (2012). Statements to measure attitudes about African swine fever were created for the purposes of this research. Participants' perceptions of game meat safety were measured with

5 statements, adopted from Corradini et al. (2022). Respondents indicated their level of agreement with each statement on a 5-point scale (1-strongly disagree, 5-strongly agree).

A univariate analysis (frequency and distribution) of the data was conducted using SPSS software, version 27, to describe the sample and examine behavior and intentions related to game meat consumption.

RESULTS AND DISCUSSION

The study included 603 respondents, 52.1% of whom were women and 47.9% men. The age structure ranged from 18 to 66 and above, with the most represented group being between 36 and 45 years old (28.2%). Respondents with a high level of education made up 70.4% of the sample, while 62.2% lived in urban areas. The monthly income of most respondents ranged between 1,500 and 2,000 euros. The dominant eating habit among respondents was omnivory (98.3%), and 66.7% of respondents were involved in hunting. – Table 1.

Table 1 – Sample Description

		N	%
Sex	Male	289	47,9
	Female	314	52,1
Age	18-25	121	20,1
	26-35	165	27,4
	36-45	170	28,2
	46-55	105	17,4
	56-65	35	5,8
	66 and above	7	1,2
		6	1,0
Education	Primary school	6	1,0
	Secondary vocational education	172	28,5
	Higher vocational degree	62	10,3
	University degree	260	43,1
	Postgraduate specialist study	22	3,6
Place of residence	Postgraduate doctoral study	81	13,4
	City	375	62,2
	Countryside	228	37,8
Monthly household income	Up to 1.000 eur	52	8,6
	1.001-1.500 eur	115	19,1
	1.501-2.000 eur	128	21,2
	2.001-2.500 eur	83	13,8
	2.501-3.000 eur	94	15,6
	3.001-3.500 eur	55	9,1
	More than 3,500 eur	76	12,6
Employment status	Student	96	15,9
	Employed	464	76,9
	Unemployed	26	4,3
	Retiree	17	2,8
Eating habits	Omnivore	593	98,3
	Vegetarian	4	0,7
	Vegan	2	0,3
	Pescetarian	4	0,7
Do you hunt?	Yes	201	33,3
	Not	402	66,7

Table 2. Attitudes towards game meat

Statements	Arithmetic mean	Standard deviation
I have positive attitudes towards game meat.	4,21	1,05
Game meat is rich in protein.	4,18	0,91

I like to eat game meat.	4,00	1,22
I like the taste of game meat.	3,98	1,20
Game meat contains small amounts of fat.	3,88	1,00
Game meat has a favorable content of essential amino acids.	3,79	0,97
Game meat is rich in vitamins.	3,72	0,98
Game meat has a positive impact on my health.	3,72	1,11
I like the smell of game meat.	3,67	1,22

Table 3. Subjective norm

	Statements	Arithmetic mean	Standard deviation
Subjective norm	My family members eat game meat.	3,70	1,28
	My friends eat game meat.	3,65	1,11
	My family members think that I should eat game meat.	3,08	1,37
	My friends think I should eat game meat.	3,05	1,27

Table 4. Perceived control of behavior

	Statements	Arithmetic mean	Standard deviation
Perceived control of behavior	It is only up to me whether I will consume game meat.	4,36	0,93
	If I wanted to, I could consume game meat.	4,25	0,96
	Nothing prevents me from consuming game meat.	4,10	1,06
	I expect to be able to consume game meat.	3,96	1,18

Table 5. Intention to consume game meat

	Statements	Arithmetic mean	Standard deviation
Intention	I intend to consume game meat in the next three months.	3,65	1,35
	I'll probably be eating game meat in the next three months.	3,60	1,38
	I have a great desire to eat game meat in the next three months.	3,33	1,43

Table 6. Perceived barriers in the consumption of game meat

	Statements	Arithmetic mean	Standard deviation
Barriers	The limited supply of game meat makes it impossible for me to eat game meat more often.	2,75	1,31
	Large seasonal variations in the offer of game meat make it impossible for me to eat game meat more often.	2,64	1,25
	Large variations in the quality of game meat make it impossible for me to eat game meat more often.	2,48	1,17

Table 7. Subjective knowledge about hunting

	Statements	Arithmetic mean	Standard deviation
Subjective knowledge about hunting	Compared to the average person, I know a lot about hunting.	3,13	1,52
	I know a lot about hunting.	2,98	1,52
	People who know me consider me an expert in hunting.	2,25	1,40

Table 8. Attitudes about hunting

	Statements	Arithmetic mean	Standard deviation
Attitudes towards hunting	Hunting helps maintain balance in nature.	3,92	1,19
	I don't see anything wrong with hunting animals for their meat as long as the animal is not endangered.	3,80	1,28
	I have a positive attitude towards hunting.	3,55	1,37
	Hunters are well trained and follow hunting regulations.	3,22	1,14

Table 9. Views on African swine fever

	Statements	Arithmetic mean	Standard deviation.
Attitudes about African swine fever	African swine fever is causing changes in the meat market.	3,93	1,08
	African swine fever threatens domestic and wild swine populations.	3,93	1,12
	African swine fever cannot be transmitted to humans.	3,37	1,27

Respondents have positive attitudes toward game meat, with an average rating of 3.90 – Table 2. The highest agreement is with the statement that game meat is rich in proteins (4.18) and that respondents enjoy eating game meat (4.00).

The average subjective norm value is 3.37 – Table 3, with respondents most agreeing that family members eat game meat (3.70).

Table 4 shows that respondents exhibit a high level of perceived control over game meat consumption (4.17). The highest agreement is with the statement that it depends on them whether they will consume game meat (4.36).

The intention to consume game meat is moderate (3.53), with respondents most likely to consume game meat in the next three months (3.65) – Table 5.

Perceived barriers are low (2.62), with the most prominent barrier being the limited availability of game meat (2.75) – Table 6.

In Table 7, respondents rate their subjective knowledge of hunting as moderate (2.78), with the least agreement on being experts in hunting (2.25).

Attitudes toward hunting are positive, with an average rating of 3.62. Respondents believe that hunting helps maintain balance in nature (3.92) – Table 8.

Respondents show a high level of agreement with statements about African swine fever, with an average rating of 3.74 (Table 9). This disease was first confirmed in Croatia in June 2023. Although research shows that African and classical swine fevers pose no risk to humans, some consumers place greater emphasis on food safety and risk perception, potentially changing their eating habits.

The analysis shows that all constructs (attitudes, subjective norms, perceived behavioral control) are significantly related to the intention to consume game meat, explaining 72.8% of the variance in intention. Attitudes toward game meat have the most positive impact, while attitudes toward African swine fever have a negative impact on the intention to consume game meat.

CONCLUSION

Research on consumer behavior in game meat consumption reveals that, despite the numerous nutritional benefits that game meat offers, its consumption in Croatia remains low. Consumers exhibit positive attitudes toward game meat, with a high level of perceived control over its consumption. However, there are still significant barriers influencing consumer decisions, including limited availability and seasonal supply variations.

Subjective norms also play a key role, with family and friends having a significant influence on respondents' food choices. Although the intention to consume game meat is moderate, consumers are aware of its positive health and nutritional effects, which may encourage increased consumption if existing obstacles are overcome. Additionally, recent food scandals, such as African swine fever, further shape risk perception, which may influence consumer habits. In light of these findings, it is important to develop strategies that will inform consumers about the nutritional benefits of game meat and improve its availability in the market, to encourage greater consumption in the future.

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IMPACT OF THE COVID-19 PANDEMIC ON IVERMECTIN SALES IN SERBIA: A STUDY OF OFF-LABEL INFLUENCES

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Summary: Ivermectin is a drug licensed for use in the veterinary medicine which is an endectocide antiparasitic drug without antibacterial or antifungal properties. The objective of this study was to investigate the differences in sales of ivermectin before, during, and after the COVID-19 pandemic in Serbia since some studies suggested the ability of ivermectin to inhibit virus replication *in vitro*, significantly increasing its off-label use. In Serbia, during COVID-19, the demand for ivermectin reached its peak in 2021, which could be a result of its use in human medicine.

Key words: ivermectin, COVID-19, Off-label

INTRODUCTION

Ivermectin is a macrolide drug (macrocyclic lactone) that belongs to the avermectin group (a group of related substances isolated from the actinomycete *Streptomyces avermitilis*). Unlike macrolide antibiotics, ivermectin does not have antibacterial or antifungal properties, but rather exclusively exerts an antiparasitic effect, functioning as an endectocide antiparasitic drug. Ivermectin is an important drug in veterinary medicine and has a broad antiparasitic spectrum of activity (it is used against nematodes and ectoparasites) in domestic animals (Stojanović and Jezdimirović, 2020; ALIMIS, 2024). Furthermore, ivermectin acts as a GABA (Gamma-aminobutyric acid) receptor agonist, enhancing the release of this inhibitory amino acid and its binding to GABA receptors on nerve cells- this leads to increased activity of ion channels and a greater influx of chloride (Cl⁻) ions from the extracellular space into the cell, which results in hyperpolarization of the cell membrane and paralysis of the parasite (Ottesen and Campbell, 1994). Ivermectin rarely causes effects on the mammalian central nervous system, as it does not cross the blood-brain barrier. In domestic animals, ivermectin is most commonly applied subcutaneously or per os (Stojanović and Jezdimirović, 2020).

In human medicine, ivermectin has been used to treat several parasitic infections, such as human onchocerciasis. Besides, it has microfilaricidal activity against *Wuchereria bancrofti*, *Brugia malayi*, *Loa loa*, and *Mansonella ozardi*. It is also effective against human strongyloidiasis, cutaneous larva migrans, and other nematodes (*Ascaris lumbricoides*, *Trichuris trichiura*, *Enterobius vermicularis*) (Ottesen and Campbell, 1994). On the other hand, ivermectin was used off-label in human medicine during the COVID-19 pandemic due to its potential antiviral effect.

The World Health Organization (WHO) Country Office in China was informed about the first cases of the disease that would eventually be named COVID-19 (Coronavirus Disease 2019) on December 31, 2019. Actually, on December 12, 2019, in Hubei province, China, in the city of Wuhan, cases of a disease characterized by atypical pneumonia that did not respond to any standard treatment were reported for the first time. So, on March 11, 2020, the WHO declared COVID-19 a pandemic while, on May 11, 2021, the COVID-19 pandemic officially ended (Sencer, 2024).

Due to the insufficient results achieved using conventional antiviral drugs during COVID-19 pandemic, some studies that were conducted during 2020 and 2021 suggested that ivermectin, an antiparasitic drug, could potentially have antiviral effects. The use of ivermectin for this purpose is referred to as off-label use. When a drug is approved, indications and dosages are part of the package insert, and any use of the drug outside of the specified situations is called off-label use. In such cases, the manufacturer is not liable for adverse effects resulting from this type of use (Furlan and Caramelli, 2021). Furthermore, Caly et al. (2020) and Chaccour et al. (2020) described that ivermectin inhibits *in vitro* replication of certain single-stranded RNA viruses (Dengue virus, Zika virus, Yellow fever virus) and is a potent inhibitor of SARS-CoV-2 (Severe Acute Respiratory Syndrome Coronavirus 2) replication in *in vitro* studies. However, the concentration of ivermectin which resulted in inhibition of viral replication *in vitro* was significantly higher (about 10 times) than the C_{max}

achievable with the highest single therapeutic dose approved by the FDA (Food and Drug Administration). The same author stated that ivermectin is not a completely safe drug due to its ability to cross the blood-brain barrier in cases of hyperinflammatory conditions. Additionally, the use of ivermectin in combination with certain antiretroviral drugs (lopinavir/ritonavir and darunavir/cobicistat) is contraindicated. Moreover, Abd-Elsalam et al. (2021) conducted a randomized study in 2021 on 164 patients diagnosed with COVID-19, dividing the patients into two groups. The first group received standard treatment lasting 14 days combined with ivermectin (12 mg, once daily for 3 days), while the second group received only standard treatment lasting for 14 days. Patients that received ivermectin recovered more quickly and had shorter length of hospital stay, but this difference was not statistically significant.

Cepelowicz Rajter et al. (2021) performed a study from March 15 to May 11, 2020, in four hospitals in Florida, involving 280 patients with confirmed COVID-19. Out of these, 173 patients were treated with ivermectin, while 107 were not. Both groups also received hydroxychloroquine and azithromycin. The analysis showed lower mortality in the ivermectin-treated group compared to the standard group (15.0% vs. 25.2%), as well as lower mortality in ivermectin-treated patients with severe pulmonary involvement compared to the same group receiving standard therapy (38.8% vs. 80.7%). Besides, Ahmed et al. (2021) conducted a randomized, double-blind, placebo-controlled trial involving 72 COVID-19 patients in Dhaka, Bangladesh. The patients were divided into three groups: one group received oral ivermectin (12 mg, once daily, for 5 days); another group received oral ivermectin in combination with doxycycline; and the third group was a placebo control group. Patients treated with ivermectin showed the earliest virological clearance compared to all three groups, while this study was described as preliminary. Furthermore, a total of 14 studies were published, involving 1,678 participants, aimed at demonstrating the efficacy of ivermectin therapy for COVID-19 compared to therapies with proven efficacy, placebo, and treatment without therapy. Most of the studies were conducted on small samples, and only a few were of high quality. Therefore, Popp et al. (2021) stated that ivermectin is not a drug that can be reliably recommended for the treatment of COVID patients.

Despite numerous studies suggesting a potential role for ivermectin in treating COVID-19, to date, no study has clearly explained the mechanism by which ivermectin could act as an antiviral drug. On the other hand, the European Medicines Agency (EMA) stated in March 2021 that it had considered and reviewed all available evidence regarding the role of ivermectin in the prevention and treatment of COVID-19 and concluded that it does not support the use of this drug for any purpose other than its established use as an antiparasitic drug. Moreover, the EMA noted that ivermectin may block the replication of SARS-CoV-2, but achieving adequate concentrations of this drug in the lungs requires much higher doses than recommended (EMA, 2021). Furthermore, the most significant adverse drug reactions (ADR) that can result from the use of extremely high concentrations of ivermectin in humans include toxic dermatoses, encephalopathies, and confusional disorders (Campillo et al., 2021).

Therefore, the aim of this study was to examine the trend in ivermectin sales in Serbia before, during, and after the COVID-19 pandemic, with a focus on pharmaceutical formulations potentially suitable for human treatment.

MATERIAL AND METHODS

Information about the consumption of ivermectin were obtained from the annual reports of The Agency of Medicines and Medical Devices of Serbia (ALIMS) entitled Sales and consumption of medicines used in veterinary medicine, for a six-year period (from 2017 to 2022) (14). Based on the submitted reports, the total consumption of ivermectin was expressed as the total amount of active substance (ivermectin) in grams. This was calculated by multiplying the total volume (ml) of the medicine solution sold, obtained from the sum of all sold pharmaceutical formulations (e.g., bottles of 100, 250, or 500 ml), by 10, since each formulation contains 10 mg/ml of the active substance. Using this method, the total quantity of ivermectin sold was obtained, expressed in milligrams (later converted to grams), for each individual year. These results are presented in Table 1. Afterwards, by comparing the total sales volume of ivermectin of each year, the sales trend of this drug in the years before (2017, 2018), during (2019, 2020, 2021), and after (2022) the COVID-19 pandemic was created (Graph 1). Regarding the pharmaceutical formulation used for this calculation, only injectable solutions (in different strengths and packaging) were considered, while premixes and oral powders were not included, as these formulations are not applicable in human medicine.

RESULTS AND DISCUSSION

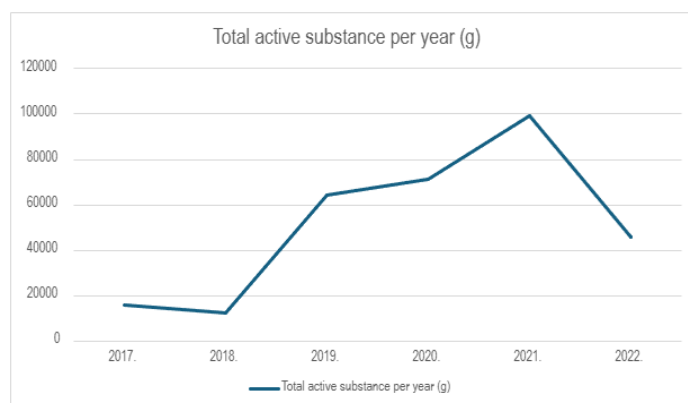
The market activity of all strength and packaging of injectable ivermectin solutions in the years 2017 to 2022 is shown in Table 1. In the years before the COVID-19 pandemic, the sales of ivermectin were 15,749 g (in 2017) and 12,344.2 g (in 2018). The first significant rise in sales of this drug occurred in 2019, when the

sales for the year increased by 420% compared to the previous year. This increase was the highest in the entire study period; however, various causes can be attributed to it, considering that the first registered case of COVID-19 was reported in December 2019. In 2020, the demand rose again, this time by 10.86% compared to the previous year, and in 2021, it increased once more by 39.65%. The increase in 2021 is the most significant data obtained in this study, as it is the year in which the highest number of studies on the potential impact of ivermectin on virus replication were published. In the same year, ivermectin in Serbia reached a record sales of 99,419.1 g, which is an increase of 705% compared to the data from 2018 (over a period of three years). Notably, in 2021, the official report from EMA was published, stating that ivermectin is not safe for use in the treatment of coronavirus, which significantly contributed to a decline of 54.007% (more than a twofold decrease) in the following year, 2022.

Table 1. Sales Volume of All Ivermectin Pharmaceutical Formulations (2017–2022)

ATCvet group	Form, strength and packaging	Number of pharmaceutical formulations sold per year					
		2017	2018	2019	2020	2021	2022
QP54AA01	injectable solution; 10mg/mL; vial; 1x100mL	2967	8123	/	/	/	/
QP54AA01	injectable solution; 10mg/mL; vial; 1x250mL	278	454	81	/	/	/
QP54AA01	injectable solution; 10mg/mL; vial; 1x500mL	754	/	/	/	/	/
QP54AA01	injectable solution; 10mg/mL; vial; 4x250mL	72	58	/	/	/	/
QP54AA01	injectable solution; 10mg/mL; small vial; 1x100mL	1467	296	51577	57040	79461	37589
QP54AA01	injectable solution; 10mg/mL; small vial; 1x20mL	8422	8020	7988	7294	6719	4646
QP54AA01	injectable solution; 10mg/mL; small vial; 1x30mL	1598	1854	2291	2008	12191	2015
QP54AA01	injectable solution; 10mg/mL; small vial; 1x50mL	6248	100	12559	16525	21370	10807
QP54AA01	injectable solution; 10mg/mL; small vial; 24x100mL	/	/	160	148	178	50
Total active substance sold per year (g)		15749	12344,2	64183,9	71155,7	99419,1	45726,2

The previously mentioned data is also presented in the form of a graph, which simplifies the representation of the trend in the sales of ivermectin during the pre-pandemic period (2017, 2018), period during the pandemic (2019, 2020, 2021), and post-pandemic period (2022). The peak in sales in 2021 is clearly shown, where the demand for ivermectin reached nearly 100,000 grams (100 kg of active substance), and in the following year, it was halved.



Graph 1. Sales Trend of Ivermectin (2017–2022)

Despite the unproven efficacy of this drug, ivermectin was significantly used as an off-label treatment in some countries during the pandemic. Similar results, but much more evident, were obtained in the study conducted in Brazil where Hentschke-Lopes et al. (2022) published a study examining the correlation between the increase in sales of so-called “COVID kits” (which included off-label use of ivermectin, azithromycin, chloroquine, and hydroxychloroquine) in Brazil during the pandemic and the rise in adverse drug reactions (ADRs) associated with these drugs. The study indicates that ADRs for all medications, except for azithromycin, were zero before the pandemic, but increased during the pandemic, with the use of the “COVID kit” resulting in an average of 1.6 ADRs per patient (a large number of ADRs went unreported due to a lack of knowledge about the process of identifying and notifying ADRs, among other issues). During the pandemic, the sales of these medications significantly increased (69.75% for azithromycin, 10,856,481.39% for hydroxychloroquine, and 12,291,129.32% for ivermectin). It is worth noting that considerably fewer ADRs were reported for ivermectin compared to azithromycin and hydroxychloroquine.

CONCLUSION

This study shows a significant change in the sales of ivermectin in three different periods: before (2017, 2018), during (2019, 2020, 2021) and following (2022) the COVID-19 pandemic. The use of ivermectin increased drastically during the pandemic, largely driven by its off-label use for the treatment of COVID-19 patients, spurred by the circulation of studies in 2020 and 2021 advocating its potential efficiency. The findings of this study underscore the profound impact of the COVID-19 pandemic on ivermectin sales, reflecting both the influence of scientific publications on drug demand and the complexities associated with off-label drug use. While the dramatic sales increase illustrates the heightened demand for potential COVID-19 treatments during a time of urgent need, it also raises critical considerations about the dissemination and interpretation of scientific data. This study emphasizes the need for rigorous, evidence-based guidelines to prevent future misapplications of veterinary drugs and protect public health.

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SCREENING OF DIFFERENT ROMANIAN HONEY SAMPLES FOR THE DIASTASE ACTIVITY

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Honey is perceived as a valuable substitute of sugar because of its therapeutic properties and ability to improve several characteristics of foods. Honey varieties differ in terms of water content, sugar and minerals content, as well as enzymes that influence the color and viscosity. Conditioning and storage conditions also influence the quality of honey. In the present study, eight samples of honey (rapeseed, black locust, sunflower, mint and marsh flowers, coriander, eucalyptus, lavender and orange honeys) were analyzed for quality indices (diastase index, refractive index and total soluble solids). The results of this study showed good quality of black locust, lavender and eucalyptus honeys. However, all tested samples had a low moisture content and an appropriate diastase index indicating a good capacity for long-term storage and intense enzymatic activity.

Key words: honey, diastase index, refractive index, total soluble solids, moisture content

INTRODUCTION

Honey represents a beekeeping product with high utility in treating various digestive (Syaripudin et al., 2024), liver (Sekar et al., 2023), respiratory (Muhamad et al., 2018), renal (Nerli et al., 2023), infectious (Albaridi, 2019), cardiovascular (Yaghoobi et al., 2008), nervous system (Fadzil et al., 2023) and skin (Alangari et al., 2017) diseases. Compared to sugar, in addition to the health benefits due to the content in bioactive compounds from flower nectar (Viteri et al., 2021), honey improves the taste and preservation of foods (Adeboye et al., 2013).

Good quality honey should have a water content of 17-18 % (EU Council, 2002). The main compounds in honey are carbohydrates, glucose and fructose (monosaccharides) (Tedesco et al., 2020), turanose, isomaltotriose, nigerose (disaccharides) (Tedesco et al., 2020) and dextrin (polysaccharide) (Sereia et al., 2017), free essential amino acids (threonine, alanine, valine, leucine, methionine) (Vrabie et al., 2019), proteins (albumins, nucleo-proteins) (Habati et al., 2017), organic acids (gluconic, acetic, lactic, and malic acids) (Mato et al., 2006), minerals (Potassium, Magnesium, Iron, Calcium) (Bogdanov, 2012), vitamins (vitamin C, B1, B2, B3, B6, B9, K) (Bogdanov, 2012), enzymes (invertase, peroxidase, lipase, oxidase) (Sereia et al., 2017).

The color of honey ranges from almost transparent light yellow to almost black (White and Doner, 1980), the color being in relation to the content of plant pigments (anthocyanins, carotenoids, chlorophylls, xanthophylls) and tannins (Owayss et al., 2004), being further influenced by storage time and crystallization process (Dettori et al., 2018).

The viscosity of honey is influenced by the water content, sugar composition (Kolayli et al., 2013) and temperature (Mehryar et al., 2013).

Before bottling, any honey needs to go through a conditioning process in order to remove impurities, to decrease the water content to below 20% and to inactivate enzymes of the microorganisms that produce the fermentation process (White and Doner, 1980). Honey should be stored in hermetically sealed containers, made of glass or other transparent materials, in clean rooms with low humidity, at temperatures under 14°C, to avoid the loss of flavor, the change in color and the potential fermentation process (Farswan, 2022).

Apart from the chemicals used in control of *Varroa destructor*, parasite mite of honey bees, pesticides used in crop protection can be stored in honey samples from agricultural areas, which can be utilized as a bioindicator of their residual presence (Balayiannis and Balayiannis, 2008).

Important criteria in assessing the quality and authenticity of honey are as follow: diastase index by measuring the activity of amylase (diastase) (Gropoșilă-Constantinescu et al., 2020), refractive index as index of purity and concentration of honey (Sereia et al., 2017), total soluble solids indicating the soluble sugars in honey (Albu et al., 2021), and the moisture content.

The aim of the present study was to determine quality indices for eight commercial honey samples of different origin, provided by Romanian manufacturers/distributors.

MATERIALS AND METHODS

In the present study, eight varieties of honey were analyzed: rapeseed (*Brassica napus* subsp. *napus*) honey, black locust (*Robinia pseudoacacia*) honey, sunflower (*Helianthus annuus*) honey, mint (*Mentha* sp.) and marsh flowers honey, coriander (*Coriandrum sativum*) honey, eucalyptus (*Eucalyptus* sp.) honey, lavender (*Lavandula* sp.) honey and orange (*Citrus* sp.) honey. The characteristics of these samples, as well as the country of origin, are presented in Table 1.

Table 1. Characteristics of honey samples included in the quality analyses

Sample	Type of honey	Organoleptic Characteristics	Country of origin
1	rapeseed	whitish color, creamy consistency, specific smell	Romania
2	black locust	almost colorless yellowish, fluid consistency	Romania
3	sunflower	golden yellow, fluid consistency, pleasant smell	Romania
4	mint and marsh flowers	reddish yellow, creamy consistency	Romania
5	coriander	reddish yellow, fluid consistency, aromatic and fragrant	Romania
6	eucalyptus	dark brown, fluid consistency, specific aroma	Paraguay
7	lavender	yellow-brown, floral smell, specific aroma	Romania
8	orange	yellow-white, creamy consistency, aromatic and fragrant	Spain

The honey solution, prepared from 5 g of honey in 50 ml distilled water, neutralized with 0.05 N sodium carbonate and brought to a volume of 100 ml with distilled water, was subjected to diastase analysis.

The diastase index was determined using the Gothe method (STAS 784/89). According to Gropoșilă-Constantinescu et al. (2020), this index gives information about the freshness and authenticity of honey. The experimental design is presented in Table 2.

The values for the diastase index were calculated by formula:

$$Diastase\ index = 10 \times \frac{5}{V}$$

Where:

V = the volume of honey in the test tube (mL)

5 = number of milliliters of 1% starch solution

10 = the dilution factor of the honey solution

Table 2. Experimental design for determination of the diastase index

Test tube no.	5% Honey (mL)	0.02 N Acetic acid (mL)	0.1 N NaCl (mL)	Distilled water (mL)	Partial volume (mL)	1% Starch (ml)	Final volume (mL)
1	10.0	0.5	0.5	–	11	5	16
2	7.7	0.5	0.5	2.3	11	5	16
3	6.0	0.5	0.5	4.0	11	5	16
4	4.6	0.5	0.5	5.4	11	5	16
5	3.6	0.5	0.5	6.4	11	5	16
6	2.8	0.5	0.5	7.2	11	5	16
7	2.1	0.5	0.5	7.9	11	5	16
8	1.7	0.5	0.5	8.3	11	5	16
9	1.3	0.5	0.5	8.7	11	5	16
10	1.0	0.5	0.5	9.0	11	5	16

All ten test tubes were kept at 45°C in a water bath for 60 min, then cooled and tested for the reaction of starch with iodine solution. In the test tubes where the blue color appeared, the starch was still present (Radu et al., 2021). In the tubes where other colors appeared (purple/red/yellowish/colorless), the starch was completely transformed in dextrin (Moise, 2019).

For the evaluation of refractive index and total soluble solids, an Abbe refractometer (Kruss AR2008) was used.

The moisture content of all samples was analyzed at 105°C, using a moisture analyzer (Mac 210/NP Radwag, Radom, Poland).

A Spearman's correlation analysis between the indices of honey quality was performed using rcorr function from Hmisc R Package Version 5.1-3 (Harrell, 2024, R Core Team, 2024), while a correlation plot was obtained by corrplot.mixed function from corrplot package Version 0.94 (Wei and Simko, 2024).

RESULTS AND DISCUSSIONS

It is considered that the value of the diastase index should be at least 10.9, with the exception of black locust honey, where the accepted minimum value is 6.5, and manna honey, where the minimum value is 13.9 (Lazăr, 2007).

The values for diastase activity of the investigated samples are comparatively represented in Figure 1a. The value for eucalyptus honey wasn't illustrated on the histogram because of the absence of a blue color in all 10 test tubes, which can be associated with an intense enzyme activity through which starch was hydrolyzed. Thus, eucalyptus honey can be considered fresh, showing good quality and no thermal or storage processes that could have affected its enzymes (Korkmaz and Küplülü, 2017, Raweh et al., 2022).

The highest determined diastase activities were found in sunflower honey and coriander honey (38.5), followed by the samples from mint and marsh flowers and lavender (29.4). These are good quality honey samples with strong enzymatic activity. The lowest values of the diastase index were found in black locust honey (6.5) and rapeseed honey (17.9), indicating the weakest enzymatic activity.

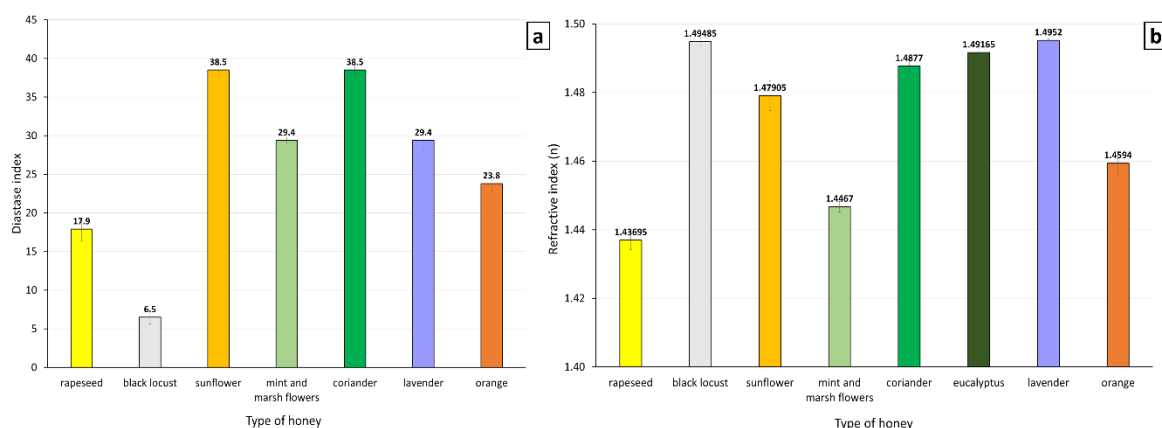


Fig. 1 Values of diastase index (a) and refractive index (b) of investigated honey samples

However, all samples had a diastase index with value above other reported ones (Lazăr, 2007), so these can be appreciated as natural, of good quality, minimally processed and stored under proper conditions.

Similar values for rapeseed, but in multifloral honey, were previously reported by Machado et al. (2022) ($13.8 \pm 2.0 - 15.8 \pm 0.5$). High values, above 30 Goethe units, were previously found in polyfloral honey samples (Zhuk et al., 2022) and in linden honey (10.1 to 45.8 units Goethe) (Lazarieva et al., 2023).

Apart from long storage time under inadequate conditions, heating and adulteration of honey, the enzyme activity can also be decreased by intensive feeding of bees with sugar syrup (Kanelis et al., 2022).

Regarding the refractive index (n), values between 1.4369 and 1.4952, were found, as observed in Figure 1b. The highest values for this index were recorded for black locust honey, followed by lavender and eucalyptus honey, these being the most concentrated samples, of good quality. On the other side, rapeseed honey and the one from mint and marsh flowers showed the lowest values of the refractive index, presenting lower nutritive values.

Similar numbers for the refractive index were found in lime honey (1.4983 to 1.4998) by Nikolova et al. (2012). In their study, it was mentioned that the value of refractive index decreases with the increase of the glucose content. From this point of view, honey samples from rapeseed, mint and marsh flowers showed higher content in glucose.

The values obtained for total soluble solids ranged between 50 and 80 °Brix, as shown in Figure 2a. The lowest values for this index were found in rapeseed honey and mint with marsh flowers sample, as a proof of a lower content in sugars and a higher moisture content, which represents a risk for a potential fermentation (Rivera-Mondragón et al., 2023). The values of these two samples (58.95 °Brix, respectively 62 °Brix), significantly below the minimum permissible value of 80 °Brix (Rivera-Mondragón et al., 2023), suggest that honey probably was diluted or adulterated (Guelpa et al., 2017). On the other hand, lavender, black locust and eucalyptus honey samples presented the highest values of total soluble solids, indicating qualitative and authentic honey with high content in nutrients.

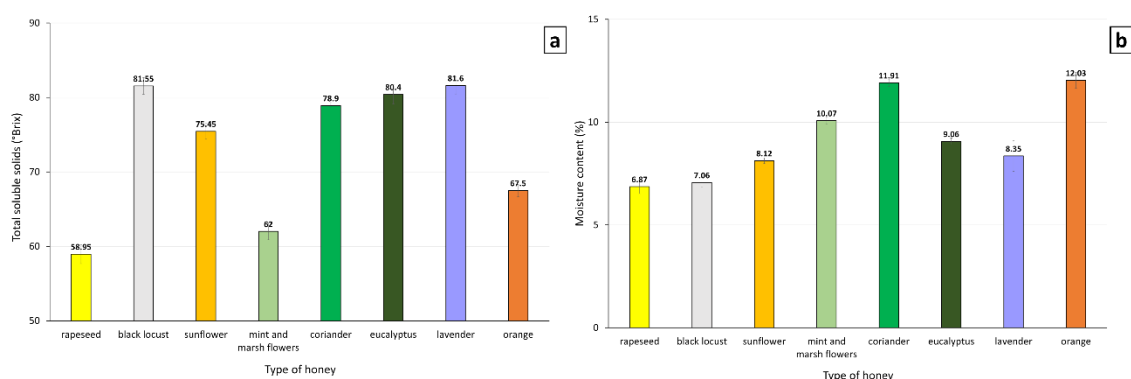


Fig. 2 Representation of total soluble solids (°Brix) (a) and moisture content (%) (b) for the analyzed honey samples

The results regarding the moisture content of the honey samples, which were situated between 6.865% and 12.032%, are illustrated in Figure 2b.

Orange and coriander honey samples had the highest water content, while the rapeseed and black locust

honey samples presented the lowest moisture. Because of the low moisture content for all the analyzed samples, there is a low risk of fermentation during storage, so the honey can retain their nutritional and therapeutic properties for a long time.

Moreover, low moisture of selected assortments will slow down the crystallization process (Dettori et al., 2018). Honey samples with moisture under 15% are rarely found and could be explained by a very dry harvesting season (Nyau et al., 2013).

Overall, black locust, lavender and eucalyptus honey samples were found of good quality, being natural, pure, rich in nutrients and with a long shelf life.



Fig. 3 Correlogram of the Spearman's correlation coefficients between the evaluated criteria of quality for the honey samples

Analyzing the relationship between the four honey quality variables, a strong positive correlation between the refractive index and total soluble solids was found ($r = 1$, $p < 0.001$) (Figure 3).

CONCLUSIONS

Honey is of particular importance in nutrition and traditional medicine, being used for digestive, hepatic, respiratory and cardiac disturbances.

The quality of honey is determined by its properties: content in water, glucose and mineral salts, as well as the storage conditions and conditioning processes.

The eight investigated honey samples showed a diastase index higher than the minimum accepted value (10.9, respectively 6.5 in the case of black locus), all those samples proving to be pure, qualitative and natural. The enzymes in the eucalyptus honey hydrolyzed the entire starch content, reflecting the highest enzyme activity among those of the other samples. Apart from this, high activity of amylase was also recorded in sunflower and coriander honey samples, which are considered fresh and properly processed honeys.

The highest refractive index was found in lavender and black locust honey samples (1.4952, respectively 1.4948), while the lowest values for total soluble solids, representing low amounts of sugars that increase the risk of fermentation, were recorded in rapeseed sample and in mint and marsh flowers honey (58.95-62 °Brix).

All the tested samples had a low moisture content, offering a longer shelf life, in which the therapeutic properties and aroma are well preserved, and the crystallization process is slowed down.

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QUALITY INDICATORS OF SELECTED TYPES OF OILSEED VARIETIES

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Oilseeds are crucial in agriculture and industry, with sunflower, rapeseed, soybean, and poppy being key crops in Slovakia. This study assessed quality indicators like moisture, fat content, acidity, fatty acid profiles, and oxidative stability. Soybean had the highest moisture (14.07–12.65%), while poppy seeds had the lowest (6.2–5.0%). Sunflower and rapeseed met minimum fat requirements, and sunflower exhibited the highest fat acidity (2.00–0.98 mg KOH/g). Rapeseed oil had the best oxidative stability (33.46–40.5 hours), whereas poppy oil had the lowest (1.39–6.36 hours), indicating suitability for different applications

Keywords: Oilseeds, fatty acids, oxidative stability, analytical methods, edible oils.

INTRODUCTION

Oilseeds represent a crucial group of commodities globally, with over 600 million metric tons produced in the 2021/2022 crop year (Dunford, 2022). Among the most widely cultivated oilseeds are palm, soybean, and rapeseed, which serve as major sources of vegetable oils. In Slovakia, sunflower and rapeseed dominate vegetable oil production, and oilseeds also provide vital nutrients such as fiber, phosphorus, iron, magnesium, and vitamins, including vitamin E, folic acid, and niacin (Lafarga, 2021). Their high phytoestrogen content, particularly isoflavones, and beneficial omega-3 fatty acids contribute to their health benefits, including the prevention of cardiovascular diseases, obesity, diabetes, and cancer (Tanwar, 2021). The quality and composition of oilseeds are influenced by factors such as genotype, growing conditions, and storage practices, and they are utilized in both food products and industrial applications (Jithender, 2019). Nevertheless, weeds pose a significant threat to oilseed crop yields, leading to potential losses of up to 77% depending on various factors, including crop type and climate (Unnikrishnan, 2022). Moreover, oilseeds are especially popular in gluten-free bakery products, enhancing their appeal to consumers with celiac disease. However, their oxidative stability and the rheological properties of the resulting baked goods must be considered (De Lamo, 2018). Poppy seeds (*Papaver somniferum* L.) have a long history of cultivation, primarily in the Mediterranean region, and are widely grown for both their alkaloid content and edible oil (Özbek, 2020). Poppy seeds are rich in fats, with linoleic acid comprising 53–74% of their fatty acid profile (Ghafoor et al., 2019). Although the presence of linoleic acid may not be ideal nutritionally, poppy seed oil contains other beneficial compounds such as phytosterols and tocopherols (Dybbrowski, 2020). In Central Europe, poppy seeds are often used in bakery products and desserts, although they may become contaminated with opium alkaloids due to improper handling (López et al., 2018). Rapeseed (*Brassica napus* L.) is another significant oilseed crop, known for its oil, which ranks third globally in consumption, following palm and soybean oil (Secchi, 2023). Rapeseed oil is valued for its low saturated fat content and high monounsaturated fat content, including omega-3 fatty acids, which promote heart health (Eskin, 2021). The cultivation of rapeseed also supports the production of biodiesel, enhancing its economic importance (Viana et al., 2022). Nonetheless, glucosinolates present in rapeseed meals can affect iodine absorption, which could lead to health concerns if not properly managed (Tanwar, 2021). Sunflower (*Helianthus annuus* L.) is another key oilseed crop, with sunflower oil being widely used due to its light flavor, high linoleic acid content, and health benefits (Tanwar, 2021). Sunflower oil is trans-fat free, making it a healthier alternative for human consumption, and it contains valuable compounds such as tocopherols and phytosterols, which protect against oxidative stress (Bernard, 2022). Sunflower cultivation also benefits animal feed production, with sunflower meal being used as a nutritious feed (Smirnova, 2021). Soybeans (*Glycine max*) are arguably the most important legumes worldwide, with approximately 80% of production concentrated in the United States, Brazil, and Argentina (Tanwar, 2021). Soybeans contain a high percentage of oil and protein, making them vital for both human consumption and animal feed. However, anti-nutritional compounds in soy, such as trypsin inhibitors, require technological processing to reduce their impact on food quality (Sech et al., 2022). Soybeans are also increasingly used as a base for meat substitutes, highlighting their versatility and growing importance in modern diets (Medic, 2014). Consequently, this study seeks to explore the quality characteristics of selected oilseeds, focusing on moisture content, fat content, fatty

acid profile, acidity of fat, and oxidative stability, which are critical to their utility in both food and industrial applications.

MATERIALS AND METHODS

Samples used for analysis were obtained from agricultural cooperatives in southern Slovakia. The study included 12 types of oilseeds from the 2022 harvest: three varieties of poppy (Ideal, Krokodil, Snow White), three varieties of sunflower (Shoumi, LG 50 635 CLP, Neoma), three varieties of rapeseed (PX 126, Architect, Parcours), and three varieties of soybean (Altona, ES Mentor, RGT Scala).

The sample for analytical purposes was conducted according to the ISO 664 standard from 2008. After removing large impurities, samples were divided using suitable tools (dividers) into an analytical sample representing the laboratory sample. The analytical sample was then transferred to a dry and clean container, hermetically sealed, and labeled. Before analysis, the sample is ground using a mechanical mill (ZM 200 Retch) to the desired particle size. Small, medium, and large seeds are ground until the largest particle size does not exceed 1 mm. For sunflower seeds, a sieve with 1.5 mm openings was used.

Determination of Moisture Content. The water content is determined gravimetrically under prescribed conditions according to the STN EN ISO 665 standard. A metal dish with a lid is dried for 1 hour at a temperature of 103 °C. After cooling, the desiccator is weighed with a precision of 0.001 g. Then, the ground sample is weighed into the desiccator, except for sunflower seeds, where moisture is determined in the whole seed. The metal dish with an open lid and the test sample is then placed in an electric drying oven (HS 61 A) (Fig. 1) set to a temperature of 103 °C ± 2 °C. After 3 hours, the sample is removed from the oven and allowed to cool in a desiccator to cool to a laboratory temperature and then reweighed. The sample is reinserted into the oven for 1 hour and then weighed under similar conditions. The determination is considered complete if the difference between the two weighings is less than or equal to 0.005 g. If the difference is greater, the sample is repeatedly dried for 1 hour in the oven until the difference between the two repeated weighings is less than or equal to 0.005 g. The moisture content is calculated using the following formula:

$$w = \frac{m_1 - m_2}{m_1 - m_0} \times 100 \%$$

Where:

- m_0 = mass of the desiccator [g]
- m_1 = mass of the desiccator with the test sample before drying [g]
- m_2 = mass of the desiccator with the test sample after drying [g]



Fig. 1 Electric dryer HS 61 A

Determination of Fat Content. The fat content in oilseeds is determined by gravimetric analysis according to the STN EN ISO 659 standard. A container that has been previously dried in an oven and cooled in a desiccator is weighed to a precision of 1 mg. A weight of 10 g ± 0.5 g of the ground sample is placed in the extraction cartridge, which is sealed with a small piece of cotton wool. The extraction flask is filled with 150 ml of petroleum ether. Subsequently, the extraction cartridge with the sample is inserted into the central part of the extraction apparatus. The Twisselmann extraction device (Fig. 2) is connected to the extraction flask, and the contents of the cartridge are extracted for 6 hours. After 4 hours of extraction, the cartridge is removed from the extraction apparatus and dried. The contents of the extraction cartridge are emptied into a cylinder of a micro grinder with steel balls and ground for 7 minutes. The residue of the crushed seeds is then transferred back into the cartridge, which is placed back into the extraction apparatus for an additional 2 hours of extraction. After two hours, the grinding process is repeated as described above, followed by another 2 hours of extraction. The petroleum ether is removed from the extraction flask in an oven at 103 °C ± 2 °C. Finally, the flask with the extract is weighed, and the result is calculated using the formula (1):

$$w = \frac{(a_2 - a_1)}{m_1} \times 100 \% \quad (1)$$

Where:

- a_2 = mass of the extraction flask with fat [g]
- a_1 = mass of the empty extraction flask [g]
- m_1 = mass of the test sample [g]



Fig. 2 Extraction apparatus according to Twisselmann and Nahita water bath

Determination of Acid Value. An extraction flask containing the extracted fat is used for the determination (Fig. 3). To a beaker, 50 ml of ethanol is added, followed by 0.5 ml of phenolphthalein indicator, and the mixture is brought to a boil. The ethanol solution is then neutralized with a 0.1 mol/l potassium hydroxide solution. The neutralized ethanol is added to the test sample in the extraction flask and thoroughly mixed. The contents are brought to a boil and titrated with potassium hydroxide solution, with vigorous stirring throughout the titration. The endpoint is reached when the addition of a single drop of KOH solution causes a color change that lasts for at least 15 seconds. The acid value of the fat, expressed as mg KOH per gram of fat, is calculated using the following formula (2):

$$w = \frac{(M_{KOH} \times c \times V)}{m} \quad (2)$$

Where:

- M_{KOH} = molar mass of KOH [g/mol]
- c = concentration of the potassium hydroxide standard solution [mol/l]
- V = volume of potassium hydroxide standard solution consumed [ml]
- m = mass of the extracted fat [g]



Fig. 3 Determination of acidity number using the hot ethanol method with indicator

Determination of Fatty Acid Profile. The extracted fat must be liquid, dry, and clean. A micro-pipette is piped 210 μ l into a test tube with a stopper. Five milliliters of n-hexane is added to the fat using a graduated pipette. The fat is dissolved by mixing. Next, 1 ml of a 2 M methanol solution of KOH is added with a graduated pipette. The contents of the test tube are shaken vigorously. The test tube is then placed in a water bath at 60 $^{\circ}$ C for 30 seconds. Afterward, the sample is allowed to stand for 1 minute, and then 2 ml of a 1 M HCl solution is added with a graduated pipette, followed by gentle mixing. After reaching equilibrium and separating the layers,

the FAME's upper layer is carefully removed with a 5 ml graduated pipette. The extracted FAME layer is filtered through anhydrous sodium sulfate into clean test tubes. The prepared FAME samples are transferred into vials. Before injecting the sample into the column, the sample is diluted in a ratio of 50 μ l of FAME solution to 950 μ l of n-hexane. A volume of 1 μ l is injected into the instrument. The fatty acid profile is determined using gas chromatography, and the area under the peak of the analyzed component is used as the quantity indicator.

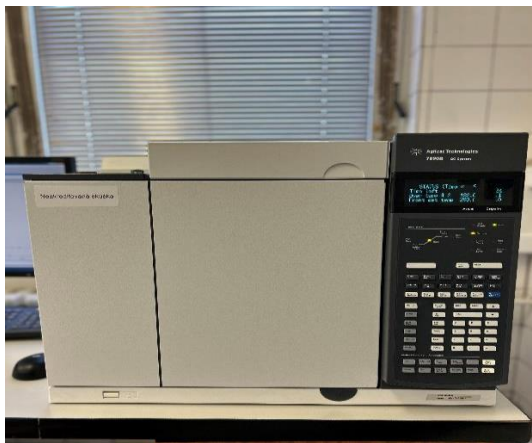


Fig. 4 Gas chromatograph Agilent

Determination of Oxidative Stability. A sample weighing 0.5 g is weighed into a test tube. The test tube is inserted into the Rancimat 892 device (Figure 5), and hoses are connected for the continuous supply of oxygen and the removal of oxidation products into a container with 60 ml of distilled water. The device begins to heat the sample, causing volatile oxidation products to be transported via the airflow from the test tube to the container containing distilled water. The device measures the conductivity of the water conductively. The results of the curves are evaluated using software.



Fig. 5 Rancimat 892 device

RESULTS AND DISCUSSIONS

Moisture Content Analysis. Based on the analysis, the moisture content of the oilseeds varied across different types. Soybeans exhibited the highest moisture content, ranging from 12.65% to 14.07%. This exceeded the standard set by STN 46 2300-7, which stipulates that the moisture content in soybeans should not exceed 12.0%. This deviation highlights a potential issue with the soybean samples, as higher moisture content can affect their quality and shelf life.

The sunflower seeds had moisture levels between 6.43% and 8.33%. According to STN 46 2300-6, sunflower seeds are categorized into two market classes, both of which require a maximum moisture content of 8.0%. The sunflower samples fell within this acceptable range. Rapeseed moisture content, measured between 5.7% and 6.1%, adhered well to the STN 46 2300-2 standard, which mandates that moisture should not exceed

8.0%. Unlike sunflower seeds, rapeseed is not divided into different market classes for moisture content, but all the rapeseed samples fell comfortably within the prescribed limit. Poppy seeds, on the other hand, exhibited the lowest moisture content, ranging from 5.0% to 6.2% (see Fig. 6). According to STN 46 2300-3, Grade 1 poppy seeds should not exceed 8.0% moisture, while Grade 2 can have a maximum of 10.0%. All the poppy seed samples from the study were well within these limits, indicating that they met the standards for high-quality seeds. The lower moisture content in poppy seeds can be advantageous for longer storage periods, reducing the risk of spoilage.

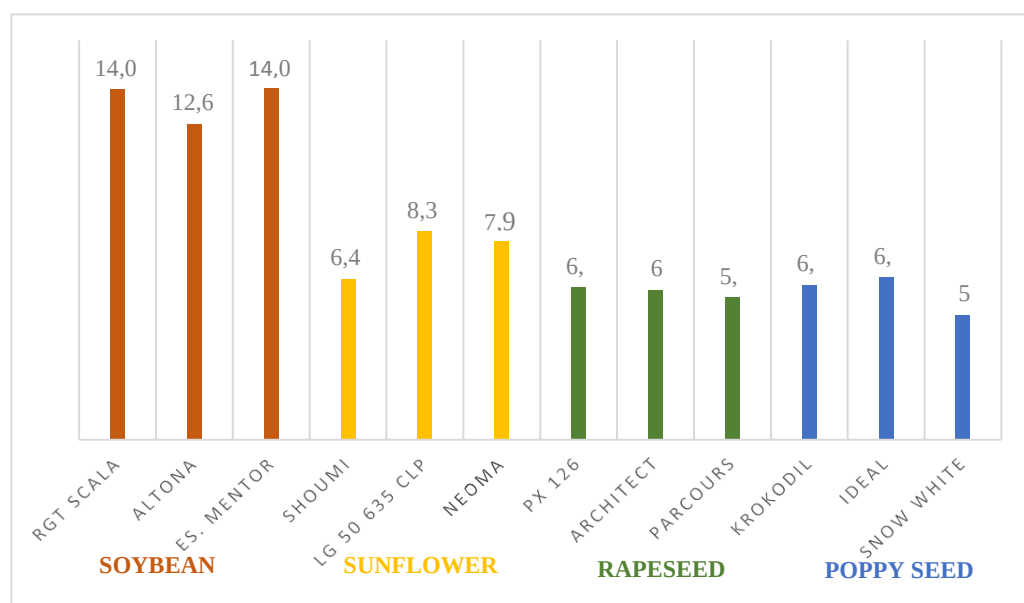


Fig. 6 Results of moisture determination for various types of oilseeds

Fat Content. Poppy seeds exhibited the highest fat levels, ranging from 49.47% to 52.0%, which aligns closely with the findings of Lančaričová et al. (2016), who reported fat content in Slovak poppy seed varieties between 45.5% and 52.4%. This consistency suggests that poppy seeds, particularly white varieties, tend to have higher fat content due to their thinner seed coats, as observed in both studies. In contrast, rapeseed samples from this study displayed fat content ranging from 49.31% to 43.83%, which exceeds the minimum requirement of 42.0% set by STN 46 2300-2. These findings indicate that the rapeseed samples meet the quality standards and surpass the expected fat content, demonstrating their viability for oil extraction. Sunflower seeds, on the other hand, exhibited lower fat content, with values ranging from 39.85% to 42.57%, which falls below the 44.0% minimum requirement established by STN 46 2300-6 (see Fig. 8). This discrepancy raises concerns about the quality of the sunflower samples, as they do not meet the required standards for fat content. This could potentially impact their market classification and usability in oil production. Soybean samples recorded the lowest fat content, with values between 22.17% and 24.73%. While these values comfortably exceed the minimum requirement of 19% specified by STN 46 2300-7, they align with the findings of Miao et al. (2019), who reported fat content in Chinese soybean samples ranging from 17.0% to 24.0%. This suggests that soybeans consistently exhibit lower fat content compared to other oilseeds, yet still meet the necessary quality standards for fat content.

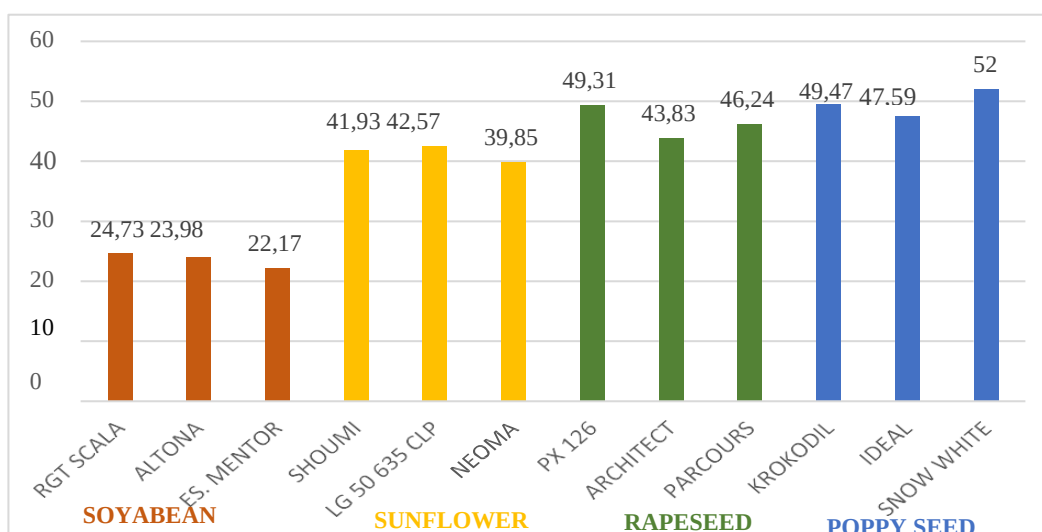


Fig. 7 Results of fat content determination for various types of oilseeds

Acid number of fats. The sunflower samples exhibited the highest values, ranging from 2 to 0.98 mg KOH/g. This aligns with the maximum allowable acid number of 3.0 mg KOH/g for sunflower seed oil as per STN 46 2300-6. However, these values are slightly lower than the average of 1.39 mg KOH/g found in crude sunflower oil based on interlaboratory tests involving 25 laboratories according to the STN 660 standard. This suggests that the sunflower oil in the current experiment falls well within acceptable quality parameters for acid number, demonstrating a lower degree of free fatty acids formed from fat hydrolysis.

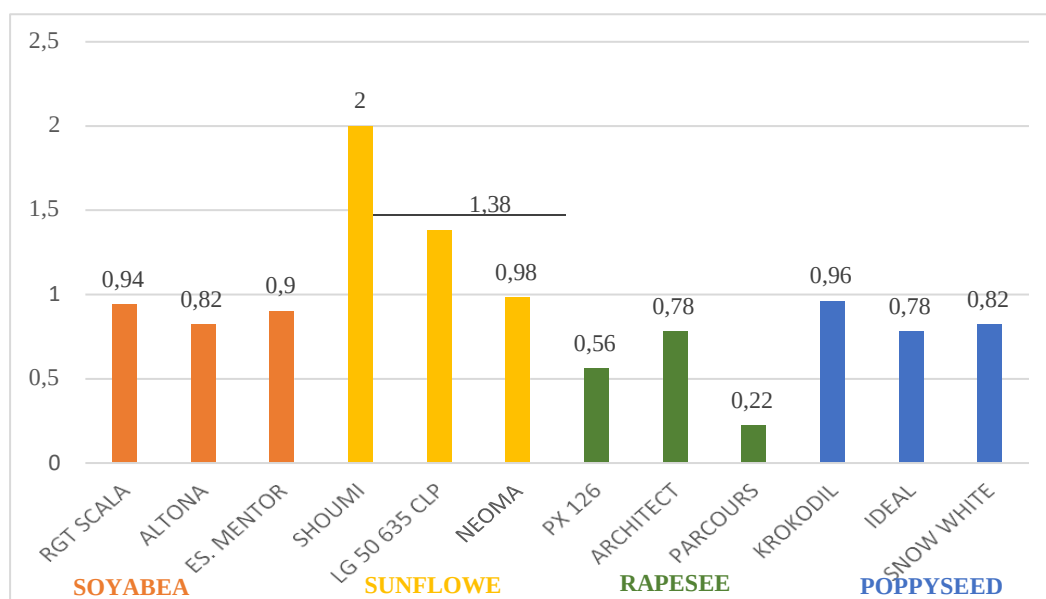


Fig. 8 Results of acid number determination for various types of oilseeds

Soybean samples, with acid numbers ranging from 0.94 to 0.82 mg KOH/g, are within the range reported by Prodhan et al. (2015), who found acid numbers in soybean oils between 0.37 and 0.75 mg KOH/g. Although the current experiment shows slightly higher values, they remain relatively low, suggesting minimal fat degradation. This is in line with expectations for good-quality soybean oil and indicates stable oil content in the tested soybean varieties. Poppy seeds demonstrated acid numbers between 0.96 and 0.78 mg KOH/g, which contrasts sharply with the much higher values reported by Dąbrowski (2020), where acid numbers ranged from 2.40 mg KOH/g for cold-pressed oil to 6.93 mg KOH/g for Soxhlet-extracted oil. The significantly lower acid numbers

in the current study suggest superior quality or minimal degradation in the poppy seed oils, likely due to different extraction methods or seed handling techniques. Finally, the rapeseed samples showed the lowest acid numbers, ranging from 0.78 to 0.22 mg KOH/g (see Fig. 8). These values are well below the 1.80 mg KOH/g threshold specified by STN 46 2300-2 and align with the findings of Hu et al. (2002), who reported values between 0.47 and 0.57 mg KOH/g for Chinese rapeseed oil. This suggests that the rapeseed oil in the current experiment is of high quality, with minimal free fatty acids indicating low levels of fat hydrolysis.

Fatty acid profile. The fatty acids were analyzed using gas chromatography. We identified the presence of ten fatty acids, specifically myristic acid, palmitic acid, stearic acid, oleic acid, linoleic acid, linolenic acid, arachidonic acid, docosahexaenoic acid, erucic acid, and lignoceric acid. The distribution of these fatty acids in individual samples is illustrated in separate graphs, where the retention time of the separated analytes serves as an indicator of quality. Soybean samples showed a predominant presence of linoleic acid, ranging from 52.87% to 55.64%, which aligns with the findings of Ivanov et al. (2011), where linoleic acid ranged from 47.57% to 68.02%. The oleic acid content in our samples was between 20.24% and 21.31%, which is consistent with both Ivanov et al. (2011), who reported values from 16.02% to 22.22%, and the standards of the Codex Alimentarius (2023), which state that oleic acid in soybean oil should range from 17.0% to 30.0%. The palmitic acid content in our samples, recorded between 10.89% and 11.44%, also falls within the Codex Alimentarius standard (8.0-13.5%), though it is lower than the higher concentration (16.95%) observed by Ivanov et al. (2011) in soybean oil. Other analyzed fatty acids were detected in amounts less than 1% (see Fig. 9)

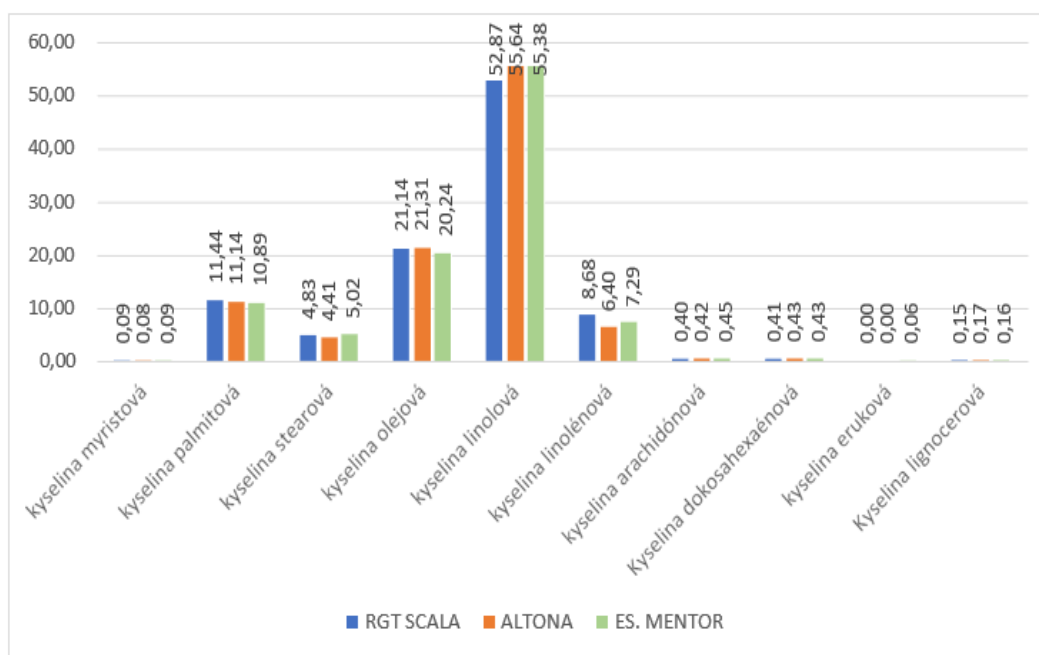


Fig. 9 Results of fatty acid profile determination in individual soybean samples

Sunflower oil from our study showed higher linoleic acid content (62.65-63.45%) and oleic acid (23.92-24.24%). This is consistent with Kostik et al. (2013), who reported linoleic acid concentrations of 52.0% to 67.0% and oleic acid values between 27.0% and 36.0% (see Fig. 10). While our findings indicate slightly lower oleic acid content compared to Kostik et al. (2013), they fit well within the Codex Alimentarius (2023) standard for sunflower oil, which specifies linoleic acid content between 45.4% and 74.0% and oleic acid between 14.0% and 43.0%. The relatively higher linoleic acid content and moderate oleic acid content in sunflower oil suggest high nutritional value and stability, as Aşkın and Kaya (2020) indicated.

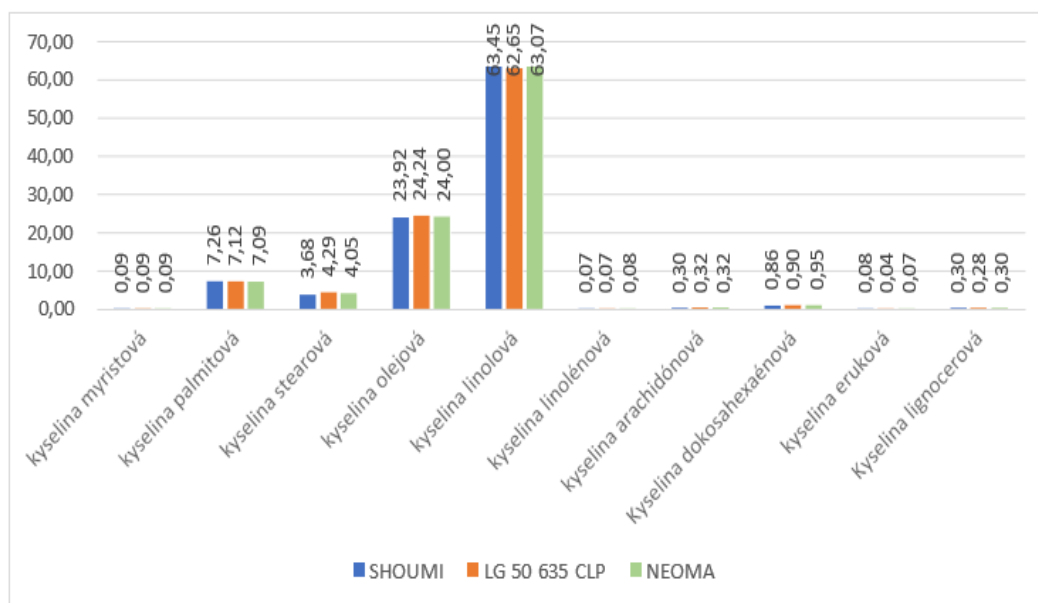


Fig. 10 Results of fatty acid profile determination in individual sunflower samples

The rapeseed oil samples from our study displayed the highest oleic acid content (62.12-62.82%), which closely matches the 65% reported by Beyzi et al. (2019) and is within the Codex Alimentarius (2023) standard of 51.0% to 70.0%. Linoleic acid levels in our samples (18.57-20.94%) also align well with the Codex standard (15.0-30.0%) (see Fig. 11). Notably, the presence of alpha-linolenic acid (8.07-9.82%) in our samples mirrors the range of 9% identified by Beyzi et al. (2019), further confirming the stability of the fatty acid profile. These results emphasize the high nutritional value of rapeseed oil, particularly due to its significant oleic acid content and the beneficial effects of its polyunsaturated fatty acids on health.

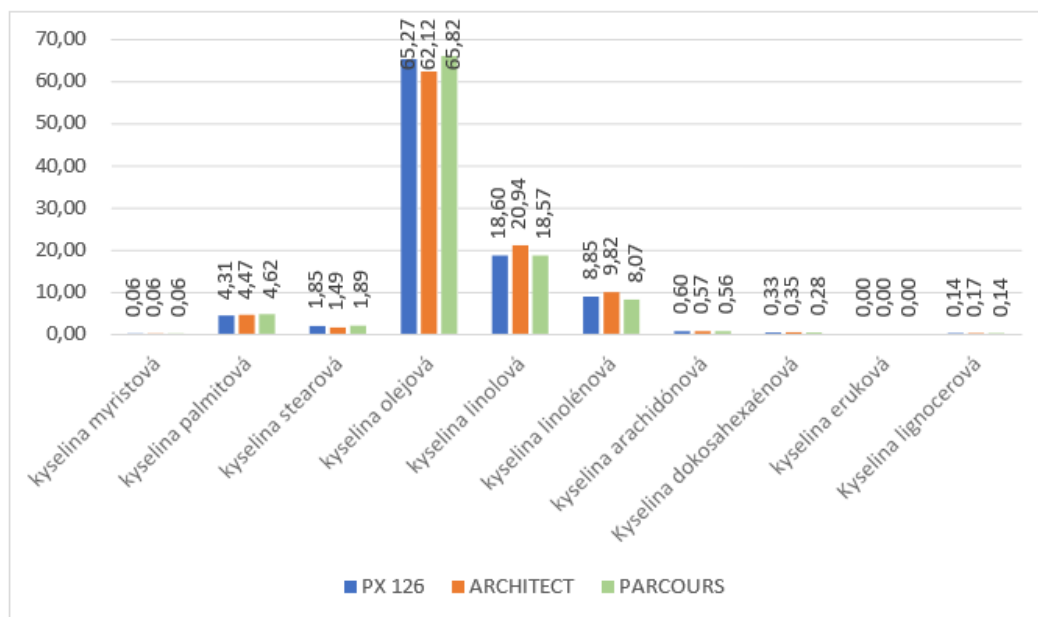


Fig. 11 Results of fatty acid profile determination in individual rapeseed samples

Poppy seed oil in our analysis exhibited the highest linoleic acid content (71.75-76.10%), surpassing the range reported by Dąbrowski (2020), who found linoleic acid content between 53% and 74%. This higher linoleic acid content also corresponds to the findings of Lančaričová et al. (2016), who reported values between 69.5% and 74.1%. Additionally, the oleic acid content in our samples (12.12-16.49%) was slightly lower than the range of 13.6% to 18.2% observed by Lančaričová et al. (2016), though still within an acceptable range for

poppy seed oil (see Fig. 12). The high linoleic acid concentration underscores the predominance of polyunsaturated fatty acids (PUFAs), making poppy seed oil a common component in diets high in n-6 fatty acids, though nutritionally imbalanced due to the low levels of other unsaturated fats like oleic acid.

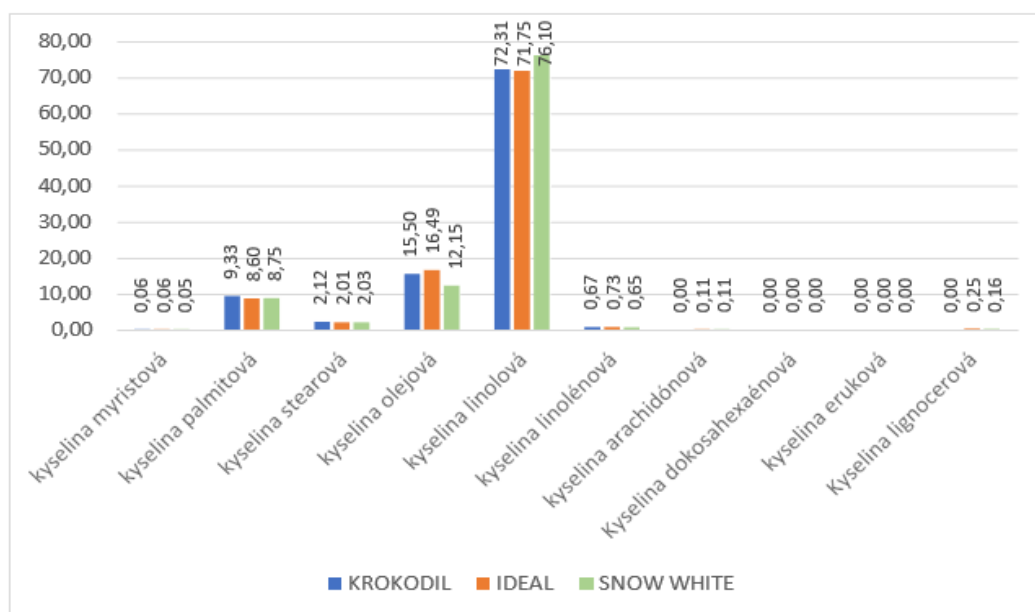


Fig. 12 Results of fatty acid profile determination in individual poppy seed samples

Oxidative stability. The oxidative stability of the oil samples varied significantly, influenced by factors such as the degree of unsaturation in fatty acids, the presence of antioxidant compounds, and external conditions. In our samples, the induction time ranged from 1.39 hours for poppy oil to 40.5 hours for soybean oil (see Fig. 13).

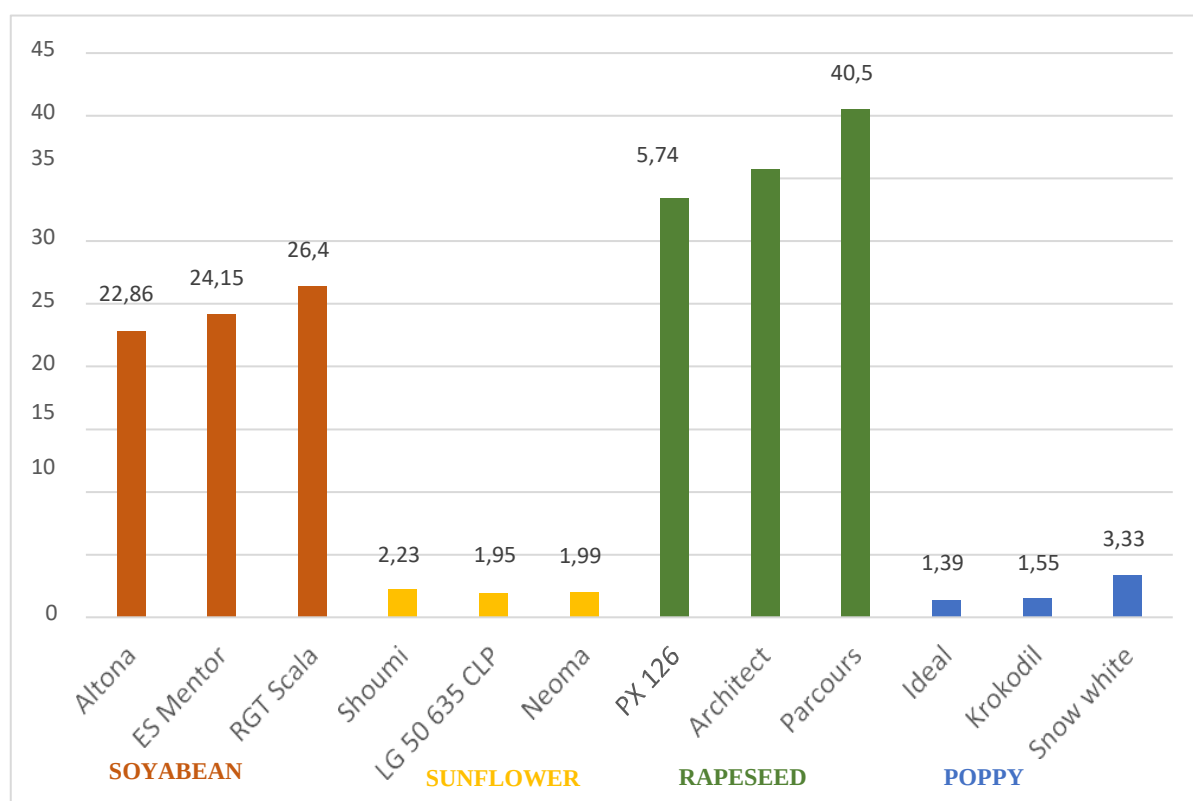


Fig. 13 Results of oxidative stability determination for various types of oilseeds

The quality and shelf life of the product are significantly affected by the oxidative stability of the oil. Autooxidation is primarily driven by atmospheric oxygen, which reacts with lipid radicals to form lipid peroxy radicals, resulting in chain reactions of free radicals. Factors such as heat, light, and metals can accelerate autooxidation. The products of secondary oxidation, including short-chain hydrocarbons, can lead to undesirable flavors (Dunford, 2015). Research by Naz et al., (2004) examined the effects of adding antioxidants to vegetable oils. They utilized coffee, vanilla, and raw tea extracts as antioxidants and analyzed their impact after frying for 30, 60, and 90 minutes at 180 °C. Their study concluded that all antioxidants effectively reduced the oxidation rate in the oil. Furthermore, they investigated changes during frying at the temperature and durations, finding that oxidative stability values increased in the following order: frying > exposure to air and light > exposure to air. Dunford (2015) analyzed sunflower samples under high-temperature conditions (180 °C). Initially, before frying, the oxidative stability values were approximately 8.93 hours. After twelve hours, the oxidative stability decreased to 4.44 hours. After 24 hours, this value further declined to 3.7 hours, and after 48 hours, the oxidative stability was reduced to 0.5 hours. Fresh rapeseed oil is characterized by a long induction time of about 10 hours at 110 °C, according to Maszewska et al., (2018). The induction time for rapeseed oil at 120 °C ranges from 4.3 to 4.7 hours. Bozan and Temelli (2008) studied the oxidative stability of various oilseeds. Based on their findings, the induction time for poppy oil ranged from 4.76 to 6.36 hours.

CONCLUSION

The study focused on three soybean varieties: Altona, ES Mentor, and RGT Scala, with moisture contents of 12.65 %, 14.07 %, and 14.02 %, respectively. According to STN 46 2300-7, the moisture content of all soybean samples exceeded the standard quality requirement of a maximum of 12.0%. However, the fat content ranged from 22.17 % to 24.73 %, meeting the required minimum of 19.0%. Soybeans had the lowest fat content among all analyzed oilseeds. The acidity numbers ranged from 0.82 to 0.94 mg KOH/g fat, indicating suitable fat acidity, like that found in poppy seeds. The fatty acid profile in all soybean varieties was comparable and met the Codex Alimentarius standards. Soybeans exhibited better oxidative stability (22.86 – 24.40 hours) than poppy seeds and sunflower seeds. For sunflower seeds, the moisture content in Shoumi (6.48 %) and Neoma (7.93 %) was within the acceptable range under STN 46 2300-6, but LG 50 635 CLP (8.33 %) exceeded the maximum allowable limit of 8.0%. The fat content in sunflower seeds ranged from 39.85 % to 42.57 %, which did not meet the minimum required value of 44.0%. The acidity numbers in sunflower oilseeds were the highest among the samples (0.98-2.0 %) but still within the allowable limit of 3 mg KOH/g fat as per STN 46 2300-6. Despite meeting fatty acid standards according to Codex Alimentarius, sunflower seeds had the lowest oxidative stability (1.95-2.23 hours), along with poppy seeds, indicating low shelf life. Rapeseed samples met the moisture and fat content requirements under STN 46 2300-2. The acidity numbers (0.22-0.56 mg KOH/g fat) were the lowest of all the analyzed samples, correlating with the highest oxidative stability (33.46 – 40.5 hours), indicating good fat durability and stability. Poppy seed samples had moisture content (5.0-6.5 %) within acceptable limits under STN 46 2300-3, with the highest fat content (47.59-52.0 %) among all samples. However, the poppy seeds exhibited the highest acidity numbers (0.78-0.96 mg KOH/g fat), confirmed by the lowest oxidative stability, suggesting that poppy oil is the most sensitive to external environmental changes. When consuming oils, the golden rule applies: "Oil is a good servant but a bad master." With proper selection and processing, oils are an essential component of a balanced human diet, providing vital nutrients and essential elements.

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POTENTIAL OF SHEA NUT SHELL BIOCHAR AS SUBSTRATE ENHANCER IN VERTICAL FLOW CONSTRUCTED WETLAND TREATING WASTEWATER FOR AGRICULTURAL REUSE

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The overall objective of the study was to treat municipal wastewater for irrigation using pilot scale VFCWs. Biochar and macrophytes were the treatments applied in the systems. Removal efficiencies for biochar-enhanced wetlands were up to 100% for Faecal Coliform and up to 98.71% for Total Coliform. For physicochemical parameters, results revealed 94.93%, 37.68%, 100%, 60.39%, 100%, 100%, and 92.17%, respectively for TSS, TDS, Turbidity, Salinity, EC, NO₃-N and NH₄-N. The study realized significant potential of utilizing sheanut shell biochar for wastewater treatment.

Keywords: Macrophytes, Sheanut Shell Biochar, Constructed Wetlands, Physicochemical Pollutants, Removal Efficiency

INTRODUCTION

Managing wastewater is a critical developmental challenge in many cities across Sub-Saharan Africa. In developing nations, three out of every four cities use untreated wastewater (Raschid-Sally & Jayakody, 2008). In some of these countries, up to 95% of untreated wastewater is discharged directly into the environment. Globally, wastewater generation is estimated at over 380 billion m³ annually, a figure expected to rise to 470 billion m³ by 2030 and 574 billion m³ by 2050 (WHO/UNICEF., 2012). According to Qadir et al. (2020), these estimates reflect an increase of 24% and 51%, respectively. The Global Water Research Group projects that by 2030, annual global water demand will exceed available resources by over 60%, assuming current consumption patterns persist (Tram et al., 2014).

The use of residential or urban wastewater for irrigation has become increasingly vital due to the agricultural sector high water demand and the nutrient content of wastewater (WW). This practice is already widespread in countries like Israel and throughout the European Union (Hochstrat et al., 2008). Utilizing wastewater for irrigation can reduce fertilizer costs, as it contains essential nutrients for plant growth (Carr et al., 2011). Beyond benefits to health and water supply, wastewater reuse has significant environmental implications. Recycling wastewater across various sectors reduces the volume released into natural water bodies, crucial for preserving these ecosystems and associated biodiversity (Kiziloglu et al., 2008).

Constructed wetlands, a decentralized treatment technology, have emerged as particularly effective, sustainable wastewater treatment options for rural and peri-urban areas due to their cost-effectiveness, low maintenance, and environmental sustainability (Parkinson and Tayler., 2003; Chirisa et al., 2017). In Ghana, however, treated wastewater reuse for agriculture remains uncommon, largely due to the high costs of centralized treatment systems and limited government focus on this area. Introduced in Europe during the 1960s, constructed wetland technology has since spread worldwide (Kadlec & Wallace, 2009). This sustainable eco-technology removes pollutants through physical, chemical, and biological processes in a controlled environment, in contrast to natural ecosystems (Kadlec and Wallace, 2009; Mitsch and Gosselink, 2015; Sandoval et al., 2019). These systems rely on three key components: microorganisms, vegetation, and porous filter media (Mitsch and Gosselink, 2015).

Biochar derived from nutshells presents an innovative approach for wastewater treatment. According to Xu et al. (2022), biochar made from walnut shells is a great adsorbent for eliminating estrogenic pollution from water. Preparing biochar from nutshells is therefore a great way to transform waste into something useful. Due to their high nutritional content, nuts account for a significant portion of agricultural production worldwide (Alouw and Wulandari, 2020). In 2020, the world produced 3.2 million tonnes of shelled almonds and 2 million

tonnes of walnuts (Massantini and Frangipane, 2022; Orooji et al., 2022). A significant quantity of nutshell waste is produced by the rising production of nuts. For instance, the shells of coconuts make up around 35% of the entire fruit (Yuliusman et al., 2020). Generally regarded as worthless household waste, nutshell accumulation can cause ecological issues.

This research explores the use of two indigenous macrophyte species (*Eragrostis curvula* and *Paspalum dilatatum*) in combination with sheanut shell biochar in a Vertical Flow Constructed Wetland (VFCW) to treat wastewater for reuse in irrigation.

MATERIAL AND METHODS

Study Area. The experiment was carried out at the WACWISA experimental site at the University for Development Studies, Nyankpala Campus, which is situated at 9° 24' 37" N and 0° 58' 53" (Figure 1). It is situated 15 kilometres from Tamale, the regional capital. The study area is within Guinea Savannah Agro-ecological Zone. The region experiences an arid climate, with unimodal rains occurring from June to November. These rains average 800-1200 mm annually, with August through September experiencing peak rain levels (Gyasi et al., 2014). The region experiences high temperatures, with minimum and maximum values of roughly 22 and 36°C, respectively, with maximum temperatures up to 45°C. The dry season typically lasts seven months, from November to June. The rate of evapotranspiration varies from 4.4 to 6.5 mm/day on average. Most of the soils are alluvial.

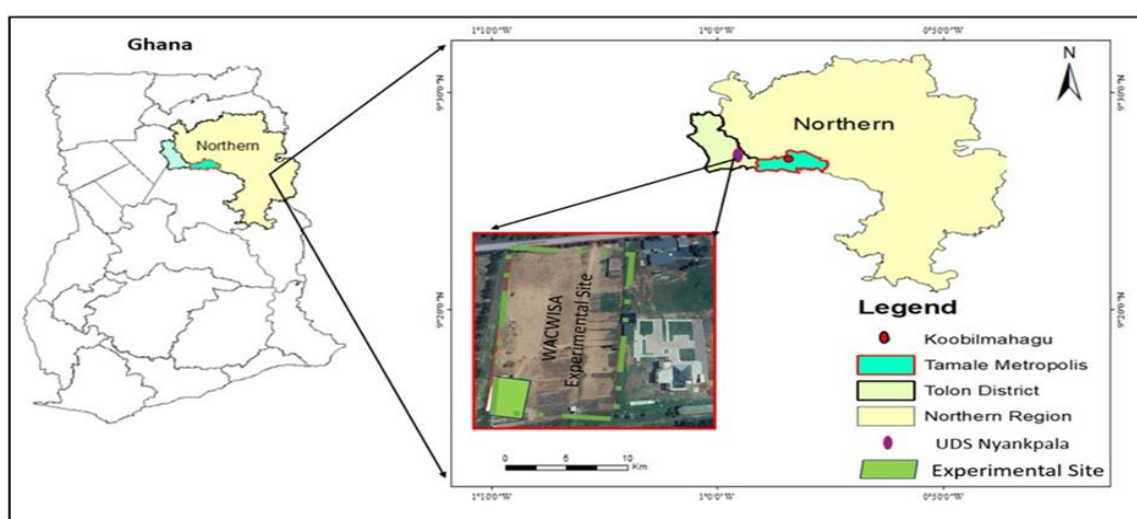


Fig. 1 Study Area

Source of Feedstock. Sheanut shells were used to prepare biochar for the VFCW experiment (Figure 2). The feedstock was sourced from Pagsung shea butter processing company in Sagnarigu community, Tamale metropolis, Ghana. Where natural and raw shea butter is produced locally by native women in the community.

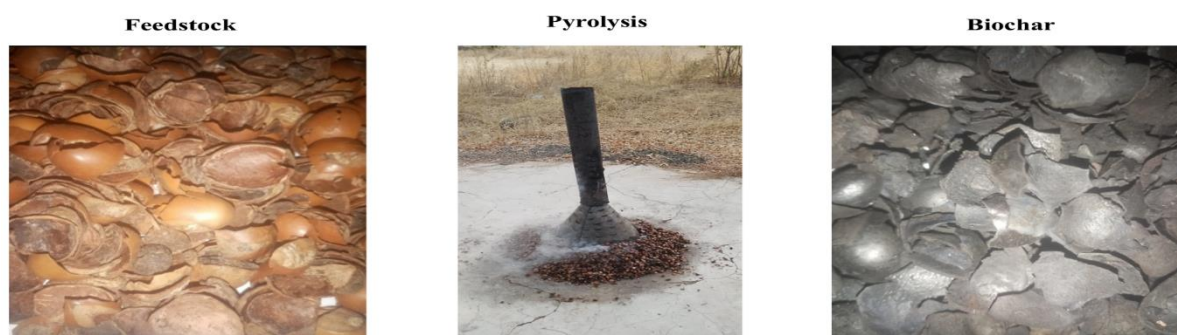


Fig. 2 Biochar Pyrolysis

Experimental Design and Layout. Twenty-two (22) experimental pilots (pots), representing VFCWs with dimension 37cm depth, 24cm width and 26cm length were constructed and all were operational Figure 3. Each pilot had a 15cm drainage layer of gravel (Ø: 9.5-19mm) at the bottom followed on top with coarse sand

(Ø: 0.08-4mm) of either 5cm depth or 15cm. A 5cm biochar layer (Ø: 4-12mm) was then spread in the middle of the 15cm coarse sand layer (Table 1). A uniform freeboard of 5cm was created for each pilot (Figure 4). Perforated tubes were then inserted into the middle of the substrate media to permit aeration within the wetlands. The tubes were wrapped with a rubber mesh to prevent blockage of the perforations during operation.

A split plot experimental design was adopted with six treatments in triplicates (18 pilots) and two controls in duplicate. Two biochar levels (BCH or No-BCH) were the main plot treatments while three macrophyte levels (*E. curvula* or *P. dilatatum* or No-Grass) were considered the sub-plot treatments. The treatments, in no particular order were: *P. dilatatum* (PD) planted with biochar, PD+BCH or No-biochar, PD+WW. Then unplanted beds with biochar, UPB+BCH or No-biochar, UPB+WW. The last treatment combinations were *E. curvula* (EC) planted with biochar, EC+BCH or No-biochar, EC+WW. As for the treatment control, CTRL, freshwater was used in irrigating the wetlands represented as PD+FW and EC+FW.

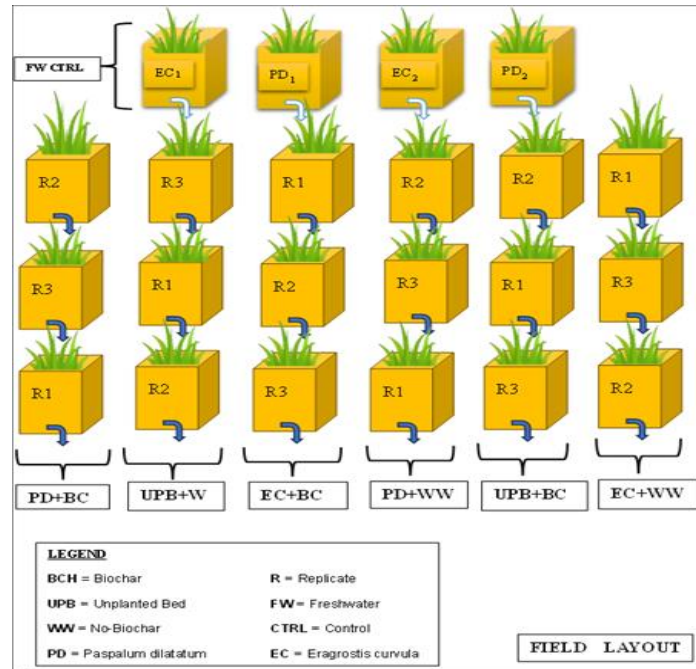


Fig. 3 Experimental Layout

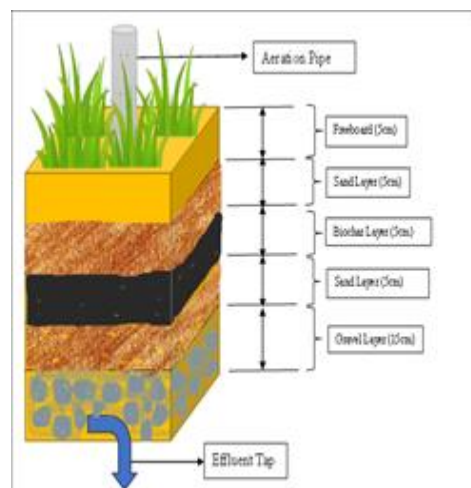


Fig. 4 VFCW Pilot

Effluent Sampling. Treated wastewater (effluent) was sampled from the wetland outlets fitted with three-quarter inch taps at the base of the pilot pots (Figure 4). Effluents from each of the eighteen (18) pilots were

collected in 500 mL sterilized plastic bottles. The collected samples were kept in ice-chests to maintain their microbial conditions and then conveyed (within 24 hours) to the laboratory for analysis. Physical, microbiological, and nutrient analysis were conducted on the samples to evaluate their removal rates from the raw wastewater (influent). Effluent sampling was done every bi-weekly with an average retention time of 5 days.

Statistical Analysis. The variation among macrophyte species average growth rate was analysed for statistical significance. Additionally, the statistical significance of biochar effect on pollutant removal was also analysed. These were determined by analysis of variance (ANOVA) at 5% significance level with the 12th edition of GENSTAT software. Macrophytes morphological properties were subjected to trend analysis and plotted with Microsoft Excel.

RESULTS AND DISCUSSION

Table 1: Influent and Effluent Concentrations from Study

TRT	RAW	EC+WW	EC+BCH	UPB+WW	UPB+BCH	PD+WW	PD+BCH	P-value	LSD
TDS (mg/L)	1235.27	1725.50	2267.25	1499.33	1961.58	1745.17	2127.25	0.006	148
TSS (mg/L)	51.00	6.92	12.42	7.50	8.42	2.58	14.75	0.005	1.9
Turbidity (NTU)	23.72	2.67	13.00	4.50	14.58	2.42	16.25	0.03	8.7
pH	8.18	7.97	7.71	8.86	8.76	8.05	7.68	0.132	0.421
Sal (ppt)	1.02	1.83	2.49	1.60	2.19	1.81	2.46	0.012	0.298
Eh (mV)	-77.00	-65.63	-47.95	-117.22	-112.19	-70.86	-49.31	0.083	19.4
EC (μ S/cm)	1928.73	3445.83	4534.17	2991.75	3927.00	3486.75	4255.00	0.005	276.9
NO ₂ -N (mg/L)	2.35	0.01	0.01	0.01	0.01	0.00	0.03	0.147	0.019
NO ₃ -N(mg/L)	7.47	0.03	0.12	0.76	0.17	0.04	0.06	0.131	0.273
NH ₄ -N(mg/L)	1.07	0.23	0.12	0.08	0.15	0.12	0.14	0.741	0.095
FC (CFU/100mL)	15.00	3.92	3.17	2.25	0.00	2.92	1.91	0.358	5.0
TC (CFU/100mL)	204.00	9.00	9.17	4.17	4.25	7.08	7.50	0.897	6.5

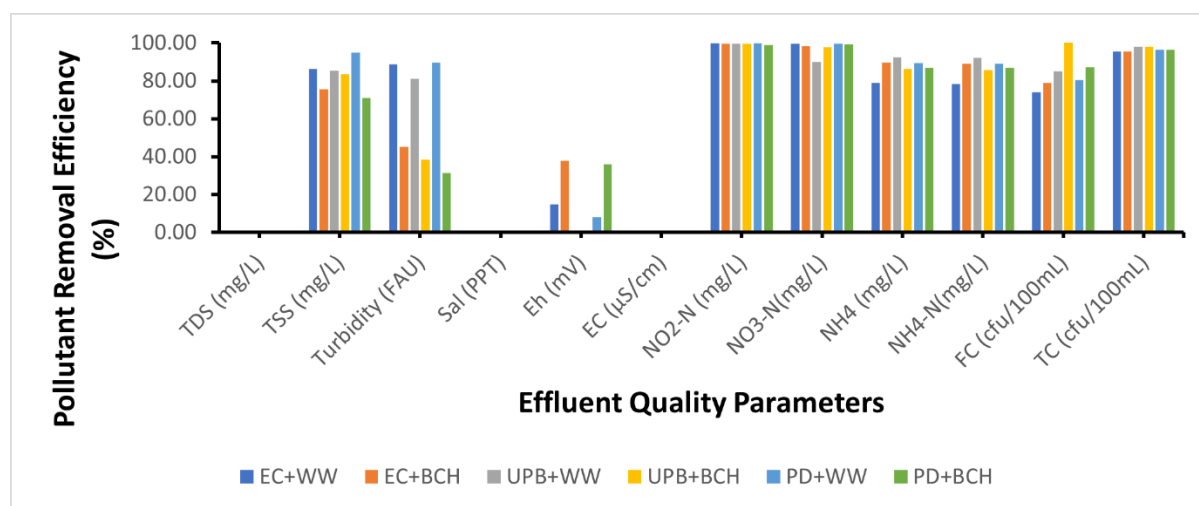


Fig. 5 Pollutant Removal Efficiencies

Table 2: Removal Efficiencies of Wetlands

TRT	EC+WW	EC+BCH	UPB+WW	UPB+BCH	PD+WW	PD+BCH
TDS (mg/L)	-39.69	-83.54	-21.38	-58.80	-41.28	-72.21
TSS (mg/L)	86.44	75.65	85.29	83.50	94.93	71.08
Turbidity (NTU)	88.76	45.19	81.03	38.51	89.81	31.49
Sal (ppt)	-79.81	-144.11	-57.48	-114.99	-77.44	-141.33
Eh (mV)	14.77	37.72	-52.24	-45.71	7.97	35.96
EC (μ S/cm)	-78.66	-135.09	-55.11	-103.61	-80.78	-120.61
NO ₂ -N (mg/L)	99.68	99.38	99.58	99.54	99.93	98.90
NO ₃ -N(mg/L)	99.62	98.37	89.85	97.72	99.45	99.16
NH ₄ -N(mg/L)	78.28	89.22	92.17	85.81	89.22	86.97
FC (cfu/100mL)	73.89	78.89	85.00	100.00	80.56	87.27
TC (cfu/100mL)	95.59	95.51	97.96	97.92	96.53	96.32

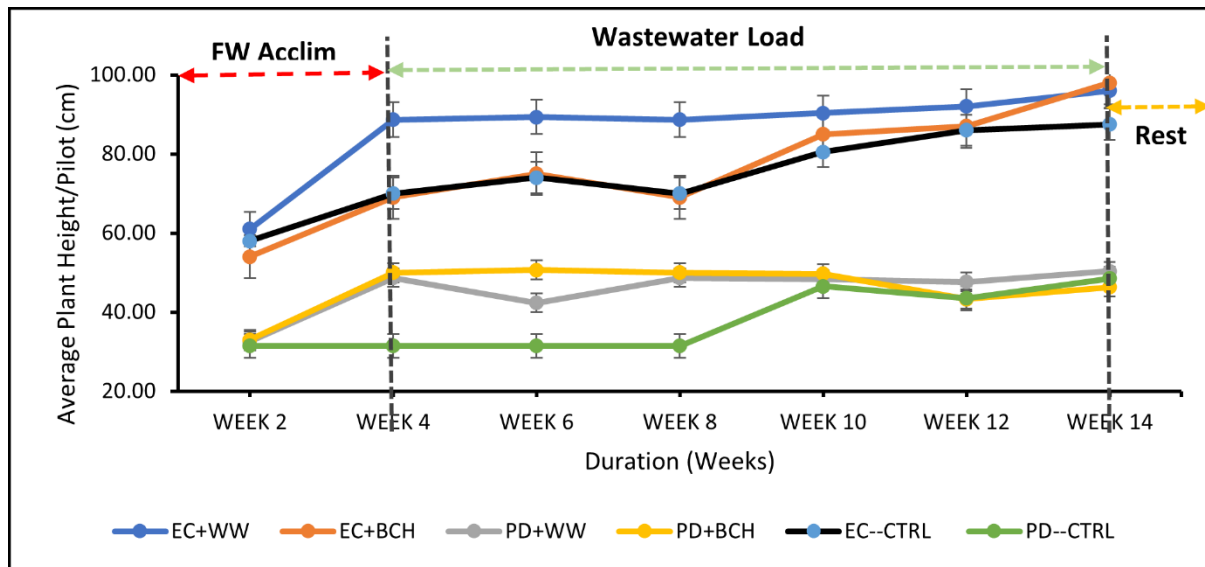


Fig. 6 Average Plant Height/Pilot over Time

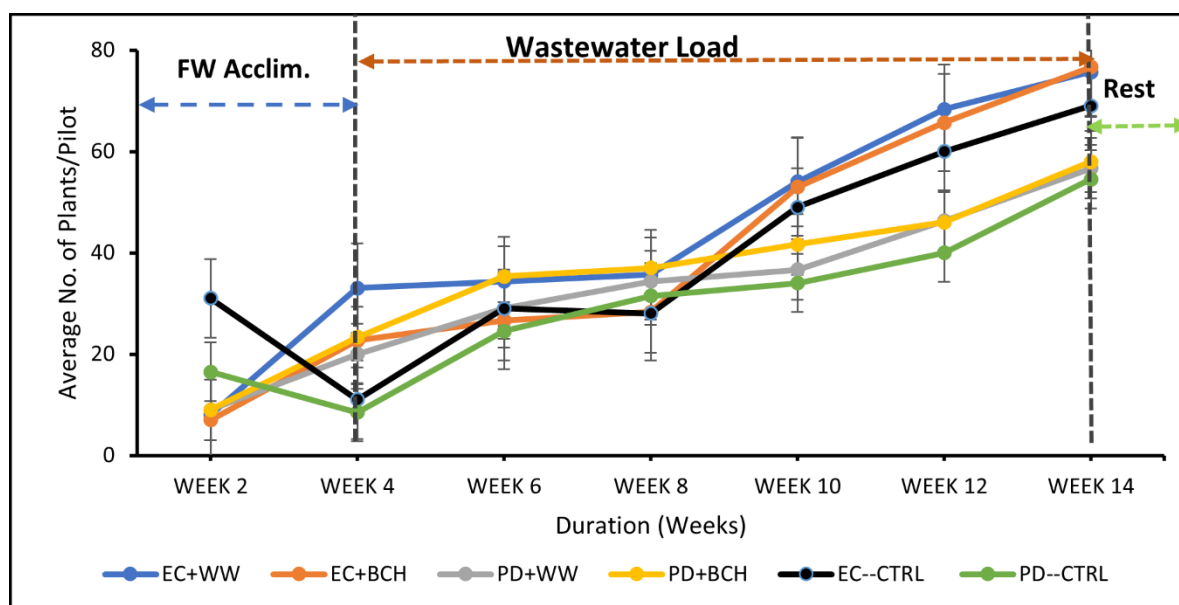


Fig. 7 Average Number of Plants over Time

CONCLUSION

The pollutant removal efficiencies were credible and the treated effluents ranged within the maximum permissible limits for irrigation. Generally, all pollutants were reduced except for TDS, Salinity and Electrical Conductivity which were measured below 30%. Removal efficiencies were as high as 99.93% for nutrients and 100% for faecal coliform for both macrophytes. However, statistical analysis showed significant effects of EC+BCH, and PD+BCH treatments on TDS, TSS, Turbidity, Salinity and Electrical Conductivity. The morphological growth of macrophytes also showed 20% increase for the same treatment wetlands.

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ASSESSMENT OF GROUNDWATER QUALITY FOR DOMESTIC CONSUMPTION IN THE TOLON DISTRICT OF NORTHERN GHANA

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This study assesses groundwater quality for drinking purposes in the Tolon district of the Northern Region, Ghana, utilizing the Water Quality Index (WQI) and Geostatistical Interpolation Tool (GIT). 97 groundwater samples were analyzed during rainy and dry seasons and for 20 physicochemical parameters. The WQI was determined to classify water quality into five categories: Excellent, Good, Poor, Very Poor, and Unsuitable for drinking while GIT was employed to interpolate the distribution of WQI values across the district. The results revealed various contaminants exceeding the standard limits. The WQI values during the rainy season indicate that 89.09% of samples were classified as Excellent, with 7.27% Good, 1.82% Poor, and 1.82% Unsuitable for drinking. In the dry season, 92.86% were Excellent, 4.76% Good, and 2.38% Unsuitable for drinking, with no samples falling into the Very Poor category. GIT analysis revealed that most parts of the study area exhibit excellent water quality.

Keywords: Groundwater, water quality index, geostatistical, physicochemical Parameters

INTRODUCTION

Groundwater plays a significant role in both environmental and human contexts. It is a vital component of the hydrological cycle, serving as a crucial source of water for domestic, industrial, and agricultural use (Findikakis & Sato, 2011). However, the vulnerability of groundwater quality to human activity and climate change is a major concern, as it has implications for human health, the sustainability of rural agricultural economies, and the costs associated with land and water management (Martin-Bordes & Gurdak, 2011). Groundwater quality is essential as it directly affects how water can be used now and, in the future, and it is susceptible to pollution and overuse (Martin-Bordes & Gurdak, 2011). In Ghana and throughout Sub-Saharan Africa, high water demand continues amidst economic growth. Millions of people still rely on potentially unsafe surface water sources, which increases their risk of waterborne diseases (water.org). In the Northern Region of Ghana, limited access to clean water is a major contributor to the prevalence of waterborne illnesses, underscoring the urgent need for improved water management (Cobbina et al., 2010; Cobbina et al., 2021). Furthermore, groundwater contamination with heavy metals in this region poses additional health risks (Asare-Donkor & Adimado, 2020; Cobbina et al., 2021). Regular groundwater quality monitoring is essential to ensure its safety for human consumption and agricultural use (Cobbina et al., 2021; Asare-Donkor & Adimado, 2020; Abagale et al., 2020). This study aims to assess the groundwater quality in the Tolon District of the Northern Region, Ghana, specifically for domestic use. It utilizes the Water Quality Index (WQI) as a comprehensive tool to quantify overall water quality based on several physicochemical parameters. Additionally, the study employs a Geostatistical Interpolation Tool (GIT) to analyze the spatial and temporal distribution of the water quality index across the district.

MATERIALS AND METHODS

Description of Study Area. The study was conducted in northern Ghana's Tolon district, one of the 261 Metropolitan, Municipal, and District Assemblies (MMDAs). Covering approximately 1,355 square kilometers, the district has a population of 118,090 (GSS, 2021). It is situated between latitudes 9° 15' and 10° 02' N, and longitudes 0° 53' and 1° 25' W. Tolon district shares its borders with the Sagnarigu District to the east, Gonja to the west, Kumbungu to the north, and Central Gonja to the south (Figure 1). The district is classified within the tropical continental or interior savannah climatic zone, as defined by Dickson and Benneh in 2004. It experiences a single rainy season from late April to late October. The topography is characterized by undulating land and scattered depressions, with elevations ranging from 87 to 225 meters above sea level, indicating the absence of significant high elevations.

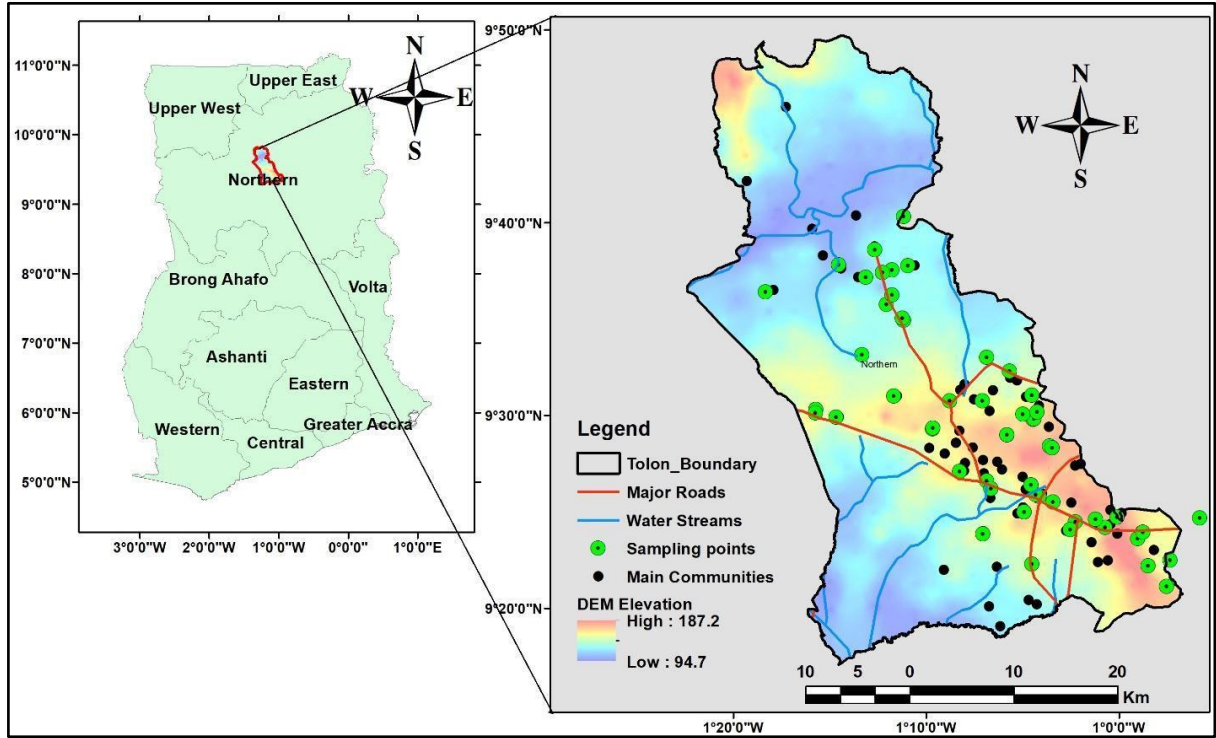


Fig. 1 Study area map

Groundwater sampling and laboratory analysis. Sampling was done in two seasons, the rainy season (August to September 2023) and the dry season (March 2024). 97 groundwater samples were collected from operational wells, with 55 samples obtained during the rainy and 42 during the dry seasons. The sampling locations were recorded using GPS mapping and plotted in Figure 1. Before sampling, aquifers were purged by continuous pumping for 5–10 minutes until stable physicochemical parameters were achieved. Samples were collected in 1-liter polyethylene plastic bottles, pre-cleaned and rinsed, and marked with masking tape for identification. 500 mL polyethylene bottles were used for heavy metal analysis, with pH adjusted to < 2 by adding 10 drops of ultrapure nitric acid (HNO_3). Samples were filtered through $0.45 \mu\text{m}$ filters to remove suspended materials before storage at 4°C in an ice chest and transported to laboratories at WACWISA and the Water Research Institute. Total of twenty hydrochemical parameters, including field measurements of pH, Electrical Conductivity (EC), and Total Dissolved Solids (TDS) using a handheld water quality meter (LAQUA WQ-330). Major cations (Ca^{2+} , Na^+ , K^+) were analyzed at WACWISA using an FP 910-5 flame photometer, while turbidity, nitrate (NO_3^-), and nitrite (NO_2^-) were measured with the hydro-test photometer HT1000. Heavy metals (Cd, Pb, Mn) were determined using the Metalyser[®] HM3000 via Single-point and Multi-point Standard Addition methods. Quick[™] Arsenic Test Kit part number (481396) method was used to test Arsenic (As), sulfate (SO_4^{2-}), fluoride (F^-), and titration method for ion magnesium (Mg^{2+}), carbonate (CO_3^{2-}), bicarbonate (HCO_3^-) and chloride (Cl^-). Total hardness was calculated according to Sawyer and McCarthy (1967), Equation (1).

$$TH \text{ as } \text{CaCO}_3 \left(\frac{\text{mg}}{\text{L}} \right) = 2.5\text{Ca}^{2+} + 4.1\text{Mg}^{2+} \quad (1)$$

The Ionic Balance Error (IBE) was utilized to ensure analytical accuracy, with all groundwater samples demonstrating an IBE within $\pm 5\%$, indicating a satisfactory balance between cations and anions as per established protocols (Friedman & Erdmann, 1982; APHA et al., 2023), Equation 2.

$$IBE = \frac{\sum \text{Cations} - \sum \text{Anions}}{\sum \text{Cations} + \sum \text{Anions}} \times 100 \quad (2)$$

Water Quality Index (WQI). As noted by Ketata et al. (2012) and Tyagi et al. (2013), the Water Quality Index (WQI) is an essential tool for evaluating the overall quality of water. It condenses extensive data into a single, easily understandable value. Numerous studies, including those by Satish Kumar et al. (2016), Khan & Jhariya (2017), Chegbele et al. (2020), and Mussa & Kamoto (2023), have utilized the WQI to determine the suitability of groundwater for drinking. WQI is calculated using the weighted arithmetic index method, emphasizing the influence of individual quality parameters. The process involves several steps:

1. Assigning Relative Weight (W_i): Each parameter is assigned a relative weight (W_i) inversely proportional to its recommended standard (S_i) (Equation 3).

$$W_i = \frac{1}{S_i} \quad (3)$$

2. Calculating Quality Rating (Q_i): The quality rating scale (Q_i) for each parameter is determined by dividing its concentration (C_i) in each water sample by its respective standard (S_i) and multiplying by 100 (Equation 4).

$$Q_i = \frac{C_i}{S_i} \times 100 \quad (4)$$

3. Calculating WQI: The overall WQI is obtained by summing the products of the quality ratings (Q_i) and their respective unit weights (W_i), then dividing by the sum of the relative weights (Equation 5).

$$WQI = \frac{\sum_{i=1}^n Q_i W_i}{\sum_{i=1}^n W_i} \quad (5)$$

The resulting WQIs are classified according to the criteria established by Sahu and Sikdar (2008) (Table 2).

The Inverse Distance Weighted (IDW) geospatial interpolation method was applied to create water quality index (WQI) maps. IDW calculates values at unsampled points by using a weighted average of values from sampled points, with weights inversely proportional to the distance from the point of interest (Isaaks & Srivastava, 1989). This method helps visualize variations in water quality across different areas, providing valuable insights for natural resource management and biological conservation (Li & Heap, 2008).

RESULTS AND DISCUSSION

Groundwater quality for drinking. The summary of hydrogeochemical data from groundwater samples collected during the rainy and dry seasons is presented in **Table 1**. The limits for individual parameters are defined based on their effects when present in excess or insufficient concentrations, as recommended by WHO. Groundwater quality assessments for drinking purposes in the district revealed several contaminants exceeding the permissible limits of WHO. Turbidity levels surpassed acceptable thresholds in 25 samples (45%) during the rainy season and 5 (12%) during the dry season. Electrical conductivity (EC) exceeded limits in 10 samples (18%) in the rainy season and 10 (24%) in the dry season. Elevated levels of calcium (Ca^{2+}) were found in three samples, sulfate (SO_4^{2-}) in three samples, and fluoride (F^-) in two samples. Nitrate (NO_3^-) levels exceeded permissible limits in two samples. Additionally, heavy metals such as arsenic (As) in one sample, manganese (Mn) in four samples, and cadmium (Cd) in two samples were also above acceptable levels.

Water quality index. The Water Quality Index (WQI) was determined to evaluate the overall groundwater quality in the study area. The evaluation considered 20 physicochemical parameters as summarized in Table 1.

WQI values are classified into five levels: Excellent ($WQI < 50$), Good ($50 < WQI < 100$), Poor ($100 < WQI < 200$), Very Poor ($200 < WQI < 300$), and Unsuitable for Drinking ($WQI > 300$). Table 2 details the WQI values alongside their respective percentages. During the rainy season, 89.09% of samples were rated as Excellent, 7.27% as Good, 1.82% as Poor, and 1.82% as Unsuitable for Drinking. In contrast, during the dry season, 92.86% were rated as Excellent, 4.76% as Good, and 2.38% as Unsuitable for Drinking. Overall, groundwater quality appears better in the dry season than in the rainy season, though it should be noted that the dry season analysis excluded heavy metals, which could influence these results. Higher concentrations of TDS, EC, alkalinity, potassium, hardness, fluoride, and chloride contributed to elevated WQI values in certain areas.

Spatial distribution maps of WQI (Fig. 1a & b), derived using Inverse Distance Weighting (IDW) interpolation, indicate that most of the study area has excellent water quality, with isolated locations showing good quality, such as TD2 (Gbulahagu), TD14 (Tolon), TD23 (Wantugu), and TD36 (Namdu), and unsuitable quality for drinking in certain southern regions, including TD42 (Woribogu Kukuo) and TD27 (Yiplagu) during the rainy season, and TD63 (WACWISA) in both seasons. Noted, sample TD42 shifted from Poor to Unsuitable for Drinking between the rainy and dry seasons. These results align with findings reported by Dandge & Patil (2022).

Table 1. Statistical summary of physicochemical parameters in groundwater

Parameter	Unit	Mean	Std. dev.	Min	Max	Mean	Std. dev.	Min	Max
Turbidity	TAU	30.60	75.80	0	520	3.73	11.97	0	65
pH		7.39	0.64	5.91	9.36	7.56	0.33	6.74	8.32
EC	$\mu\text{S}/\text{cm}$	891.18	628.32	24.00	2580.00	1614.75	3872.23	30.4	25700
TDS	mg/L	446.16	314.63	12.04	1290.00	932.16	2079.94	14.90	12870.00
Ca^{2+}	mg/L	7.03	4.68	0.00	16.00	49.40	150.19	4.50	960.00
Na^{+}	mg/L	96.20	64.41	1.00	291.00	163.50	156.22	2.00	853.00
Mg^{2+}	mg/L	23.24	29.62	1.27	157.00	47.65	60.59	1.20	321.50
K^{+}	mg/L	12.41	17.46	0.10	67.00	3.78	2.45	1.00	12.90
CO_3^{2-}	mg/L	45.77	80.12	0.03	488.58	34.60	26.57	0.00	117.00
HCO_3^{-}	mg/L	242.95	197.92	12.20	805.00	403.80	267.85	12.80	1003.70
Cl^{-}	mg/L	52.74	71.89	4.00	424.00	425.02	1336.22	9.70	6277.00
F^{-}	mg/L	0.24	0.38	0.00	1.30	0.36	0.46	0.00	2.08
SO_4^{2-}	mg/L	93.98	106.92	8.70	587.00	26.11	36.50	0.00	136.00
NO_3^{-}	mg/L	12.18	19.23	0.00	79.60	3.12	5.62	0.00	26.40
NO_2^{-}	mg/L	0.06	0.19	0.00	1.14	0.07	0.26	0.00	1.53
TH	mg/L	384.49	256.46	46.16	1561.70	577.54	512.53	9.94	3067.74
As	mg/L	0.001	0.002	0	0.010	0.003	0.008	0	0.045
Mn	mg/L	0.037	0.060	0	0.251	-	-	-	-
Cd	mg/L	0.001	0.004	0	0.024	-	-	-	-
Pb	mg/L	0.003	0.004	0	0.013	-	-	-	-

Table 2. Water quality classification based on WQI value and their respective percentages.

WQI value	Water quality class	Rainy season		Dry season	
		No. of samples	Percentage %	No. of samples	Percentage %
<50	Excellent	49	89.09	39	92.86
50 – 100	Good water	4	7.27	2	4.76
101 – 200	Poor water	1	1.82	-	-
201 – 300	Very poor water	-	-	-	-
>300	Unsuitable for drinking	1	1.82	1	2.38

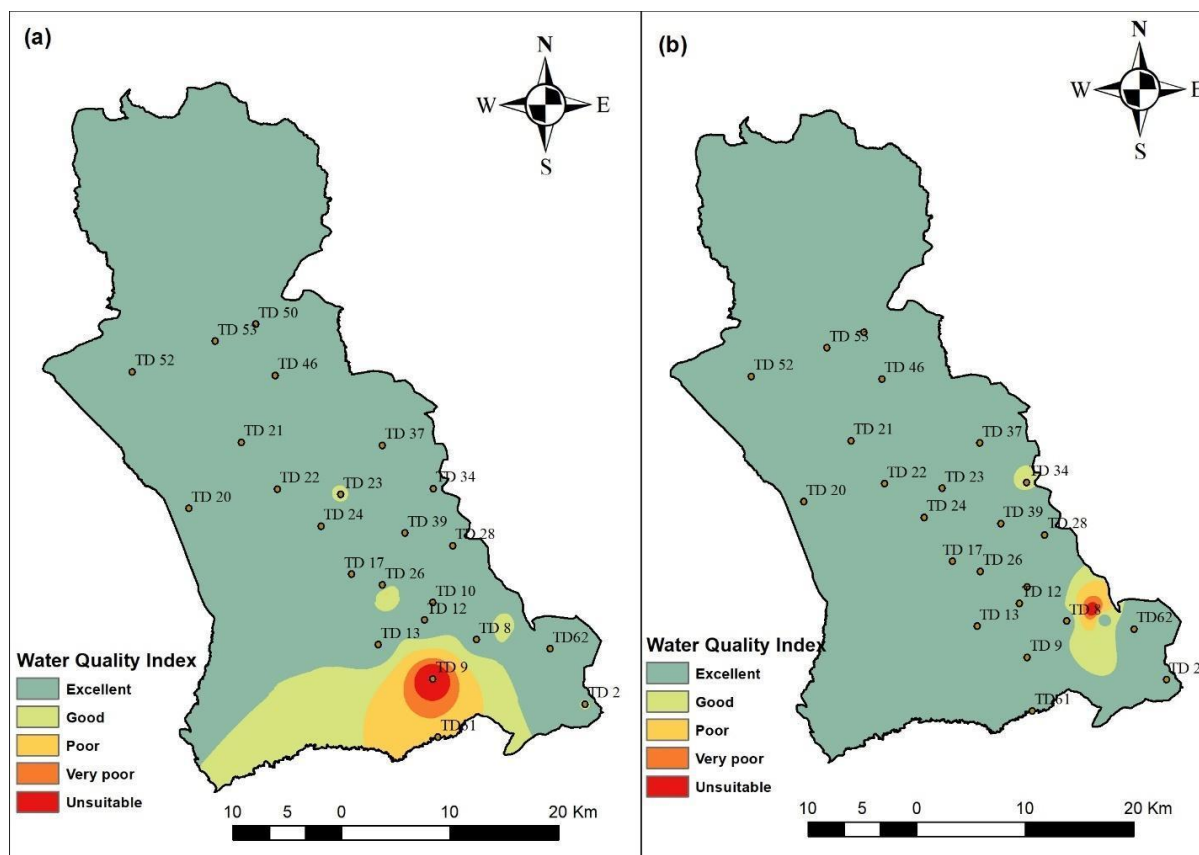


Fig. 2 Spatial distribution of water Quality Index maps (a) rainy season, (b) dry season

CONCLUSION

This study was conducted in the Tolon district of the Northern Region, Ghana, to assess groundwater quality for drinking purposes using the Water Quality Index (WQI) and Geostatistical Interpolation Tool (GIT). 97 groundwater samples were collected from various locations during rainy and dry seasons and analyzed for 20 physicochemical parameters. The WQI was calculated based on Turbidity, pH, EC, TDS, Na^+ , Ca^{2+} , K^+ , Mg^{2+} , CO_3^{2-} , HCO_3^- , F^- , Cl^- , SO_4^{2-} , NO_2^- , NO_3^- , TH, Cd, Mn, Pb, and As to classify water quality into five categories: Excellent, Good, Poor, Very Poor, and Unsuitable for drinking. GIT was employed to interpolate the distribution of WQI values across the district. Limits for individual parameters were established based on limits from World Health Organization (WHO) and Bureau of Indian Standards (BIS) guidelines. Groundwater quality assessments revealed various contaminants exceeding these limits. The WQI values during the rainy season indicate that 89.09% of samples were classified as Excellent, with 7.27% Good, 1.82% Poor, and 1.82% Unsuitable for drinking. In the dry season, 92.86% were Excellent, 4.76% Good, and 2.38% Unsuitable for drinking with no samples falling into the Very Poor category. GIT analysis revealed that most parts of the study area exhibit excellent water quality. In contrast, locations such as Gbulahagu, Tolon, Wantugu, and Namdu exhibit good quality. They are unsuitable for drinking, particularly in southern areas like Woribogu, Kukuo, and Yiplagu during the rainy season. This comprehensive assessment contributes valuable insights for sustainable groundwater resource management and public health protection in the Tolon district, emphasizing areas requiring attention for water quality improvement and management strategies.

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STINK BUG EGG PARASITOIDS FOUND IN NOVI SAD PARKS

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Besides being unpopular in urban areas, due to their smell, stink bugs cause severe damage in agricultural production and overwinter in urban areas. In 2024, stink bug eggs were collected weekly from four parks in Novi Sad. Collection was made by visual inspection in 60 minutes per location. In total, 319 egg masses were collected out of which 221 parasitised. The highest number of eggs was collected in Limanski park, followed by Železnički park, and the same pattern was observed for parasitised eggs. Results of this work indicate potential for biological control of stink bugs in urban areas.

Key words: stink bug eggs, parasitoids, plant hosts, parasitism rate, urban area

INTRODUCTION

The Pentatomidae family encompasses many species that are significant for both ecological and agricultural reasons. Among these, *Halyomorpha halys* (Stål) (the brown marmorated stink bug - BMSB) and *Nezara viridula* L. (the green vegetable stink bug) are important pests in agricultural production and urban areas (Garipey et al., 2014). The brown marmorated stink bug originates from east Asia (Hoebeker and Carter, 2003) and has rapidly spread to the United States and Europe. Its presence in Serbia was for the first time recorded in 2015 near Vršac (Šeat, 2015). Due to its significant polyphagy, over 300 plant species have been identified as hosts for BMSB (Nielsen et al., 2009). The spread of this species is closely linked to globalization and trade, which have facilitated its transport to new regions (Lee et al., 2013). Another invasive species, *N. viridula* is also one of the most important and widely distributed polyphagous species (Todd, 1989). It can be found in tropical and subtropical areas around the world, including the Americas, Asia, Australia and Europe (Waterhouse, 1998). This pest arrived in Serbia in 2008, and significant spread and damage were reported in 2011 in the vicinity of Belgrade, Novi Sad and Sombor (Kereši et al., 2012). The species was described in 1776, and its wide range of hosts has allowed it to adapt and rapidly spread. *Nezara viridula* has numerous hosts from dicotyledonous families as well as several monocotyledonous plants (Kiritani et al., 1965; Todd and Herzog, 1980). Many studies indicate that *N. viridula* primarily targets specific legumes, particularly soybean and pea, which negatively impacts the quality and quantity of these crops (Jones and Sullivan, 1982). This research involves monitoring of the stink bugs egg masses from the family Pentatomidae, focusing specifically on the species *H. halys* and *N. viridula*, at the same time examining the bugs' selection of perennial plant species for egg laying in urban and semi-urban environments. The goal of this research was to investigate the presence of parasitoids in stink bug egg masses and their potential role in controlling the stink bug populations. This analysis aims to enhance our understanding of the interactions between stink bugs and their natural enemies, which can serve as biological control agents for pest populations in urban environments.

MATERIAL AND METHODS

During 2024, four localities with diverse plant species were selected in the territory of Novi Sad: Limanski park, Železnički park, Futoški park, and Park Novo naselje, in order to investigate the presence of stink bug egg masses and their natural enemies. From May to September, weekly visual inspections were conducted on randomly selected flowering or fruiting woody and bushy plants to collect stink bug egg samples by examining the undersides of leaves, branches, and trees. Special attention was paid to the species *H. halys* and *N. viridula*, as well as other species of stink bugs from the family Pentatomidae. The research was carried out by one or two observers using a visual method for 60 minutes between 8:00 and 12:00 o'clock AM, according to a pre-defined protocol. Egg masses were collected by hand and placed into sterile containers, with each egg mass placed in a separate container. The egg masses were kept in the entomology laboratory at the Faculty of Agriculture, University of Novi Sad, until the hatching of stink bugs or the emergence of parasitic wasps. This procedure allowed the stink bug species identification based on the morphological characteristics of the eggs (Reeve and Haye, 2021), and the monitoring of any potential parasitoid emergence. The total number of egg masses was counted, with special attention to damaged masses by various predator species and dark (parasitised) egg masses, in all four parks in Novi Sad. Each sample was carefully labeled with the date of collection, locality, and plant species on which the sampling was performed.

RESULTS AND DISCUSSION

The results of the study described showed that the total number of found egg masses in the four parks in Novi Sad was 319. The highest number of stink bug egg masses was recorded in Limanski park, with a total of 111, accounting for 35% of the total number of egg masses found. The number of egg masses in Železnički park was 102, or 32%, in Park Novo naselje there were 60 egg masses recorded, which represents 19%, while the lowest number of egg masses was found in Futoški park, with a total of 46 masses, or 14%, of the total number of egg masses found in 2024 (Figure 1).

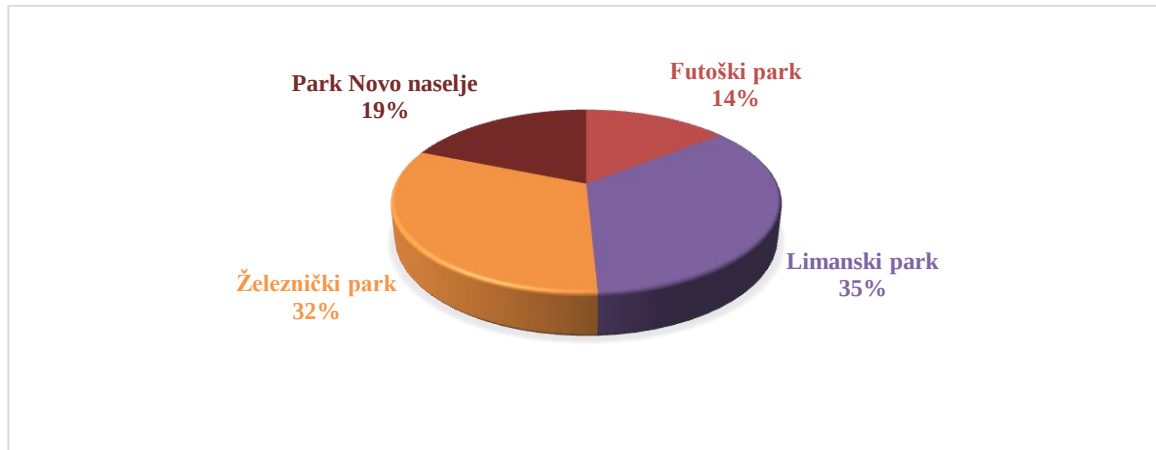


Fig. 1 The percentage of stink bug eggs found in Novi Sad parks from May to September 2024.

The analysis of the sampled stink bug egg masses revealed that 69% of the total number of egg masses were parasitised, indicating a substantial presence of parasitoids within these populations (Figure 2). Vital egg masses, from which stink bugs hatched in the laboratory, accounted for only 18%. The remaining 13% of egg masses were either consumed by predators or found to be damaged and parasitised at the same time, resulting in neither stink bugs nor parasitoids hatching from them.

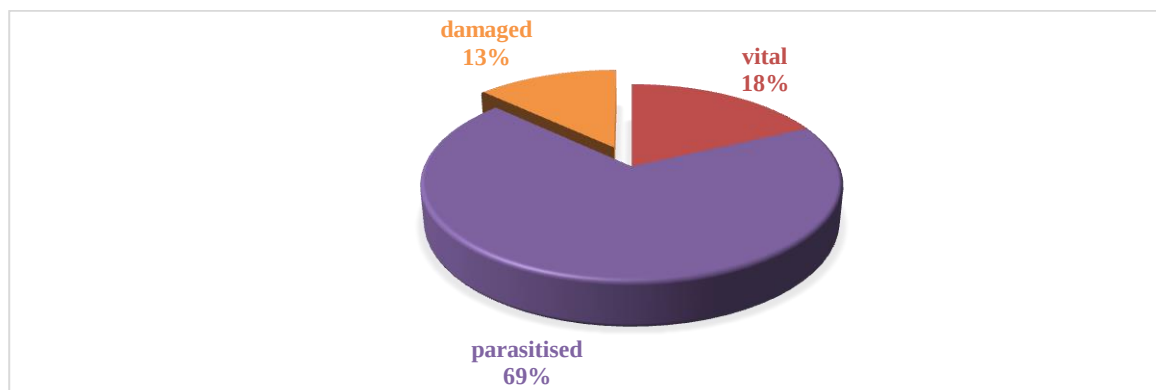


Fig. 2 Percentage of parasitised, viable and damaged egg masses in the total number of eggs found during 2024.

The highest number of egg masses was recorded in Limanski park, where 111 egg masses were identified, of which 86 were parasitised, 14 were viable, and 11 were damaged. (Figure 3). This may indicate specific ecological conditions that attract a higher number of parasitoids to this park. The second park with the highest number of egg masses was Železnički park, where 102 egg masses were found, 54 of which were parasitised, 29 were viable, and 19 were damaged. The parasite-to-stink bug ratio in this park suggests a lower number of parasitoids, as it had the highest number of viable egg masses among all four parks examined. This situation

may imply that, despite favorable conditions for stink bug development, there are unfavorable conditions for parasitoid development. Park Novo naselje had five times more parasitised egg masses than viable ones (45 and 8, respectively), and the same was observed in Futoški park, where a total of 36 parasitised and only 7 viable egg masses were found. The number of damaged egg masses in both parks was fewer than 10 (3 damaged egg masses were found in Futoški Park and 7 in Park Novo Naselje). We can conclude that the number of viable egg masses is more than five times smaller than the number of parasitised egg masses in all parks except for Železnički park, where the ratio favors viable egg masses. Given the dominance of parasitised egg masses over viable ones, this suggests that egg parasitoids contribute to reducing stink bug populations in the urban environment of Novi Sad. These findings highlight the need for additional research to gain a better understanding of the interactions between parasites and hosts, as well as the influence of ecological factors on stink bug populations.

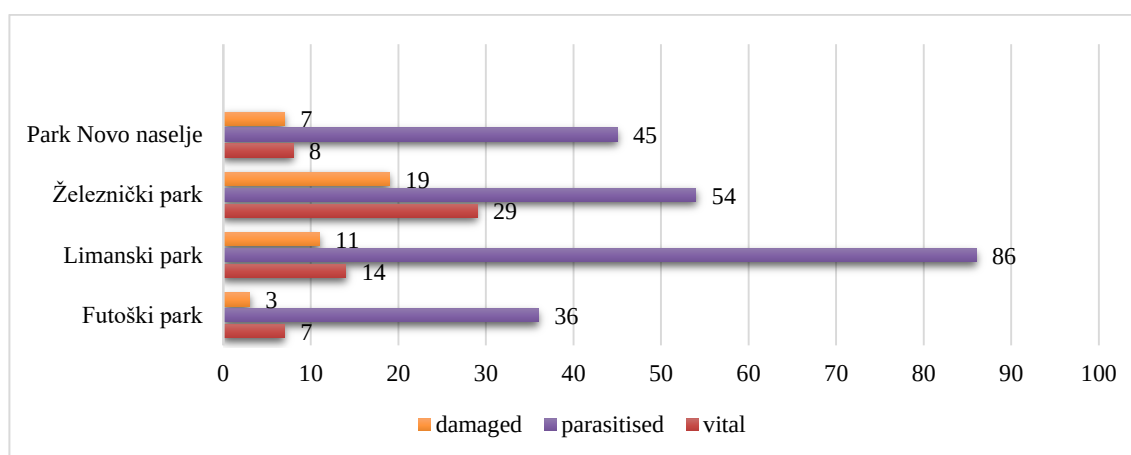


Fig. 3 Abundance of parasitised, viable and damaged egg masses in four parks in Novi Sad.

Among the sampled stink bug egg masses, eggs of *H. halys* were the most frequently recorded, accounting for 73% of the total egg masses found. As expected, this species dominated in all four analyzed locations in Novi Sad. The second most abundant were the egg masses of *N. viridula*, which comprised 19% of the total sample. Other stink bug species, such as *Rhaphigaster nebulosa* (Poda), *Piezodorus lituratus* (Fabricius), and *Palomena prasina* (L.), were recorded with a percentage share of less than 10%.

The egg mass sampling results confirmed that *H. halys* dominated in all four parks, particularly in Limanski park (25%) and Železnički park (24%) (Figure 4). Less number of egg masses of this species was found in Park Novo naselje (15%) and Futoški park (9%), indicating a less frequent presence of this species in these two parks. A similar distribution pattern was observed for other stink bug species, whose distribution, although in lower numbers, closely mirrors that of *H. halys*.

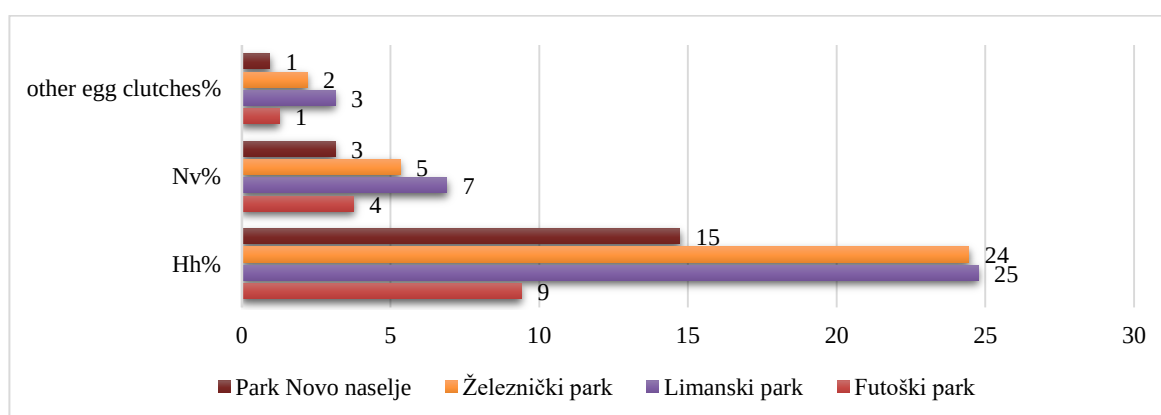


Fig. 4 Percentage distribution of egg masses from specific stink bug species in relation to the total number of egg masses collected in Novi Sad parks.

Stink bug egg masses were found on a total of 23 perennial plant species, belonging to 10 different plant families. The highest number of egg masses was laid on plants from the Fabaceae family, accounting for more than 53% of the total number (Figure 5). Among the plant hosts from this family, the following species were the most frequent: *Styphnolobium japonicum*, *Cercis siliquastrum*, *Cercis canadensis*, and *Robinia pseudoacacia*. These data highlight the importance of the Fabaceae family as a preferred host and habitat for stink bug reproduction, particularly for *H. halys*, which can be found on both woody and cultivated plants from this family, as it is observed in previous studies (Lee et al., 2013). Given that the greatest economic damage occurs on soybeans and various legumes from the Fabaceae family (Owens et al., 2013), these findings regarding the urban environment are crucial for formulating effective pest management and control strategies also for cultivated plants.

Egg masses of stink bugs were recorded on plants from the Sapindaceae family in 16% of cases, while only 10% of egg masses were found on plants from the Oleaceae family. Egg masses of stink bugs on plants from other plant families (Bignoniaceae, Rosaceae, Malvaceae, Cannabaceae, Fagaceae, Betulaceae, Magnoliaceae) were found in negligible numbers, suggesting these plant species are less attractive to stink bugs. These results are consistent with the findings from studies in Switzerland and Slovenia (Haye et al., 2014; Rot et al., 2021), where *H. halys* was also identified on host plants from the *Prunus* genus (family Rosaceae), *Acer* genus (family Sapindaceae), as well as on the plants from the Oleaceae, Bignoniaceae and Malvaceae families.

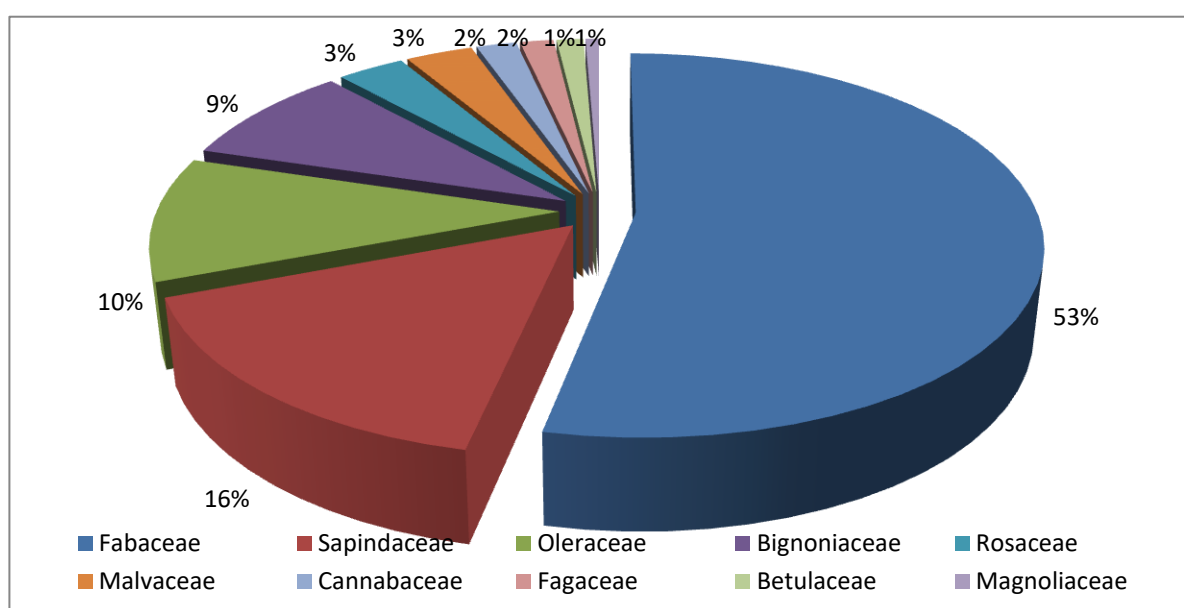


Fig. 5 Percentage distribution of stink bug egg masses on plant host families

The data obtained from this study indicate a significant potential for parasitoids to regulate stink bug population density in Novi Sad. Specifically, the number of parasitised egg masses exceeds that of viable egg masses. Limanski Park has the highest number of egg masses and also the highest level of parasitism, suggesting that specific ecological conditions in this area favor both the pest and its parasitoids. In contrast, Železnički Park, while it has the highest number of viable egg masses, also records a considerable number of parasitised eggs. This indicates that the conditions that support stink bug development also tend to promote the presence of parasitoids. Park Novo naselje recorded the highest rate of parasitism in relation to the total number of egg masses, while Futoški Park showed a similar trend, although it had fewer samples. These data again indicate that egg parasites might play a significant role in regulating the stink bug population. However, additional detailed studies on the developmental dynamics and diversity of parasitoids are necessary to gain a clearer understanding of their interactions with stink bugs and the ecological factors that influence this relationship.

CONCLUSION

The analysis of collected stink bug egg masses from four parks in Novi Sad revealed a total of 319 egg masses belonging to the family Pentatomidae. Of these, 69% were parasitised, indicating a relatively high

presence of parasitoids within the stink bug populations in the urban environment of Novi Sad. Viable egg masses comprised 18% of the sample, while also 40 damaged egg masses, accounting for 13%, were found. This damage not only suggests the presence of parasitoids but also indicates that predators are active in these parks.

Limanski park stands out as the location with the highest number of egg masses, as well as the highest number of parasitoid eggs (of 111 masses, 86 were parasitised 14 were viable, and 11 were damaged). It is followed by Železnički park with 102 egg masses (54 parasitised, 29 were viable, and 19 were damaged), then Park Novo naselje with 60 egg masses (45 parasitised, 8 viable, and 7 were damaged), and Futoški park with 46 egg masses (36 parasitised, 7 viable, and 3 were damaged).

In terms of species, the dominant species in the parks was *H. halys*, whose egg masses accounted for 73% of the total number of collected samples, while *Nezara viridula* ranked second with a 19% presence. Other species from the Pentatomidae family, such as *Rhaphigaster nebulosa*, *Piezodorus lituratus*, and *Palomena prasina*, were found in only 8% of the collected samples. The dominance of *H. halys* in all parks, with the highest percentage in Limanski (25%) and Železnički park (24%), suggests that common ecological factors in the urban environment influence the distribution of these species.

Stink bug egg masses were identified on 23 perennial plant species, distributed across 10 plant families. The highest percentage of egg masses (53%) was found on host plants from the Fabaceae family. Plants from the Sapindaceae family hosted 16% of the egg masses, while those from the Oleaceae family hosted only 10%. Other families, including Bignoniaceae, Rosaceae, Malvaceae, Cannabaceae, Fagaceae, Betulaceae, and Magnoliaceae, were less favorable for egg laying, with less than 10% of the total egg masses found on these plants.

This study confirms the presence of parasitoids in the parks of Novi Sad, which likely plays a significant role in the decline of stink bug populations in urban areas. These findings provide valuable insights that can foster the development of effective biological control strategies for managing pest species in both urban and agricultural areas. Moreover, they offer basic guidance for creating management plans aimed at controlling pest stink bug populations and enhancing our understanding of the ecological dynamics within the ecosystems that these species inhabit.

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COMPATIBILITY OF THIAMETOXAM WITH NATIVE STRAINS OF ENTOMOPATOGENIC NEMATODES *HETERORHABDITIS BACTERIOPHORA* AND *STEINERNEMA FELTIAE*

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This study evaluates the compatibility of native entomopathogenic nematodes, Heterorhabditis bacteriophora and Steinernema feltiae, with the insecticide thiamethoxam. Conducted under laboratory conditions, nematode viability and mortality were assessed at intervals of 1, 3, 24, 48, and 72 hours after incubation with thiamethoxam at recommended (100%) and elevated (120%) concentrations. Both tested nematode species showed compatibility with thiamethoxam, suggesting their potential to be tank mixed.

Key words: Actara 25 WG, *Heterorhabditis bacteriophora*, *Steinernema feltiae*, tank mix

INTRODUCTION

Pest resistance to insecticides is recognized as a significant global threat, highlighting the necessity for alternative pest management approaches (Negrisoli et al., 2010). Due to a significant availability of chemical pesticides, ecologically sustainable methods became the most important, such as the use of entomopathogenic nematodes (EPNs). Today, we know about the nine families of entomopathogenic nematodes, which are microscopic organisms. However, only certain species within the families Heterorhabditidae, Mermithidae and Steinernematidae are capable of infecting insects (Oštreč, 2001). The genus *Steinernema* includes 113 species, while *Heterorhabditis* includes 21 species, and the number is constantly increasing (Pūža and Machado, 2024). Entomopathogenic nematodes are associated with symbiotic bacteria from the family Enterobacteriaceae, including *Xenorhabdus* in Steinernematidae and *Photorhabdus* in Heterorhabditidae (Koppenhöfer, 2007). Insect larvae infected by *S. feltiae* are identifiable by their beige or brown coloration, while *Heterorhabditis* infective larvae exhibit a red-orange hue that darkens over time (Gulcu et al., 2017; Sagun et al., 2015). Thiamethoxam is a globally utilized neonicotinoid insecticide known for its systemic and contact activity, enhancing its effectiveness against diverse insect pests across multiple agricultural systems. It belongs to the neonicotinoid group of insecticides, and it has been banned in Croatia since 2023. It targets various important insect pests from orders Lepidoptera, Hemiptera, Coleoptera, Thysanoptera and Diptera (Tomlin, 2000; Uneme, 2011). The EPNs are compatible to tank mix with different chemical pesticides (Laznik and Trdan 2014). The aim of this study is to determine the impact of the insecticide thiamethoxam on the viability and mortality of entomopathogenic nematodes *Heterorhabditis bacteriophora* and *Steinernema feltiae*.

MATERIAL AND METHODS

The study examined the effects of the thiamethoxam-based insecticide (Actara 25 WG) on two native entomopathogenic nematode species, *H. bacteriophora* ISO9 (MG944244) and *S. feltiae* ISO18, which were Croatian strains reared in the Nematology Laboratory at the Faculty of Agrobiotechnical Sciences, Osijek. The infective juveniles (IJs) were stored at 4 °C at a density of 2000 IJ ml⁻¹. The concentration of the IJs suspension used in the experiment was calculated according to Laznik et al. (2010). The nematode viability was determined prior to the initiation of the experiment, and only nematode stocks with >95 % survival were used. The experiment was conducted under controlled laboratory conditions, with three treatment groups and four replicates. Control treatments were carried out using distilled water. The nematodes were exposed to insecticide concentrations at 100% and 120% of the recommended dose. Nematode viability and mortality were assessed by extracting a 50 µm droplet from each treatment and replicate using an automatic pipette, with four replicates per sample. Viability and mortality were monitored at 1, 3, 24, 48, and 72 hours after the nematodes were mixed with the insecticide. To assess the infectivity and pathogenicity of the nematodes, surviving individuals were cleansed of insecticide residues. Solutions from Petri dishes were transferred to Falcon tubes and centrifuged at 2200 RPM for 2 minutes. Excess water was removed, and distilled water was added, repeating this process twice more. The final solution was stored at 4°C until application to insects. Nematodes used in the experiment were

used to infect two *G. mellonella* larvae. The mortality of the larvae was monitored for seven days post treatment. The experiment was conducted in darkness at room temperature, and nematodes were counted under a stereomicroscope (Olympus SZX 16). Data were analyzed using GLM and ANOVA, with means compared using Tukey's test at $\alpha = 0.05$.

RESULTS AND DISCUSSION

The mortality rates of *H. bacteriophora* and *S. feltiae* depending on the insecticide concentrations and exposure durations, is shown in Table 1. The results indicate that significant differences in mortality were primarily observed at extended exposure intervals (72 h post mixing).

Table 1. Mean mortality rates of *H. bacteriophora* and *S. feltiae* mixed with thiametoxam at different insecticide concentrations

EPN Species	Insecticide Concentration(% Recommended dose)	Incubation (h)				
		1	3	24	48	72
<i>Heterorhabditis bacteriophora</i> ISO9	0	6.25 a	0.00 a	5.55 a	6.25 a	0.00 a
	100	20.70 a	13.22 a	12.65 a	2.50 a	18.81 b
	120	11.23 a	7.04 a	5.40 a	2.72 a	0.00 a
<i>Steinernema feltiae</i> ISO18	0	4.55 a	14.77 a	22.88 a	5.00 a	10.50 a
	100	4.55 a	17.50 a	21.27 a	18.45 b	27.45 b
	120	9.55 a	5.00 a	24.40 a	15.39 a	23.18 a

Values in the column for each nematode species with different letters indicate statistical significance ($P < 0.05$).

The mortality rate for *H. bacteriophora* ranged from 3.16% to 17.91%, while for *S. feltiae*, it varied from 5.79% to 23.36%. For *H. bacteriophora*, the mean nematode mortality rates showed no significant differences across insecticide concentrations during the first 48 hours. However, after 72 hours, a statistically significant increase in mortality was observed, with the 100% pesticide concentration treatment differing notably from the other treatments.

Regarding *S. feltiae*, no statistically significant differences were observed between insecticide concentration treatments after 1, 3, and 24 hours of incubation. However, at 48 and 72 hours, a statistically significant increase in mortality was detected in the 100% pesticide concentration treatment compared to the others. At 48 hours, the 100% concentration resulted in a mortality rate of 18.45%, versus 5% in the control treatment. After 72 hours, mortality at 100% concentration increased to 27.45%, compared to 10.50% in the control. The natural mortality of *S. feltiae* (up to 22.88%) was considerably higher than that of *H. bacteriophora* (up to 6.25%).

Monitoring the viability and mortality of the two nematode species showed that the insecticide Actara 25 WG stimulated the movement of surviving infectious larvae of *H. bacteriophora*. In contrast, *S. feltiae* nematodes were more active in the control treatment than when combined with the insecticide, while *H. bacteriophora* exhibited the opposite effect.

The both greater wax moth larvae died in the pathogenicity bioassay from survived nematodes incubated in thiamethoxam. This proved that thiametoxam did not influence on EPNs infectivity and pathogenicity. Our research demonstrated that there is a difference between EPN species in terms of their compatibility when mixed with thiamethoxam, depending on the exposure duration to the insecticide. After 72 hours, both entomopathogenic nematodes exhibited statistically significant increases in mortality at the 100% pesticide concentration.

Several authors have investigated the compatibility of different insecticides on *H. bacteriophora* and *S. feltiae* to enhance integrated pest management. Erdem (2024.) specifically studied the compatibility of these two entomopathogenic nematode species with four commonly used insecticides: flubendiamide, deltamethrin, betacyfluthrin and spinetoram. The findings indicated that deltamethrin and flubendiamide demonstrated the highest compatibility among insecticides with EPNs, showing no significant negative effects on their survival, infectivity, or reproductive capacity. Yan et al. (2019) mixed several low-toxicity insecticides with *S. feltiae*,

finding that fourteen different insecticides had no significant effect on its mortality or infectivity. Synergism was observed when imidacloprid was combined with *S. feltiae* against *Bradysia odoriphaga* larvae, resulting in successful suppression of the larvae population and a significant increase in crop yields. Koppenhofer et al. (2000) studied the interaction between imidacloprid and three nematode species in the laboratory to control root-knot nematodes, revealing that *H. bacteriophora* exhibited the highest compatibility for mixing with this insecticide. On the other hand, Wu et al. (2017) tested six EPNs for their virulence against *Bradysia odoriphaga* larvae and examined their interactions with thiamethoxam under laboratory and greenhouse conditions, highlighting that the synergistic combination of *S. feltiae* with thiamethoxam could be a promising strategy for integrated pest management. In contrast, our research suggests that *H. bacteriophora* is also compatible with thiamethoxam, demonstrating that both *S. feltiae* and *H. bacteriophora* can be effectively integrated with this insecticide for enhanced pest control. In contrast to the findings of our research, which suggest that higher insecticide concentrations had a weaker effect on EPNs compared to lower concentrations, Atwa et al. (2017) reported that the mortality rate of *H. bacteriophora* increased progressively with both the concentration and exposure duration to insecticides. This difference may be attributed to the use of different insecticide types or EPN strains.

CONCLUSION

This study reveals that *H. bacteriophora* and *S. feltiae* showed different levels of compatibility with thiamethoxam, with both species showing increased mortality at prolonged exposure, particularly at 100% pesticide concentration after 72 hours. Although *H. bacteriophora* had lower natural mortality than *S. feltiae*, maximum mortality for both species remained within an acceptable range. This research suggests that both nematode species can be mixed with thiamethoxam (Actara 25 WG); however, additional field studies and laboratory tests on various pest species are recommended to validate these results. These findings emphasize the importance of selecting compatible EPN-insecticide pairings to optimize pest control while minimizing chemical use in integrated pest management (IPM) programs.

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ANTIMICROBIAL ACTIVITY OF OLIVE OIL (*OLEA EUROPAEA*) SUPPLEMENTED WITH FIG (*FICUS CARICA*) FRUIT

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The latest progress in the modern food industry strives for natural. Nowadays, antimicrobial activities of extracted olive oil in combination with another type of fruit, are the subject of many research studies. Their possible use as natural food additives is particularly prominent. This study analyzed the influence of extracted extract of olive oil in combination with dried fruits of fig on the growth of pure bacterial culture using the disk diffusion method. Escherichia coli, Staphylococcus aureus and Salmonella spp. were used as test organisms for antimicrobial susceptibility testing. The results showed that this type of oil inhibits bacterial growth, although the inhibition rate varies between different bacterial species. It certainly depends on the type of plant used for oil extraction. Studies have shown that extracted olive oil mixed with dried fruits of fig can potentially be used in treating diseases caused by these bacteria.

Keywords: antimicrobial activity; olive (*Olea europea*), fig (*Ficus carica*), human pathogens

INTRODUCTION

Extracted extracts of olive oil has been used for years in alternative medicine (Boesl and Saarinen, 2016), In the last decades, the interest in extracted oil in the combination with another plant, mainly fruits, has been increased (Chouhan *et al.*, 2017).), mainly due to the large number of biologically active phytochemicals in their composition. Nowadays, many research shows that this type of extracted oil have antimicrobial activity and are used for the prevention of many infectious diseases as alternative medicines.

The food industry is putting more effort into finding all health benefits that mixture of oils is offering including their antimicrobial activity.

Protecting food from oxidation and preventing microbial proliferation has become a burning issue. Therefore, research has been directed towards extracted oils and plant extracts, as potential natural additives in food (Bellou *et al.*, 2016). Phytochemical studies on the leaves and fruits of the most fig species have shown that they are rich in phenolics, organic acids, and volatile compounds contrary to roots and stems which are not abundant with these compounds. The dried fruits of *Ficus carica* have been reported as an important source of vitamins, minerals, carbohydrates, sugars, organic acids, and phenolic compounds (Jeong WS *et al.*, 2001). Phytochemical studies on *F. carica* revealed the presence of numerous bioactive compounds such as phenolic compounds, phytosterols, organic acids, anthocyanin composition, triterpenoids, coumarins, and volatile compounds such as hydrocarbons, aliphatic alcohols, and few other classes of secondary metabolites from different parts of *F. carica* (Mawa *et al.*, 2013).

Phenolic acids such as 3-O- and 5-O-caffeoylquinic acids, ferulic acid, quercetin-3-O-glucoside, quercetin-3-O-rutinoside, psoralen, bergapten, and organic acids (oxalic, citric, malic, quinic, shikimic, and fumaric acids) have been isolated from the water extract of the leaves of *F. carica* L. (Tavares *et al.*, 2009). The leaves of *F. carica* consist of various volatile compounds which are identified and distributed by distinct chemical classes, such as aldehydes: methyl-butanal, 2-methylbutanal, 2-pentanal, monoterpenes such as β -caryophyllene, β -cubebene, aromadendrene, α -caryophyllene, and miscellaneous compounds such as psoralen.

Many higher plant species have been used for spice and medical application; however, recent research have addressed the application of extracted oils and spice extracts in control of diseases caused by pathogens (Cui *et al.*, 2018; Teles *et al.*, 2019) and improvement of food safety.

Its fruit, root, and leaves are used in traditional medicine to treat various ailments such as gastrointestinal, respiratory and cardiovascular disorders and as anti-inflammatory and antispasmodic remedy (Collins *et al.*, 1993).

MATERIAL AND METHODS

The study analyzed the effect of extracted extract of olive oil in the combination with dried fruits of fig on the growth of bacteria from the *Enterobacteriaceae* (*Escherichia coli* and *Salmonella* spp.) and the *Staphylococcaceae* family (*Staphylococcus aureus*). Oil extraction from olive oil in combination with dried fig fruits was carried out in the Microbiology Laboratory at the Faculty of Agriculture and Food Sciences in Sarajevo.

So, for this type of study, extra virgin oil was used which was placed in a sterile jar, previously cleaned with distilled water, in which were added selected parts of dried fig fruits. The sterile jar stayed in the dark room at the temperature of 5°C for forty days. During that period, plant extracts have been extracted. Later, with maceration technique, oil was extracted through a filter which was previously sterilized in the autoclave. The final extract was analyzed later for the antimicrobial susceptibility testing of extracted oil on the growth of *S. aureus*, *E. coli* and *Salmonella* spp.

Pure bacterial cultures were inoculated on Miller-Hinton agar, on which was with sterile tweezers, applied sterile paper discs previously immersed in extracted oil.

Incubation was carried out at a temperature of 37 for 24 hours for *Escherichia coli*, *Salmonella* spp. and *Staphylococcus aureus*

Antimicrobial susceptibility testing of extracted oil on the growth of *S. aureus*, *E. coli*, and *Salmonella* spp., was done using the disk diffusion method. This method is based on the diffusion of extracted oil through a filter paper applied on the surface of the own substrate. Mueller-Hinton's (MH) agar was used as the experimental medium. 18- hour cultures of *S. aureus* (ATCC 25923), *E. coli* (ATCC 25922) and *Salmonella* spp. were used for the experiment, which were seeded in a nutrient medium and incubated at 37°C. Petri dishes with MH medium were inoculated with 0.1 ml of bacterial suspension. Filter paper sterile disks with a 6 mm diameter were placed on the surface of prepared Petri dishes using aseptic technique. Each disc was then immersed with 10 µl of essential oil. The experiment was performed in six replications. Petri dishes were incubated for 24 h at 37 °C. The results were examined after 24 hours by measuring the diameter of inhibition zones. Afterward, the effect of the extracted oil was determined.

RESULTS AND DISCUSSION

Research was done to analyze the antimicrobial activity of extracted olive oil enriched with dried fruits of fig on three human pathogens using the disk diffusion method. The results have been established using the standard values for the inhibition zones of netilmicin and tetracycline antibiotics for *Staphylococcus aureus*, *Escherichia coli*, and *Salmonella* spp. (Table 1).

Table 1: Standard values for the inhibition zones of netilmicin and tetracycline antibiotics for *Staphylococcus aureus*, *Escherichia coli* and *Salmonella* spp.

Human pathogens	Antibiotics	Inhibition zone (mm)		
		S	I	R
<i>Escherichia coli</i>	Netilmicin	≥15	13-14	≤12
	Tetracycline	≥15	12-14	≤11
<i>Staphylococcus aureus</i>	Netilmicin	≥15	13-14	≤12
	Tetracycline	≥19	15-18	≤14
<i>Salmonella</i> spp.	Netilmicin	≥15	13-14	≤12
	Tetracycline	≥15	12-14	≤11

Legend: S – sensitive; I – moderately sensitive; R – resistant

For *E. coli*, extracted olive oil has an average inhibition zone of 11 mm (Table 2), and therefore belongs to category I (moderately sensitive) compared to both types of antibiotics. Correspondingly with the results, extracted olive oil shows a medium antimicrobial and therapeutic effect. In comparison with netilmicin antibiotic *E. coli* is resistant to netilmicin and correspondingly the extracted olive oil has less therapeutic effect. In comparison with the tetracycline, this type of oil belongs to the R category (resistant) which leads to the conclusion that *E. coli* is resistant to extracted olive oil and does not have the same effect as the afore-mentioned antibiotic in an increased dosage. Extracted olive oil has a strong inhibitory effect on *S. aureus* with an average

value of the inhibitory zone of 17.33 mm, which is included in category S (sensitive) for netilmicin, and tetracycline belongs to category I (moderately sensitive). For *Salmonella* spp., extracted olive oil enriched by dried fig fruits has an average inhibition zone of 17.16 mm and therefore belongs to S category (sensitive) for both antibiotics. This type of oil mixture has strong potential to inhibit the growth of *Salmonella* spp. and act therapeutically.

Table 2. Impact of extracted olive oil enriched with dried fig fruits on the growth of *Escherichia coli*, *Staphylococcus aureus* and *Salmonella* spp.

Human pathogen	Extracted olive oil	<u>Inhibition zone (mm)</u>						AVG ±SD
		<u>Discs</u>						
		1	2	3	4	5	6	
<i>E. coli</i>	Extracted olive oil with dried fruits of fig	10	9	8	11	10	18	11±3.58
<i>S. aureus</i>	Extracted olive oil with dried fruits of fig	12	22	24	13	13	20	17.33±5.28
<i>Salmonella</i> spp.	Extracted olive oil with dried fruits of fig	12	22	23	13	13	20	17.16±5.04
Legend: AVG – average; SD - standard deviation								



Fig. 1 Results of the action of extracted olive oil mixed with dried fig fruits on *E. coli*

CONCLUSIONS

The disk diffusion method confirmed that extracted extract of olive oil mixed with dried fig fruits has antimicrobial activity and that in certain conditions they prevent the growth of some human pathogens. Correspondingly with the results, extracted oil shows a medium antimicrobial and therapeutic effect on *E. coli*. Extracted olive oil has a potential to inhibit the growth of *Salmonella* spp. and act therapeutically.

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CHARACTERIZATION OF BACTERIAL STRAINS ISOLATED FROM LETTUCE

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*This study examines bacterial strains isolated from lettuce to identify pathogens affecting its quality. Two bacterial groups, *Pseudomonas* spp. and *Xanthomonas* spp., were isolated from symptomatic lettuce samples. Biochemical and molecular tests, including REP-PCR, classified the pathogens, confirming their potential impact on lettuce health. The findings underscore the need for effective management of these bacterial pathogens to prevent crop losses and maintain quality.*

Key words: Lettuce, *Pseudomonas*, *Xanthomonas*, REP-PCR, 16S rRNA.

INTRODUCTION

Lettuce (*Lactuca sativa*) is one of the most commonly grown vegetable crops around the world. Nutritiously rich composition, availability throughout the year and low price are the features that make consumers choose to buy and use lettuce on a daily basis.

Lettuce is a plant that belongs to the Asteraceae family, a large botanical family whose species are found all over the world. This family has almost 30 000 species, which is why it is considered one of the most numerous botanical families (Bremer et al., 1992). Based on its morphometric characteristics, the genus *Lactuca* is included in the subfamily Cichorioideae. According to the latest data (FAO, 2022), annual production of lettuce worldwide reaches about 27 million tons. More than half of world's lettuce is produced in China, with approximately 14,9 million tons harvested in 2022. Following China, the world's leading lettuce producers are the USA, Spain, India, Italy and Japan.

The nutritional value of lettuce largely depends on the variety that is consumed. Ascorbic acid (vitamin C), vitamin A, vitamin K, carotenoids (β -carotene, zeaxanthin and lutein) and folates are just some of the compounds that salad is rich in. Green salad is the most commonly consumed leafy vegetable. With a water content of up to 95% and its low-calorie nature, lettuce is particularly well-suited for various dietary plans. While other vegetables may contain higher nutrient levels, the fresh consumption of lettuce often offers greater health benefits compared to thermally processed ones (Still, 2007).

The structure, shape, and color of lettuce leaves vary greatly due to years of breeding that have led to the development of multiple varieties. Leaf types are the most variable, with a large number of combinations in color, shape, size, texture, and leaf margin. The most common type is head lettuce (*L. s. var. capitata*), which is divided into butterhead or butter types, characterized by very tender leaf veins, and crisphead (commonly known as Iceberg) with prominent leaf veins. Butterhead varieties form spherical heads weighing between 200 and 400 grams. Their leaves are pliable and appear to have a fatty texture. Iceberg lettuce is recognizable by its thick and crunchy leaves that also form a spherical head weighing between 800 grams and one kilogram. Romaine lettuce (*L. s. var. romana* subvar. *longifolia*) forms elongated heads, averaging around 750 grams, with upright leaves that are much longer than they are wide. In addition to the aforementioned varieties, there are also looseleaf lettuce (*Lactuca sativa*), cutting lettuce (*L. s. var. acephala*), asparagus lettuce (*L. s. var. angustana* subvar. *asparagina*) and others (Ryder, 1999; Rodenburg & Basse, 1960). There are also varieties of lettuce cultivated for producing fruit, or seeds, and for extracting oil from them (Simko et al., 2014).

The importance of knowing the biotic factors that threaten lettuce production is crucial for vegetable growing. Various microorganisms can be pathogenic to lettuce, including fungi, viruses, bacteria and phytoplasmas. The most important fungi that cause lettuce diseases are: *Botrytis cinerea*, *Erisiphae cichoracearum*, *Rhizoctonia solani*, *Bremia lactucae*, *Fusarium oxysporum* f. sp. *lactucae*, *Verticillium dahliae*, *Sclerotinia sclerotiorum* and *S. minor*, as well as pseudofungi such as *Phytophthora cryptogea*, *P. drechleri*, *P. lactucae*, *P. pseudolactucae*, *Pythium aphanidermatum*, *P. megalacanthum*, *P. polymastum*, *P. spinosum*, *P. tracheiphilum*, *P. ultimum* and *P. uncinulatum*. Viral diseases of lettuce are caused by *Cucumber mosaic virus* (CMV), *Lettuce big vein associated virus* (LBVaV), *Lettuce mosaic virus* (LMV), *Turnip yellows virus* (TuYV), *Lettuce necrotic yellows virus* (LNYV), *Lettuce mottle virus* (LeMoV) and other viruses (APS, 2017).

Among the most significant bacteria that affect lettuce are: *Xanthomonas hortorum* pv. *vitians*, *Pectobacterium carotovorum* subsp. *carotovorum*, *Pseudomonas cichorii* and *Pseudomonas viridiflava*. Considering the nature of pathogens and their destructive power, the damage caused by their infection can be absolute and lead to great economic losses. Depending on the conditions of the external environment, the stage of development of the host plant, as well as the nature and aggressiveness of the pathogen, the symptoms can vary greatly, from leaf spotting to the complete decay of the heads, as a result of which this crop loses its market value completely (Mijatović et al., 2007).

MATERIAL AND METHODS

On two occasions, butterhead lettuce has been bought on marketplace in Belgrade, in 2020 and 2023. The appearance of the head of lettuce indicated the presence of the pathogen. On the upper side of the leaves, small necrotic lesions might be observed, while on the lower side, numerous water-soaked spots. Dark necrosis spread from the edge of the leaf towards the center (Figure 1). Strains of *Pseudomonas syringae* pv. *syringae* (RKFB 1064), *P. viridiflava* (KFB 012), *P. cichorii* (KFB 026), and *Xanthomonas hortorum* pv. *vitians* (KFB 031) were used as references to verify the cultural, biochemical, and physiological characteristics of the isolates, as well as positive controls in the pathogenicity test.



Fig. 1 Variety of symptoms observed on infected lettuce leaf; A) upper side, B) lower side

Symptomatic leaves were washed with tap water and disinfected with 96 % ethanol. Fragments of infected leaf tissue were macerated with 1 ml of sterile distilled water and spread on nutrient agar medium (NA). Different type of colonies were picked and purified. The appearance of different strains on Kings medium B and Yeast dextrose CaCO₃ (YDC) medium was also noticed.

Pathogenicity test of isolated strains was conducted in greenhouse. The seedlings of lettuce variety Genesis were transplanted into pots and used for pathogenicity test. Bacterial strains grown on NA medium for 48 hours were used for preparing bacterial suspensions in sterile distilled water with concentration of 10⁸ colony forming units per milliliter (cfu/ml). Five week old lettuce plants were sprayed with bacterial suspensions of different strains and positive controls using hand-held sprayer. The test plants were covered with plastic bags to maintain high humidity for 24 to 48 hours.

Gram reaction was studied through the ability of the cell wall to degrade under the influence of strong bases using a test with 3% KOH. A sample of a pure 24 hours old culture was resuspended in a drop of 3% KOH using a sterile wooden toothpick on a microscope slide. Under the influence of a strong base, the cell wall of Gram-negative cells dissolves, causing DNA to leak from the bacterial cell, resulting in the drop acquiring a slimy consistency, and when the toothpick is lifted, the appearance of an elastic thread is observed. In the case of Gram-positive bacteria, slimy thread formation is absent (Arsenijević, 1997). As a positive control, a strain of Gram-negative bacteria - *Pseudomonas fluorescens* (B130) was used, and as a negative control, a strain of Gram-positive bacteria - *Clavibacter michiganensis* subsp. *michiganensis* (KFB 0243).

The application of catalase test determines the presence of the enzyme catalase, which decomposes hydrogen peroxide (H₂O₂). A sample of bacterial colonies grown for 24 hours is transferred with a sterile wooden toothpick to a microscope slide and resuspended in a drop of 3% H₂O₂. A positive reaction is indicated by the formation of gas bubbles, resulting from the decomposition of hydrogen peroxide and the release of free oxygen, which occurs if the studied strain possesses the enzyme catalase (Arsenijević, 1997). As a positive

control, the strain *Pseudomonas syringae* pv. *syringae* (KFB 0103) was used.

O/F test studies the oxidative-fermentative metabolism of glucose, specifically the ability of bacteria to decompose glucose in the presence or absence of oxygen. For this test, it is necessary to prepare Hugh-Leifson medium with the final concentration of glucose of 1% (Lelliot & Stead, 1987). Two test tubes are used for each strain, and in one of them, sterile liquid paraffin is added to ensure anaerobic conditions. The medium was inoculated by stabbing using pure cultures.

The ability of bacterial strains to hydrolyze starch is tested on nutrient agar plates containing 0.2% starch (Lelliot & Stead, 1987). Bacterial cultures are inoculated in a single stroke across the surface of the medium in Petri dishes. The inoculated plates are incubated for five days at a temperature of 27°C, after which they are flooded with Lugol's solution. The positive control used is the strain of the bacterium *Xanthomonas campestris* pv. *campestris* (KFB 105).

The study of the ability of bacterial strains to hydrolyze gelatin is conducted using a medium composed of yeast extract, peptone, gelatin, and distilled water. The medium is poured into test tubes and inoculated by resuspending a pure culture in the medium. If the medium does not solidify, it indicates a positive reaction, meaning gelatin hydrolysis has occurred. A negative reaction indicates the absence of gelatin hydrolysis, reflected in the medium remaining solid (Lelliot & Stead, 1987). As a negative control, uninoculated media were used, while *P. s. pv. syringae* (KFB0103) was utilized as the positive control.

LOPAT test is used to identify pathogenic varieties of *Pseudomonas syringae*. The name LOPAT is acronym for the initial letters of the tests included: Levan production (L), oxidase activity (O), potato rot (P), arginine dihydrolase activity (A) and tobacco hypersensitivity reaction (T) (Schaad et al., 2001).

The molecular analysis in this study involved sequencing and PCR techniques. To explore differences within bacterial populations, the widely used REP-PCR method, which targets repetitive sequences within the genome, was conducted according to the protocol by Schaad et al. (2001). Using the Rep 1R-I (5' - III ICG ICG ICA TCI GGC - 3') and Rep 2-I (5' - ICG ICT TAT CIG GCC TAC - 3') primers, REP-PCR enabled the examination of genetic diversity among bacterial strains isolated from lettuce and facilitated comparison with reference strains.

The ribosomal RNA gene (16S rRNA) sequence, known for its evolutionary stability, offers a straightforward yet reliable basis for phylogenetic studies across bacterial species. With universal primers fD1 and rP2 (Weisburg et al., 1991), a ribosomal gene sequence of approximately 1400 bp was amplified in the „*Pseudomonas* group” strains RKFB 1346, RKFB 1349, RKFB 1352, RKFB 1522, and RKFB 1528.

For the „*Xanthomonas* group”, three strains were analyzed via PCR using B162t7 (5' - CGG CCG AAT AAT CAA GAG CA - 3') and B162r (5' - TTG TCC GTT ACG ATT GGT CG - 3') primers (Barak et al., 2001). The amplified fragments were compared with the reference strain *Xanthomonas hortorum* pv. *vitiens* (KFB 031).

RESULTS AND DISCUSSION

After 48 h, numerous colonies developed on the plates. In total, 18 bacterial strains were isolated. All strains caused hypersensitive reaction of tobacco leaves. Based on the appearance of colonies, they were divided into two groups. The first group consisted of 15 strains, whose colonies were varying from beige to pale yellow on NA medium, and fluorescent on King's medium B. Those features are characteristic of *Pseudomonas* spp. Three strains from the second group grew yellow colonies on NA medium, and formed yellow mucoid colonies on yeast dextrose CaCO₃ (YDC) medium – characteristic for *Xanthomonas* spp.

Ten days after the inoculation, the first water-soaked lesions appeared on the lower side of the lettuce leaves. Symptoms were developing and small necrotic spots could be seen on the upper side of the leaves fourteen days after inoculation. To fulfill Koch's postulates, bacteria were reisolated from infected test plants two weeks after inoculation. The reisolated strains also caused a hypersensitive reaction in tobacco leaves. REP-PCR analysis confirmed that each original strain and its corresponding reisolate had identical genetic profiles.

The isolated strains were identified as Gram-negative, catalase-positive, and utilize glucose oxidatively. They were able to hydrolyze gelatin but not starch. According to the LOPAT test results, the strains within the “*Pseudomonas* group” were classified into three distinct subgroups of *Pseudomonas syringae*: I b, II, and III (Table 1).

Table 1. LOPAT test results for isolated “*Pseudomonas*” strains

Strain	L	O	P	A	T	Group
RKFB 1344	-	-	-	-	+	Ib
RKFB 1345	-	-	-	-	+	
RKFB 1346	-	-	-	-	+	
RKFB 1347	-	-	-	-	+	
RKFB 1348	-	-	-	-	+	
RKFB 1349	-	+	+	-	+	III
RKFB 1351	-	-	-	-	+	Ib
RKFB 1352	-	-	+	-	+	II
RKFB 1522	-	-	-	-	+	Ib
RKFB 1523	-	-	-	-	+	
RKFB 1524	-	-	-	-	+	
RKFB 1526	-	-	-	-	+	
RKFB 1527	-	-	-	-	+	
RKFB 1528	-	-	+	-	+	II
RKFB 1530	-	-	-	-	+	Ib

In this study, both main isolated bacterial groups, *Pseudomonas* spp. and *Xanthomonas* spp., exhibited typical symptoms on lettuce leaves, including necrotic spots and water-soaked lesions.

These findings align with prior research on these pathogens, which have been widely documented as common culprits in lettuce disease. Results suggest a level of consistency in the symptoms, colony morphology and biochemical and physiological characteristics across different studies, reinforcing that these bacterial groups exhibit similar mechanisms of pathogenicity that could be influenced by genetic and environmental factors (Cottyn et al., 2009; Ozyilmaz & Benlioglu, 2018; Lipps & Samac, 2022).

REP – PCR analysis differentiate strains into several groups, based on genetic profile similarity, confirming the diversity of this population indicated by LOPAT tests. Five strains from “*Pseudomonas* group” were used as representative for conducting 16S rRNA analysis, based on their genetic profiles. All of the three strains from “*Xanthomonas* group”, RKFB 1525, RKFB 1529 and RKFB 1531, had identical genetic profile.

16S rRNA analysis proposed differentiation in “*Pseudomonas*” group. 16S rRNA gene sequencing analysis was applied to 6 strains (RKFB 1346, RKFB 1349, RKFB 1352, RKFB 1522 and RKFB 1528) selected as representatives of groups based on certain biochemical-physiological, and molecular characteristics. Forward and reverse sequences were processed using the MEGA 9 software package. They were further compared with sequences available in the NCBI database using the BLAST algorithm. The results showed that the strains RKFB 1522, RKFB 1528 and RKFB 1346 belong to the species *Pseudomonas viridiflava* given that the percentage of identity was over 99%. Strain RKFB 1349 showed 99,39 % similarity with species *Pseudomonas cichorii*. Strain RKFB 1352 showed 100 % similarity with certain *Pseudomonas* sp. strains. Based on PCR analysis using B162t7 (5’ - CGG CCG AAT AAT CAA GAG CA - 3’) and B162r (5’ - TTG TCC GTT ACG ATT GGT CG - 3’) primers amplifying DNA fragment of 700 bp, we concluded that three strains from “*Xanthomonas* group” belong to *Xanthomonas hortorum* pv. *vitians*, as compared to the reference strain KFB 031.

The REP-PCR and 16S rRNA analyses performed in this study provided valuable insights into the genetic diversity and relationships between the *Pseudomonas* and *Xanthomonas* strains. Specifically, REP-PCR allowed for the grouping of strains based on genetic profile similarity, identifying distinct subgroups within *Pseudomonas* spp. Furthermore, the 16S rRNA analysis revealed high similarity (over 99%) for several strains within the *Pseudomonas* group, identifying them specifically as *Pseudomonas viridiflava* and *Pseudomonas cichorii*. These findings not only validated the accuracy of the molecular techniques used but also emphasized the genetic variability within pathogenic *Pseudomonas* populations in lettuce. This variability could indicate adaptation processes that enable certain strains to survive better under specific conditions or host responses. Additionally, the consistency of genetic profiles among the *Xanthomonas* strains (identified as *Xanthomonas hortorum* pv. *vitians*) suggests a potentially more stable population structure for this group, which may have implications for its management and predictability as a pathogen. This detailed genetic characterization provides a basis for future comparative studies and could inform targeted interventions against these specific bacterial groups.

CONCLUSION

The occurrence of mixed infections in vegetable crops is common, given the broad host range and natural dissemination of certain pathogenic bacteria. Lettuce, in particular, provides ideal environment for the postharvest growth of these pathogens. Their presence and activity can significantly reduce product marketability.

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PHYSIOLOGICAL PERFORMANCE IN PHOTOSYNTHESIS ASPECT OF *AESCULUS* × *CARNEA* AND *ROBINIA PSEUDOACACIA* 'GLOBOSA' IN URBAN AREA OF NOVI SAD, SERBIA

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Nowadays, numerous tree species are planted as roadside trees in urbanized areas. However, urban environments are different from natural environments, and tree species growing on urban streets are much more exposed to multiple environmental stressors such as excessive heat accumulation, high light doses due to reflection, soil salinity due to winter deicing, severe drought and soil compaction, all of them challenging the growth and vitality of urban trees. Therefore, special attention should be focused on defining and planting of tree species that can tolerate the adverse environmental factors in an urban environment, thrive and maintain a capacity to deliver wide range of ecosystem services. With that in mind, the aim of this study was to investigate the physiological performance of *Aesculus* × *carnea* and *Robinia pseudoacacia* 'Globosa' in the urban environment in Novi Sad, Serbia. For this purpose, Chlorophyll *a* fluorescence parameters were chosen in combination with relative chlorophyll content measurement, as they provide useful quantitative information about the state of plants and their physiological processes associated with the light phase of photosynthesis. The results of this study indicate that under various environmental stressors occurring in cities, *Aesculus* × *carnea* exhibited superior physiological performance and showed better adaptation in comparison to *Robinia pseudoacacia* 'Globosa'.

Key words: Chlorophyll *a* fluorescence; *Robinia pseudoacacia* 'Globosa'; *Aesculus* × *carnea*; Physiological performance; Urban forestry

INTRODUCTION

Urban settlements are becoming ever increasingly populated spaces, with more than 4 billion people living in cities. This means that over half of the world's population lives in urban settings nowadays, and this proportion is predicted to rise to 68% by 2050 (Antrop, 2004). In terms of environment, one of the most important effects of urbanization is observable in the increase of CO₂ emissions and the urban heat island effect, leading to harsher conditions in urban areas and a decrease in thermal comfort in comparison with natural surroundings (Ahongshangbam, 2023).

Cities are complex habitats for trees to live in, as trees in urban conditions are challenged by many factors, which come from all around them acting simultaneously (Lüttg and Buckeridge, 2020). Initially, there are certain factors that are exclusive to urban environments. Much of this has to do with the fact that humans are sharing the land with trees. These factors are related to dense building, predominant use of artificial building materials and heavy traffic (Kronenberg, 2015). Generally, dense building practices and the abundance of pavements may cause conditions in urban areas similar to arid environments with increased temperatures and decrease in air humidity. Additional urban ecological limitation factors, affecting the growth and vitality of urban trees include: high soil pH and soil compaction, de-icing with salts, limited space for root growth, limited space for upper branch growth, limited access to sunlight, excessive heat accumulation and vandalism (Steenberg et al., 2017). These urban factors, combined with natural stress from changing weather patterns (higher temperature), air pollution, drought, and increased levels of CO₂ concentration, may cause a slow and steady decline of the urban trees.

In such heterogenous conditions, it has been noted that the performance of plants is divergent, due to differences in their leaf morphology and physiology (Vastag et al., 2020; Wang et al., 2019). Even though many studies have been conducted regarding growth performance, the *in situ* physiological performance of trees in urban environments has been less researched (Swoczyna et al., 2014). Improving our understanding of the physiological responses of trees under urban conditions can provide important insights that may help the future selection process and tree improvement efforts (Cregg et al., 2023).

As the most important elements of urban green areas, urban trees are of key importance in offsetting anthropogenic CO₂ emissions and alleviating the urban heat island effect, given their potential for carbon sequestration and storage, as well as in cooling their surroundings through transpiration. Taking into consideration that the effects of drought are predicted to be even more severe in the future, the cooling effect of urban trees may provide vital service in helping to mitigate the negative effects of the predicted climate changes (Gill et al., 2007). Furthermore, urban trees provide additional ecosystem services, including: improving air quality, reducing noise pollution, acting as wind buffer, providing shade, aesthetic function and recreation for the dwellers (Roy et al., 2012).

An important challenge in urban forestry is to define tree species that can tolerate adverse environmental factors in urban environments, thrive and maintain capacity to deliver wide range of ecosystem services since their provision is reliant on their health (Vogt et al. 2017). In this context, the introduction of non-native species should be incorporated in new planting programs, alongside the use of drought-tolerant autochthonous ones, as high species diversity is positively correlated with the resilience of urban forests. Another strategy important to be included in future planting programs is to promote increased diversity, the palette of trees available for urban planting, as it has been shown that high diversity of species contributes to increasing resilience and reducing the maintenance requirements of urban forests under abiotic and biotic challenges (Sjöman et al., 2024).

In accordance with that, the objectives of this study were to: (1) evaluate the *in situ* physiological performance and adaptation potential of *Aesculus* × *carnea* and *Robinia pseudoacacia* 'Globosa', two ornamental tree species planted in urban environments and (2) to provide valuable guidelines and information regarding the selection of tree species for future decision making and planning programs. Tree responses were determined through accurate, reliable and non-invasive tools, including chlorophyll *a* fluorescence (Strasser et al., 2004) and assessment of the relative chlorophyll content.

MATERIAL AND METHODS

Site selection and plant material. In the present experiment five trees of Red horse-chestnuts (*Aesculus* × *carnea*) and five of Black locust (*Robinia pseudoacacia* 'Globosa') were studied, located on the Boulevard of Cara Lazara (45°14'55"N, 19°51'05"E) in Novi Sad, Serbia. At the studied site, the chosen trees were planted in a continuous row along the sidewalk at both sides of the street, with 5 m distance between each tree. The mean morphometrical characteristics of the studied trees are shown in table 1.

The morphometrical characteristics were estimated on site and were the following; Height of tree (m), First branch height (m), Diameter of the crown (m), Diameter at breast height (DBH, cm), Vitality and Decorative values. Tree height and first branch height were measured using visual alignment. The crown diameter was calculated as the average of two values measured along two perpendicular directions (Anastasijević, 2007). Furthermore, the vitality (vit) and decorative values (dec) were assessed regarding scale indices from 1 (poor) to 5 (excellent) (Kostić et al., 2019)

In order to assess the differences in physiological performances in the same environment between the two above mentioned species, measurements of chlorophyll *a* fluorescence and SPAD vales were made during on 19th of November, 2024.



Fig. 1 The location of the studied trees

Table 1. Mean morphometrical characteristics of the studied trees.

Species	Height/m	First branch height/m	Diameter of the crown/m	DBH /cm	vit (1-5)	dec (1-5)
<i>Robinia pseudoacacia</i> 'Globosa'	5.89±0.75	2.05±0.43	5.85±1.68	16.18±6.15	3	4
<i>Aesculus × carnea</i>	7.59±0.78	2.56±0.26	8.19±0.75	25.02±2.73	4	5

Assessment of chlorophyll *a* fluorescence. The photosynthetic activity of PSII was measured with a PAM-2500 portable chlorophyll fluorometer (Walz, Germany) by a non-destructive method in a time scale between 9:00 AM and 11:00 AM, due to midday depression of photosynthesis. Before the measurements, the leaves were dark adapted for 15 min with light exclusion clips. This is of high importance as dark-adaptation allows PSII reaction centers to open and the electron transport chain to be oxidized. At each tree, three fully expanded and healthy leaves, from the same orientation and at the same height (2.5-3 m) were chosen for the measurement (Jardi et al., 2020). All the parameters measured were displayed on the screen of the fluorometer, automatically recorded, calculated, and stored and the records of data were downloaded via a computer connection. Relevant fluorescence parameters (Table 2.) were derived from the recorded data.

Table 2. List of derived chlorophyll *a* fluorescence parameters

Definition of the and Literature Source of parameters	Formula
Effective photochemical quantum yield of PS II (Genty et al., 1989)	$Y(II) = (F_m' - F')/F_m'$
Coefficient of photochemical fluorescence quenching (Schreiber et al. 1986 as formulated by van Kooten and Snel, 1990)	$qP = (F_m' - F')/(F_m' - F_o')$
Coefficient of photochemical fluorescence quenching assuming interconnected PS II antennae (Kramer et al. 2004)	$qL = qP * F_o' / F'$
Coefficient of photochemical fluorescence quenching (Schreiber et al. 1986 as formulated by van Kooten and Snel, 1990)	$qN = 1 - (F_m' - F_o') / (F_m - F_o)$
Stern-Volmer type nonphotochemical fluorescence quenching (Bilger and Björkman, 1990)	$NPQ = F_m / F_m' - 1$
Quantum yield of non-light induced non-photochemical fluorescence quenching (Kramer et al. 2004)	$Y(NO) = 1 / (NPQ + 1 + qL * (F_m / F_o - 1))$
Quantum yield of light-induced (Δ pH and zeaxanthin-dependent) non-photochemical fluorescence quenching (Kramer et al. 2004)	$Y(NPQ) = 1 - Y(II) - Y(NO)$
Electron transport rate (Xu et al. 2009)	$ETR = PAR * 0.84 * 0.5$

Assessment of the relative chlorophyll content. The same leaves, chosen for measurement of chlorophyll *a* fluorescence were used for estimation of relative chlorophyll content (SPAD values). SPAD values were estimated *in situ*, by a non-destructive method using a portable chlorophyll meter (SPAD-502DL Plus, 160 Minolta; Spectrum Technologies Ltd., Plainfield, IL, USA). This instrument provides an estimation of the relative amount of leaf chlorophyll content by measuring the red and infrared regions (650 nm and 940 nm wavelength, respectively) in leaf tissue. By measuring three times on a leaf (from the tip and middle of the leaf, and the part near the petiole), the relative chlorophyll content was estimated and the average SPAD values are shown.

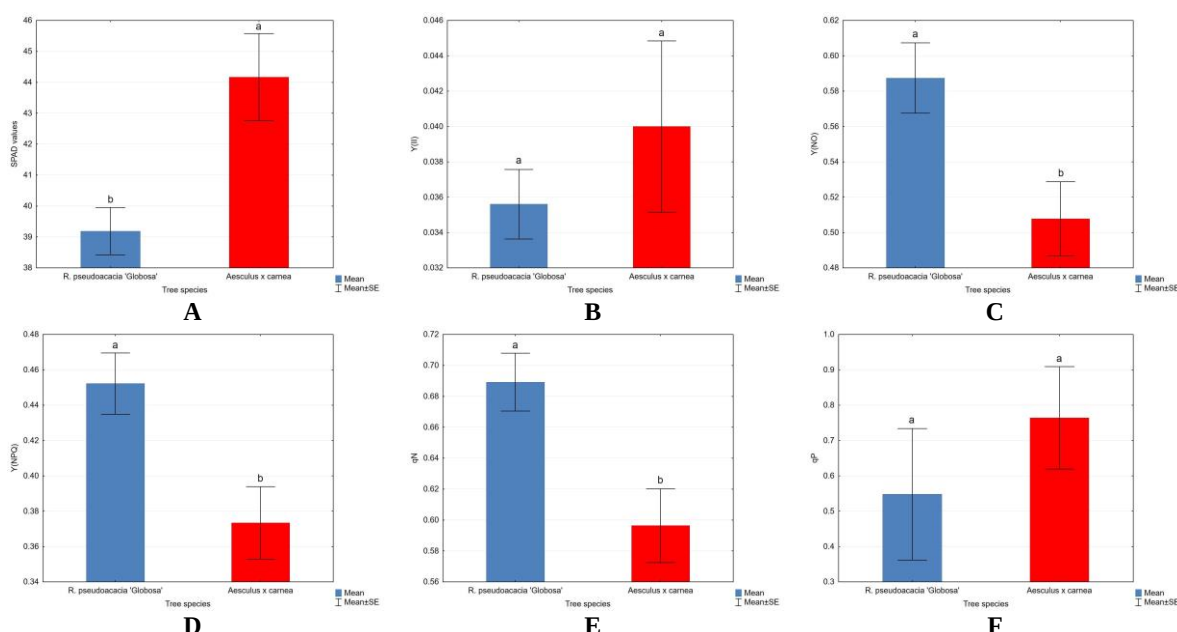
Statistical analysis. Statistical analysis of chlorophyll *a* fluorescence and SPAD values were conducted using Statistica v. 13 (TIBCO Software Inc. 2017) for Windows. To assess the differences between the observed species, one-way ANOVA was applied, followed by Tukey's honestly significant difference (HSD) test. For better visualization of these data, the above-mentioned traits were shown in the form of a boxplot diagram.

RESULTS AND DISCUSSION

The results of the present study evidenced a significant difference between the two observed tree species in terms of five parameters, namely Y(NO), Y(NPQ), qN, NPQ, SPAD value parameters. On the other hand, no significant differences were found in case of Y(II), qP, qL, ETR values (Table 3.). Under the same urban conditions, five parameters of *A. × carnea*, namely: SPAD value, Y(II), qP, qL, and ETR, were higher than those of *R. pseudoacacia* 'Globosa'. In contrast to this, the average values of four parameters, namely: Y(NO), Y(NPQ), qN, NPQ were lower in *A. × carnea* compares to *R. pseudoacacia* 'Globosa' (Figure 2.).

Table 3. Results of the one-way analyses of variance for assessed physiological traits of *Robinia pseudoacacia* 'Globosa' and *A. × carnea*.

Effect		SPAD values	Y(II)	Y(NPQ)	Y(NO)	NPQ	qN	qP	qL	ETR
Species	F	9.664	0.708	8.572	7.595	6.214	9.364	0.841	0.853	0.025
	P	0.014	0.425	0.019	0.025	0.037	0.016	0.386	0.383	0.878



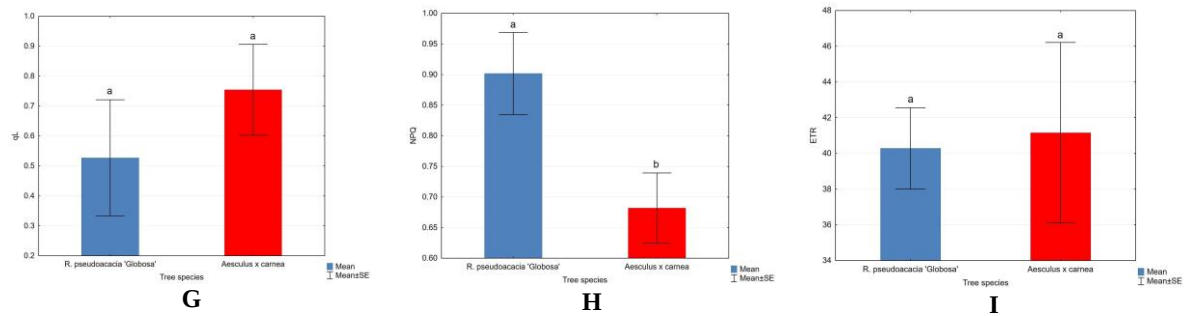


Fig. 2 Physiological parameters measured in *R. pseudoacacia* 'Globosa' and *A. × carnea*. A- relative chlorophyll content (SPAD value, relative units); B-Effective photochemical quantum yield of PS II(Y(II), relative units); C-Quantum yield of non-regulated heat dissipation and fluorescence emission(Y(NO, relative units), relative values; D-Quantum yield of light-induced non-photochemical fluorescence quenching (Y(NPQ), relative values); E-Coefficient of non-photochemical fluorescence quenching(qN, relative units); F-Coefficient of photochemical quenching (qP, relative units); G-Coefficient of photochemical fluorescence quenching assuming interconnected PSII antennae(qL, relative units); H-Stern-Volmer type of non-photochemical quenching(NPQ, relative units); I-Electron transport rate (ETR, $\mu\text{mol m}^{-2} \text{s}^{-1}$).

In terms of Y(NO), the values of *R. pseudoacacia* 'Globosa' ranged between 0.532 and 0.651 with an average of 0.587. Observing the same parameter for *A. × carnea* we observed a min value of 0.455, a max value of 0.582 and an average of 0.508. Concerning Y(NPQ), a min value of 0.388, max of 0.485 and an average of 0.452 was detected in *R. pseudoacacia* 'Globosa', while this parameter ranged between 0.305 and 0.418 with an average 0.373 in *A. × carnea*. Observing qN, in the present study we evidenced a min value of 0.636 and max value 0.739 with an average 0.689 in terms of *R. pseudoacacia* 'Globosa'. The values of the previously mentioned parameter ranged between 0.526 and 0.640 with an average of 0.596 for *A. × carnea*. The values of NPQ in case of *R. pseudoacacia* 'Globosa' were between 0.667 and 1.067 with an average of 0.902, while in *A. × carnea* were between 0.468 and 0.786 with an average of 0.682. SPAD values ranged between 37.133 and 40.867 with an average of 39.187 in *R. pseudoacacia* 'Globosa', and 39.400 and 47.300 with an average 44.167 in *A. × carnea*. In terms of Y(II) min values were estimated to be 0.030 and max values 0.040 with an average of 0.036 in *R. pseudoacacia* 'Globosa'. Observing the same parameter in *A. × carnea*, a min value of 0.023, and max value of 0.050 and average of 0.040. qP parameter values in *R. pseudoacacia* 'Globosa' were between 0.192 and 1.000 an average of 0.548, while in *A. × carnea* were between 0.264 and 1.000 and an average of 0.764. Concerning qL, in terms of *R. pseudoacacia* 'Globosa' we evidenced a min value of 0.159 and max of 1.000 with an average of 0.527, while in case of *A. × carnea* the min value was 0.225, the max 1.000 and the average 0.755. Observing ETR, the values ranged between 33.200 and 46.100 with an average of 40.280 in *R. pseudoacacia* 'Globosa', while in *A. × carnea* values for this parameter were between 23.300 and 51.500 with an average of 41.160.

Table 4. Descriptive statistics of estimated physiological parameters in *R.pseudoacacia* 'Globosa' and *A. × carnea*.

Species	Variable	Mean	Minimum	Maximum	Std.Dev.	Coef.Var.	Standard Error
<i>R. pseudoacacia</i> 'Globosa'	Y(II)	0.036	0.030	0.040	0.004	12.340	0.002
	Y(NPQ)	0.452	0.388	0.485	0.039	8.592	0.017
	Y(NO)	0.587	0.532	0.651	0.045	7.578	0.020
	NPQ	0.902	0.667	1.067	0.150	16.633	0.067
	qN	0.689	0.636	0.739	0.042	6.075	0.019
	qP	0.548	0.192	1.000	0.416	75.912	0.186
	qL	0.527	0.159	1.000	0.435	82.514	0.194
	ETR	40.280	33.200	46.100	5.081	12.614	2.272
<i>A. × carnea</i>	SPAD value	39.187	37.133	40.867	1.707	4.356	0.763
	Y(II)	0.040	0.023	0.050	0.011	27.099	0.005
	Y(NPQ)	0.373	0.305	0.418	0.046	12.308	0.021
	Y(NO)	0.508	0.455	0.582	0.047	9.215	0.021

NPQ	0.682	0.468	0.786	0.128	18.810	0.057
qN	0.596	0.526	0.640	0.053	8.944	0.024
qP	0.764	0.264	1.000	0.324	42.400	0.145
qL	0.755	0.225	1.000	0.340	45.015	0.152
ETR	41.160	23.300	51.500	11.313	27.486	5.060
SPAD value	44.167	39.400	47.300	3.149	7.130	1.408

The present study showed the adaptation potential and physiological performance of *Aesculus × carnea* and *Robinia pseudoacacia* 'Globosa', two ornamental tree species planted in the urban environment of Novi Sad. Under the same environmental conditions, *A. × carnea* exhibited statistically higher values of relative chlorophyll content (SPAD values) in comparison to *R. pseudoacacia* 'Globosa'. As chlorophylls play a vital role in photosynthesis, plant species characterized by higher chlorophyll content tend to have higher photosynthetic activity, show higher growth rate, accumulate more biomass and sequester more carbon (Richardson et al., 2002). It has been well documented that differences between features of a plant, including physiological features as well, depends both on their genetic structure (Taner and Sade, 2005) and on the environmental factors, such as drought (Urban et al., 2023) and heat wave (Kim et al., 2024).

Trees possess the ability to develop structural, morphological, and physiological adjustments to harsh conditions prevailing in urban settlements (Kjelgren and Clark, 1992). As roadside trees are planted on both sides of streets, they are inevitably imposed to air pollution, soil heavy metal pollution and other factors that are different from the natural environment. In terms of physiological adjustment, the chlorophyll *a* fluorescence technique has been recognized as a commonly used nondestructive diagnostic tool in photosynthesis researches for detecting species tolerant or susceptible to various environmental stresses (Maxwell and Johnson 2000; Vastag et al., 2019). Namely, under favorable ecological conditions, most of the absorbed light energy is used for photochemical reaction (about 80%), that is, for energy transfer in the antennas of the photosystem and the initiation of the electron transport chain, resulting in high values of Y(II) and ETR (Strasser et al., 2004). In case of our study, showing elevated values of Y(II) and ETR, it can be concluded that *A. × carnea*, exhibited a higher efficiency in transporting electrons and potential which could fix more CO₂ in comparison to *R. pseudoacacia* 'Globosa'. Higher levels of the above-mentioned parameters were observed to be good indicators of drought tolerance, as well (Vastag et al., 2020). Lower values of these parameters observed in *R. pseudoacacia* 'Globosa' indicate a decrease in the carboxylation efficiency, which affects the rate of consumption of ATP and NADP (Queiroz-Alves et al., 2019).

However, under urban stress condition such as excessive heat accumulation, water deficit, which is often accompanied by high light intensity, a significant proportion of the reaction centers of PSII are closed, resulting in conversion most of the energy to fluorescence and heat, while part of the energy used for photochemical activity is significantly decreased (Maxwell and Johnson, 2000). As a response to stress conditions, plants perform non-stomatal regulation by non-photochemical quenching (NPQ) to protect their photosynthetic apparatus from photodamage and heat-induced impairment (Maxwell and Johnson 2000; Strasser et al., 2004). In accordance with that, we observed statistically higher values of parameters related to the heat dissipation of light energy (qP, NPQ, Y(NO), Y(NPQ)) in *R. pseudoacacia* 'Globosa', indicating that a significant proportion of the reaction centers of PSII were reduced, resulting in the dissipation most of the energy into the form of fluorescence and heat, while part of the energy used for initiation of the electron transport chain decreased. Studying the physiological performance of two *Magnolia* species, Vastag et al., (2019) observed the same photoprotective mechanism in *Magnolia × soulangeana* under severe drought occurring in the urban area of Novi Sad. Statistically lower values of parameters related to the heat dissipation of light energy (qP, NPQ, Y(NO), Y(NPQ)) observed in *A. × carnea* indicate that photosystem PSII hasn't experienced any photochemical stress over the studied period and that the most of the energy absorbed by chlorophyll molecules was still for the CO₂ assimilation.

In terms of vitality and decorative value, *A. × carnea* showed slightly higher values in comparison to *R. pseudoacacia* 'Globosa'. In accordance with our findings, *A. × carnea* has been noted as a desirable tree species in urban environments due to its attractive deep red flowers and straight trunk. (Marinšek et al., 2002).

Our results highlight the overall higher *in situ* physiological performance of *A. × carnea* compared to *R. pseudoacacia* 'Globosa' in the urban environment of Novi Sad. The observed superior physiological performance of *A. × carnea* in the urban area is in accordance with the results of Marinšek et al., (2002) which noted that among the benefits of the species are its good resistance to drought, to compacted soil and to the horse-chestnut leaf miner (*Cameraria ohridella*). Furthermore, these results are in line with the description provided from in Citree, the urban tree database, stating high drought tolerance, good heat and late frost tolerance and overall good performance of *A. × carnea* in urbanized area.

CONCLUSION

According to the overall results of this study, under the same urban conditions *Aesculus × carnea* showed higher activity of primary photochemical reaction in PSII and had higher values of relative chlorophyll content, indicating an overall better physiological performance in comparison to *Robinia pseudoacacia* 'Globosa'. As the climate change scenarios predict more severe and long-lasting drought periods, which are expected to be even more pronounced in an urbanized area, due to higher radiation absorbance of urban structures, the selection of suitable tree species will become even more important. Therefore, this study provides valuable information and insight into the selection of suitable tree species based on evaluation of *in situ* physiological performance and supports decision making processes for choosing of tree species in light of the urban stressors and predicted climate changes.

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LEAF GAS EXCHANGE CHARACTERISTICS OF SILVER LINDEN TREES (*TILIA TOMENTOSA* MOENCH) IN URBAN ENVIRONMENT

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*This study investigates the physiological traits of *Tilia tomentosa* (silver linden) in an urban environment to evaluate their potential for urban greening. We focused on analyzing leaf gas exchange parameters and chlorophyll content among five different genotypes of silver linden trees. Against the backdrop of climate change, understanding these physiological responses is increasingly crucial, as shifts in temperature, precipitation, and CO₂ levels pose new challenges to urban trees. The results showed significant differences among genotypes in photosynthetic performance and water management strategies. Genotypes with higher chlorophyll content and photosynthetic rate demonstrated stronger environmental adaptability, while those with higher water use efficiency exhibited greater drought tolerance. These findings indicate that physiological differences among genotypes influence the adaptability of silver lindens to urban environments under changing climate conditions, providing scientific guidance for the selection and management of resilient urban greening species.*

Key words: Silver linden, urban ecosystem, leaf gas exchange, physiological parameter

INTRODUCTION

As global urbanization accelerates, environmental challenges such as air pollution, and the urban heat island effect have emerged as critical concerns (Mirzaei et al., 2010). Urban green tree species play a vital role in mitigating these issues, which directly influence urban air quality and microclimate regulation (Pataki et al., 2021). Lindens, as valued trees, are widely distributed in many countries across Europe, usually used for urban vegetation in public parks and streets. In Europe, there are mainly four species of Linden: *Tilia.cordata* Mill, *Tilia platyphyllos* Scop, *Tilia.tomentosa* Moench and *Tilia.dasystyla* Steven. In Serbia, the most common tree species is *T. tomentosa* Moench. (Benedetti et al., 2022). Although urban environments pose unique challenges for trees, the resilience of linden trees makes them particularly valuable for improving air quality and providing shade (Chang et al., 2024).

Climate change exacerbates urban challenges through rising temperatures, increased frequency of extreme weather events, and altered precipitation patterns, further stressing urban trees. Examining leaf gas exchange and physiological traits, including chlorophyll content, in silver linden enhances understanding of its responses to climate change impacts. (Kasper et al., 2023). Gas exchange, chlorophyll fluorescence, and antioxidant levels were utilized as bioindicators to assess heavy metal air pollution. Findings indicated a marked reduction in photosynthetic efficiency and elevated lipid peroxidation, along with significant alterations in antioxidant enzyme activity under pollutant exposure (Hassan et al., 2013). Similarly, a study on *Quercus mongolica* leaves exposed to elevated O₃ levels in an urban environment reported consistent declines in photosynthetic parameters across the entire growing season (Wang et al., 2009). Another study investigating aphid colonization in urban shrubs observed no significant effects on photosynthetic pigments; however, gas exchange parameters, particularly nitrogen allocation and the C/N ratio, were notably affected (Leopold et al., 2023).

Chlorophyll content is a critical parameter that influences photosynthetic efficiency and serves as an indicator of a tree's physiological health (Li et al., 2018). The leaf gas exchange characteristics and chlorophyll content of silver linden are significantly affected by various environmental conditions, including increased carbon dioxide concentrations, light intensity, temperature, humidity, and levels of air pollution (Niglas et al., 2017). For example, since the growth of urbanization intensity, the density of potentially toxic elements in tree leaves increased a lot, which caused leaf injuries, and enhanced the maximum photosynthetic rate but decreased

the transpiration rate and stomatal conductance (Petrova et al., 2023). Understanding how silver linden adapts its physiological traits under these variable urban conditions is essential to maximize its ecological benefits (Marinova et al., 2023).

In this study, we focus on silver linden trees in a specific urban environment to analyze how their gas exchange characteristics respond to local conditions. We examined the physiological differences in leaf traits among five genotypes of silver linden, all growing under identical urban environmental conditions. The aims of our study were to assess the variation in leaf gas exchange parameters across these genotypes. By investigating physiological differences in leaf gas exchange of silver lindens in a particular urban setting, this study aims to provide insights that support more targeted selection and management of tree species for urban greening initiatives.

MATERIAL AND METHODS

Five *Tilia tomentosa* (Silver linden) trees planted along Antona Čehova Street in Novi Sad, Serbia, were selected for this study. To assess the variability in leaf gas exchange for silver linden, measurements were conducted on October 15, 2024. Fully expanded leaves, all oriented in the same direction, were chosen for consistency in measurement.

Assessment of leaf gas exchange and chlorophyll content. Leaf gas exchange parameters—including net photosynthesis (A [$\mu\text{mol m}^{-2} \text{s}^{-1}$]), transpiration rate (E [$\text{mmol m}^{-2} \text{s}^{-1}$]), stomatal conductance (g_s [$\text{mmol m}^{-2} \text{s}^{-1}$]), water use efficiency (WUE [mmol mol^{-1}]) and intercellular CO_2 concentration (C_i [$\mu\text{mol mol}^{-1}$])—were measured using a CIRAS-3 portable photosynthesis system (Amesbury, MA, USA) equipped with red and blue LED light sources. Measurements were conducted between 9:00 AM and 11:00 AM under a photosynthetic active radiation (PAR) level of $1000 \mu\text{mol m}^{-2} \text{s}^{-1}$, with concurrent monitoring of humidity, air temperature, and ambient CO_2 concentration.

Chlorophyll content in the leaves of *Tilia tomentosa* was measured using a Konica Minolta SPAD-502 portable chlorophyll meter. This non-destructive method provides a reliable estimate of chlorophyll concentration by assessing the absorbance of light at specific wavelengths associated with chlorophyll a and b.

The measurements were conducted on fully expanded leaves to ensure consistency and accuracy. For each leaf sample, three readings were taken from different areas to obtain an average value. The SPAD readings were recorded during the same time frame as the leaf gas exchange measurements to minimize the effects of diurnal variations.

The data obtained from the chlorophyll meter was used to correlate chlorophyll content with gas exchange parameters, facilitating a comprehensive understanding of the physiological status of the trees in the urban environment. This method allows for efficient monitoring of plant health and can serve as an indicator of stress responses related to environmental factors.

Data were collected from one fully expanded, disease-free leaf per genotype, with each parameter recorded in quintuplicate per leaf. Leaves selected were located in the lower canopy (2–3 m above ground level) and exposed to full sunlight. This approach aligns with methodologies from Herrick and Thomas (1999) and Tissue et al. (2001).

Statistical analysis. The statistical evaluation of differences among the observed genotypes of *Tilia tomentosa* (Silver linden) was conducted using Analysis of Variance (ANOVA). To assess the significance of the differences, a post-hoc analysis was performed utilizing the Least Significant Difference (LSD) test for mean comparison. Additionally, the variations among the different genotypes were visually represented in graphical forms to enhance interpretation of the data. All statistical analyses were executed using Statistica 13.3 (TIBCO Software, Inc.), ensuring rigorous evaluation of the results.

RESULTS

There are significant differences in chlorophyll content between Silver Linden genotypes. The highest value was observed for genotype 3 ($57,01 \pm 0,95$) while the lowest value was detected for genotype 2 ($50,91 \pm 0,95$). There are no differences between genotype 4 ($53,03 \pm 0,95$) and 5 ($53,58 \pm 0,95$) as well as between genotype 1 ($52,18 \pm 0,95$) and genotype 2 ($50,91 \pm 0,95$) (Figure 1).

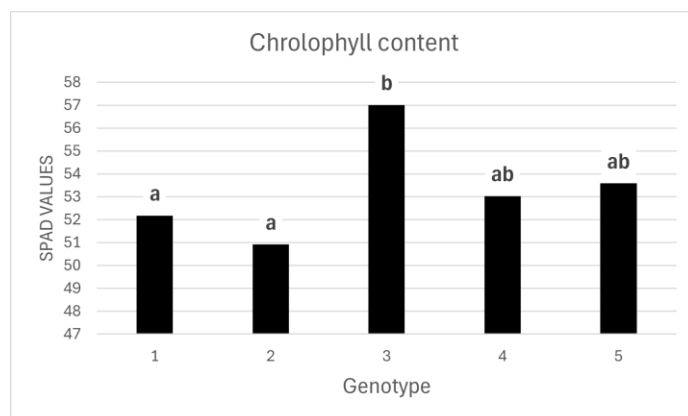


Fig. 1 Chlorophyll content of silver linden trees in urban area. Bars with the same letters were not different according to Tukey's test at $p < 0.05$.

The highest A value was observed in genotype 3 ($2,30 \pm 0,18 \mu\text{mol m}^{-2} \text{s}^{-1}$) and genotype 5 had the lowest value ($1,43 \pm 0,18 \mu\text{mol m}^{-2} \text{s}^{-1}$). There are no significant differences between genotype 1 ($2,19 \pm 0,18 \mu\text{mol m}^{-2} \text{s}^{-1}$) and 3 ($2,30 \pm 0,18 \mu\text{mol m}^{-2} \text{s}^{-1}$) as well as between genotype 2 ($1,64 \pm 0,18 \mu\text{mol m}^{-2} \text{s}^{-1}$) and 4 ($1,71 \pm 0,19 \mu\text{mol m}^{-2} \text{s}^{-1}$). The highest Ci value was detected for genotype 2 ($244,67 \pm 15,34 \mu\text{mol mol}^{-1}$) while the lowest value for genotype 1 ($166,89 \pm 15,34 \mu\text{mol mol}^{-1}$). Genotype 5 ($241,78 \pm 15,34 \mu\text{mol mol}^{-1}$), 4 ($206,13 \pm 16,27 \mu\text{mol mol}^{-1}$), 1 ($166,89 \pm 15,35 \mu\text{mol mol}^{-1}$) were ranked in descending order into different homogenous groups. The highest E value was observed in genotype 2 ($1,46 \pm 0,01 \text{mmol m}^{-2} \text{s}^{-1}$) and genotype 1 had the lowest value ($1,37 \pm 0,01 \text{mmol m}^{-2} \text{s}^{-1}$). There are no significant differences between genotype 2 ($1,46 \pm 0,01 \text{mmol m}^{-2} \text{s}^{-1}$) and 3 ($1,43 \pm 0,01 \text{mmol m}^{-2} \text{s}^{-1}$) as well as between genotype 1 ($1,37 \pm 0,01 \text{mmol m}^{-2} \text{s}^{-1}$), 4 ($1,37 \pm 0,01 \text{mmol m}^{-2} \text{s}^{-1}$) and 5 ($1,39 \pm 0,01 \text{mmol m}^{-2} \text{s}^{-1}$). The highest gs value was observed in genotype 2 ($18,89 \pm 0,52 \text{mmol m}^{-2} \text{s}^{-1}$) and genotype 4 had the lowest value ($16,25 \pm 0,55 \text{mmol m}^{-2} \text{s}^{-1}$). There are no significant differences between genotype 1 ($17,00 \pm 0,52 \text{mmol m}^{-2} \text{s}^{-1}$) and 3 ($17,78 \pm 0,55 \text{mmol m}^{-2} \text{s}^{-1}$) as well as between genotype 4 ($16,25 \pm 0,55 \text{mmol m}^{-2} \text{s}^{-1}$) and 5 ($16,56 \pm 0,55 \text{mmol m}^{-2} \text{s}^{-1}$). The highest WUE value was observed in genotype 1 and 3 ($1,60 \pm 0,13 \text{mmol CO}_2 \text{mol}^{-1}$) and genotype 5 had the lowest value ($1,03 \pm 0,13 \text{mmol CO}_2 \text{mol}^{-1}$). There are no significant differences between genotype 1 and 3 ($1,60 \pm 0,13 \text{mmol CO}_2 \text{mol}^{-1}$) as well as between genotype 2 ($1,13 \pm 0,13 \text{mmol CO}_2 \text{mol}^{-1}$) and 4 ($1,25 \pm 0,13 \text{mmol CO}_2 \text{mol}^{-1}$).

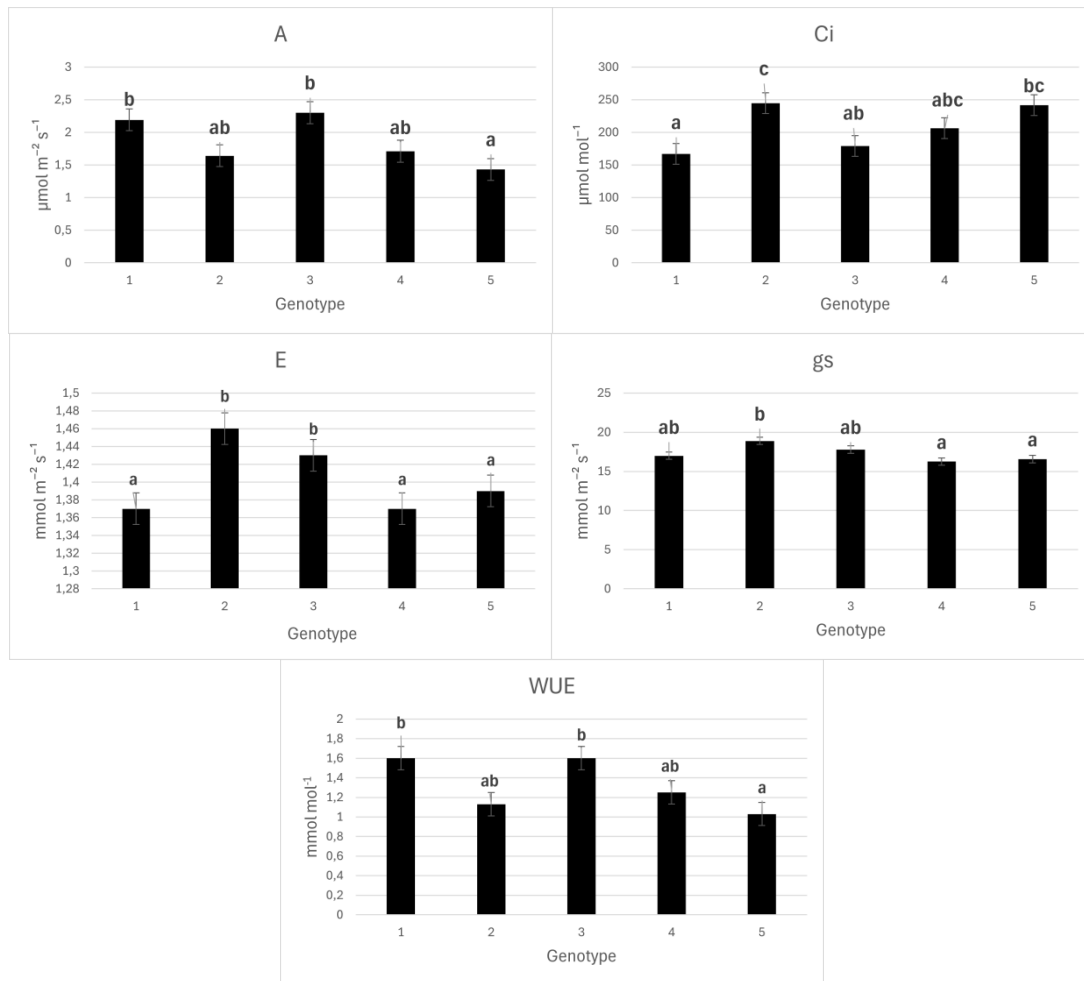


Figure 2. Differences between leaf gas exchange (net photosynthesis (A), rate of transpiration (E), stomatal conductance (gs), intracellular concentration of CO₂ (Ci) and water use efficiency (WUE)) of silver linden trees. Bars with the same letters were not different according to Tukey's test at $p < 0.05$.

DISCUSSION

Urban-planted trees exhibited higher maximum photosynthetic rates, along with elevated overall photosynthesis and stomatal conductance throughout the summer, compared to suburban or rural wild-type trees. However, urban wild-type trees demonstrated greater water use efficiency as air temperatures rose and maintained similar overall photosynthetic rates to cultivated varieties in mid to late summer (Lahr et al., 2018). This suggests that urban environmental conditions may enhance certain physiological traits, while the wild-type trees adapt by optimizing water use under higher temperatures.

Transpiration rate is influenced by a range of environmental factors, such as photosynthetically active radiation (PAR), ambient temperature, air movement, and relative humidity (RH). Elevated levels of PAR stimulate stomatal opening, which increases CO₂ uptake for photosynthesis and concurrently enhances water vapor loss through transpiration (Zhu et al., 2022). Higher ambient temperatures further accelerate transpiration by increasing the vapor pressure deficit between the leaf and surrounding air, which drives water out of the leaf (Urban et al., 2017a). Additionally, increased air velocity around the leaf surface disrupts the boundary layer, promoting more efficient water vapor exchange (Lawson and Vialet-Chabrand, 2019). Conversely, as relative humidity rises, the vapor pressure gradient between the leaf interior and surrounding air decreases, thereby reducing transpiration rates (Ferrante et al., 2018). This inverse relationship with RH highlights the importance of external vapor conditions in regulating transpiration dynamics, as plants adjust their responses to maintain internal water balance (Zhu et al., 2022).

Water use efficiency (WUE) is closely linked to net photosynthesis and transpiration rates. At the leaf level, increased CO₂ concentration enhances WUE up to a certain temperature threshold, after which WUE begins to decline due to heat stress, as stomatal closure and reduced photosynthetic efficiency limit CO₂ uptake (Hatfield and Dold, 2019). Under drought stress, plants exhibit varying WUE responses, which depend on species,

physiological state, and stress intensity. For example, drought-resistant species are often able to maintain higher WUE by efficiently managing water loss and optimizing photosynthetic activity under water-limited conditions (Ullah et al., 2023).

With increased CO₂, stomatal openings are reduced, leading to lower water loss through transpiration, while photosynthesis is maintained or even enhanced, resulting in higher WUE (Lawson and Vialet, 2019). However, under high-temperature conditions, stomatal closure can restrict CO₂ intake, negatively impacting photosynthetic efficiency and WUE (Driesen et al., 2020). Additionally, during drought stress, plants may adjust resource allocation to reduce transpiration, resulting in different WUE responses depending on species, physiological state, and stress intensity (Driesen et al., 2020). Ullah et al. (2023) demonstrated that an increase in intercellular CO₂ concentration is linked to improved water use efficiency (WUE) and enhanced drought resistance in bread wheat. This adaptation allows wheat to maintain photosynthetic activity while minimizing water loss during periods of water scarcity. Similarly, Li et al. (2015) showed that the application of exogenous abscisic acid (ABA) on ginger leaves led to an increase in intercellular CO₂ concentration, which subsequently promoted stomatal closure. This stomatal response reduces transpiration and thus improves water conservation, especially under stress conditions.

In trees, similar mechanisms are observed. For instance, elevated CO₂ concentrations often lead to reduced stomatal conductance, which helps trees conserve water while still maintaining sufficient photosynthetic rates. Studies on various tree species, such as poplar (*Poplar ssp.*) and loblolly pine (*Pinus taeda*), have shown that increased CO₂ can enhance water use efficiency by reducing stomatal opening, thereby minimizing water loss without significantly impairing carbon uptake (Hatfield and Dold, 2019). Additionally, under drought stress, trees like oak and pine exhibit varying stomatal responses depending on species and environmental conditions. While some species can maintain high photosynthesis and WUE under drought, others exhibit more significant reductions in stomatal conductance to conserve water, thereby limiting growth (Lawson and Vialet, 2019). This dynamic highlights the importance of intercellular CO₂ concentration and stomatal regulation in improving drought resilience in trees, especially as climate change drives shifts in environmental conditions.

Stomatal conductance tends to increase under high temperature conditions, although leaves can mitigate heat stress by enhancing transpiration rates and maintaining CO₂ concentrations (Urban et al., 2017a). In addition, the researchers evaluated the physiological responses of *Populus* species (poplar) and *Pinus taeda* (loblolly pine) and observed similar patterns. Even though elevated CO₂ concentrations typically induce stomatal closure, the general trend of stomatal conductance rising with increasing temperature remains consistent across these species (Urban et al., 2017b). This suggests that temperature-induced increases in stomatal conductance may be a common response in plants, potentially driven by the need to maintain internal water balance and gas exchange under varying environmental conditions.

CONCLUSION

This study demonstrates that different genotypes of silver lindens exhibit significant physiological differences under the same environmental conditions, affecting their adaptability and potential functions in urban landscapes. Genotypes 1 and 3 show the highest net photosynthetic rates (A) and water use efficiency (WUE), exhibiting greater resilience to water stress, while genotype 2 makes the best performance in transpiration rate (E), stomatal conductance (gs) and intercellular CO₂ concentration (Ci), indicating better tolerance under heat stress. These findings suggest that selecting genotypes with specific adaptive traits can enhance the resilience and ecological value of urban green spaces. Overall, this study provides a foundation for making informed choices in urban forestry to address environmental challenges in a changing climate.

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TESTING AND ANALYSIS OF THE ANTIBACTERIAL EFFECT OF THE CHAMOMILE, IMMORTELLE AND CALENDULA ESSENTIAL OIL

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In contemporary times, the protection of fresh fruits and vegetables from contamination and spoilage is most commonly achieved through the use of various chemical treatments and preservatives. However, some of these substances, after entering the food chain, can cause complications and health issues for consumers. As a result, there has been a growing need for the development of new, more effective, and safer natural alternatives, such as biopreparations and preservatives of plant origin. Given that bacterial species like Escherichia coli and Enterococcus faecalis, due to improper agrotechnical practices during production, frequently contaminate fresh agricultural products, current research is focused on identifying essential oils and plant extracts with the potential to inhibit the growth of these bacteria. The aim of this study was to examine the antimicrobial effects of essential oils from Helichrysum italicum (immortelle) and Calendula officinalis (calendula) on the growth of Escherichia coli and Enterococcus faecalis under in vitro conditions. The antimicrobial activity of these oils was assessed using the disk diffusion method on Mueller-Hinton agar. Paper disks, 6 mm in diameter, impregnated with the selected essential oils, were placed at designated spots on the nutrient agar, which had been previously inoculated with pure bacterial cultures using the swabbing method. Incubation was carried out at 37°C for 24 hours. The results were interpreted according to the EUCAST (European Committee on Antimicrobial Susceptibility Testing) and CLSI (Clinical and Laboratory Standards Institute) standards. The obtained results showed that the essential oils, extracted using the water distillation method according to Clavenger, exhibited a certain inhibitory effect on the growth of the tested bacterial species.

Keywords: essential oils, disk diffusion method, antimicrobial activity

INTRODUCTION

The plant world is a rich source of bioactive substances with significant antimicrobial potential. Essential oils (EOs) extracted from plant materials have long been valued for their antimicrobial and antioxidant properties. Since the Middle Ages, EOs have been widely applied for bactericidal, virucidal, fungicidal, antiparasitic, insecticidal, medicinal, and cosmetic purposes. Today, they are extensively utilized in the pharmaceutical, sanitary, cosmetic, agricultural, and food industries. These oils are obtained from various parts of plants through distillation, extraction, or cold pressing. EOs are complex natural mixtures of volatile, lipophilic, and aromatic compounds produced as secondary metabolites by plants. Their chemical composition encompasses hundreds of organic compounds. While they may contain numerous components, one or a few typically dominate, imparting distinct characteristics. The composition of these components depends on various factors, including geographic origin, the plant part utilized, vegetative stage, plant and fruit maturity, the method of EO isolation, and the plant's chemotype. The antimicrobial activity of EOs is influenced by their chemical makeup, functional groups, and potential synergistic interactions between different components (Peruč et al., 2018).

This study focuses on essential oils extracted from specific plant species within the Asteraceae family and pure bacterial cultures from the Enterobacteriaceae and Enterococcaceae families. The primary aim was to investigate the antimicrobial potential of essential oils derived through hydrodistillation from *Helichrysum italicum* (immortelle) and *Calendula officinalis* (calendula) against these bacterial species. Owing to their chemical composition, essential oils exhibit therapeutic properties and frequently possess antimicrobial activity. Numerous studies have confirmed their effectiveness against a range of pathogenic bacterial and fungal species. This research hypothesizes that the tested bacterial species will exhibit varied sensitivities to the extracted oils and that the oils will demonstrate distinct inhibitory effects on the same bacterial strain. It is expected that the inhibition zones produced by the oils will be smaller or similar to those produced by the control antibiotic, Gentamicin.

MATERIALS AND METHODS

The study was conducted in 2022 at the Microbiology Laboratory and the Crop Science Laboratory of the Faculty of Agriculture and Food Sciences, University of Sarajevo. The research materials included dried aerial parts of *Helichrysum italicum* (immortelle) and *Calendula officinalis* (calendula). The antimicrobial effects of essential oils were tested on pure cultures of *Escherichia coli* and *Enterococcus faecalis*.

Extraction method. Essential oils were extracted through hydrodistillation using a Neo Clevenger apparatus. The dried aerial parts of the plants were ground and distilled using water, a commonly employed method for isolating essential oils. While effective, this method requires extended processing time, increasing costs and potentially causing hydrolysis of components, thereby reducing yields and volatile compound retention (Pingret et al., 2014). The Clevenger apparatus comprises a flask, condenser, and separator for automatic separation of distilled water, with steam preventing oil degradation (Mikulić, 2022).

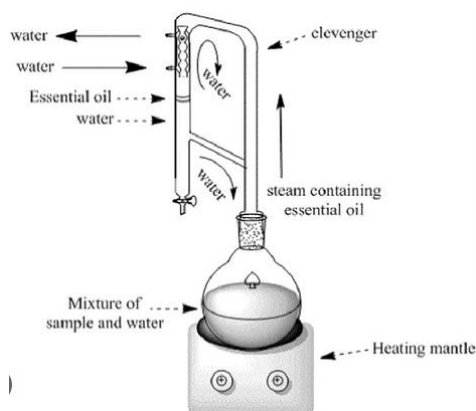


Fig. 1 NeoClevenger Apparatus (Photo sourced from: https://www.researchgate.net/figure/Hydro-distillation-Clevenger-apparatus-system_fig1_308574938)

Preparation of growth medium and sterilization. The antimicrobial potential of immortelle and calendula essential oils against *Escherichia coli* and *Enterococcus faecalis* was tested on Mueller-Hinton agar, a standard medium for antimicrobial sensitivity testing. Laboratory glassware and tools were sterilized at 121°C, while the nutrient medium was prepared by mixing the appropriate quantity of agar and distilled water, followed by heating until fully dissolved. The solution was autoclaved at 121°C and 1.5 bar for 15 minutes and then poured into sterile Petri dishes under aseptic conditions. Additional reagents included ethanol and distilled water, and the equipment utilized comprised a precision balance, autoclave, laminar airflow cabinet, incubator, micropipettes, and other laboratory tools.

Preparation of pure bacterial cultures. A "pure bacterial culture" refers to a population originating from the replication of a single bacterial cell. Pure cultures were prepared by transferring 3–5 colonies of target bacteria into tubes containing physiological saline. These were incubated at 37°C for 24 hours to achieve microbial suspensions corresponding to 0.5 McFarland standard (approximately 1×10^8 CFU/ml).

Inoculation of growth medium with test bacteria. Using sterile swabs, test bacterial cultures were streaked onto the prepared agar plates in a zigzag pattern under aseptic conditions to ensure confluent growth. Before inoculation, the suspensions were allowed to rest at room temperature for 15 minutes, and any excess liquid was removed by pressing the swab against the tube wall. After inoculation, the agar plates were left for 15 minutes to absorb the inoculum.

Diffusion test method (Kirby-Bauer Test). Sterile paper disks (6 mm in diameter) impregnated with the tested essential oils were placed at designated spots on the inoculated agar plates. The plates were then incubated at 37°C for 16–24 hours. The essential oil diffuses from the disks, inhibiting the growth of bacteria sensitive to its action. The diameter of the inhibition zones, measured in millimeters, reflects the degree of microbial sensitivity. The results were compared against standard values for the tested bacteria and the control antibiotic, Gentamicin.

RESULTS AND DISCUSSION

The interpretation of results was carried out according to EUCAST (The European Committee on Antimicrobial Susceptibility Testing) and CLSI (Clinical and Laboratory Standards Institute) standards.

According to these standards, the effect of essential oils on the tested microorganisms was classified into three categories:

- **S** – Sensitive
- **I** – Intermediate (Moderately Sensitive)
- **R** – Resistant

Based on the measured inhibition zones of the tested oil samples and bacterial species, a comparative analysis was performed between the obtained results and the standard values for the antibiotic Gentamicin against *Escherichia coli* and *Enterococcus faecalis*. The results are presented in both tabular and graphical formats. The standard values for Gentamicin for *Escherichia coli* and *Enterococcus faecalis* are shown in the following tables.

Table 1. Antibigram for *E. coli* CISI M100-S21 Performance Standards for Antimicrobial Susceptibility Testing. EUCAST

<i>Escherichia coli</i>	Inhibition zone diameter		
	S	I	R
Gentamicin	≥ 15	13-14	≤ 12

Table 2. Antibigram for *Enterococcus faecalis*

<i>Enterococcus faecalis</i>	Inhibition zone diameter		
	S	I	R
Gentamicin	≥ 10	7-9	≤ 6

The incubation lasted for 16 to 24 hours at a temperature of 37°C for *Escherichia coli* and *Enterococcus faecalis*. The oil diffuses from the disks into the medium, and if effective, it prevents bacterial growth. During the incubation, bacterial colonies grew at a certain distance from the disk, creating clear areas, known as inhibition zones. After incubation, the diameter of the inhibition zone was measured and presented in the table, expressed in millimeters.

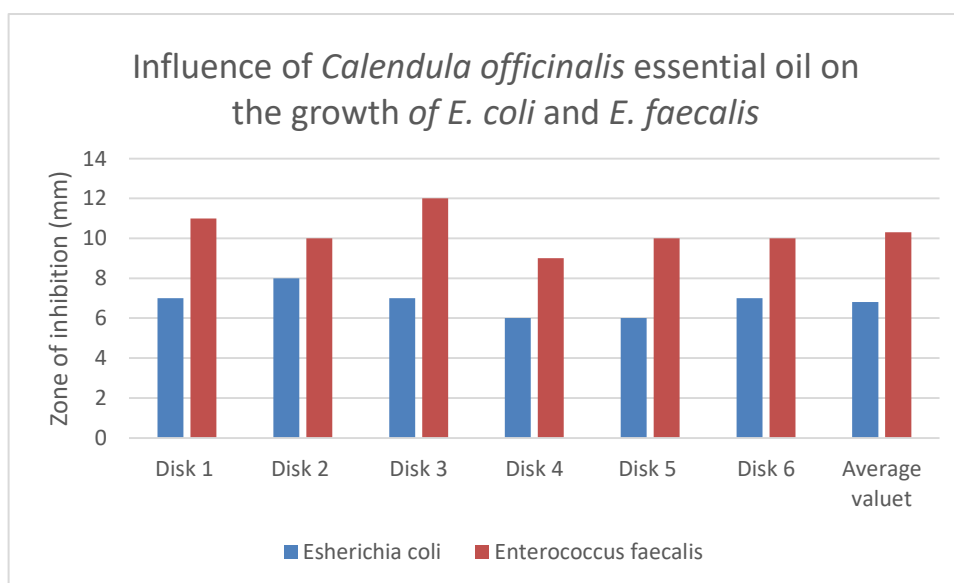


Fig. 2 Results of inhibition zone measurement for Calendula essential oil against *E. coli* and *E. faecalis*

According to the results shown in Table 5, the essential oil of calendula does not exhibit an inhibitory effect on *Escherichia coli*. The average diameter of the inhibition zone for calendula essential oil was 7 mm for *E. coli*, indicating insufficient antimicrobial activity when compared to the reference values for the antibiotic

gentamicin, indicating insufficient antimicrobial activity when compared to the reference values for the antibiotic gentamicin. Therefore, according to the EUCAST standard, the tested strain is classified as **R** (Resistant).



Fig. 3 Presentation of the results of the influence of EU marigolds on *Escherichia coli*

Contrary to the results obtained in this study, research by Gazim et al. (2018) observed the largest inhibition zone diameter (13.31 ± 1.24 mm). This is attributed to the chemical components of the plant, which make calendula effective against Gram-negative bacteria. They found that *Calendula officinalis* essential oil exhibits a higher level of antibacterial activity against Gram-negative compared to Gram-positive bacteria ($P < 0.05$). Earlier data showed that *C. officinalis* leaf extracts were significantly effective against both Gram-positive and, in particular, Gram-negative organisms. Gazim et al. (2018) tested the in vitro activity of *Calendula officinalis* essential oil using the disk diffusion technique. The antifungal test results demonstrated that the essential oil has good potential antifungal activity, being effective against 23 clinically tested fungal strains.

In contrast to the effect of calendula oil on *Escherichia coli*, the average value for *Enterococcus faecalis* places it in the S (Sensitive) category, compared to the average value for gentamicin. This suggests a high probability of therapeutic success following the administration of standard doses in the usual manner.



Fig. 4 Presentation of the results of the influence of EU marigolds on *Enterococcus faecalis*

The extract of calendula flowers exhibits antibacterial activity against various bacteria. In vitro, the essential oil of calendula flowers inhibited the growth of Gram-positive bacteria, including *Staphylococcus aureus* and *Bacillus subtilis*, as well as Gram-negative bacteria, including *Pseudomonas aeruginosa* and *Escherichia coli*, showing maximum inhibition against *Pseudomonas aeruginosa*. The reproductive parts of calendula show less antibacterial activity compared to the petals. The flower decoction and methanol extract exhibited antibacterial activity against various facultative aerobic and obligate anaerobic periodontal bacteria, including *Fusobacterium nucleatum*, *Porphyromonas gingivalis*, *Capnocytophaga gingivalis*, *Prevotella* spp., *Veillonella parvula*,

Peptostreptococcus micros, *Eikenella corrodens*, and *Actinomyces odontolyticus*. Extracts of calendula leaves, stems, roots, and flowers demonstrated antimicrobial activity against various human pathogenic microbes: *Candida albicans*, *Candida parapsilosis*, *Pseudomonas aeruginosa*, *Escherichia coli*, coagulase-positive *Staphylococcus* spp., *Enterococcus* spp., and coagulase-negative *Staphylococcus* spp., as reported by Nelofer et al. (2017).

Resistant microorganisms can withstand the effects of antimicrobial agents such as antibacterial drugs (e.g., antibiotics), antifungals, antivirals, and antimalarials, making standard treatments ineffective, while infections persist and increase the risk of spreading to others. The emergence of antimicrobial resistance has led chemical and pharmaceutical companies to explore new sources for antibiotic production. The use of medicinal plants for producing antimicrobial agents has been extensively studied for this reason (Chaleshtori et al., 2016).

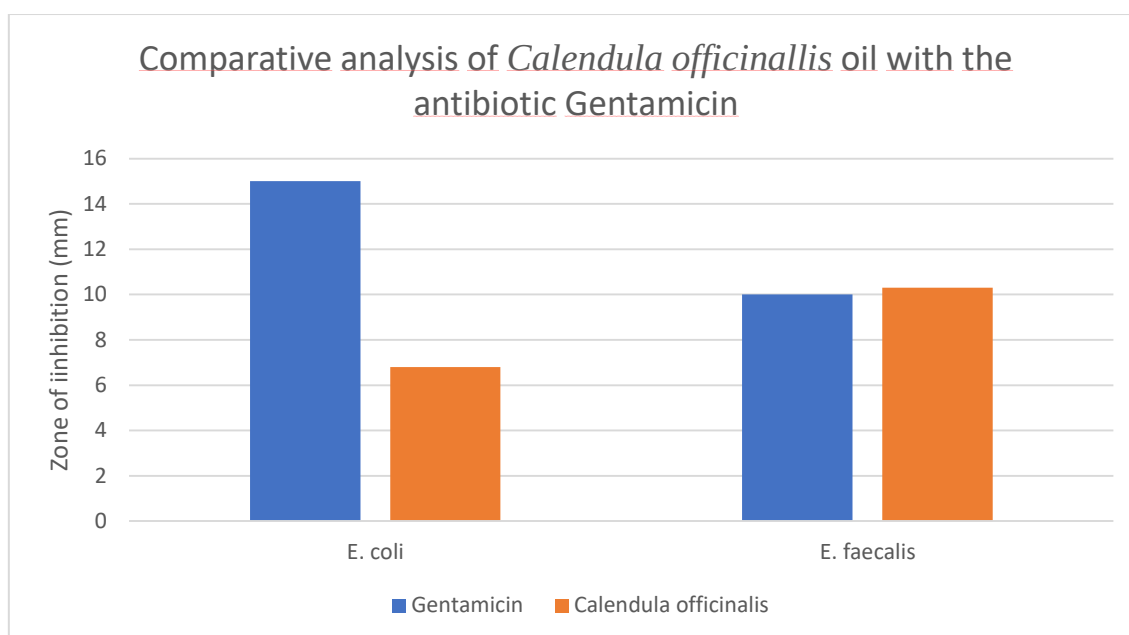


Fig. 5 Graphic representation of the comparative analysis of calendula oil with the antibiotic Gentamicin

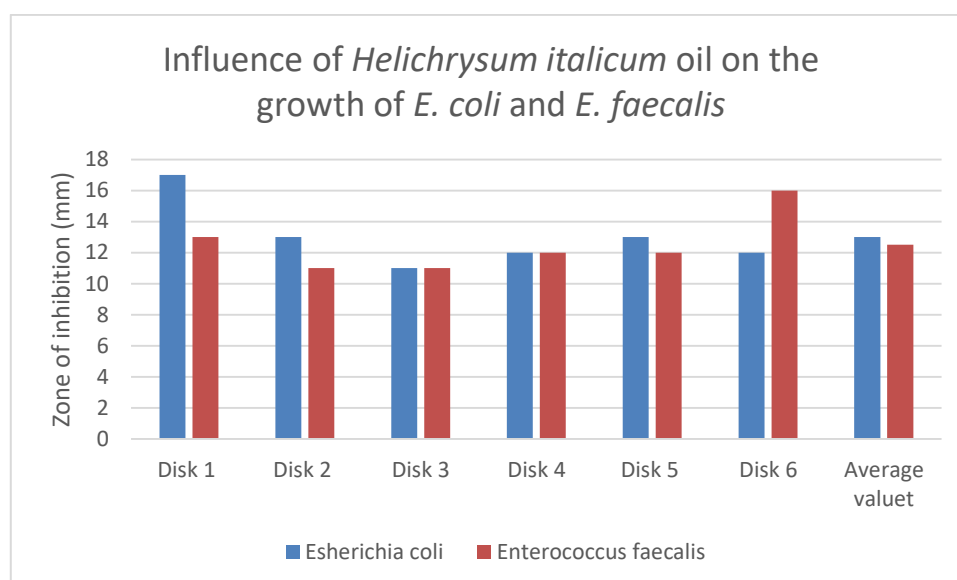


Fig. 6 Results of measurement of inhibition zones of immortelle essential oil for bacteria *E. coli* and *E. faecalis*

The results presented in the table show that immortelle essential oil has a strong effect on the tested bacterial species *E. coli* and *E. faecalis*. The average inhibition zone diameter for immortelle essential oil was

13.3 mm for *E. coli* and 12.5 mm for *E. faecalis*. When compared to the reference values for the antibiotic gentamicin, according to the EUCAST standard, the tested strains were classified as I (intermediate/moderately sensitive) for *E. coli*, and S (sensitive) for *E. faecalis*.



Fig. 7 and 8 (right) Results of the impact of EU immortelle on *Escherichia coli*

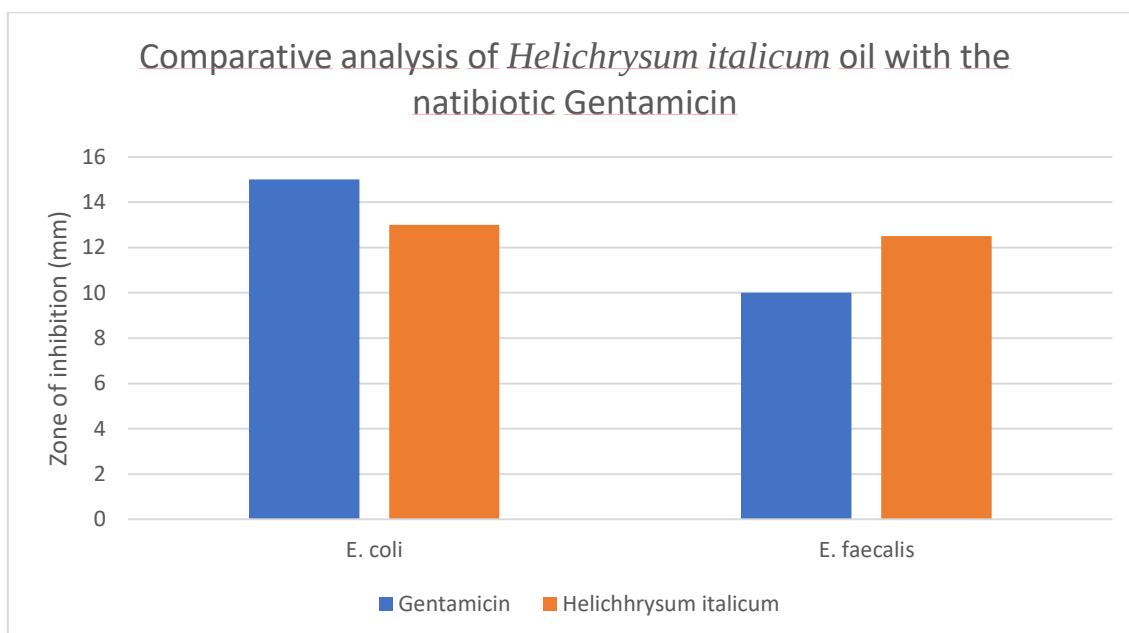


Fig. 9 Graphical representation of comparative analysis of immortelle oil with the antibiotic Gentamicin

Numerous studies in the literature have investigated the antimicrobial effects of *Helichrysum italicum* essential oil, focusing on its impact on Gram-positive and Gram-negative bacteria as well as various fungi. Baniček (2019) states that *H. italicum* essential oil significantly reduces the resistance of *Enterobacter aerogenes*, *E. coli*, *P. aeruginosa*, and *Acinetobacter baumannii* to multiple antimicrobial agents.

According to Zheljazkov et al. (2022), fungi were the most resistant class of microorganisms to the presence of *H. italicum* essential oil, which aligns with the findings of Djihane et al. (2017), who studied the antimicrobial activity of *H. italicum* essential oil against Gram-positive (G+) and Gram-negative (G-) bacteria as well as fungi. Gram-positive bacteria were more sensitive to the presence of the essential oil than Gram-negative bacteria or fungi. The antibacterial test showed that the most resistant strains were *E. coli*, *K. pneumoniae*, and *L. monocytogenes* to all concentrations. In contrast, *M. luteus* and *S. aureus* were the most sensitive. Šćepanović et al. (2019) demonstrated that *Staphylococcus aureus*, *Listeria monocytogenes*, and

Bacillus subtilis were the most susceptible bacteria. Servi (2022) reported that although the essential oil was inactive against *E. coli*, it showed promising activity against *S. aureus*, *P. aeruginosa*, *B. cereus*, and *B. subtilis*.

Aćimović et al. (2021) tested *H. italicum* essential oil grown in Serbia on 16 bacterial isolates using an antibacterial test on a microtiter plate, employing resazurin as a cell growth indicator. Gentamicin strip tests were used as controls to determine the minimum inhibitory concentration. The oil exhibited low or no activity against the tested bacteria. Mollova et al. (2020) found that *H. italicum* essential oil inhibited only Gram-positive bacteria and had antifungal effects, noting that the oil from Bosnia showed inhibitory activity only against *Aspergillus brasiliensis*. According to Chinoua et al. (1996), oxidized monoterpenes (geraniol, geranyl, and neril acetate) are most likely responsible for the antimicrobial activity.

CONCLUSION

Based on the conducted research, the following conclusions can be drawn:

The effect of essential oils on the inhibition of bacterial growth was influenced by both the plant species from which the oil was extracted and the type of bacteria tested.

Immortelle essential oil showed the strongest inhibitory effect on the growth of *E. coli* (13.3 mm), while calendula oil had the weakest inhibitory effect on this bacterium.

The most significant inhibitory effect on the growth of *Enterococcus faecalis* was shown by immortelle essential oil (12.5 mm). Calendula oil showed slightly weaker, but consistent inhibitory effects.

Based on the zones of inhibition and their comparison with the standard values for the antibiotic gentamicin against *Enterococcus faecalis*, it can be concluded that this bacterial species is sensitive to the 2 tested essential oils, classifying it as S (sensitive).

Comparing the diameters of the inhibition zones of immortelle essential oil with the reference values for gentamicin, *E. coli* was classified as I (intermediate/moderately susceptible), while calendula oil showed no effect, classifying *E. coli* as R. (resistant).

Based on these findings, further research into the effectiveness of these essential oils in preventing fecal bacterial contamination of certain vegetable crops is recommended.

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PHYLOGENETIC ANALYSIS OF THE SERBIAN VIOLA WHITE DISTORTION ASSOCIATED VIRUS ISOLATE FROM PANSY VIOLETS

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*In 2019, greenhouse-grown pansies with bleaching and leaf deformation were observed in the Bečej locality. Symptomatic plants were sampled and mechanically transmitted on *Nicotiana benthamiana* which produced mild leaf deformation of the upper leaves, indicating infectious nature of the disease. RT-PCR confirmed presence of one of ilarviruses, while sequence of the partial 2a and MP gene of a selected isolate showed the highest similarity to viola white distortion associated virus (VWDaV). Further phylogenetic analysis confirmed grouping of Serbian isolate among VWDaV group.*

Key words: *Viola x wittrockiana*, ilarviruses, VWDaV, phylogenetic analysis

INTRODUCTION

Purple, pink, yellow, white or blue are just some of the colors that make plants from the family *Violaceae* one of the most beautiful and frequently used ornamental plants in many gardens. In addition to this role, these plants are also used as a decorative component in culinary dishes. *Viola sororia*, *Viola pedata*, *Viola odorata*, *Viola cornuta*, *Viola palmensis*, *Viola hirsutula*, *Viola riviniana* are the best-known species, that make up only a small part of the large genus *Viola*, which comprises around 500 species. In addition, *Viola x wittrockiana* that is commonly known as the pansy violet, has attracted a lot of attention in the recent years. The main reason for increasing interest in this ornamental plant is its high nutritional potential and its property as a source of bioactive compounds, such as flavonoids and anthocyanins (Kaisoon et al., 2012). Cardoso et al. (2011) suggested that bioactive compounds present in the diet exert a synergistic effect on free radicals and consequently a protective effect on health. A recent study has shown that the petals of *Viola x wittrockiana* have a high content of carbohydrates and proteins, so that these flowers can be used as a protein supplement in food (La da Silva et al., 2020). Apart from its interesting chemical composition and bioactive compounds, this semi-evergreen annual plant is one of the most popular bedding plants, often growing in early spring and late autumn offering winter color in cool weather in many areas. Its cultivation is neither difficult nor expensive to cultivate as they are tolerant of growing conditions (Kuehny and Morales., 1998).

Despite their adaptability to varying climatic factors, it is well-known that the pansies are a natural host for many pathogens, but plant viruses are among the most serious constraints to their production worldwide. Data on viruses infecting plants of the genus *Viola* are scarce but cucumber mosaic virus (CMV, *Cucumovirus CMV*), tobacco rattle virus (TRV, *Tobravirus tabaci*), bean yellow mosaic virus (BYMV, *Potyvirus phaseoliteum*) and *Viola* isolate of tulip virus X (TVX-Vi, *Potexvirus ecstulipae*), synonyms viola mottle virus (VMoV) have been reported infecting pansy plants (Cardin and Moury, 2007; Ciuffo et al., 2014; Matsumoto et al., 2021; Bak et al., 2024).

In addition, a new syndrome on pansies characterised by white deformation was noticed at the end of 2000 and it is caused by viola white distortion associated virus (VWDaV). The virus has been reported in Italy on *Viola x wittrockiana* (Ciuffo et al., 2014) and in the United Kingdom on *Viola tricolor* (Fox et al., 2016). Pansy plants infected with VWDaV exhibit typical white patterns and bleaching, accompanied by deformation and leaf curling. The infected plants does not change in the number, size or color of flowers (Ciuffo et al., 2014).

In Serbia, VWDaV was first recorded on *Viola x wittrockiana* in 2019. After detection, the main objective of this research was to determine the genetic relationship of the Serbian VWDaV isolate with previously reported VWDaV isolates from GenBank as well as with sequences of other ilarviruses. This information is needed as a first step in developing and implementing appropriate control measures since intensive international trade from distant regions favoring long-distance movement of the virus.

MATERIAL AND METHODS

Samples collection. In September 2019, pansy violets (*Viola x witrockiana* cv. unknown) with virus-like symptoms were observed in a greenhouse in the Bečej locality, South Bačka District, Serbia. Samples prepared from symptomatic leaves from five plants (one sample per plant) were collected, transported to the laboratory at the Faculty of Agricultural in Belgrade and stored at 4°C until tested.

Mechanical transmission. To prove the infectious nature of the disease, sap extracted from collected samples were mechanically inoculated onto five *Nicotiana benthamiana* plants. Symptomatic leaves were ground in 0.01 M phosphate buffer (pH 7) and inoculated to test plants using silicon carbide abrasive. Inoculated plants were maintained in a greenhouse at a temperature between 22 and 25°C and were inspected regularly for symptom development during the period up to one month post-inoculation.

Serological testing. In the further analysis, the collected samples as well as mechanically inoculated *N. benthamiana* plants, were analysed by double-antibody sandwich (DAS)-ELISA using a commercial kit (DSMZ, Braunschweig, Germany) against four viruses known to infect viola species (Ciuffo et al. 2014): CMV, TRV, BYMV and VMoV. Plant tissue samples were homogenized with extraction buffer at a ratio of 1:10 (weight/volume) using a mortar and pestle. Absorbance at 405 nm was measured spectrophotometrically using an ELISA microplate reader (DAS srl, Italy) after incubating the mixture with p-nitrophenyl phosphate (Sigma-Aldrich, St. Louis, MO) at 23°C for 1-2 hours in the dark. Samples were considered positive if their mean absorbance value was two times higher than that of the negative control. Commercial positive and negative controls were included in each ELISA.

Molecular detection. The samples were also screened by reverse transcription (RT)-PCR using the One-Step RT-PCR Kit (Qiagen, Hilden, Germany) and ilarvirus-specific primers Ilar2F5/Ilar2R9, designed to amplify fragments of 380 bp of all ilarviruses (Untiveros et al., 2010). Total RNAs from all pansy samples and test plants were extracted from 100 mg of symptomatic leaves using RNeasy Plant Mini Kit (Qiagen), according to the manufacturer's instructions. The RT-PCR reaction mixture included 400 µM of each of the four dNTPs, 0.6 µM of the viral sense and complementary sense primers, 5 µl of 5x Qiagen OneStep RT-PCR Buffer, 1 µl of RT-PCR enzyme mix and 1 µl of extracted RNA. The volume of the mixture was adjusted to 25 µl using RNase-Free Water. One-step RT-PCR was performed in a thermal cycler (Applied Biosystem 2720 Thermal Cycler) under the following conditions: 30 min at 50°C for reverse transcription, 15 min at 94°C for initial denaturation, followed by 40 cycles of denaturation for 30 s at 94°C, annealing for 1 min at 44°C and extension for 1 min at 72°C, with a final extension step for 10 min at 72°C. The size of the amplified products was determined on 1% agarose gel in TBE buffer and stained with Midori Green Advance DNA Stain (Nippon Genetics Europe, GmbH).

Sequencing and phylogenetic analysis. For specific detection of VWDaV, previously described primers Viola-For/Viola-Rev (Ciuffo et al., 2014) designated to amplify a 482 bp amplicon of partial MP gene were used. The components of the RT-PCR reactions were as previously described above, while amplification was performed under the following conditions: reverse transcription at 50°C for 30 min, initial denaturation at 95°C for 15 min, followed by 35 cycles of denaturation at 94°C for 1 min, annealing at 57°C for 1 min, and extension at 72°C for 1 min. The final extension step was performed at 72°C for 10 min. The size of the amplified products were determined as described earlier.

Amplified product of one selected isolate, 211-19, was purified (QIAquick PCR Purification Kit, Qiagen), Sanger sequenced in MacroGen-Europe Laboratory (Amsterdam, the Netherlands), and deposited in GenBank. The sequence obtained in this study was compared with previously identified VWDaV sequences from other parts of the world deposited in GenBank using the BLAST algorithm (<http://www.ncbi.nlm.nih.gov/BLAST/>).

A phylogenetic tree was constructed using the neighbor-joining method implemented in MEGA X (Kumar et al., 2018), using 22 MP gene sequences of different ilarviruses. These included the sequence generated in this study, two previously sequenced VWDaV isolates, as well as sequences of other ilarviruses retrieved from GenBank (Table 1). The sequence of MP gene of brome mosaic virus (GenBank Acc. No. AB183261) serving as an outgroup. The reliability of the obtained tree was evaluated using the bootstrap method based on 1000 replicates, and bootstrap values <50% were omitted.

Table 1. MP gene sequences of ilarviruses used in the phylogenetic analysis

Virus	Country	GenBank Accession number
viola white distortion associated virus (VWDaV)	Serbia	MT551219
viola white distortion associated virus (VWDaV)	Italy	GU168941
parietaria mottle virus (PMoV)	USA	U35145
tobacco streak virus (TSV)	Netherlands	X00435
asparagus virus 2 (AV2)	Mexico	X86352
citrus leaf rugose virus (CiLRV)	USA	U17390
citrus variegation virus (CVV)	USA	U17389
elm mottle virus (EMoV)	USA	U85399
lilac ring mottle virus (LiRMoV)	USA	U17391
spinach latent virus (SpLV)	USA	U93194
tulare apple mosaic virus (TaMV)	USA	AF226162
apple mosaic virus (APMV)	USA	U15608
prunus necrotic ringspot virus (PNRSV)	USA	U57046
fragaria chiloensis latent virus (FCiLV)	USA	AY707772
american plum line pattern virus (APLPV)	USA	AF235166
humulus japonicus latent virus (HJLV)	USA	AY500238
bacopa chlorosis virus (BaCV)	USA	FJ607142
blackberry chlorotic ringspot virus (BCRV)	United Kingdom	DQ091195
strawberry necrotic shock virus (SNSV)	USA	AY363228
lilac leaf chlorosis virus (LLCV)	Canada	FN669169
prune dwarf virus (PDV)	USA	L28145

RESULTS AND DISCUSSION

Symptoms and virus detection. During the visual inspection of ornamental plant production in 2019, virus-like symptoms were observed in a greenhouse in the Bečej locality. Infected plants exhibited symptoms in the form of bleaching and leaf distortion (Figure 1) and the disease incidence was estimated to be about 40%. Sap prepared from symptomatic leaves of five pansy violets was mechanically inoculated onto *N. benthamiana* plants, which developed mild leaf deformation of the upper leaves 14 days post-inoculation, indicating the presence of an infectious agent. Mock inoculated *N. benthamiana*, which was used as a negative control, showed no symptoms.

Figure 1. Symptoms of VWDaV infection: **a** bleaching and leaf deformation; **b** leaf distortion

Further, leaves sampled from five symptomatic pansy plants as well as five mechanically inoculated *N. benthamiana* were analysed by DAS-ELISA for the presence of four viruses known to infect viola species (Cardin and Moury, 2007; Ciuffo et al., 2014; Bak et al., 2024) and no tested viruses were found.

The samples were also screened by RT-PCR using the ilarvirus-specific primers Ilar2F5/Ilar2R9 (Untiveros et al., 2010). The result of the RT-PCR testing revealed that an amplicon of the expected size (380 bp) was obtained from all pansy samples and symptomatic *N. benthamiana* plants.

In addition, RT-PCR with the specific primers for the detection of VWDaV, Viola-For/Viola-Rev (Ciuffo et al., 2014), revealed that all symptomatic pansy violet and *N. benthamiana* plants yielded an amplicon of approximately 480 bp that confirming the presence of the virus. No amplification product was recorded in the negative control.

In the last 20 years, a new syndrome - white deformation, were often noticed on pansies in different parts of the world. The first characterisation of the infectious agent associated with this syndrome was carried out in

Italy, where a new ilarvirus, viola white distortion associated virus, was identified (Ciuffo et al., 2014). Subsequently, the virus was reported in nurseries across England (Fox et al., 2016).

Sequence analysis and phylogenetic tree. To characterise selected isolate 211-19, the RT-PCR products were directly sequenced and obtained sequence were deposited in GenBank under the accession number MT551219. BLAST results revealed that MP sequence of the Serbian VWDaV isolate shared the highest nucleotide identity of 98.91% and 94.10% with two VWDaV isolates available in GenBank (GU168941 and KX196164, respectively).

Phylogenetic tree (Figure 2) showed clustering of the selected isolates according to virus species as previously indicated by Ciuffo et al. (2014). The Serbian VWDaV isolate 211-19 was grouped together with previously reported VWDaV isolates.

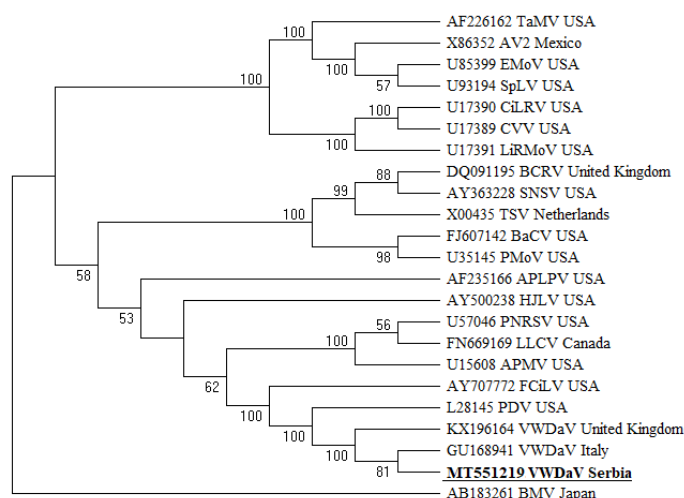


Figure 2. Phylogenetic tree based on 22 partial MP nucleotide sequences of ilarviruses using MEGA X and bootstrap analysis with 1000 replicates. The viola white distortion associated virus isolate from this study is underlined and bolded.

Considering the easy transmission of ilarviruses through seed and pollen (Mink, 1993), VWDaV could have serious limitations in the production of pansies and further monitoring of this disease is required. In addition, further research should focus on understanding the basic epidemiological characteristics of this new virus. This is essential to prevent spread of the virus to other, yet unknown host plants.

CONCLUSIONS

Following the previously reported outbreak of VWDaV in pansies (*Viola x wittrockiana*), this study represents the first characterisation by sequencing and phylogenetic analysis of this virus in Serbia. The Serbian isolate originating from pansy violets was clustered with previously reported VWDaV isolates. Considering that VWDaV is a new virus species, further investigations should focus on clarifying some unknown epidemiological aspects such as seed and pollen transmission and the discovery of new host plants, which is crucial for the development and implementation of effective control measures.

ACKNOWLEDGEMENTS

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EFFECTIVENESS OF *BACILLUS* STRAINS ON PEPPER SEED GERMINATION

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Pepper stands out as a crucial vegetable plant, and its successful cultivation relies on the use of certified seeds. Prior to sowing, it is imperative to assess the energy and final germination potential of pepper seeds. Among the notable promoters of seed germination and plant growth are bacteria belonging to the genus Bacillus. Bacillus spp. facilitates plant growth through mechanisms such as phytohormone production, nitrogen fixation, and phosphate solubilization. In this study, pepper seeds were inoculated with two mixtures of Bacillus spp.: Mixture No. 1 and Mixture No. 2. The first mixture contained three strains of Bacillus subtilis (20.10, 57.2, and 65.5) and two strains of Bacillus amyloliquefaciens (28.3 and 21/IV). The second mixture consisted of four B. subtilis strains (B5, B7, B13, and B32). Pepper seed inoculated with Mixture No. 1 or Mixture No. 2 showed higher values of final germination (91.33% and 91.66%) compared to the control (83%). Also, by measuring the morphological parameters, a statistically significant increase in the shoot length of the seedlings was obtained using Mixture No. 2 (2.49 g) compared to the control (1.67 g). Furthermore, with the application of Mixture No. 1, a statistically significant increase in root length (6.61 cm) was recorded compared to the control (2.52 cm). Also, both mixtures provided a statistically significant effect on increasing the dry mass of shoot and root (0.039 g) compared to the control treatment (0.028 g). These findings indicate the potential of the tested Bacillus spp. mixtures in promoting both pepper seed germination and the initial plant growth.

Key words: stimulation, seed germination, mixtures, *Bacillus* spp.

INTRODUCTION

Pepper is an annual herb belonging to the *Solanaceae* family and *Capsicum* genus, which consists of more than 31 different species, including five domesticated species: *Capsicum baccatum*, *C. annuum*, *C. pubescens*, *C. frutescens*, and *C. chinense* (Batiha et al., 2020). Among these, sweet pepper (*C. annuum*) stands out for its diverse purpose and application. Recognized for its high nutritional value, pepper is rich in phytochemicals with antioxidant properties, featuring substantial quantities of vitamins A and C. The consumption of 60-80 g of pepper per day can fulfil 100% and 25% of the recommended daily intake of these vitamins, respectively (Palma et al., 2020). Additionally, sweet pepper contains phenolic compounds, carotenoids, tocopherols, and capsaicin (Thuphairo et al., 2019). Pepper is cultivated globally on 2.020.816 ha, yielding 18.29 t/ha (FAO, 2022). In Serbia, pepper cultivation spans over 9.915 ha with an average yield of 12.0 t/ha according to 2023 data (<https://data.stat.gov.rs/>).

Bacillus spp. represents a diverse group of Gram-positive, motile, aerobic or facultative anaerobic bacteria with rod-shaped cells, ubiquitous in nature (Zeigler and Perkis, 2008). Certain members of this genus producing phytohormones such as auxins, gibberellins, and cytokinins, positively influence seed germination, growth, and development of plant organs (Tsotetsi et al., 2022). In addition, plant hormones indirectly regulate crucial processes such as pathogen resistance, antioxidant functions, and improved absorption of water and minerals from the soil. Through this mechanism, *Bacillus* spp. increase plant biomass, consequently impacting crop yield. By assimilating nutrients through nitrogen fixation, phosphate solubilization, and siderophore production, *Bacillus* spp. indirectly influences the concentration of phytohormones in plants, as the hormonal status is contingent upon the availability of mineral nutrients in the soil (Kudoiarova et al., 2019).

Rapid seed germination and seedling emergence are prerequisites for successful crop establishment and efficient production, for which seed treatment with *Bacillus* strains could have an important role. Therefore, the aim of this study was to investigate the effect of two mixtures of *Bacillus* spp., consisting of proven effective strains and newly isolated strains, on seed germination and the initial growth of pepper.

MATERIAL AND METHODS

Experimental design. Pepper seeds, variety NS Prva (Institute of Field and Vegetable Crops, Novi Sad - IFVCNS), were first disinfected in 70% ethyl alcohol (30 s), then immersed in 2.5% NaOCl (30 s) and rinsed three times with sterile distilled water (SDW). The seeds were inoculated by immersion in a 10 ml suspension containing two mixtures of isolates of *Bacillus* spp. in a concentration of 10^8 CFU ml⁻¹. The first mixture (Mixture No. 1) contained three strains of *Bacillus subtilis* (20.10, 57.2, and 65.5) and two strains of *Bacillus amyloliquefaciens* (28.3 and 21/IV). The second mixture (Mixture No. 2) included four strains of *B. subtilis* (B5, B7, B13, and B32). Strains from Mixture No. 1 originated from the rhizosphere soil of different agricultural crops, while strains from Mixture No. 2 are from the Collection of Section for Microbiological Preparations, IFVCNS. In the control treatment, pepper seeds were immersed in 10 ml of nutrient broth (NB), a liquid medium that was used for the cultivation of the abovementioned *Bacillus* strains.

The inoculated seeds were placed on sterile filter paper at the bottom of a Petri dish (R = 90 mm). An additional filter paper moistened with SDW was placed over the seeds, and the Petri dishes were incubated in a germination chamber at a temperature range of 30 °C, ensuring complete saturation of the air with moisture (70.7%). The test involved germinating 100 seeds in three replicates. Germination energy and final germination of pepper seeds were evaluated after 7 and 14 days, respectively. After germination assessment, 10 seedlings were randomly selected from each replicate for measurement of shoot and root length (cm), as well as shoot and root dry weight (g) (ISTA, 2022).

Statistical analysis. Experimental data were processed using one-way analysis of variance (ANOVA). All analyses were performed in the program STATISTICA 10, for a confidence level of 95% (StatSoft Inc., USA).

RESULTS AND DISCUSSION

The influence of *Bacillus* spp. isolates on the germination of pepper seeds. Pepper seed inoculated with Mixture No. 2 showed a higher value of germination energy (87.66%) in comparison with the Mixture No. 1 (69.33%) and the control (72.33%) (Figure 1). However, by reading the final germination, similar values were recorded for Mixture No. 1 (91.33%) and the Mixture No. 2 (91.66%) compared to the control (83%) (Figure 2). Liang et al. (2022) reported a similar final germination percentage for pepper seeds in their study, where the germination value of seeds treated with *B. subtilis* Ydj3 reached 91.7%, compared to the control treated with distilled water (66.7%). In pots containing nutrient soil, Shen et al. (2023) observed the germination of pepper seeds treated with a suspension of *Bacillus* spp. FZB42, HAB-5, and HAB-2, achieving germination percentages of 96%, 98%, and 90%, respectively, compared to the control. Yildirim et al. (2021) also supported the ecological approach of priming pepper seeds with beneficial bacteria such as *Bacillus* and *Pseudomonas*, emphasizing its efficacy in promoting rapid seed stimulation and uniform germination.

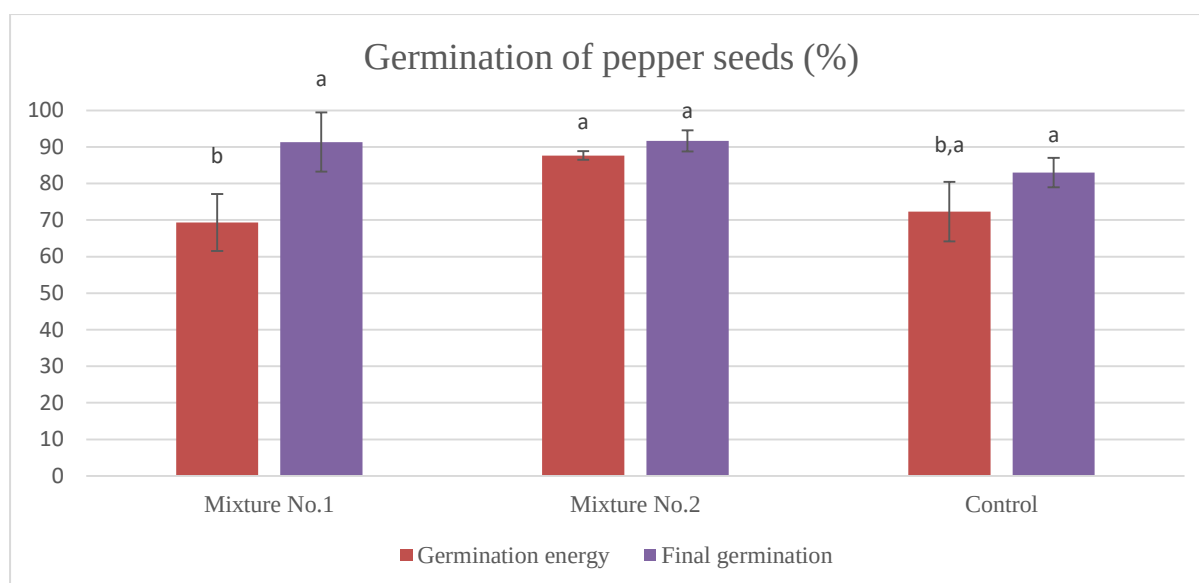


Fig. 1 The germination energy and final germination (%) of pepper seeds with standard deviation (SD) are shown. The values with the same letters did not differ significantly ($p < 0.05$) according to ANOVA

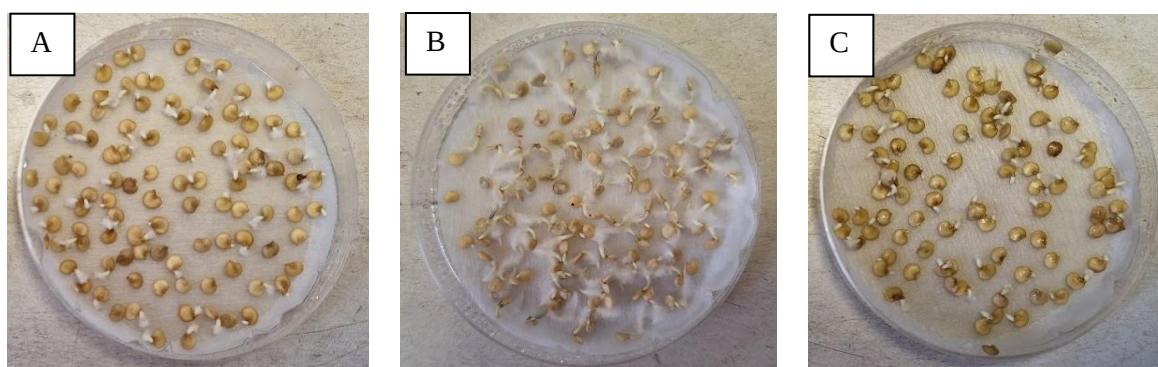


Figure 2. Germination energy of pepper seeds treated with the Mixture No. 1 (A); Mixture No. 2 (B); and control (C)

The impact of *Bacillus* spp. isolates on the morphological parameters of pepper seedlings. A statistically significant increase in the shoot length of pepper seedlings was achieved with Mixture No. 2 (2.49 cm) compared to the control (1.67 cm) (Table 1). Also, Mixture No. 1 (2.24 cm) led to the increase in the shoot length of the seedlings compared to the control, but it was not statistically significant (Table 1). The root length was affected by both mixtures, while a statistically significant increase was obtained with Mixture No. 1 (6.61 cm) compared to the control (2.52 cm) (Table 1). Both mixtures statistically significantly increased the dry weight of the shoot and root (0.039 g) compared to the control (0.028 g). In support of these findings, **Liang et al. (2022)** noted that the application of strain *B. subtilis* Idj3 stimulates the growth of pepper shoots. This highlights the potential of *Bacillus* spp. isolates in promoting positive morphological changes in pepper seedlings.

Table 1. Morphological parameters of pepper seedlings

Treatment	Shoot length (cm) $\pm$ SD	Root length (cm) $\pm$ SD	Dry weight of shoot and root (g) $\pm$ SD
Mixture No. 1	2.24 ^{ab} $\pm$ 0.17	6.61 ^a $\pm$ 1.05	0.039 ^a $\pm$ 0.002
Mixture No. 2	2.49 ^a $\pm$ 0.45	5.52 ^{ab} $\pm$ 1.13	0.039 ^a $\pm$ 0.002
Control	1.67 ^b $\pm$ 0.16	2.52 ^b $\pm$ 1.45	0.028 ^b $\pm$ 0.004

Note. The morphological parameters of pepper seedlings with standard deviation (SD) are shown. The values followed by the same letter in the columns did not differ significantly ($p < 0.05$) according to ANOVA

CONCLUSION

In this work, both mixtures increased germination energy and final germination compared to the control.

A statistically significant increase in the shoot length of pepper seedlings was achieved with Mixture No. 2, whereas Mixture No. 1 led to a significant increase in the root length as compared to control.

Both mixtures had a positive effect and significantly increased the dry weight of the shoot and root.

Together, these studies highlight the potential of the tested mixtures of *Bacillus* spp. that not only stimulated pepper seed germination but also encouraged the initial plant growth.

Acknowledgements

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APPLICATION OF AN ALGORITHM FOR VARIABLE NITROGEN FERTILIZATION BASED ON VEGETATION INDICES IN MAIZE CULTIVATION

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The research conducted during the 2023/2024 growing season on a 1.66-hectare plot in Ravno Selo, South Bačka, aimed to examine the effects of different nitrogen doses on maize yield using precision agriculture tools. The plot was divided into six distinct management zones, with base nitrogen application rates of 0, 70, and 140 kg ha⁻¹. Top-dressing of maize in the V7-V8 growth stage was carried out using the Plant-O-Meter multispectral sensor and the defined POM vegetation index. An algorithm was developed and applied for optimizing nitrogen top-dressing in real-time, resulting in improved nitrogen application efficiency.

Keywords: precision agriculture, optical sensors, variable rate technology, maize, nitrogen.

INTRODUCTION

Agricultural practice typically involves supplying soil with essential nutrients to enhance crop yield. The uncontrolled application of fertilizers, particularly nitrogen (N), should be avoided to minimize environmental impact. In fact, although nitrogen fertilization promotes plant development, the loss of huge N promotes the concerns due to the phenomenon of leaching and gasification into the atmosphere (Mosier et al., 1998). However, even if the use of nitrogen fertilizers can be reduced without significantly impacting final yields (Sehy et al., 2003), farmers often apply excessive nitrogen supplies to avoid any deficiency and ensure profitability at the end of the season (Schroder et al., 2000). The term Precision Agriculture (PA) is generally defined as the process of taking the right action in the right place at the right time; therefore, PA is not just a technology but rather a management philosophy enabled by new technologies (Kitchen et al., 2005; Yost et al., 2019). Advances in remote sensing, machine control systems, crop modeling, weather monitoring, decision-making, cloud computing, and big data analytics are leading PA into a new agricultural revolution known as smart farming (Fastellini & Schillaci 2020). Tools that have begun to be implemented in agriculture contribute to improving the accuracy of input application, allowing for enhanced production practices through the ability to predict the onset of nutrient deficiency stress or water shortage stress and to make real-time decisions. The tool known as Variable Rate Technology (VRT) is a method employed for the precise application of various agricultural inputs, including seeds, irrigation water, fertilizers, and tillage, tailored to the specific requirements of different management zones within agricultural fields. The proper assessment of required amount of nutrient for particular crop at different locations is important, which will assist to make customized dose of fertilizer that should be applied in a precise quantity, at required location. Nowadays, VRT is viewed as pivotal for the precise application of inputs (Wang et al., 2015; Späti et al., 2021). The advent of VRT enables the implementation of precise fertilizer application methods tailored to address variations within agricultural fields (Al-Gaadi et al., 2015). Maize is one of the most important grain crops in the world for feed, food, and industrial uses (Mitrović et al., 2020; Revilla et al., 2022). Even with the obvious benefits of newly developed varieties, the productive capacity of a maize genotype and its yield potential are greatly influenced by a specific combination of various factors, including climate conditions, pests, soil properties, sunlight, crop cultivation practices, seed quality, and the genetic potential of the maize hybrid (Tandzi and Mutengwa 2020). Given the pressure on modern crop production to maximize yields while minimizing inputs, timely monitoring of crop growth status early in the growing season could be vital for in-season site-specific crop management and grain yield estimation, especially in regions characterized by climatic uncertainties (Henik 2012; Ljubičić et al., 2018). Recent studies suggest that remote sensing has become an important tool in modernizing field-based high-throughput plant phenotyping,

including improvements in yield potential, adaptation to abiotic stress, biotic limitations, and grain yield estimation (Araus and Cairns 2014; Pandžić et al 2019). In comparison to remote sensing, proximal sensing, also known as ground-based remote sensing, is conducted using sensors positioned close to the plants. A large number of spectral reflectance indices calculated based on proximal sensing data have become valuable tools for monitoring agriculture and assessing crop variability at the parcel level, which can support informed decision-making in field management.

This paper presents a study on the application of Variable Rate Technology (VRT) for fertilizer application, utilizing multispectral sensors to enable variable-rate fertilization in maize crops, aiming to optimize nutrient distribution based on specific field conditions.

MATERIAL AND METHODS

The research was conducted during the 2023/2024 production season on a 1.66-hectare plot located in Ravno Selo, South Bačka District (latitude: 45.463662, longitude: 19.615545). The plot is characterized by a chernozem soil type and is used for cultivating field crops in rotation, including soybeans, maize, and wheat, with each crop grown in different production seasons. In this research, maize was cultivated following wheat as the preceding crop. A FAO 390 maturity group maize hybrid was planted with a row spacing of 19 cm, aiming for a planting density of 75,000 plants per hectare. During planting, nitrogen fertilizer was applied in strips beside the row, allowing the implementation of varying nitrogen amounts across the plot. These variations were crucial for further analysis and for monitoring their impact on maize growth and development. The plot was divided into six management zones to monitor differences in crop status. Although initial soil analyses indicated that variations were not significant, the plot was divided into zones with different nitrogen levels: 0 kg ha⁻¹ (control zone), 70 kg ha⁻¹, and 140 kg ha⁻¹, to study the effects of different nitrogen doses on maize yield. The crop status was monitored using the Plant-O-Meter multispectral sensor, which records data across six wavelengths (Blue (455 nm), Green (528 nm), Red (657 nm), Red Edge (740 nm), NIR1 (810 nm), and NIR2 (940 nm)). The data obtained was used to generate vegetative indices, allowing for quantification of crop status and nitrogen requirements of the plants. Based on previous research (Tagarakis, 2019), vegetative indices showing the strongest correlation with maize yield and nitrogen requirements were identified, leading to the development of an index known as the POM index, which defines optimal nitrogen levels for maize. Nitrogen application was divided into two phases: the first phase was performed before planting, and the second phase occurred during side-dressing when the maize reached the eight-leaf stage, ensuring that agricultural machinery could carry out agronomic operations without interruption. A Deutz-Fahr 5115 tractor equipped with a precision guidance system was used, enabling accurate application of liquid nitrogen fertilizer in the zone beneath the plant roots. Two multispectral sensors mounted on a 1-meter-long frame on the front of the tractor recorded the crop condition in real-time. The controller in the tractor cab collected sensor data and calculated the POM index, which determined the nitrogen rate to be applied during side-dressing. The fertilizer application system was customized for experimental needs, including a modified inter-row cultivator and an additional 320 liter spray tank. The liquid nitrogen applicators were specially designed to ensure nitrogen distribution in the root zone, at a depth of 10 cm and 5 cm away from the plants. For the base dose, UREA fertilizer containing 46.2% nitrogen was used, while UAN liquid nitrogen fertilizer, containing 30% total nitrogen (7.5% nitrate nitrogen, 7.5% ammonium nitrogen, and 15% amide nitrogen), was applied for side-dressing.

The aim of this research was to validate the accuracy of an algorithm developed for optimal nitrogen fertilization in real-time using the Plant-O-Meter multispectral sensor. The algorithm is designed to operate in real-time, enabling precise determination of the nitrogen required by the plants based on generated vegetative indices that represent the crop condition in the field.



Fig. 1 a) process of maize sowing, b) Top-dressing of maize through the integration of the Plant-O-Meter multispectral sensor

RESULTS AND DISCUSSION

The spatial variability of crop conditions on the experimental plot is clearly visible on the variability rate map shown in Figure 2.

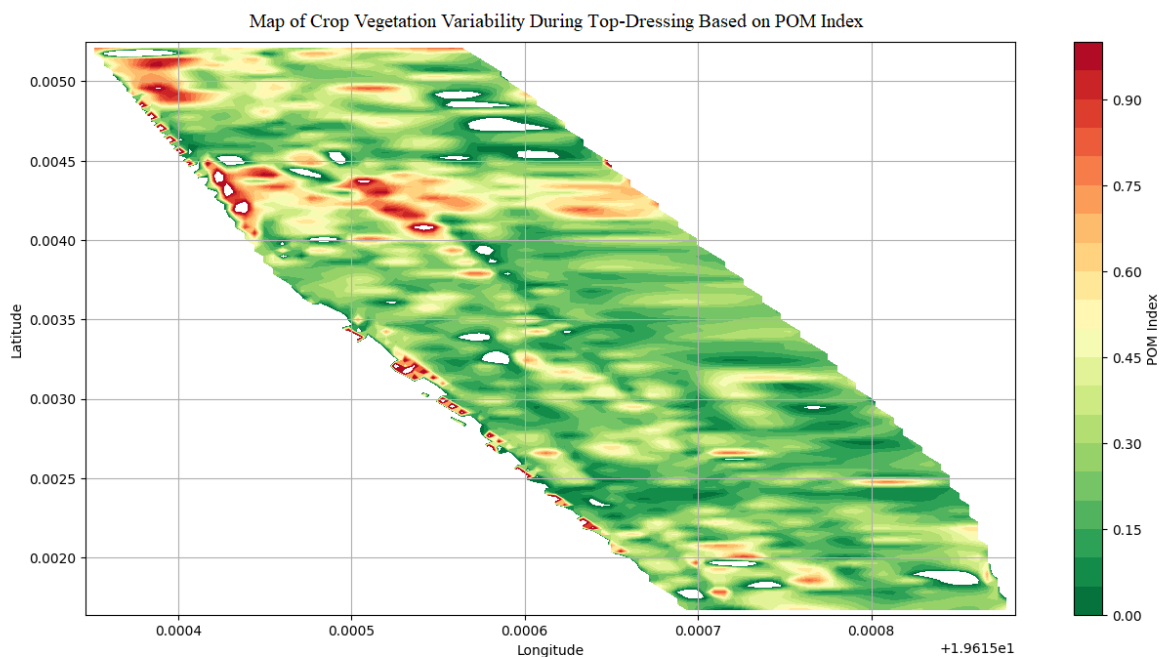


Fig. 2 Map of crop variability during top-dressing

This map illustrates the crop conditions on the plot during nitrogen application at the V7-V8 growth stages, where the heterogeneous state of the crop can be observed. The POM index, which integrates data from the multispectral sensor to assess crop health, varied significantly across the entire plot. The color gradient on the map, ranging from red (low index values) to green (high index values), highlights the unevenness in crop conditions. This variability underscores the need for precise nutrient management, particularly in fields with differing soil characteristics and variable crop status. By applying the algorithms, real-time assessments of

nitrogen needs were conducted, allowing for adjustments in nitrogen quantities based on the specific requirements of different zones within the plot. As a result, areas with lower POM index values, indicating nutritional deficiencies, received higher nitrogen doses, while areas with optimal POM index values required fewer inputs. This approach not only directly influenced the optimized use of nitrogen but also reduced the risk of over-application, minimizing potential negative environmental impacts.

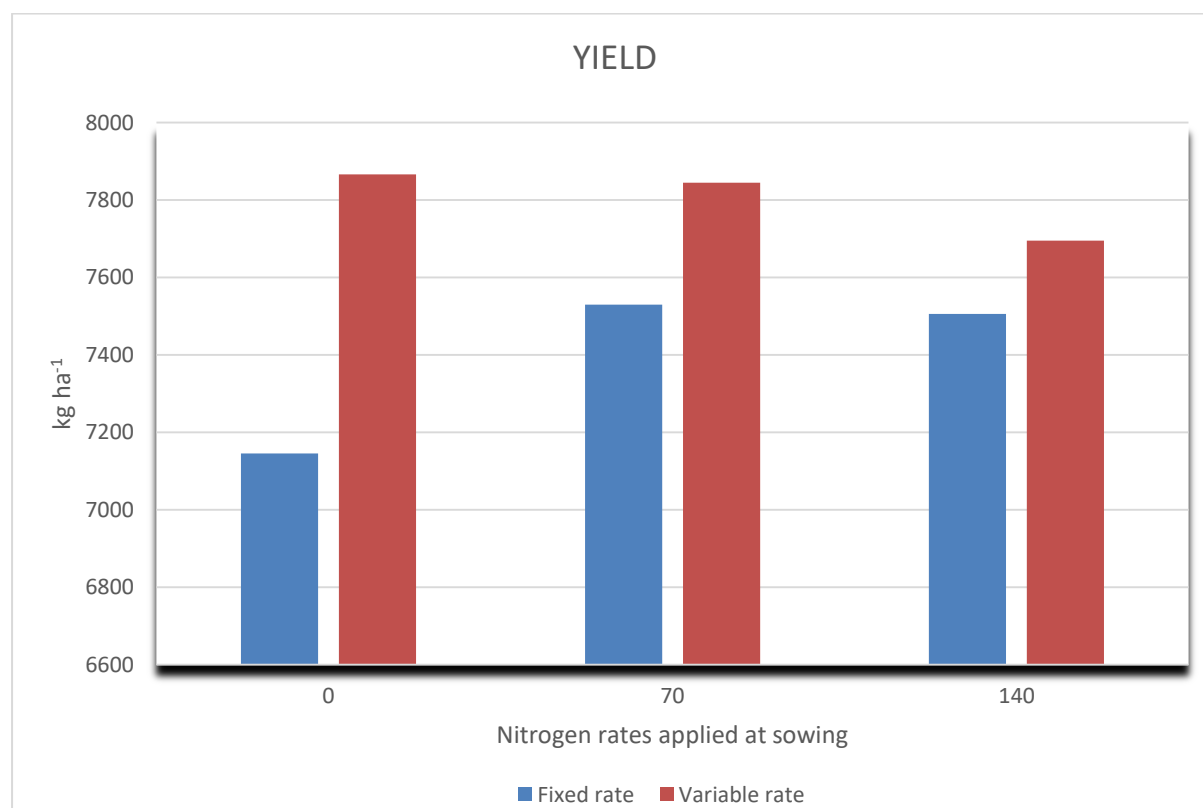


Figure 3. Grain maize yield by plots with fixed and variable nitrogen rates

The comparative analysis of maize yields under fixed and variable nitrogen application rates clearly demonstrates the effectiveness of the variable application approach. As shown in Figure 3, the application of the nitrogen fertilization algorithm resulted in higher yields compared to fixed rates, particularly at doses of 0 kg ha⁻¹ and 70 kg ha⁻¹. In control plots without a base nitrogen application (0 kg ha⁻¹), the largest difference was observed between the two fertilization technologies: the standard technology with a fixed nitrogen dose and the VRT (variable rate technology). In this case, VRT technology achieved a 10.08% higher yield compared to the traditional method with a fixed nitrogen dose. In plots where the base nitrogen amount was 70 kg ha⁻¹, the difference between the two technologies was 4.18%. This difference indicates that the initial nitrogen amount was sufficient for normal crop growth, but the fixed-dose fertilization applied a larger amount of nitrogen compared to VRT technology. Despite this, the higher nitrogen amount did not result in a higher maize yield; on the contrary, the yield was lower, suggesting that more efficient resource use is possible with VRT technology. The difference between traditional and VRT technology in plots with a base nitrogen dose of 140 kg ha⁻¹ was 2.52%, but both cases' yields were lower than those of plots treated with 70 kg ha⁻¹ of nitrogen. This indicates that this nitrogen amount was excessive and that additional nitrogen application under such production conditions did not lead to a significant increase in yield. It is evident that plants were provided with sufficient nitrogen, but under these production conditions, they were unable to fully utilize it for growth and achieving higher yields. Nitrogen deficiency is evident in the reduction of light interception due to a decrease in leaf area index, which results in lower grain yield (Uhart and Andrade 1995). However, grain yield of hybrid maize positively increases up to 300 kg ha⁻¹ (Khaliq 2008; Khaliq et al., 2009). Similarly, lower yields were observed in maize subjected to high doses of nitrogen (Ma B.L. et al., 1996; Mansouri-Far et al., 2010). Therefore, judicious nitrogen management optimizes grain yield and nitrogen use efficiency while also reducing the potential for nitrogen leaching beyond the root zone (Moser et al., 2006; Raun and Johnson 1999). The results of

the analysis of the collected data indicate that the variable nitrogen application method, guided by the POM index algorithm, can effectively assess crop status and optimize side-dressing practices.

CONCLUSIONS

The research results demonstrated significant potential in the application of multispectral sensors and algorithms for variable nitrogen applications. The developed algorithm successfully adjusted nitrogen doses based on the actual crop conditions, leading to improved yields and more efficient resource use. Given the unfavorable weather conditions during the 2023/2024 production year, characterized by high temperatures and prolonged dry periods, nutrient uptake efficiency was significantly hindered. Nevertheless, the application of variable rates through the algorithm helped mitigate these stresses, ensuring that crops received adequate nutrition according to their actual needs. The yield increase in the variable application zone confirms the reliability of the POM index and its practical application in real-time. The integration of multispectral sensor technology and algorithms for variable nitrogen application in maize cultivation can effectively assess the required nitrogen quantities, aiming to develop sustainable agricultural practices. Given the higher yield achieved through VRT technology under unfavorable maize production conditions, it is necessary to conduct multi-year studies in different climatic conditions and assess their impact on nitrogen absorption to develop more accurate models for nitrogen application using VRT technology tools. Further studies could explore the long-term impacts of this technology on soil health and the economic benefits from reduced input costs.

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POPULATION OF LADYBUGS (COCCINELLIDAE) IN THE VINEYARDS OF ILOK IN 2024.

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The aim was to determine the population size as well as species richness of the adult ladybugs in different types of vineyards during 2024 vegetation season in vineyards of Ilok in 2024. Five yellow sticky traps were placed in each vineyard and changed every two weeks from June to September. A total of 698 individuals belonging to the six species were collected and the highest population (51.58%) was recorded for the species *Propylea quatuordecimpunctata*. The uncultivated vineyard had the highest number of specimens, accounting for 50.85% of the total. These results indicate that although chemical treatments reduce the number of specimens, they do not appear to change the diversity of species.

Key words: Coccinellidae, vineyard, yellow sticky boards, species

INTRODUCTION

Ladybugs (Coccinellidae) are important predatory insects that play a key role in pest control. They inhabit all terrestrial ecosystems, and species richness is of exceptional importance for agricultural production (Kundoo and Ali Khan, 2017). They are among the most important predators both in the field and in protected areas (Dixon, 2000). The first checklist of ladybugs in Croatia was made by Koren et al. (2012), which confirmed a total of 78 species. The latest publications indicate the presence of 94 species of ladybugs in Croatia, and *Coccinella septempunctata* L. 1758 is the most common species of ladybug in Croatia (Horvatić, 2017). Franin et al. (2014) investigated the ladybug fauna in the vineyards of Dalmatia (Croatia), where they found that the most abundant species appearing in all localities were *Hippodamia variegata* Goeze, 1777 (41.1 %) and *Scymnus* spp. (23.8 %). *Coccinella septempunctata*, *Psyllobora vigintiduopunctata* L. 1758 and *Stethorus punctillum* Weise, 1891 were less represented. The frequency of a certain species varies during the growing season (Honek et al., 2015) but also between sites and weed species (Franin et al., 2014). *Harmonia axyridis* Pallas, 1773, the harlequin ladybug, is an aphidophagous coccinellid native to Central and East Asia. It is a significant invasive species worldwide and has also been detected in Croatia (Mičetić Stanković et al., 2011; Ivezić et al., 2011). *Harmonia axyridis* poses a threat to native species by competing for food resources. Its broad dietary range can lead to unintended consequences as it attacks beneficial and non-target insects, which is a major problem (Koch, 2003). Our aim was to determine the population size as well as species richness of the adult ladybugs in different types of vineyards during 2024 vegetation season in vineyards of Ilok in 2024.

MATERIAL AND METHODS

During the vegetation season from June to September 2024, 180 yellow sticky boards were located in the vineyards near the town of Ilok in Vukovar-Srijem County. Yellow sticky boards were placed in six different vineyards: four cultivated grapevine (*Vitis vinifera* L.) vineyards (RZ, RV, RA and R), one non-cultivated (abandoned -OG) and one vineyard with rootstocks of American *Vitis* sp. hybrids (M). Cultivated vineyards were protected against harmful organisms in accordance with the principle of integrated pest management (IPM). In each vineyard, five yellow sticky boards were placed and changed every two weeks. Every two weeks the yellow sticky boards were collected, wrapped in transparent material and labelled. Species determination was done in the Laboratory of entomology Faculty of Agrobiotechnical Sciences Osijek under a stereo microscope and the species were determined with the following key (Freude et al. 1967). The data were analyzed to calculate dominance and similarity index as well as species richness in different vineyards. The dominance coefficient and the Sørensen index were calculated. Categories of dominance are shown in the Table 1. Dominance coefficient is defined as the percentage of a species in a single habitat. It is calculated by dividing

the total number of individuals of a species in a habitat by the total number of individuals of all species in that habitat. The resulting quotient is then multiplied by 100 (Durbešić, 1988). The formula for dominance index is:

$$D(\%) = \frac{n}{N} \times 100$$

where: D = dominance of a species

n = total number of individuals of a species

N = total number of individuals of all species in that habitat

Table 1. Tischler's scale (1949) for dominance

Categories of dominance	
eudominant	10 % – 100 %
dominant	5 % – 10 %
subdominant	2 % – 5 %
recent	1 % – 2 %
subrecent	< 1 %

The Sørensen Index (SQ) was used as a measure of similarity, and it was calculated by the following formula:

$$SQ = \frac{2J}{a + b}$$

where:

J = number of similar species in both communities

a = total number of species in community A

b = total number of species in community B.

The SQ index ranges from 0 to 1, where 0 indicates no similarity and 1 indicates complete similarity; thus, higher values reflect greater similarity (Southwood and Henderson, 2000).

RESULTS AND DISCUSSION

In our research, we were focused only on species of the family Coccinellidae, subfamily Coccinellinae. During the vegetation a total of 698 individuals belonging six species was collected (Table 2). *Propylea quatuordecimpunctata* L. 1785 (51.58%) and *Psyllobora vigintiduopunctata* (34.10%) were the most abundant species, found in all six investigated vineyards. The lowest abundance had *C. septempunctata* (0.86%) (Table 2). Although in vineyards is often a high population of *Scimnus* spp. (Fratin et al., 2014; Taranto et al., 2022) here in this research we have focused only on the Coccinellinae subfamily. Considering the Coccinellinae subfamily, *C. septempunctata* is often mentioned in vineyards as the most common species (Fratin et al., 2014; Taranto et al., 2022). However, in our research, the most common species was *P. quatuordecimpunctata* and similar results were confirmed by Canovai et al. (2017).

Table 2. Dominance coefficient of ladybugs in vineyards 2024

Ladybugs		Date						Σ	D%
Species	Tribe	06/28	07/12	07/26	08/09	08/23	09/06		
<i>Propylea quatuordecimpunctata</i>	Coccinellini	33	41	16	25	133	112	360	51.8%
<i>Coccinella septempunctata</i>	Coccinellini	4	0	0	0	0	2	6	0.86%
<i>Psyllobora vigintiduopunctata</i>	Halyziini	3	25	39	61	75	35	238	34.10%
<i>Hippodamia variegata</i>	Coccinellini	2	0	1	8	34	34	79	11.32%
<i>Harmonia axyridis</i>	Coccinellini	3	2	0	2	1	0	8	1.15%
<i>Vibidia duodecimguttata</i>	Halyziini	0	0	0	3	1	3	7	1.00%

In the habitats from this research *P. quatuordecimpunctata* (51,58%), *P. vigintiduopunctata* (34,10%) and *H. variegata* (11,32%) were eudominant, and *V. duodecimguttata* (1,00%) and *H. axyridis* (1,15%) were recent as we can see in table 2. *Coccinella septempunctata* had the lowest dominance coefficient at 0.86% (subrecent), although *C. septempunctata* and *H. axyridis* are numerically important species in agricultural and urban habitats (Lucas et al., 2007; Roy and Migeon, 2010; Evans et al., 2011). Horvatić (2017), based on a list made by using available college, museum and private collections, and available references, stated how *C. septempunctata* is the most common ladybug species in Croatia. Study of ladybug populations conducted in Nova Gradiška, determined the population of ladybugs in different ecosystems. *Coccinella septempunctata* was the most dominant species in orchards (54%), while in urban areas, *H. axyridis* was the most dominant species (Ribarić, 2020). In our study, this both species were in low population.

The vineyard with the highest number of ladybugs was uncultivated (OG - 355) (Table 3). Number of ladybugs in cultivated vineyards was similar as well as species richness. Total number of ladybugs in four cultivated vineyards (234) were smaller than number of ladybugs in one uncultivated vineyard. This data shows us about negative impact of pesticide use in vineyards with intensive production. Vineyard with rootstocks of American *Vitis sp.* hybrids (M) had also a higher number of ladybugs compared to the single cultivated vineyard. Many other studies indicate a greater diversity of species within organic or uncultivated vineyards compared to conventional (Canovai et al., 2017; Taranto et al., 2022), but one other study show opposite situation, pest control was reduced by sown cover-crops and organic farming. They assume that this is related to high fungicide inputs (Reiff et al., 2021).

Table 3. Number of species in different vineyards in 2024

Ladybug species	Vineyard					
	RZ	RV	RA	R	OG	M
<i>Propylea quatuordecimpunctata</i>	32	49	31	22	129	97
<i>Coccinella septempunctata</i>	1	0	2	3	0	0
<i>Psyllobora vigintiduopunctata</i>	3	1	2	1	219	10
<i>Hippodamia variegata</i>	17	21	13	18	1	1
<i>Harmonia axyridis</i>	2	1	0	3	1	1
<i>Vibidia duodecimguttata</i>	0	0	1	1	5	0
Σ	55	82	49	48	355	109

It is important to point out that vineyard OG has the greatest number of specimens found (355), that number corresponds to 50.85% of all specimens found in all vineyards, however that was expected since it was uncultivated. The other five vineyards, which were treated with chemicals and cultivated regularly, collectively have less (49.15%) than OG. With this it is somewhat proven that chemical treatment and human influence play a significant role in the population of ladybugs.

Each insect species reaches its peak at a certain moment during the growing season. In Figure 1 we can see a population of most dominant ladybug species every two weeks in the vineyards of Ilok during 2024. We recorded the species *P. quatuordecimpunctata* during the entire growing season, but peak was detected in the second half of August while the second most abundant species *P. vigintiduopunctata* appeared in constant numbers throughout the growing season (Table 2). Abiotic and biotic factors affect population size and occurrence of some species, genetic predisposition also has a significant influence. In the study of Taranto et al. (2022) most specimens were concentrated in July. September had the lowest abundance and richness which is contrary to our data, where we determined the lowest number of ladybugs in June (45), while in August (343) and September (186) the abundance was the highest.

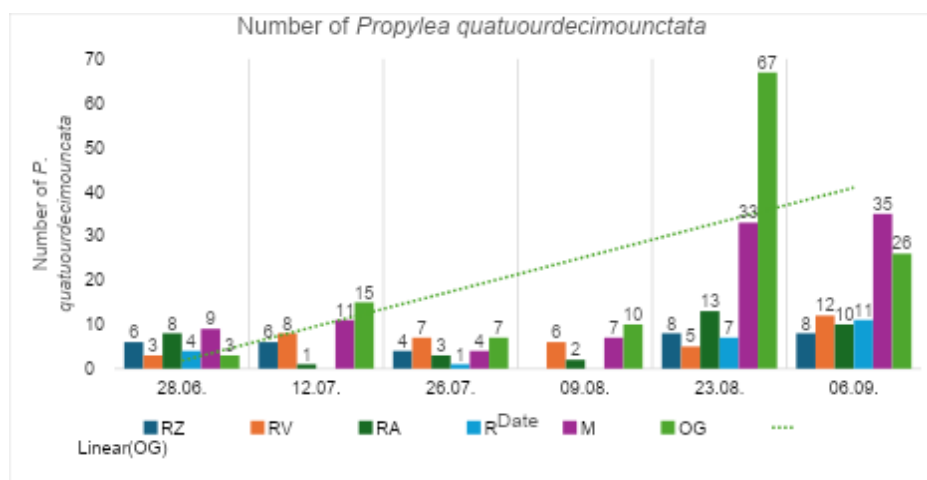


Fig. 1 Population of *P. quatuordecimpunctata* in vineyards of Ilok in 2024

With data provided by calculating the Sørensen index we can notice that the similarities between the uncultivated vineyard (OG) are high with other vineyards as shown in table 4. The highest Sørensen index, 0.91, is observed between OG and R, but the index values of all vineyards are approximately equal. When we compare Sørensen index to the fact that OG has half of the population of all six vineyards, we can notice that chemical treatments and human influences do in fact influence the quantity of population but not its diversity.

Table 4. Sørensen index of ladybug populations in vineyards of Ilok with the uncultivated vineyard (OG)

Vineyard	OG - M	OG – RZ	OG-RV	OG - RA	OG– R
SQ	0.89	0.80	0.90	0.80	0.91

CONCLUSION

In this research we found six species of ladybugs. A total of 698 individual specimens of ladybugs was sampled and determined. In our research *H. axyridis* is a recedent species with dominance index 1.00%. The species with the highest dominance index, 51.58%, was *P. quatuordecimpunctata* and it was found in all six vineyards. The uncultivated vineyard had the highest number of specimens, accounting for 50.85% of the total. These results indicate that although chemical treatments reduce the number of specimens, they do not appear to change the diversity of species.

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MONITORING OF THE CODLING MOTH (*CYDIA POMONELLA* L.) USING PHEROMONE TRAPS

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Harmful insects such as the codling moth cause great economic damage in apple orchards in many parts of the world. The codling moth (*Cydia pomonella* L.) represents the most important technological pest of apples, which can destroy 30% to 50% of the expected yield in the form of fruit worminess. There are many ways to reduce the damage caused by the codling moth in modern apple production. Pheromone baits or pheromone traps are increasingly being used, where timely treatment with phytopharmaceuticals is determined based on monitoring the mass flight of insects. In this paper, it was examined how external factors can influence the occurrence of a mass flight of the codling moth. The pheromone traps were placed in two different locations in Bosnia and Herzegovina, with different altitudes and a different set of apple varieties. During the flight period of the first generation of pests, average daily temperatures, weather conditions, as well as the number of caught individuals of the codling moth on pheromone traps were recorded daily. In the mentioned locations, there is a different set of apple varieties, of different ages. The results showed that the attack of the codling moth was less in the second location, where older indigenous apple varieties are represented, in contrast to the first location, where younger consumer apple varieties are present. Also, the results of the work showed that the pest does not fly in the period of unfavorable weather conditions, which was observed by the number of moths on the pheromone traps. Based on the sum of the effective temperatures, the timely treatment with phytopharmaceuticals was determined at the investigated locations. Higher altitude, greater resistance of indigenous apple varieties, as well as worse weather conditions during the flight of the first generation of the codling moth, resulted in the appearance of fewer individuals of the codling moth on the pheromone traps, and therefore less incidence of fruit worming during the apple harvest.

Keywords: apple, codling moth, pheromone traps

INTRODUCTION

The apple ranks third in world production, and in Bosnia and Herzegovina it holds second place, right behind the plum (Čivić, et al., 2017). Apple producers in the territory of Bosnia and Herzegovina are often faced with the damage of apple fruits caused by the pest *Cydia pomonella* L. This pest is the most significant technological pest of apples, which, apart from this species, can cause damage to some other fruit species as well. If protection measures are not implemented, this pest, in the form of fruit worminess, can destroy 30% to 50% of the expected yield, and in some cases damages of over 80% have been recorded (Šubić, et al., 2015).

Cydia pomonella is a pest that rapidly develops resistance to chemicals from different groups of insecticides due to the presence of several generations per year (Pajač & Barić, 2012). Due to issues with the application of insecticides, specifically the number of treatments against the codling moth, higher tolerance levels have been accepted, meaning that in many European countries, up to 1% of the fruit can be infested, which partially simplifies the control of the codling moth (Ciglar, 1998). In addition to phytopharmaceutical products, alternative approaches are used for protection, such as the application of biological insecticides, confusion or disruption techniques (Tomaš, et al., 2015), and the use of pheromone traps. These traps are most used to determine the presence, abundance, and optimal timing for controlling certain pest insect species (Čandić, 2018).

Pheromones are packaged in small ampoules or capsules made of various synthetic materials, each containing a specific amount of pheromone. Typically, a pheromone trap is set by placing the pheromone inside a capsule or ampoule, which is then placed in a ferotrap, or a standard trap, such as a triangular sticky trap or a trap with a sticky base (Pest Control Catch It, 2016). Males attracted by the scent of the pheromone become trapped on the sticky bottom of the trap, allowing for the determination of the start, duration, and end of the flight period of *C. Pomonella*. If the pest appears in a high population, above the critical threshold, the optimal timing for control is determined based on the accumulation of average effective temperatures. Effective temperature is the temperature at which the insect performs its life activities, i.e., completes its biological development cycle. For the codling moth, the effective temperature is above 10°C (Ciglar, 1998). The flight and

emergence of this pest are influenced by many external factors. For example, this pest becomes active at temperatures above 15°C, and its flight ceases at lower temperatures, with stronger winds, or during the rain (Salkić, 2015).

This paper presents a relatively simple method for determining the timely application of chemical treatments, based on experimental monitoring of the flight of the first generation of insects using pheromone traps, as well as the impact of external factors on the occurrence of mass flight of *C. Pomonella*. The experiment was conducted at two different locations to compare results and determine how climatic conditions, or external factors, can be decisive in the final outcome of the pest's activity.

MATERIAL AND METHODS

To obtain the results in this study, pheromone traps from two different manufacturers were used. At location 1, a pheromone trap from the manufacturer Biobest Sustainable Crop Management, Belgium, was used, while the flight of the codling moth population at location 2 was monitored using a pheromone trap from the manufacturer Csalomon, Hungary. For recording daily temperatures, mercury thermometers from the manufacturer Kenzo Garden, Netherlands, were used.

Experimental monitoring of *C. Pomonella* flight was conducted during the 2021 growing season at two different locations, in private orchards. The first location was in the town of Srebrenik (Location 1), at 44° 41' 59.4" North latitude and 18° 28' 52.6" East longitude, while the second location was in the municipality of Bratunac (Location 2), at 44° 11' 23.4" North latitude and 19° 17' 58.4" East longitude (Google Maps, 2021). The first location was at an altitude of 213 m, while the second location was at an altitude of 245 m (Google Earth, 2021). The flight of the *C. pomonella* population was monitored at the two different locations using pheromone traps, set near the end of flowering, which occurred in late April at these locations. The traps were placed in the middle of the orchard, in the upper third of the tree. The sticky base of the traps was replaced twice during the monitoring of the first generation's flight, and the number of captured moths was recorded daily. According to Ciglar (1998), the critical number for codling moths is considered to be five moths per trap cumulatively, after which the incubation period (i.e., the duration of embryonic development) and the beginning of caterpillar emergence are calculated.

At location 2, the pheromone trap for monitoring the flight of the codling moth was placed in an orchard of indigenous apple varieties, primarily Kolačuša (Ramička), Bjelunika, and Samoniklica. Their age is estimated to be over twenty-five years, and the insufficient isolation of the orchards at both locations was also considered when preparing this study. On the other hand, at location 1, there is an orchard of various commercial apple varieties (Gala, Red Delicious, Golden Delicious). The age of the trees is also estimated at around 20 years, and the seedlings of these trees were purchased from a registered fruit nursery, grafted on the M9 rootstock.

To determine the accumulation of effective temperatures for establishing the timing of phytopharmaceutical applications, daily temperatures at the specified locations were monitored. The thermometer was placed in the orchard at an approximate height of 2 meters. The average daily temperature was calculated using the formula: $T1 = (T7+T14+2 \times T21)/4$ (Burggen, et al., 2015). To determine the timing of codling moth larval control, the average daily temperatures above 10°C were summed from the day the critical number of moths was caught. When the accumulated effective temperature reached 90°C, the period for the first chemical treatment was determined.

RESULTS AND DISCUSSION

The pest has two generations per year (Maceljski, 1999), and the flight of the moths of the first generation begins in the spring when the last petals of the apple fall (the fruits are already set), most often at the beginning or in the middle of May (Salkić, 2015), and the moths of the second generation fly from the middle July until mid-August (Tomaš, 2015). Pheromone trap at location 1 was placed on April 28th 2021, and it was used to monitor the flight of the first generation of codling moth until June 6th 2021. Temperatures were recorded using a thermometer, and the daily temperatures are shown in Table 1. The inspection of moth catches was carried out every day, and the dynamics of moths catches is also shown in Table 1.

A critical number of moths at location 1 was caught three days after the placement of the pheromone trap, i.e. on 3rd of May, when 5 moths were caught on the trap. The next day, 5 moths were also caught. In the next few days, fewer moths were caught than in the first two days, so that on May 9th and May 10th a critical number of moths reappeared. On the 17th, 18th, and 25th of May a critical number of moths also appeared. Between those dates, the number of moths was variable, mostly lower than critical, and due to the bad weather conditions that prevailed in the mentioned location, the results show that the flight of moths was disabled, and that the number of moths caught on the mentioned days was equal to zero. Moths of the first generation of the codling moth at location 1 appeared until 31nd of May, after which the number of moths caught on the pheromone trap was zero

for several consecutive days. According to Salkić (2015), the moth is active at a temperature above 15°C, and some moths do not show up at lower temperatures, stronger wind and rain. Because of the above, bad weather conditions, insufficient isolation of the plants, as well as possible deterioration of the glue on the underside of the pheromone trap should be taken into account. The total number of moths caught during the monitoring period of the first generation was 71, and on 10th of May the largest number of moths was caught.

At location 2, a pheromone trap was installed on April 28th 2021 and used to monitor the flight of the codling moth until 4th of June, 2021. Temperatures were recorded using a thermometer, and the mean daily temperatures are shown in Table 1. The inspection of the moth catch was carried out every day, and the dynamics of the moth catch is also shown in Table 1.

In contrast to location 1, the flight dynamics at location 2 showed that the only critical number of moths was caught on 6th May, and it was 5 moths in the pheromone trap. The next day, three moths were caught, and the following days, the number of moths was no more than one. Due to bad weather conditions, there were days when the number of caught moths was zero. The smaller number of caught moths during the day indicates the fact that in this area the quality of the fruits is much better, which was also confirmed by the quality of the harvested fruits during the harvest season. After May 26th, several days were observed without a single caught moth, so the monitoring of the first generation of the codling moth at this location was also completed. The sum of the first generation caught moths was 19, and on the 26th May the largest number of moths was caught. At location 1, as shown in Table 1, the first critical number of moths was caught on 3rd May, and the day after, a critical number of moths was caught again. From that day, the sum of the effective temperatures was calculated and the first deadline for treating the apple trees was determined. The sum of the effective temperatures was 93°C, and the first treatment deadline, based on this, should be on 14th May. Because of that, a critical number of moths appeared the very next day, it is assumed that the application of insecticides does not need to be repeated because the applied insecticide would also work on the caterpillars of *C. pomonella* which would develop the following day. The next critical numbers of moths were observed on 9th and 10th of May. Based on the sum of effective temperatures, which was 91°C, the next treatment date should be on 18th of May. Then, a critical number of moths was observed on 17th and 18th of May, based on the sum of the effective temperatures (95°C), the next treatment should have been on 27th of May. In both of these cases, because of that the critical number of moth appears two days in a row, it is assumed that only one treatment with the appropriate insecticide is sufficient. The last critical number of moths was observed on 25th of May. From this date until the potential date of insecticide treatment (6th of May), the sum of the effective temperatures was 94°C. After this date, the number of caught moths was significantly lower than the critical one, and in the last days of flight monitoring, the number of caught moths was even zero, on the basis of which it is assumed that, in this location, it is not necessary to carry out additional treatments to control the first generation of pests.

Table 1. Dynamics of *Cydia pomonella* L. moth catches at location 1 and location 2.

DATE	LOCATION 1		LOCATION 2	
D	N1	T1 (°C)	N2	T2 (°C)
28.04.	0	8	0	-
29.04.	0	9	0	-
30.04.	1	12	0	12
01.05.	1	12	3	13
02.05.	1	14	3	12
03.05.	0	8	5	7
04.05.	0	5	5	2
05.05.	0	11	3	7
06.05.	5	10	1	7
07.05.	3	9	0	8
08.05.	0	5	2	7
09.05.	1	2	5	3
10.05.	1	6	7	6
11.05.	0	10	1	10
12.05.	0	11	0	13
13.05.	1	11	1	11
14.05.	1	10	2	12
15.05.	0	9	1	10

16.05.	0	12	0	11
17.05.	1	10	5	10
18.05.	0	9	5	5
19.05.	1	11	0	11
20.05.	0	10	0	9
21.05.	0	10	4	5
22.05.	0	9	4	11
23.05.	1	11	0	11
24.05.	0	13	1	10
25.05.	0	13	5	11
26.05.	1	12	2	6
27.05.	0	13	4	6
28.05.	0	11	0	11
29.05.	0	12	0	6
30.05.	0	12	2	9
31.05.	0	10	0	8
01.06.	0	10	0	8
02.06.	0	10	0	5
03.06.	0	9	0	7
04.06.	0	9	0	8
05.06.	-	-	0	9
06.06.	-	-	0	13

D – Moth flight tracking date

N1, N2 - Number of moths caught per day at location 1 (N1) and location 2 (N2)

T1, T2 - Average daily temperature at location 1 (T1) and location 2 (T2)

At location 2, the critical number of caught moths was almost negligible compared to location 1. The first and only critical number of caught moths was on 6th of May. Since that date, the sum of effective temperatures have been summed up and reached 95°C on 16th of May. This is also an indicator that the treatment with phytopharmaceuticals could be carried out on 16th May. After the mentioned date the number of caught moths per pheromone trap did not increase (on some days it was zero), it can be concluded that in this case only one would be sufficient treatment with an appropriate insecticide to control the first generation of the codling moth.

The apple trees at both locations were not previously treated with phytopharmaceuticals. Even after determining the critical number of moths and the optimal treatment time, again the trees were not treated with phytopharmaceuticals. Considering the worse weather conditions during the period of monitoring the flight of the codling moth at both locations, the results indicate a less massive flight of the first generation of the pest. According to Tomaš et al. (2015), rain is the most common cause of mortality of the first developmental stage of the codling moth caterpillars before boring into the fruit, and the flight of the pest stops completely at lower temperatures (Maceljski, 1999). The results at location 2 indicate that the higher altitude affected the occurrence of colder days in the mentioned period, and therefore the occurrence of less mass flight of pests. Because of that, some authors (Korićanac, et al., 2020; Vojinović, et al., 2021) stated that the high tolerance of indigenous varieties to diseases and pests, resistance to drought and adverse environmental conditions, the result was also influenced by the composition of trees on the location 2, where exclusively indigenous apple varieties are represented. A possible cause of this phenomenon is the influence of climate changes and the specificity of the area on the genetic structure of the codling moth, as shown by the latest genetic research (Pajač & Barić, 2012). After the treatment deadlines have been determined, various phytopharmaceuticals can be used for protection purposes. How many treatments will be carried out to protect the fruits depends on the number of the pest population, the variety, gender, etc. (Festić, et al., 1981). The basic strategy for controlling the codling moth is to suppress the first generation or reduce it to a lower number, and for this reason, the emphasis in this paper is on monitoring the flight of the first generation of pests. This rule also applies to smaller orchards, but due to pests flying in from the side, care must also be taken of the second generation, especially in August, when the protection is interrupted due to the withdrawal period (Injac, et al., 2005).

CONCLUSION

After conducting the monitoring of the first generation of the codling moth at two different locations, the timely application of phytopharmaceutical treatments can be easily determined. By using pheromone traps, plant treatments are performed only when necessary and on a clearly defined date, thus adhering to the principles of integrated agricultural production. Considering the obtained results, the location and microclimatic conditions can have a significant impact on the final outcome. Therefore, when monitoring the flight of the codling moth, all external environmental factors must be taken into account. The higher altitude and the specific microclimate at location 2 resulted in a less intense flight of the codling moth, and only one treatment with phytopharmaceutical products was needed to control the first generation of the pest. The different composition of apple varieties in the orchard, according to the obtained results, can also influence the final outcome, especially when it comes to indigenous apple varieties, which are particularly adapted to the climatic conditions in Bosnia and Herzegovina and show high resistance to disease and pest attacks.

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THE INTERACTION BETWEEN THE ISOLATE OF *TRICHODERMA ATROVIRIDE* AND *FUSARIUM GRAMINEARUM*

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Trichoderma atroviride is an effective biocontrol agent against plant pathogens due to its parasitism, antibiosis, and competition mechanisms. This study aimed to evaluate the Tav1 isolate's impact on controlling *Fusarium* seedling blight in wheat, caused by *Fusarium graminearum* (Fg1). Under controlled conditions, Tav1 did not significantly reduce Fg1 infection, with the best growth observed in the control group without the pathogen. These findings suggest a need for further research to optimize *T. atroviride* isolates for effective pathogen control in wheat.

Keywords: *Trichoderma atroviride*, *Fusarium graminearum*, antagonism, wheat

INTRODUCTION

This study aimed to investigate the effects of *Trichoderma atroviride* isolate Tav1 on the development of wheat seedling blight during artificial infection with *Fusarium graminearum* (Fg1). Previous studies have shown that the application of the isolate Tav1 improves the qualitative properties of the oil (unpublished results).

T. atroviride is a saprobic fungus that serves as a biocontrol agent to suppress plant pathogens and promote plant growth and occurs naturally in soil and on plant debris (Singh et al., 2013). This fungus produces enzymes such as chitinases, beta-glucanases, and proteases that degrade the cell walls of pathogens, thus facilitating biocontrol (Li, 2003). In addition, *T. atroviride* produces antibiotic compounds and secondary metabolites that inhibit the growth of pathogenic fungi such as *Fusarium*, *Aspergillus*, and *Phytophthora* that cause damage to crops. (Stracquandano et al., 2020).

In addition to its use as a biological antifungal agent, *T. atroviride* also acts as a biofertilizer by improving nutrient uptake and promoting plant growth as a biostimulant (Petrović et al., 2023). It shows antagonistic effects against plant pathogens through competition, parasitism and antibiosis. *T. atroviride* rapidly colonizes the environment, occupying space and nutrients that would otherwise be available for pathogens. It also acts as a mycoparasite, absorbing nutrients from pathogens and producing antibiotics that inhibit their growth (Harman et al., 2004).

The genus *Fusarium* comprises over 70 species of fungi that are responsible for various plant diseases such as rots, necroses, and molds. *F. graminearum*, a common pathogen of late blight in wheat, reduces crop yields and can contaminate crops with toxins that are harmful to humans and animals (Figure 1) (Ćosić et al., 2006).



Fig. 1 Symptoms of wheat infection with *F. graminearum* at the milk ripening stage (Source: Ćosić et al., 2013)

MATERIAL AND METHODS

The *T. atroviride* isolate (Tav1) was propagated on potato dextrose agar (PDA) under sterile conditions, where it was inoculated with a mycelial disc from a pure culture (Figure 2). The cultures were kept at 22°C and 70% humidity to ensure optimal growth. After two weeks of incubation, the developed mycelium was transferred to sterile water. Polysorbate 80 was added to prevent aggregation of the spores in the water suspension. To obtain an exact spore concentration, the suspension was diluted with distilled water in a ratio of 1:500, whereby 1 ml of the Tav1 suspension was added to 499 ml of distilled water. The spore concentration was determined using a Neubauer counting chamber under a light microscope at 20*15 magnification. Spores touching the upper and left edges of a grid square were counted, while those touching the lower and right edges were excluded. The resulting suspension was adjusted to a concentration of 108 spores/ml and this was used as the Tav1 treatment solution for the wheat seeds.

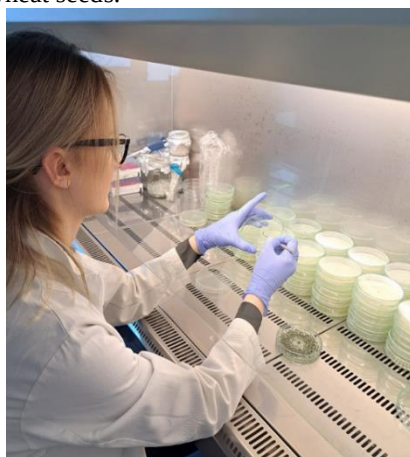


Fig. 2 Inoculation of the fungus *T. atroviride* on a nutrient medium (Source: Ćirić, K.)

The *F. graminearum* isolate (Fg1) was similarly prepared. A mycelial disc from a pure Fg1 culture was inoculated onto PDA in a laminar flow cabinet and incubated under the same conditions of 22°C and 70% humidity. After two weeks, the developed Fg1 mycelium was transferred into sterile water, creating a spore suspension. This Fg1 suspension was used to contaminate sterile sand.

The experimental setup involved four treatments, each with three replicates, with 15 seeds per container. Treatments included:

- (1) untreated wheat seeds sown in sterile sand (K)
- (2) Tav1-treated seeds sown in sterile sand (Th)
- (3) untreated seeds sown in Fg1-contaminated sand (SK)
- (4) Tav1-treated seeds sown in Fg1-contaminated sand (FgTh)

To prepare the seeds, 2.41 ml of the Tav1 suspension was added to 7.59 ml of sterile water. This resulted in a 10 ml solution, which was used to soak 90 wheat seeds for four hours. For the control groups, the seeds were soaked in sterile water under the same conditions. After soaking, the seeds were sown in the containers with moistened, sterile sand. The containers were covered with aluminum foil to retain moisture and incubated for 48 hours before being transferred to a growth chamber with a temperature of 22 °C, a 12-hour light/dark cycle and a relative humidity of 70%. The seedlings were watered with distilled water as required.

After 10 days, the seedlings were carefully removed from the containers, washed to remove sand from the roots, and examined for their germination rate. The severity of infection was scored on a scale from 0 (healthy) to 5 (severe infection with no visible shoot) based on the McKinney index. Each seedling was photographed for digital measurement of root and shoot length. Fresh mass was measured separately for roots and shoots using an analytical balance. Subsequently, the dry mass was determined by freeze-drying: Seedlings were frozen and then placed in a vacuum chamber to sublimate the moisture, yielding the dried tissue for mass analysis.

Statistical analysis was performed using SAS software 9.1.4 (SAS Institute Inc., 2003). Data were analyzed using ANOVA followed by Duncan's Multiple Range Test to determine significant differences between treatments.

RESULTS AND DISCUSSION

This study aimed to preliminarily investigate the effect of the isolate of *T. atroviride* (Tav1) on wheat seed rot caused by *F. graminearum* (Fg1). Previous research has shown that Tav1 has a positive effect on olive oil quality (unpublished results), which prompted us to investigate its potential to reduce *F. graminearum* infection in wheat seedlings.

As shown in Figure 3, the highest germination rate was observed in the K (93.33%), while the lowest germination rate (17.78%) was recorded in the SK treatment, indicating a significant effect of Fg1 on germination.

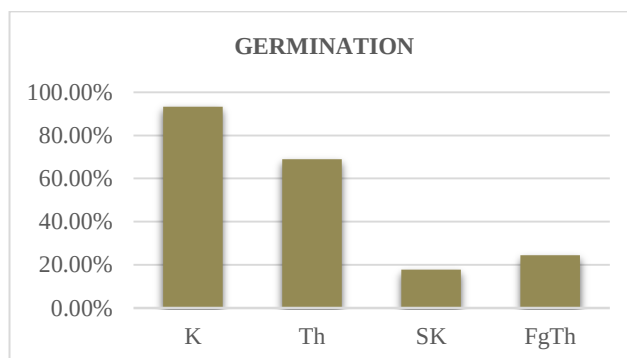


Fig. 3 The Effect of the *T. atroviride* Isolate on seed germination

Fresh root weight showed the highest mass in Th (0.654 g), with no significant difference compared to K and FgTh. Figure 4 shows a positive effect of the *T. atroviride* isolate on the mass of lyophilized plants, as there is a statistically significant difference between the treatment where the seed was inoculated with Tav1 compared to the treatments SK (disinfected seed in Fg1 contaminated sand) and FgTh (Tav1 seed in Fg1 contaminated sand).

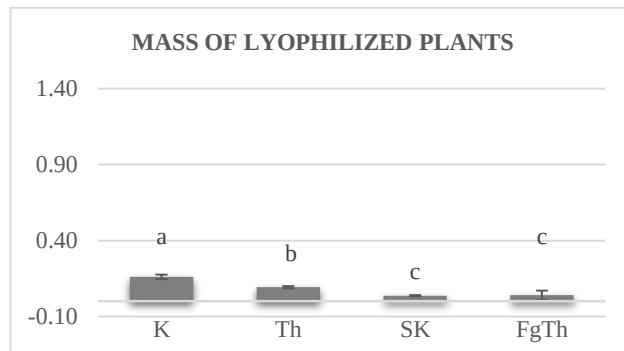


Fig 4. The effect of *T. atroviride* isolates on the mass of lyophilized plants (a, b, c, d – different letters indicate statistically significant differences according to Duncan's multiple range test at the level of $P \leq 0.95$).

No significant differences were found in the fresh weight of the plants between the treatments. However, lyophilized plant weight was highest in the K group (0.162 g) and significantly lower in SK and FgTh, indicating that Tav1 was not effective in preventing weight loss due to Fg1 infection.

Analysis of root length showed the longest average length in the control group (78.308 mm), with no significant effect of Tav1 in Th, SK, and FgTh treatments. Plant length in Th (55.529 mm) also showed no positive results compared to the K.

Infection intensity was assessed using the McKinney disease index (Table 1). The lowest disease index was found in the K (5.56%), while Tav1 did not significantly reduce infection in the SK and FgTh treatments, with indices exceeding 65%.

Table 1. Impact of treatments on disease intensity (disease index according to McKinney)

TREATMENT		DISEASE INDEKS
K	Disinfected seed / sterile sand	5,56
Th	Tav1 seed / sterile sand	26,3
SK	Disinfected seed / Fg1 sand	70,74
FgTh	Tav1 seed / Fg1 sand	65,93

The intensity of infection was assessed using a scale from 0 to 5 (0 = healthy seedling; 5 = high level of infection, no germination), and the disease index was calculated according to McKinney (1923).

The disease index is calculated using the following formula:

$$I = (\Sigma (n \times k)) / (N \times K) \times 100$$

n – number of plants per category

k – number of individual categories

N – total number of examined plants

K – number of established categories

Despite the favorable effects of Tav1 in olive, these results indicate that Tav1's effectiveness in *suppressing F. graminearum* in wheat was limited, potentially due to varying ecological and physiological conditions across plant species.

In the Republic of Croatia, biological agents are available for controlling diseases caused by pathogens from the *Fusarium* genus. According to the Phytosanitary Information System (2024.), four registered agents are listed, one of which contains bacteria from the *Pseudomonas* genus, commercially known as Proradix. Additionally, the product Serenade Aso is available, functioning as both a biological bactericide and fungicide. Two other agents are based on fungi from the *T. harzianum* genus and are commercially known as Trianum-G and Trianum-P. These agents prevent and control soilborne diseases while simultaneously promoting plant growth and uniform development.

Globally, there are commercially available products for plant protection, such as Trichostar®, Trichodex, Supresvit, Root Pro, Promote, Trichopel, Trichobject, and RootShield. These products, derived from fungi of the genus *Trichoderma*, not only suppress various species of soilborne phytopathogenic fungi and pseudofungi, including *Fusarium*, *Rhizoctonia*, and *Pythium*, but some also act as biofertilizers, enhancing plant growth and soil health (Petrović et al., 2023).

CONCLUSION

The isolate *T. atroviride*, although previously shown to be beneficial for improving the quality characteristics of olive oil, did not show the expected positive effect on the germination and quality of wheat infected with *F. graminearum*. According to the results, the control variant (disinfected grain in sterile sand) showed the best germination, while treatment with the isolate Tav1 resulted in reduced germination and an increase in the disease index. These results suggest that isolate Tav1 was ineffective in disease suppression, which could indicate specific characteristics of this isolate that make it less efficient compared to other *T. atroviride* isolates.

In addition, certain wheat cultivars may be more sensitive to *T. atroviride*, which could lead to opposite effects compared to the control. It is important to note that research on this topic is still ongoing and that the negative effects of *T. atroviride* on wheat seedlings are not always present. In numerous other studies, *T. atroviride* is often used as a beneficial fungus to control pathogenic fungi, demonstrating its potential to protect plants from fungal diseases.

Biological pest control, including the use of microorganisms such as *T. atroviride*, is emerging as a key component of sustainable agricultural systems due to the reduced impact on the environment and human health and the reduced risk of resistance development in pest organisms.

With increasing demands for sustainable production and protection of agroecosystems, further research is needed to identify the most effective isolates, optimize application conditions, and recognize situations where biological control can achieve the best results. It is crucial to conduct research that deepens our understanding of the mechanisms of action of microorganisms and develops strategies to improve the effectiveness of biological control.

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ALLELOPATHIC AND HERBICIDAL POTENTIAL OF LAVENDER (*LAVANDULA* SPP.)

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*This paper analyzes the allelopathic and herbicidal potential of lavender (*Lavandula* spp.) as a natural weed control agent. Data from key studies investigating the impact of lavender essential oil and its main bioactive components (linalool, camphor, and terpinen-4-ol) on the germination and growth of various weed species, including *Amaranthus retroflexus*, *Chenopodium album*, and *Setaria viridis*, were utilized. The results indicate that lavender essential oil, when applied at different concentrations, can significantly reduce the germination and growth of targeted weed species, demonstrating their potential application in organic agriculture. Given the rapid degradation of these compounds in the environment, lavender presents a sustainable alternative to synthetic herbicides. Further research is needed to optimize dosage and application methods in various agroecological conditions.*

Keywords: lavender, allelopathy, essential oil, bioherbicides, weeds, organic agriculture

INTRODUCTION

In modern agriculture, weed management represents one of the greatest challenges, as weeds reduce crop yield and complicate food production. The traditional approach to weed control involves the use of synthetic herbicides, which, although effective, have a number of negative effects on the environment and human health. Due to the continuous use of synthetic herbicides, there has been an emergence of resistant weeds, soil and water pollution, and a reduction in biodiversity. To mitigate these risks, in recent decades, increasing attention has been directed towards the research of alternative methods for weed control, including the use of plants with allelopathic properties.

Aromatic plants, such as lavender, rosemary, mint, and thyme, have the ability to produce bioactive substances that can act as natural herbicides. These plants contain essential oils and other secondary metabolites that inhibit the growth and development of other plant species. Such allelopathic effects make them potentially useful tools for sustainable weed management in organic and integrated farming systems. Lavender, as one of the most widespread aromatic plants, has shown significant potential in this area. Its essential oil, rich in compounds such as linalool, linalyl acetate, camphor, and terpinen-4-ol, has proven toxicity against many weed species.

Allelopathy is a term that describes the effect of one plant species on the growth and development of another species through the release of chemical substances, known as allelochemicals. These substances can be synthesized in different parts of the plant, including leaves, roots, flowers, and seeds, and can affect the germination, growth, and physiological processes of other plants. In the case of lavender, active ingredients such as phenols, monoterpenes, and sesquiterpenes act as natural herbicides by inhibiting cell division and disrupting the synthesis of essential proteins in target plants. Previous research has shown that lavender oil can reduce the germination of weeds such as redroot pigweed (*Amaranthus retroflexus*), common lambsquarters (*Chenopodium album*), green foxtail (*Setaria viridis*), etc.

Interest in the use of natural herbicides arises from their ecological advantages. Lavender essential oil decomposes rapidly in the environment, reducing the risk of toxic residues accumulating in the soil and water. Additionally, these natural herbicides have fewer negative effects on non-target organisms, such as beneficial insects, microorganisms, and wild plants. For this reason, essential oils and other plant extracts are increasingly being researched as part of integrated weed management systems that combine biological, mechanical, and chemical methods to achieve sustainable agricultural management.

However, although research on the allelopathic potential of lavender is on the rise, several challenges limit its wider application. For example, the concentration of essential oil required to achieve effective weed control may vary depending on the type of target weed, environmental conditions, and plant genotype. Moreover, the effects of lavender allelochemicals on crops grown nearby have not been sufficiently studied, raising questions about their selectivity. Some studies have shown that high concentrations of lavender oil can negatively affect the germination and growth of crops such as corn and wheat, which necessitates careful dosing and planning.

The aim of this paper is to investigate the allelopathic and herbicidal potential of lavender as a sustainable alternative to synthetic herbicides. The presented analysis includes a review of previous research on the effects of lavender essential oil on the germination, growth, and development of various weed species, as well as an examination of the ecological advantages and limitations of its use. Special attention will be given to identifying

key components of lavender essential oil that affect target plant species, as well as to exploring the possibilities of applying lavender in integrated weed management system.

LITERATURE REVIEW

Lavender (*Lavandula spp.*) is known for its essential oil, which is used in various industries due to its therapeutic and aromatic properties. However, its potential as a natural herbicide has been less explored. The allelopathic effects of lavender, which refer to its ability to inhibit the growth and development of other plants through chemical substances, make it an interesting plant for weed control in organic agriculture. The main active components of lavender essential oil include linalool, linalyl acetate, camphor, terpinen-4-ol, lavandulol, and other terpenes, which have toxic effects on target weed species (Petrova et al., 2015; Nikolova et al., 2021).

Effects of Lavender on Different Weed Species and Its Active Components. One of the key challenges in the use of synthetic herbicides is the development of herbicide resistance in weeds, which necessitates the identification of new agents with different modes of action. In this context, lavender shows promising results due to the presence of various allelopathic compounds. According to the study by Nikolova et al. (2021), linalool, the primary component of lavender essential oil, acts as a growth inhibitor for weed plants, including *Amaranthus retroflexus* (redroot pigweed) and *Chenopodium album* (common lambsquarters). The research demonstrated that applying 1% concentration of linalool reduced the germination of *Amaranthus* by more than 60%, while combining it with terpinen-4-ol further reduced germination by up to 75%. These results indicate that linalool has a strong inhibitory effect on weeds and can be utilized as a component of natural herbicides.

In addition to linalool, terpinen-4-ol has proven to be a key factor in inhibiting the germination and early growth of weeds. In a study conducted by Ghosh et al. (2020), the application of terpinen-4-ol at a concentration of 0.5% reduced the germination of the weed species *Setaria viridis* (green foxtail) by 70%. This effect is associated with the ability of terpinen-4-ol to disrupt the cell membrane and interfere with the synthesis of essential proteins required for root and shoot development. Similarly, camphor, which is present in significant concentrations in lavender oil, has shown inhibitory effects on a wide range of weeds, including *Lolium multiflorum* (Italian ryegrass) and *Digitaria sanguinalis* (crabgrass), reducing their germination by more than 50% (Nikolova et al., 2021).

Effects of Lavender Essential Oil on the Suppression of Various Plant Species. Lavender essential oil has shown high potential in suppressing weed species under laboratory conditions, as well as in *in vivo* experiments. According to Nikolova et al. (2021), treatment with lavender oil at a 1% concentration in Petri dish bioassay experiments resulted in a significant reduction in germination and inhibited the growth of weeds such as *Cyperus rotundus* (nutsedge) and *Echinochloa crus-galli* (barnyardgrass).

Similarly, Matković et al. (2019) conducted a greenhouse experiment to investigate the effects of different concentrations of lavender essential oils on weeds under controlled conditions. The application of a 1% concentration of oil on *A. retroflexus* (redroot pigweed) and *Setaria viridis* (green foxtail) led to a significant reduction in the above-ground biomass of the weeds by more than 70% over a period of two weeks.

The use of lavender oil as a bio-herbicide offers several advantages, including rapid environmental degradation, minimal risk to beneficial organisms (Ghosh et al., 2020), and reduced toxicity to soil and water. However, key challenges remain, such as optimizing concentrations and application methods, as well as improving the understanding of oil selectivity on different plant species to avoid negative impacts on neighboring crops (Hazrati et al., 2018).

Comparison with Other Bio-Herbicides. In comparison with other bio-herbicides, such as rosemary, oregano, or mint oils, lavender oil exhibits similar, and in some cases, even superior results in weed suppression. For example, Vasilakoglou et al. (2013) examined the effects of rosemary and oregano oils on weeds such as *C. album* (common lambsquarters) and *A. retroflexus* (redroot pigweed). Although both oils demonstrated certain inhibitory effects, lavender oil showed more pronounced results at lower concentrations, which was attributed to its higher content of linalool and terpinen-4-ol.

When lavender oil is used in combination with rosemary and mint oils, a synergistic effect occurs, enhancing the inhibitory action on target weeds (Matković et al., 2019). The combined application of lavender and rosemary oils at a concentration of 0.5% for each component led to a reduction in the germination of *S. viridis* (green foxtail) by more than 75%, which was significantly higher than when the oils were used individually (Nikolova et al., 2021). The synergistic effect of these oils can be explained by the complementary action of their active compounds on different metabolic pathways in the plant. This characteristic of combined bioherbicides opens new possibilities for using lavender oil as a component in natural herbicides formulations.

In conclusion, lavender essential oil, due to its high content of bioactive compounds such as linalool, terpinen-4-ol, and camphor, exhibits strong inhibitory effects on a wide range of weeds. Its potential is particularly pronounced when combined with other plant oils, achieving greater efficacy and reducing the risk of resistance development in target weed species (Vasilakoglou et al., 2013; Matković et al., 2019). Although the

effects are promising, further research is needed to optimize the doses and application conditions in various agroecological settings.

Herbicidal potential of lavender. To investigate the allelopathic effects and herbicidal potential of lavender, researchers employed various experimental methods under laboratory and field conditions. Two main approaches used were Petri dish bioassay tests under controlled laboratory settings and *in vivo* testing in greenhouses and open fields. These methods allow the quantification of the effects of different concentrations of lavender essential oils and the identification of key inhibitors of germination and growth in targeted weed species. Each method has its specific advantages and limitations, depending on the weed species, cultivation conditions, and concentrations of the bioactive compounds applied.

Petri dish Bioassay Method. The Petri dish bioassay is a standard laboratory method used to evaluate the allelopathic potential of plant extracts and essential oils. In this method, the seeds of targeted weed species are placed on filter paper in Petri dishes, which have been previously saturated with different concentrations of lavender essential oil. Conditions such as temperature, humidity, and light are strictly controlled to ensure a consistent environment for all treatments (Petrova et al., 2015).

For example, in the study conducted by Petrova et al. (2015), different concentrations of lavender essential oil (0.1, 0.5, 1 and 2%) were applied in order to investigate their effects on the germination of *A. retroflexus* (redroot pigweed) and *C. album* (common lambsquarters) seeds. Germination was monitored over a seven-day period, and additional parameters such as seedling length and total above-ground biomass were measured. The results showed that concentrations above 0.5% significantly inhibited seed germination by more than 60%, while higher concentrations (1% and 2%) completely inhibited germination. These findings confirm that lavender oil acts on the early developmental stages of plants, inhibiting key metabolic processes necessary for germination.

Greenhouse Testing. Following initial laboratory tests, the effects of lavender essential oil were further investigated under greenhouse conditions to identify their potential effectiveness in real cultivation scenarios. Greenhouses allow for a larger number of weed plants to be treated with various concentrations of oil, applied either directly to the leaves or to the soil. For instance, Matković et al. (2019) tested the effects of lavender essential oil on the growth of *Echinochloa crus-galli* (barnyardgrass) in a greenhouse setting. In the experiment, concentrations of 0.5% and 1% were applied directly to the soil around the weeds during the germination stage. The results showed a reduction in plant height and biomass by 45% in treated plants compared to control samples.

Greenhouse conditions enable the monitoring of effects in a controlled yet more realistic environment compared to laboratory tests. This allows the quantification of long-term effects and assessment of bioherbicide phytotoxicity on non-target species. Key parameters measured include plant height, total biomass, vitality index, and changes in root morphology. In studies like this, additional factors such as alterations in photosynthetic activity and plant respiration are often recorded, providing a deeper understanding of the oil's mechanisms of action (Matković et al., 2019).

Field Experiments. In the final phase of research, the bioherbicidal effects of lavender are tested in open field conditions to determine its efficacy in actual agroecosystems. Field experiments are particularly valuable as they encompass variable factors such as climate, soil composition, and the presence of other plants, which can significantly influence treatment effectiveness. In the study by Nikolova et al. (2021), lavender oil was applied to field plots at concentrations of 1% and 2%, where its impact on weeds such as *Cynodon dactylon* (bermudagrass) and *Sorghum halepense* (johnsongrass) was evaluated. The results showed that treated plots had a 70% reduction in weed coverage, while plants treated with synthetic herbicides showed similar results.

Although field experiments are the best indicators of bioherbicide efficacy, they also have limitations regarding result reproducibility and condition control. The effects of lavender oil application can vary significantly depending on weather conditions, the presence of other plant species, and soil type. Therefore, it is necessary to combine these experiments with laboratory and greenhouse tests to gain a comprehensive understanding of lavender's potential as a bioherbicide.

In conclusion, different approaches to the methodology of testing lavender's bioherbicidal properties allow for a thorough assessment of its effects on the germination, growth, and development of targeted weed species. When used in combination, these methods provide accurate information on the efficacy, selectivity, and safety of bio-herbicide application in agricultural systems.

Allelopathic effects of lavender. Experimental research has shown that lavender essential oil has a significant allelopathic effect on the germination and growth of various weed species. Different concentrations of the oil, including its main active ingredients such as linalool, terpinen-4-ol, and camphor, have been tested to identify their potential as natural herbicides (Ghosh et al., 2020; Campiglia et al., 2007).

Effects of different oil concentrations on weed germination and growth. In a study conducted by Matković et al. (2019), the effect of different concentrations of lavender essential oil (0.1, 0.5, 1, and 2%) on the germination of weed plants such as *Amaranthus retroflexus* (redroot pigweed), *S. viridis* (green foxtail), and *C. album* (common lambsquarters) was tested under controlled laboratory conditions. The results showed that

concentrations of 0.5% and above significantly reduced germination in all tested weeds, with 1% oil concentration completely inhibiting the germination of *C. album*. The 2% concentration caused total inhibition of growth in all targeted species during the early developmental stages, indicating the high toxicity of lavender oil under such conditions.

Similar findings were recorded in the study by Nikolova et al. (2021), where different concentrations of lavender oil (0.1% to 3%) were tested on *C. dactylon* (bermudagrass) and *S. halepense* (johnsongrass). Germination significantly decreased with increasing oil concentrations. For example, a 1% oil concentration reduced *S. halepense* germination by 65%, while in *C. dactylon*, germination decreased by 45%. At higher concentrations (2% and 3%), germination inhibition was over 90% for both weeds, indicating the potential use of lavender oil as a foliar herbicide for the control of invasive weeds.

Table 1: Effects of Different Concentrations of Lavender Oil on Weed Seed Germination (Petrova, Valcheva, & Velcheva, 2015)

Plant Species	Concentration of Aqueous Extract (%)	Germination (%)	Germination Compared to Control (%)	Length (cm)	Weight (g)
Wheat (<i>Triticum aestivum</i> L.)	0	86	-	19.7	118.2
Wheat (<i>Triticum aestivum</i> L.)	1.25	53	62	7.9	45.1
Wheat (<i>Triticum aestivum</i> L.)	2.5	12	14	2.3	11.6
Wheat (<i>Triticum aestivum</i> L.)	3.75	0	-	-	-
Wheat (<i>Triticum aestivum</i> L.)	5	0	-	-	-
Johnson grass (<i>Sorghum halepense</i> L.)	0	88	-	4.1	35.7
Johnson grass (<i>Sorghum halepense</i> L.)	1.25	22	25	1.9	12.2
Johnson grass (<i>Sorghum halepense</i> L.)	2.5	0	-	-	-
Johnson grass (<i>Sorghum halepense</i> L.)	3.75	0	-	-	-
Johnson grass (<i>Sorghum halepense</i> L.)	5	0	-	-	-
Twitch (<i>Cynodon dactylon</i> L.)	0	93	-	3.6	29.4
Twitch (<i>Cynodon dactylon</i> L.)	1.25	71	76	3.1	22.2
Twitch (<i>Cynodon dactylon</i> L.)	2.5	34	37	2.4	11.8
Twitch (<i>Cynodon dactylon</i> L.)	3.75	0	-	-	-
Twitch (<i>Cynodon dactylon</i> L.)	5	0	-	-	-
White pigweed (<i>Chenopodium album</i> L.)	0	85	-	3.5	26
White pigweed (<i>Chenopodium album</i> L.)	1.25	11	13	1.1	8.3
White pigweed (<i>Chenopodium album</i> L.)	2.5	0	-	-	-
White pigweed (<i>Chenopodium album</i> L.)	3.75	0	-	-	-
White pigweed (<i>Chenopodium album</i> L.)	5	0	-	-	-
Curly dock (<i>Rumex crispus</i> L.)	0	98	-	6.8	66.5
Curly dock (<i>Rumex crispus</i> L.)	1.25	60	61	4.4	33.8
Curly dock (<i>Rumex crispus</i> L.)	2.5	32	33	2.7	14.9
Curly dock (<i>Rumex crispus</i> L.)	3.75	1	1	1.4	2.2
Curly dock (<i>Rumex crispus</i> L.)	5	0	-	-	-

Inhibitory Effects of Individual Compounds. Different compounds within lavender essential oil have specific modes of action on target plants. The main component, linalool, which makes up more than 30% of the oil content, acts as an inhibitor of cell division, disrupting protein synthesis and essential growth hormones. According to the study by Ghosh et al. (2020), linalool affects the cell membrane, causing changes in permeability and reducing transpiration, which leads to seedling desiccation. This effect was particularly pronounced in *A. retroflexus*, where 1% concentration of linalool reduced seedling length by more than 70% compared to the control.

Terpinen-4-ol, the second most important component in lavender oil, has a specific mechanism of action that inhibits chlorophyll synthesis, reducing the plant's ability to perform photosynthesis. In the study conducted by Singh et al. (2006), terpinen-4-ol was responsible for a significant reduction in green biomass in *Setaria viridis*, demonstrating selective action on broadleaf weed species. Terpinen-4-ol affected the microtubular structure of weed cells, causing microtubule aggregation and inhibition of root cell elongation, which resulted in the shortening of root and above-ground plant parts.

Camphor, which is present in lower concentrations, acts as a secondary inhibitor. Although it does not have strong inhibitory effects on its own, its combination with linalool enhances the toxic effects on the weeds, indicating a synergistic interaction between these compounds (Nikolova et al., 2021). Studies have shown that a 1:1 ratio of linalool and camphor significantly reduces seedling vitality in *C.album*, causing oxidative stress in plant cells, which eventually leads to programmed cell death (Nikolova et al., 2021).

DISCUSSION AND IMPLICATIONS

The results confirm that lavender essential oil, due to its specific bioactive components such as linalool, terpinen-4-ol, and camphor, can act as a natural herbicide with selective effects on weed species. The strongest inhibitory effects were observed with linalool and terpinen-4-ol, while camphor showed weaker effects but with a potential for synergy (Ghosh et al., 2020; Singh et al., 2006). The main advantage of lavender oil lies in its ability to inhibit germination during the early developmental stages, reducing the need for synthetic herbicides.

Comparison with other bioherbicides shows that lavender may have similar, and in some cases, even stronger effects than rosemary and oregano oils, especially at lower concentrations (Vasilakoglou et al.2013; Matković et al,2019). This characteristic makes it suitable for use in organic agriculture, where products with low environmental risk and quick degradation pathways are preferred. However, the effectiveness of lavender oil depends on numerous factors, including the type of target weed, genetic variability, and environmental conditions, requiring further research under real agroecological conditions.

These findings suggest that lavender essential oil can be an effective natural herbicide for controlling problematic weeds with minimal environmental risks. Future research should focus on optimizing dosages and application methods to ensure maximum efficacy without negative impacts on non-target plant species (Nikolova et al., 2021).

CONCLUSION

The results of this research indicate that lavender essential oil has significant allelopathic and herbicidal potential due to the presence of bioactive compounds such as linalool, terpinen-4-ol, and camphor. Different concentrations of lavender oil can successfully reduce the germination and growth of a wide range of weed species, including *Amaranthus retroflexus*, *Chenopodium album*, and *Setaria viridis*. These findings suggest that lavender can be an effective alternative to synthetic herbicides, particularly in organic agriculture, where maintaining ecological balance is crucial.

Lavender oil as a bio-herbicide has multiple advantages, such as breaking down quickly in the environment, being safer for helpful organisms, and having lower toxicity levels for soil and water. However, challenges still exist, including finding the right concentrations and methods of application, and gaining a better understanding of how the oil affects different plants to avoid harming nearby crops.

Further research is needed to determine the long-term effects of lavender oil application under various agro-ecological conditions and to develop strategies for its integrated use with other natural agents (Matković et al., 2019). Future studies should focus on testing the effectiveness of lavender oil in field conditions and identifying optimal combinations with other bioherbicides, such as rosemary, oregano, and mint, in order to achieve maximum herbicidal effect (Vasilakoglou et al., 2013; Nikolova et al., 2021). The use of lavender oil as a part of an integrated weed management strategy can contribute to the development of sustainable agricultural practices and reduce the ecological consequences associated with synthetic herbicide use.

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REUSE OF BOTANICAL WASTE

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*This research explores the possibility of using yarrow hydrolate (*Achillea millefolium*) as natural herbicide for possible weed control. Yarrow, widely known for its medical use, also carries bioactive combinations with allelopathic effects that could hold back growth of weed and unwanted plants. The study explores the efficacy of yarrow's hydrolate in controlling agricultural weeds by assessing its impact on their growth.*

Keywords: yarrow hydrolate, reuse, weeds

INTRODUCTION

Yarrow (*Achillea millefolium*) is a herb known for its wide medical use and ecological significance. It is found across the Northern Hemisphere. Yarrow belongs to the Asteraceae family, this species can be found in many habitats, demonstrating its ability to adapt to numerous conditions.

Throughout history it has been used as medicine as anti-inflammatory, and in wound healing processes. In addition to this, yarrow's ecological role as pollinator-friendly plant and its role in sustainable agriculture further highlights its significance.

In spite of historical use, many aspects of this herb remain unknown. This study seeks to contribute to the existing body of knowledge by using hydrolate in treating weeds.

MATERIAL AND METHODS

Yarrow's hydrolate has been the main material used in this research. Yarrow continues to be utilized in various forms, one of which is as a hydrolate. Hydrolates unlike essential oils are composed of mainly water soluble particles, making them milder but still bio active formulation.

Yarrow hydrolate contains a small amount of evaporative compounds such as sesquiterpenes, flavonoids, and other biologically active molecules. It is made through process called steam distillation and it extracts water soluble components. Steam distillation is the most common method used, and it yields both the hydrolate and the essential oil simultaneously. Yarrow itself, as a plant, has been used in companion planting because of its ability to attract beneficial insects and improve soil health, but the hydrolate form is generally reserved for topical uses rather than for weed management.

In this experiment we have been treating *Portulaca oleracea*, *Amaranthus retroflexus*, *Chenopodium album* with this hydrolate.

Portulaca oleracea is a succulent plant that is regarded as edible and as weed. It is found in many parts of the world, and has been used for medical purposes for years. It is packed with nutrients that include Omega 3, vitamins, antioxidants. It is easy to grow, and it can be started from seeds. It thrives in poor soils and it can be invasive so it needs to be monitored.

Amaranthus retroflexus is a species of annual flowering plant in the Amaranthaceae family. It is considered a common weed in many parts of the world. It can grow to 90 cm in height. The plant thrives in disturbed soil and is often found in fields and gardens. It is very adaptable and can grow in a wide range of environments and it makes it a determined weed in many areas. Its ability to produce a great amount of seed contributes to its invasiveness. Each plant can produce thousands of seeds which are very easily spread. *Amaranthus retroflexus* can be difficult to control once it becomes established, particularly in agricultural settings.

Chenopodium album is a plant from the Amaranthaceae family, it is a common and wide spread weed. It can grow up to 180 cm in height and it is very common in disturbed soil. The plant contains oxalic acid which, if binded with calcium, can cause kidney stone formation. *Chenopodium album* is very adaptable and can grow in a variety of environments, where it takes away supplements from soil away from crops.

The task of this study was to analyse the seed germination, germination dynamics, and initial growth of the tested weed species, namely red-root amaranth (*Amaranthus retroflexus*), common purslane (*Portulaca oleracea*), and lamb's quarters (*Chenopodium album*), under laboratory conditions, following the treatment with different concentrations of yarrow hydrolate (10, 20, 50 and 100%) in controlled environmental conditions.

The objective of the paper was to determine the effect of the yarrow hydrosol on the germination and initial growth of the seedlings of the tested plants, in order to assess its potential for its practical application in weed control.

On a filter paper in every Petri dish with 8 cm diameter 25 seeds of the tested plants were placed, which was done in four replicates, after which the seeds were treated with 10 ml of different concentrations of the yarrow hydrolate (10, 20, 50 and 100%), i.e. distilled water in case of the controls. The seeds were kept in a climate chamber for 10 days under the conditions of 22°C/20°C with 8/16-hour photoperiod and 60±1% humidity. The seed germination was monitored daily, and upon completion of the experiment, the length of the hypocotyls and epicotyls of all the tested plants were measured.

RESULTS AND DISCUSSION

The yarrow hydrolate had a significant effect on the reduction of seed germination of *A. retroflexus*. The lowest hydrolate concentration reduced the germination of *A. retroflexus* seeds by 99% compared to the control, while the higher concentrations led to complete inhibition (Figure 1). In the case of *C. album*, the 10% hydrolate concentration inhibited the seed germination by 39%, while the 20% and 50% hydrolate concentrations led to the inhibition of 24% and 92%, respectively. The inhibition of germination for *P. oleracea* was complete when 100% hydrolate concentration was applied, while in case of 20% and 50% hydrolate concentrations the noted inhibition was only 9% and 17%, respectively.

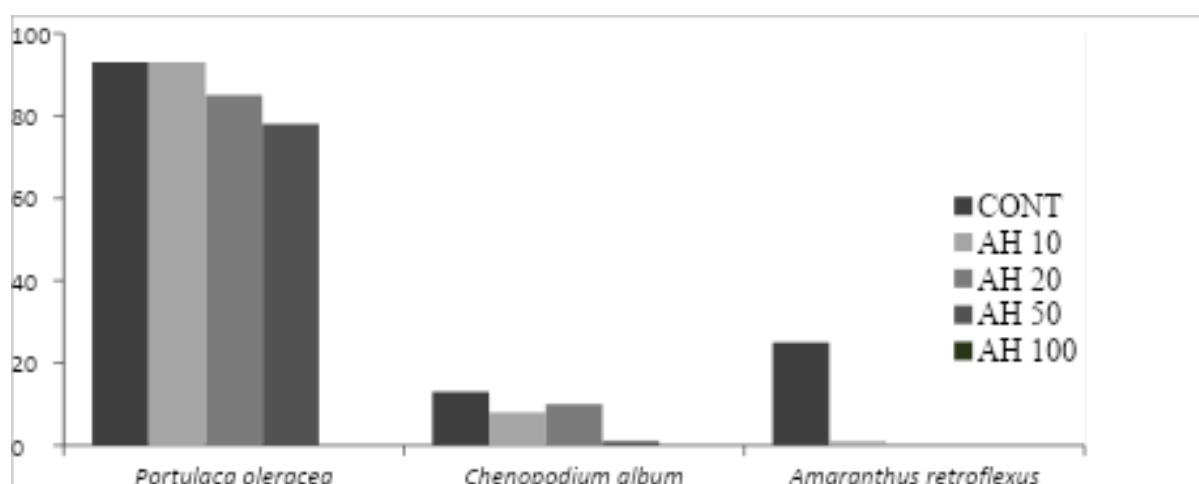


Fig. 1 The effect of different concentrations of yarrow hydrolate on the germination (%) of the tested weed species: common purslane, lamb's quarters and red-root amaranth.

The Duncan's post-hoc test showed that there is a statistically significant difference (at a significance level of $p < 0.05$) in the growth of the common purslane's hypocotyls between the control and the variants treated with 10, 20 and 50% yarrow hydrolate concentration. In the case of hypocotyls, a statistically significant difference was found in the variants treated with 10% and 50% hydrolate concentrations, compared to the control.

Regarding the lamb's quarters, a statistically significant difference was identified in the epicotyls length between the control and the variants treated with 20% and 50% hydrolate concentration. The growth of the hypocotyls was stimulated by the lowest concentration, while the 20% and 50% hydrolate concentrations inhibited the root development. The highest applied hydrolate concentration completely inhibited the lamb's quarters seed germination (Table 1).

Table 1. The effect of yarrow hydrolate on the growth of the hypocotyls and epicotyls of the seedlings of common purslane and lamb's quarters.

Yarrow hydrolate concentration (%)	<i>Portulaca oleracea</i>		<i>Chenopodium album</i>	
	Epicotyl length (mm)	Hypocotyl length (mm)	Epicotyl length (mm)	Hypocotyl length (mm)
10	7.67±2.10 ^b	18.59±5.00 ^a	10.25±2.43 ^a	13.75±6.02 ^a
20	6.51±1.52 ^c	15.53±4.53 ^b	7.30±2.91 ^b	5.60±3.50 ^b
50	6.49±1.76 ^c	7.15±3.26 ^c	5.00±0.00 ^b	2.00±0.00 ^b
100	-	-	-	-
Control	8.77±1.94 ^a	14.52±6.07 ^b	13.85±4.93 ^a	13.08±5.25 ^a

In the case of red-root amaranth, a statistically significant difference was observed in the inhibition of both the epicotyls and hypocotyls growth between the variants treated with 10% hydrolate concentration and the control. The other concentrations completely inhibited the germination of the red-root amaranth's seeds (Table 2).

Table 2. The effect of yarrow hydrolate on the growth of the hypocotyls and epicotyls of the red-root amaranth's seedlings.

Yarrow hydrolate concentration (%)	<i>Amaranthus retroflexus</i>	
	Epicotyl length (mm)	Hypocotyl length (mm)
10	14.00±0.00 ^b	20.00±0.00 ^b
20	-	-
50	-	-
100	-	-
Control	19.27±4.12 ^a	26.89±6.31 ^a

During the testing all of the weeds have been treated with 10%,20%,50% and 100% of the hydrolate. During the time period of 10 days there have been remarkable changes.

On the first and second none of the weeds treated have shown any sign of growth. On the third day *Portulaca oleracea* started to grow in 10% and 20% of hydrolate, meanwhile the others have shown no growth.

The fourth day the *Portulaca* has continued its growth and started to extend on to 50% of hydrolate.

The next control point *Chenopodium album* has grown in 50%, 20%, 10% and *Amarantus retroflexus* has also grown in 50%, 20%, 10%.

CONCLUSION

Treating weeds with yarrow hydrolate is an encouraging approach in organic agriculture. Although it does not fully chemical herbicides it has shown its potential, that it can be used in weed management while reducing chemical inputs and promoting biodiversity. Yarrow hydrolate shows remarkable potential for the future of organic agriculture, with its bioactive compounds that can interfere with the growth of weeds. The studies show its effectiveness against *Amarantus retroflexus* and *Chenopodium album*. Further research should improve its concentration and approach. Taking the matter that this is eco-friendly and one of possible replacements for chemical herbicides, yarrow hydrolate offers an encouraging solution for reducing chemicals in agriculture and promotes ecological sustainability.

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REPELLENT EFFICACY OF ICARIDIN ON *RHIPICEPHALUS SANGUINEUS* (ACARI: IXODIDAE) NYMPHS

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The aim of this study was to test the repellent efficacy of 15% icaridin based product on *Rhipicephalus sanguineus* nymphs. The goal was to compare the repellent efficacy in the function of time, measuring the percentage of repellence immediately after the application of the product, and then after 1, 2, 4, and 8 hours. The obtained results were statistically analyzed using ANOVA and Fisher's LSD test. The results showed that the highest repellent efficacy was observed in the fifth replicate (100.0%), and the average repellent efficacy calculated for five replicates was 95.6%. The applied statistical analysis highlighted a high statistical significance in the repellence rate between the test groups and the control group, and a statistically significant difference when comparing the fifth replicate to the control. However, the effect of time in the experiment was not statistically significant. The results demonstrated that 15% icaridin based product had a very high repellent efficacy against *Rhipicephalus sanguineus* nymphs, with efficacy remaining high (96.0%) up to four hours after the repellent application. However, repellent activity decreased over time due to the evaporation of the preparation, reaching 90.0% after eight hours. Studies aimed to test repellent efficacy of various active substances have significant implications, both, in theory and in practice, and require continuous, multidisciplinary research and methodological improvements.

Keywords: repellent efficacy, ticks, *Rhipicephalus sanguineus*, nymphs, icaridin,

INTRODUCTION

Ticks (Acari: Ixodidae) are obligatory hematophagous arthropods that feed on mammals, birds, reptiles and amphibians as ectoparasites. The significance of ticks lies in their role as vectors and reservoirs of various infectious pathogens important for human and animal health. They play a role in the maintenance and transmission of infectious diseases such as Lyme borreliosis, Q fever, tick-borne encephalitis, human granulocytic anaplasmosis, Crimean-Congo hemorrhagic fever, and many others.

Urban and rural habitats often consist of a large number of green areas such as lawns, parks, recreational sites, meadows, forest, sport fields, dog parks, and riverbanks. These areas are covered with various plants, and due to their specific floristic composition and structure, presence of different animal species as the potential hosts, and favourable microclimatic conditions, many of them represent ideal habitats for various tick species.

Rhipicephalus sanguineus (Latreille, 1806) is an endophilic tick species (Dantas-Torres and Otranto, 2010), and adapts very easily to the closed spaces, so it often forms numerous populations in kennels or people's homes. Females are good climbers, so they usually lay eggs in the walls crack and ceilings. The most common hosts of *Rhipicephalus sanguineus* are dogs. According to Borges et al. (2015), *Rhipicephalus sanguineus* easily shifts from three-host to one-host life cycle model when found on dogs. Nevertheless, this species can successfully parasite on cattle, cats, rodents, and birds (Dantas-Torres and Otranto, 2010). Adults are not usually found on humans, but nymphal or larval stages could be frequent. Human infestations are more frequent during the summer (Uspensky, 2009), and according to Dantas-Torres and Otranto (2010), people are considered as an accidental host.

The majority of tick species are active from mid-March to mid-October in the continental climate conditions characteristic for our country. During this period, ticks usually find adequate hosts. After taking a sufficient amount of a blood meal, fertilized females oviposit the eggs. Ticks have two juvenile or pre-reproductive stages: larvae and nymphs, and both seek potential hosts to feed on them. After feeding, nymphs moult to adults, thus completing the reproductive cycle. Ticks' seasonal activity is greatly influenced by the activity of their hosts. Therefore, what is very important to emphasize, is that seasonal peaks of high tick activity coincide with the frequent presence of people in nature, especially during spring months (recreation, agricultural activities, and sports) (Petrović et al., 2021). To ensure the safest possible time spent outdoors, whether for work or recreation, preventive measures have been promoted to protect humans against ticks. These measures include choosing appropriate clothing and footwear (light-coloured clothes, long sleeves and pants, closed shoes), avoiding areas where ticks are likely to be present (tall grass, bushes, unmaintained areas), and using repellents.

Repellents are substances whose volatile components repel insects and/or ticks from a host (human or animal) as a potential source of food (Pagés et al., 2014). Repellents do not kill insects and ticks but repel them, and this mode of action distinguishes them from the insecticides. Insecticides and acaricides, designed to kill insects and ticks, are not cosmetic products and should not be applied to humans or animals, whereas repellents are cosmetic products intended for personal protection against insects and ticks. One of those substances is icaridin.

Icaridin (2-(2-hydroxyethyl)-1-piperidinecarboxylic acid 1-methylpropyl ester) is a piperidine derivative, developed in the laboratory of Bayer AG (Germany) as a result of a long-term research. Initial studies on this substance demonstrated its high effectiveness. Nevertheless, another decade of intensive research period followed in order to confirm its safety (Vasiljević et al., 2008). Numerous studies have confirmed that icaridin is an effective and safe repellent that provides several hours of protection and is recommended for malaria prophylaxis (Orshan et al., 2010; Semmler et al., 2010; Büchel et al., 2015; Abdel-Ghaffar et al., 2016; Sangha et al., 2016). Today, this active substance is registered for use in the human population in over 40 countries (WHO, 2009). According to Vasiljević et al. (2008), in terms of effectiveness, it can be compared to DEET, but it is safer and has better dermal tolerance.

While many tick repellents demonstrate effectiveness in the short term, there's limited research on their long-term efficacy in real-world conditions. Furthermore, little is known about tick potential to develop resistance to commonly used chemicals or natural compounds. The aim of this study was to evaluate the repellent efficacy of 15% icaridin based product in laboratory conditions on *Rhipicephalus sanguineus* nymphs in the function of time.

MATERIALS AND METHODS

For the purposes of the experiment, ticks were collected from the private apartments, buildings and yards in Novi Sad, after veterinarians and homeowners notification. To prevent desiccation, collected ticks were placed in perforated plastic containers with a piece of water-soaked cotton in order to provide sufficient amount of air and humidity before their use in the experiment. Collected ticks and their stages were identified as *Rhipicephalus sanguineus* species using identification keys proposed by Walker et al. (2007) and Estrada-Peña et al. (2004). For this experiment, only nymphs of *Rhipicephalus sanguineus* were used.

Repellent efficacy of the 15% icaridin based product was calculated according to methods proposed by Kröber et al. (2013) and Adenubi et al. (2018). The experiment consisted of five replicates with the applied 15% icaridin based product and the control group which was treated with distilled water.

In order to test the repellent efficacy of the preparation, the Petri dish bottoms (90 x 14.5 mm in diameter) were used. The filter paper was placed at the bottom of each bottom, after which the Petri dish bottoms were topped by cotton gauze. Petri dishes were sealed with Parafilm in order to prevent the nymphs from escaping. The inserted filter papers were marked into three zones with a graphite pencil: zone without the product, neutral or starting zone for nymphs and the zone with the applied product (20 µl / 1 cm²).

Ten *Rhipicephalus sanguineus* nymphs were placed in six Petri dishes (five containing the product, while the sixth was the control group without the product). The ticks were placed in the neutral (starting) zone, and their movement was tracked based on the moving behaviour. The following grades were assigned to each nymph: "A" if they moved away from the zone with the repellent product, "B" if they approached the zone with the repellent product, "C" if they walked along the neutral zone, and "D" if they were not moving, and stayed at the initial zone.

The exposure time was three minutes, measured with a stopwatch. Efficacy was calculated only for ticks graded with "A", and expressed in percentages based on the formula proposed by Thorsell et al. (2006) and Tunón et al. (2006), immediately after the product application, and again after 1, 2, 4, and 8 hours. After each three-minute test cycle, the nymphs were removed, and the Petri dishes were stored in the laboratory at a constant temperature of 20-22°C and relative humidity of 40-43%.

The obtained results were statistically analyzed using ANOVA and post hoc Fisher's LSD test with Statistica 14.0.0.15 software (TIBCO, University License). Statistical significance was observed for p<0.05 and interpreted as a high significance, and p<0.01 and interpreted as a very high significance.

RESULTS

The percentage of average repellent efficacy calculated from five replicates across five time intervals is given in Table 1. The highest repellent efficacy was observed in the fifth replicate (100.0%). The average repellent efficacy across all five replicates was 95.6%.

Table 1. Average repellent efficacy of the 15% icaridin based product on *Rhipicephalus sanguineus* nymphs over 8 hours bioassay

Replicates	Time					Average
	3 minutes	1 hour	2 hours	4 hours	8 hours	
1	100,00%	100,00%	100,00%	90,00%	90,00%	96,00%
2	100,00%	100,00%	90,00%	100,00%	80,00%	94,00%
3	100,00%	90,00%	100,00%	100,00%	90,00%	96,00%
4	90,00%	90,00%	100,00%	90,00%	90,00%	92,00%
5	100,00%	100,00%	100,00%	100,00%	100,00%	100,00%
average	98,00%	96,00%	98,00%	96,00%	90,00%	95,60%
control	20,00%	10,00%	10,00%	20,00%	20,00%	16,00%

One-way ANOVA highlighted high statistical significance in obtained values between the replicates and the control group ($p=0.000000$ for $p<0.01$) (Graph. 1). Statistically significant difference was noticed when the fifth replicate was compared to the control group ($p=0.034047$ for $p<0.05$). However, the time as a predictor was not statistically significant in the experiment ($p=0.997457$ for $p<0.05$), although a slight decreasing trend in efficacy was observed. Fisher's LSD test emphasized statistical significance between the testing groups (Table 2).

Table 2. Fisher's LSD test results

LSD test; Probabilities for Post Hoc Tests Error: Between MS = 31.667, df = 24.000

Replicates	{1} 9.8000	{2} 9.4000	{3} 9.800	{4} 9.6000	{5} 10.000	{control} 2.2000
1		0.579360	1.000000	0.272171	0.272171	0.000000**
2	0.579360		0.579360	0.579360	0.104783	0.000000**
3	1.000000	0.579360		0.272171	0.272171	0.000000**
4	0.272171	0.579360	0.272171		0.034047*	0.000000**
5	0.272171	0.104783	0.272171	0.034047*		0.000000**
control	0.000000**	0.000000**	0.000000**	0.000000**	0.000000**	

Statistically significant differences are marked with *

Highly statistically significant differences are marked with **

DISCUSSION

The results showed that the 15% icaridin based product had a very high repellent efficacy on *Rhipicephalus sanguineus* nymphs in all replicates, maintaining 98.0% up to two and 96.0% efficacy up to four hours after its application on the filter paper. However, repellent activity slightly decreased over time due to product and its active substance evaporation, resulting in 90.0% efficacy after eight hours.

Büchel et al. (2015) conducted experiments comparing the repellent efficacy of 10% IR3535, 10% icaridin, and 20% DEET on *Ixodes ricinus* and *I. scapularis* nymphs, following EPA guidelines. Despite being performed on human subjects, their findings align with our results, confirming that icaridin retains efficacy beyond four hours against tested tick species.

Repellents do not kill ticks but instead disrupt the function of their sensory organs, preventing them from finding the appropriate host for a blood meal (Carroll et al., 2010). According to Koloski et al. (2019), two theories explain the repellents' mode of action. The first theory suggests that repellents mask the host's attractants, confusing ticks and insects, while the second theory assumes that arthropods perceive these active substances as harmful and therefore try to avoid them. In case of ticks, repellent mode of action are still under investigation regardless different active substances, as these ectoparasites possess a unique sensory organ for

locating potential hosts - the Haller's organ, located on the tarsi of the first pair of legs. This organ is essential for host detection, infrared radiation detection, body orientation, and mating (Carr et al., 2017; Josek et al., 2018). Additionally, palps, located in the mouth area, have a chemosensory function (Renthal et al., 2017). Despite extensive research, there is no adequate explanation on how ticks detect repellents or the exact role of chemoreceptors in this processes (Sonenshine and Roe, 2014; Bissinger and Roe, 2014; Carr et al., 2017).

The further researches should be aimed to standardize the variety of tests used for assessing repellent efficacy. The European Chemicals Agency (ECHA) and the U.S. Environmental Protection Agency (USEPA) have published guidelines for testing biocidal products, recommending the use of nymphs and adults as test organisms in experiments with ticks, as different life stages exhibit distinct behaviours (Kulma et al., 2019). One of the reasons for this is the tick tolerance to desiccation and loss of necessary moisture incorporated into the cuticle of these organisms. For example, according to Daniel et al. (2015), tick adults can survive on higher vegetation than nymphs, consequently, the period of their greatest activity during diurnal cycle also varies. On the other hand, nymphs can feed on various host species, while adults show a certain degree of preference towards certain species, and most often prefer to feed on large mammals (Grech-Angelini et al., 2016). For this reason, Kulma et al. (2019) have pointed out that the developmental stages of ticks significantly influence the results of experiments when testing repellents, as nymphs proved to be significantly more sensitive. In addition, different tick species may respond differently to repellents, therefore the future studies should be aimed to examine species - specific repellency profiles. Given the rise of tick-borne diseases like Lyme disease, babesiosis, and anaplasmosis, it's essential to understand how different repellents affect various tick species.

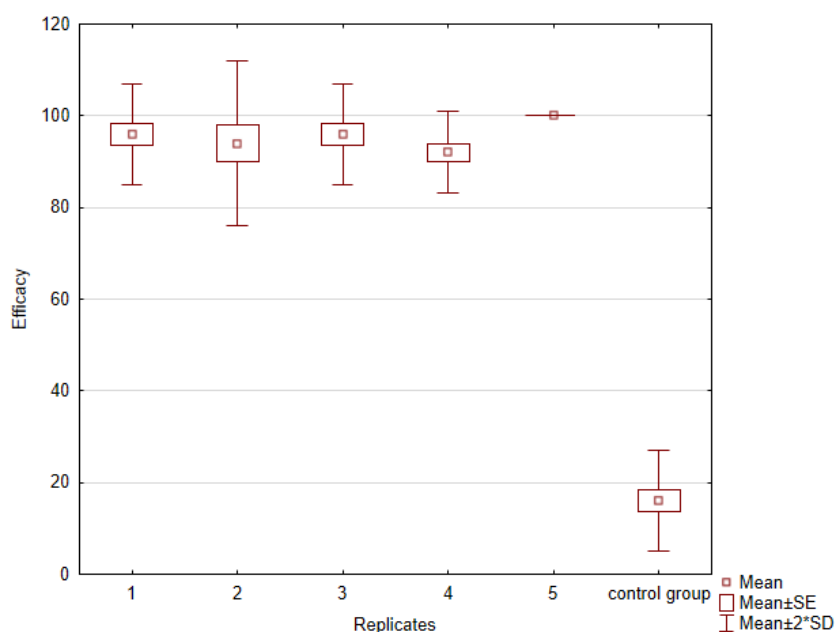


Fig. 1 Repellent efficacy of 15% based icaridin product regarding the replicates and the control group

Studies aimed to test repellent efficacy of various active substances have significant implications, both, in theory and in practice, and require continuous, multidisciplinary research and methodological improvements. This would ensure the development of new synthetic and/or natural substances, appropriate testing protocols, and the implementation of economically viable commercial products that are safe for humans, especially children. However, even with effective repellents, public awareness of tick-borne diseases and proper repellent use is often lacking. Improving education on tick prevention strategies can be as important as the repellent products themselves.

CONCLUSION

The results of this bioassay aimed to determine the repellent efficacy of 15% icaridin based product on *Rhipicephalus sanguineus* nymphs, showed that icaridin exhibited very high repellent efficacy in all replicates, remaining effective up to four hours after the application (96.0%). Repellent efficacy decreased slightly over time due to the product evaporation, with 90.0% efficacy after eight hours. Research on the effectiveness and duration of repellent action for various repellent compounds requires ongoing, multidisciplinary studies and

improvements in methodology to keep pace with advancements in synthetic and natural substances development, suitable test protocols for assessing repellency, and consequently, the development of economically viable commercial products that are safe for human health, especially for children.

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APPLICATION SWQI METHOD FOR ANALYZING THE WATER QUALITY OF THE DANUBE RIVER AT NOVI SAD AND BEZDAN STATIONS

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Due to the increasing need for water, which is becoming more polluted primarily due to human influence, monitoring water quality parameters is gaining importance recently. In the paper, we will present the water quality of one of the most important rivers in Europe - the Danube, from 2019 to 2021, at the Novi Sad and Bezdan stations. We arrive at the assessment of water quality by examining various parameters embedded in the SWQI method, namely: temperature, electrical conductivity, pH value, oxygen saturation, BOD5, suspended matter, orthophosphates, total nitrogen oxides, ammonium, and coliform bacteria. The evaluation method of SWQI is divided into 5 categories, based on which a result is obtained that integrates many parameters. The score ranges from 0 to 100. The results of the SWQI obtained at the Novi Sad station were 83, 84, and 84 for 2019, 2020 and, 2021 respectively; while for the Bezdan station results were 86, 87, 84 for the years 2019, 2020 and 2021, respectively. Through evaluation, we concluded that the water quality at these stations is very good. Based on these results, it can be said that we treat water well, but it is known that there are sources of pollution in the Danube basin that need to be systematically resolved.

Keywords: SWQI, Danube, water quality, Novi Sad, Bezdan

INTRODUCTION

The most fundamental substance for life on Earth is water. It constitutes the building block of all living organisms and serves as aquatic organisms'. Water covers 71% of the Earth's surface, with the majority is comprised of salt water - 97.5%, while only 2.5% is freshwater. Of all the freshwater available, only 0.4% is found in surface waters habitat (Shiklomanov and Rodda, 2004). The Danube is the main waterway through the territory of the European Union and provides new opportunities for the development of transportation, trade, and other economic activities. With its tributaries and the network of canals in its basin, it is of great importance for the development of agriculture, energy, fisheries, construction, chemical industry, etc. (Dragin et al., 2009). Serbia accounts for 10% of the total area of its basin, while the length of the river course is 588 km. Along the course of the Danube through Serbia, there are nine monitoring stations for water quality control.

Due to the increasing water demand, which is becoming more polluted primarily caused by human influence, monitoring water quality parameters is becoming increasingly important today. There are numerous parameters that could be monitored, but the most convenient is the use of index which integrates more parameters. In Serbia, Serbian Water Quality Index (SWQI) is used (Milijašević-Joksimović, et.al 2018). The research was conducted to investigate and assess the water quality of the Danube at two monitoring stations (Novi Sad and Bezdan) by the application of the SWQI.

MATERIAL AND METHODS

The course of the Danube begins with the confluence of two small rivers, the Brigach (47.6 km) and the Breg (41.7 km), which originate below the Black Forest mountains (Figure 1). They join together near the town of Donaueschingen, located in the Black Forest region of Germany. The Danube enters Serbia at the 1,488 river-kilometer from its mouth, at the point where the borders of Hungary, Serbia, and Croatia meet. Bezdan is the first station (out of the current 9) for monitoring water quality, located on the left bank of the Danube River, and in this area, it has the characteristics of a plain river. Novi Sad is the sixth station for measuring water quality and is located on the right bank of the Danube, approximately 70 km away from Bezdan.

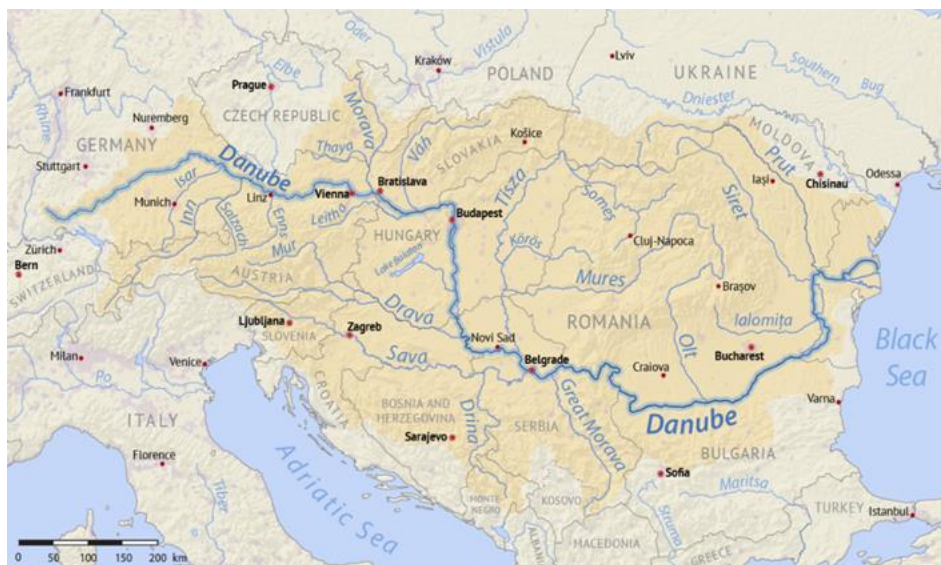


Fig. 1 The course of the Danube

For the process of analyzing water quality at the Novi Sad and Bezdan stations, data was obtained from the website of the Environmental Protection Agency from the hydrological yearbooks for the periods of 2019, 2020, and 2021.

The parameters being analyzed are: Water temperature ($^{\circ}\text{C}$), pH value, Electrical conductivity - EC ($\mu\text{S}/\text{cm}$), Oxygen saturation (%), Biochemical oxygen demand - BOD_5 (mg/l), Suspended solids (mg/l), Total nitrogen oxides (mg/l), Orthophosphates (mg/l), Ammonium (mg/l) and Coliform bacteria (MPN per 100 ml). Location of the two investigated stations is presented in Figure 2.



Fig. 2 The Pannonian section of the Danube through Serbia with Bezdan and Novi Sad stations

SWQI – Serbian water quality index. The water quality index model is a popular tool for assessing the quality of surface waters. It has been applied to evaluate water quality based on local water quality criteria. Ten parameters represent the characteristics of surface waters, reducing them to a single index number. The contribution of each parameter does not have the same significance on overall quality, as each has been assigned its weight (w_i) and score. By summing the products ($q_i \times w_i$), an index of 100 is obtained as the ideal total contribution of the quality of all parameters (House, M. A. (1989); Milijašević-Joksimović et al., 2018). The SWQI method includes five indicators of surface water quality, classified according to the intended use and purity of the water (Table 1): EXCELLENT (90-100) – Waters that, through filtration or disinfection, can be used for water supply; VERY GOOD (84-89) or GOOD (72-83) – Waters that, in their natural state, can be used for swimming and recreational activities. With modern treatment methods, they can be used to supply

settlements with drinking water; BAD (39-71) – Waters that can be used for irrigation, and after modern treatment methods, they can also be used in industry, except in the food industry.

Table 1. Classification according to the SWQI (*Source: SEPA*)

SWQI 1st class		SWQI 2nd class		SWQI 3rd class	SWQI 4th class
85 – 84		74 – 69		56 – 44	51 – 35
100 – 90	89 - 84	83 – 72		71 – 39	38 - 0
Excellent	Very good	Good		Bad	Very bad

Serbian Water Quality Index (SWQI)

Monitoring results for 10 parameters at two stations for the period 2019-2020 are shown in Tables 2-5.

Table 2. Parameter values for the Bezdan station of the year 2020

Bezdan 2020	T	pH	EC	O2 sat.	BPK5	SUS	NOx	PO4	NH4	Colif. Bact.
January	3,2	8,09	453	98	1,2	9	3	0,046	0,06	
February	5,3	8,18	510	97	1,3	11	3	0,027	0,03	
Mart	7,4	8,2	445	97	1,3	20	2,8	0,031	0,02	
April	16,5	8,33	384	115	2,5	26	1,4	<0.01	0,02	
May	18,4	8,39	348	115	1,5	10	1,2	<0.01	<0.02	300
June	22,8	7,97	329	92	1	35	1,8	0,049	<0.02	
July	23,8	8,16	361	98	1,1	9	1,6	0,033	<0.02	2700
August	21,2	8,02	353	96	1,1	14	1,6	0,043	<0.02	800
September	16,2	8,15	375	94	1	27	1,9	0,035	<0.02	
October	11,6	8,09	418	95	1	39	2,4	0,04	<0.02	
November	5,2	8,11	467	94	1,4	19	2,8	0,05	0,05	

Table 3. Parameter values for the Novi Sad station for the year 2020

Novi Sad 2020	T	pH	EC	O2 sat.	BPK5	SUS	NOx	PO4	NH4	Colif. Bact.
January	2,4	8,2	500	95	2,1	7	2,7	0,088	0,15	
February	6,4	8	442	96	2,7	26	2,9	0,095	0,04	
Mart	9	8,23	444	101	2,2	25	2,9	0,084	0,04	
April	18	8,34	366	105	3,1	20	1,2	0,077	0,07	
May	18,4	8,22	338	100	2	33	1	0,047	0,04	140
June	22	8	330	90	2,5	28	1,3	0,107	0,08	600
July	23,8	8,03	353	86	2,3	38	1,6	0,212	0,07	
August	17,6	8	365	89	2	13	1,2	0,089	0,05	2000

September	11,6	7,97	375	89	2,8	16	2,3	0,123	0,06	600
October	6,6	7,86	429	93	2,5	<4	2,4	0,098	0,04	
November	5,4	8,1	450	94	2,5	20	2,2	0,114	0,11	

Table 4. Parameter values for the Bezdan station for the year 2021

Bezdan 2021	T	pH	EC	O2 sat.	BPK5	SUS	Nox	PO4	NH4	Colif. Bact.
January	3,7	7,98	488	90	1	15	2,8	0,054	0,06	
February	3,5	8,04	505	92	2,1	71	2,8	0,043	0,05	
Mart	7	8,13	493	105	2,2	27	3,3	0,033	0,02	
April	10,6	8,64	476	112	3	27	2,2	<0,01	<0,02	300
May	15,5	8,6	442	121	3,4	16		<0,01	0,02	
June	21,4	8,26	341	106	1	34	1,2	0,013	0,02	1400
July	23,7	8,2	343	106	1,9	33	1,4	0,032	<0,02	
August	22	7,83	367	91	1,7	74	1,5	0,042	0,03	2500
September	18,2	8,02	373	98	3,4	115	1,4	0,035	0,03	270
October	16,7	8,09	442	95	1,2	28	1,8	0,032	<0,02	
November	10,7	8,62	477	114	2,9	19	1,4	<0,01	<0,02	
December	6,3	8,1	486	101	1,9	5	0,5	0,032	0,02	

Table 5. Parameter values for the Novi Sad station for the year 2021

Novi Sad 2021	T	pH	EC	O2 sat.	BPK5	SUS	NOx	PO4	NH4	Colif Bact.
January	3,6	8,1	500	94	1	7		0,092	0,06	
February	5,4	8,1	476	96	2,7	10		0,105	0,06	
Mart	6,4	8,5	475	106	3,4	<4		0,056	0,02	
April	12,6	8,4	437	109	2,8	11		0,058	0,05	280
May	16	7,96	331	83	2,3	47		0,087	0,06	
June	26	8,2	308	101	2,1	27	1	0,085	0,08	200
July	23	7,81	326	82	2,1	63	1,3	0,098	0,07	700
August	23	8	357	88	2,4	37	1,3	0,106	0,05	210
September	17,8	8,28	418	107	2,2	16	1,2	0,063	0,04	1000
October	11,4	8,27,	442	99	2,4	133	1,5	0,086	0,04	
November	8,2	8,37	473	103	2,7	21	1,7	0,083	0,16	400
December	6,8	8	474	94	2,1	6	1,9	0,126	0,14	

Description of parameters for water quality ranking.

Water temperature (°C). Temperature is a very important parameter for determining water quality. It affects chemical, physical, and biological processes in water. As the temperature increases, electrical

conductivity also increases, dissolved oxygen (O₂) and carbon dioxide (CO₂) solubility in water decreases, and the rate of chemical reactions increases.

pH value. The pH value represents the acidity or alkalinity of the water environment. The pH scale ranges from 1 to 14, where a neutral environment is indicated by a pH of 7. Values below 7 are acidic, while values above 7 represent an alkaline environment. A high or low pH value is associated with a deficiency of nutrients, toxicity of metals, and other issues. The optimal pH value for the aquatic world is between 7.2 and 8.7, and a danger to living organisms exists when the pH is greater than 9. To determine the pH value, litmus paper, sensors, and indicator solutions are used.

Electrical conductivity (μS/cm). The ability of water to conduct electrical current. Dissolved substances in water (in this case, salts) are positively and negatively charged (cations and anions). By measuring electrical conductivity, we can detect the presence of pollution, but it does not identify the specific pollutant. The cleaner the water, the lower its ability to conduct electricity.

Dissolved Oxygen in Water (mg/l). Oxygen enters water through diffusion from the air, as a product of photosynthesis by photosynthetic organisms, and through aeration. The concentration of dissolved oxygen depends on temperature, salinity, and atmospheric pressure. At lower temperatures, a greater amount of oxygen dissolves, while at higher temperatures, its concentration is lower. Dissolved oxygen values below 5 mg/l can negatively affect the functioning of the aquatic biocenosis, while values below 2 mg/l can lead to death.

Five-day biochemical oxygen demand (BOD5) (mg/l). This is the amount of oxygen required for the breakdown of organic matter by aerobic microorganisms, during which a significant amount of oxygen is consumed. BOD is measured over any number of days. When BOD is less than 2, the water is considered clean, while a value greater than 10 indicates that the water is extremely polluted.

Suspended solids (SS) (mg/l). These are particles that do not pass through the filtration process; they enter the water through soil washing or water erosion in waterways. These substances negatively affect organisms. When they settle on the bottom, they threaten the habitats of fish and other organisms. The concentration of suspended solids in water is determined by the photometric method.

Nitrogen oxides. If there is too much nitrogen in the water, it leads to excessive growth and blooming of phytoplankton, aquatic algae, and plants. This results in eutrophic processes and changes in the structure of the biocenosis (increased dominance of certain plant and animal species). In nature, the level of ammonia and nitrates usually does not exceed 1 mg/l, while in wastewater, it can reach up to 30 mg/l. Total nitrogen oxides, according to the original algorithm for calculating the total SWQI water quality value, are calculated as the sum of nitrites and nitrates (Scottish Development Department, 1976).

Total phosphorus. Phosphate salts in water range from 0.05 mg/l to 0.07 mg/l. The blooming of phytoplankton and the accelerated process of eutrophication are consequences of the presence of phosphorus in water, and a sudden increase in phosphorus levels indicates pollution of surface waters. In marine waters, there is much more phosphorus, but it is not available to plants.

Ammonium ion. Elevated concentrations of ammonia can lead to massive fish kills or the death of aquatic invertebrates, invertebrates and later the growth of algae and aquatic plants. Additionally, at high ammonium concentrations, bacteria convert NH₄⁺ to NH₃ during the nitrification process, consuming dissolved oxygen in the process. The toxicity of ammonia increases with rising temperature and pH levels. Measurement is done using the photometric method and is expressed in mg/l. According to EPA standards, ammonia levels exceeding 0.025 mg/l are considered toxic to fish.

Coliform bacteria. Total coliform bacteria refer to a large group of gram-negative rod-shaped bacteria (thermotolerant coliforms, fecal origin bacteria, and bacteria that can be isolated from the external environment). The presence of coliform bacteria may or may not necessarily indicate fecal contamination.

RESULTS AND DISCUSSION

Figure 3 shows the water quality rating of the Danube at the Novi Sad and Bezdan stations using the SWQI method for the period from 2019 to 2021. Based on the obtained results on water quality, the average values for the Danube at the Novi Sad station during the analyzed period ranged from 83 to 84, which corresponds to the descriptive indicators "good" and "very good," placing it in the first and second classes. For the Bezdan station, the average values for the Danube ranged from 84 to 87, which corresponds to the descriptive indicator "very good" and belongs to first class.

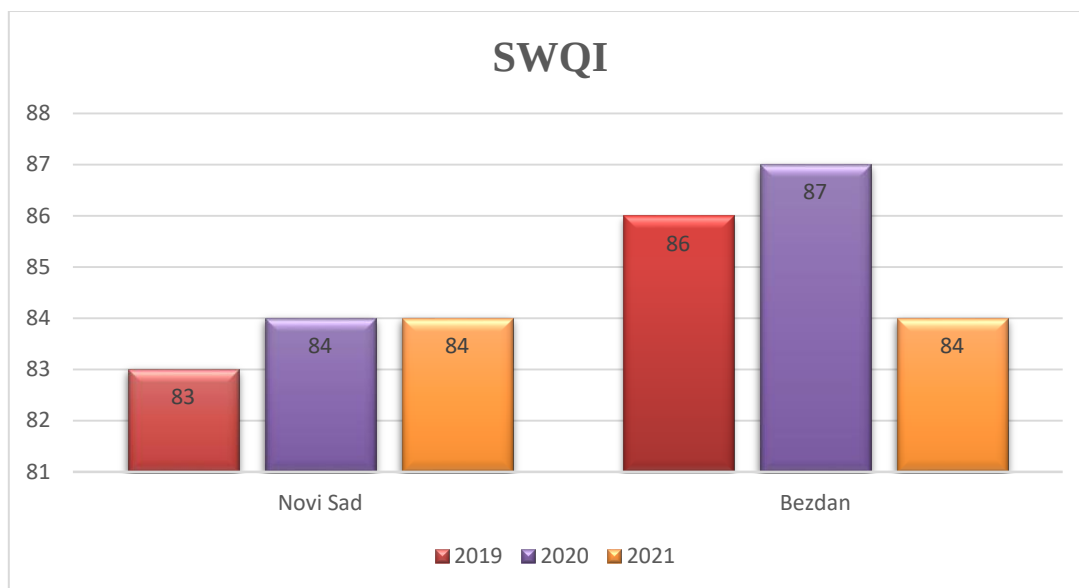


Fig. 3 Average multiyear values of the SWQI Index for the stations Novi Sad and Bezdan

CONCLUSION

The aim of the research was to analyse the Danube water quality at the Novi Sad and Bezdan stations for the period from 2019 to 2021. The water quality at the Novi Sad and Bezdan stations was determined using the SWQI method. The SWQI indices for the Novi Sad station ranged from 83 to 84, corresponding to the descriptive indicators – very good and good; the results belong to classes I and II. At the Bezdan station, the WQI indices ranged from 84 to 87, corresponding to the descriptive indicator – very good, and the results belong to class I. The presented results indicate that the application of SWQI provides a comprehensive overview of the state of surface water quality.

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MORPHOMETRIC ANALYSIS OF CARNIOLAN HONEY BEE (*APIS MELLIFERA CARNICA* POLL.) QUEENS

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This study presents a morphometric analysis of Carniolan honey bee (Apis mellifera carnica) queens, focusing on key reproductive and physical traits essential for colony productivity. Parameters measured include queen weight, thorax width, number of ovarioles, ovariole weight, and semen motility. Results show an average queen weight of 0.20 g, thorax width of 0.52 cm, and a mean ovariole count of 129.90. Ovariole weight averaged 0.06 g, and semen motility was 66%, with significant variability observed. These findings indicate substantial individual differences in queen morphology, which may impact reproductive potential and colony performance. The observed morphometric variation supports the importance of selective breeding to optimize queen quality and enhance colony resilience.

Key words: honey bee, queen, morphometric analysis, ovarioles, spermatheca

INTRODUCTION

The Carniolan honey bee (*Apis mellifera carnica* Poll.) is one of the most widespread subspecies of honey bees due to its desired traits, including non-aggressive behavior, efficient honey production, and strong overwintering capabilities (Ruttner, 1988). Due to these qualities, the Carniolan honey bee has gained popularity across Europe and other parts of the world, making it an important subject of study.

Morphometric analysis has long been an essential method for identifying and distinguishing honey bee subspecies and strains, with specific focus on traits like wing venation, body size, and coloration (Rinderer et al., 1993). In recent years, morphometrics has also been applied to assess variability within subspecies, supporting conservation efforts and selective breeding programs (Puškadija et al., 2020 Lukić et al., 2024). Studies indicate that morphological characteristics are not only vital for identifying subspecies but also for evaluating genetic diversity within populations, which is essential for sustainable breeding programs (Lukić et al., 2024).

Carniolan queens are of particular interest, as the morphometric traits of queens can influence colony performance and survival. Parameters such as thorax width, wing length, and overall body size have been shown to correlate with a queen's reproductive potential, affecting the colony's brood-rearing capacity and population dynamics (Cobey et al., 2012). Morphological variations in queens also reflect environmental and genetic factors, providing insights into adaptability and health (Hatjina et al., 2014).

Given the importance of queens in maintaining colony vigor and adaptability, it is important to study morphometrics of queens to evaluate their quality. This includes measurements on queen body such as queen weight and number of ovarioles. High-quality queens should fulfill all criteria in order to meet the conditions for the development of the honey bee colony. The goal of this study was to examine the quality of young mated queens in a commercial apiary by measuring morphometric parameters of the queens.

MATERIALS AND METHODS

A total of 26 Carniolan honey bee queens were collected for morphometric measurements. Each queen was anesthetized with CO₂ and weighed to determine mass. After weighing, each queen was placed in a covered Petri dish. The queens were then re-anesthetized for dissection, during which the head, wings, and legs were removed. An entomological pin was inserted into the thorax to fix the queen to the substrate in a Petri dish prepared with a wax base in physiological solution. Abdominal dissection was conducted using FST 91501-09 dissection scissors, providing a clear view of internal organs.

The ovaries (Figure 1) were gently removed, weighed, and placed in a 1.5 ml tube containing 1 ml of 10% formalin for further microscopic analysis. Each ovary was photographed, and the ovarioles were separated and counted individually. The weight of ovary and number of ovarioles was measured on the left ovary. The spermatheca (Figure 2) was carefully removed and preserved in liquid to prevent desiccation.

To measure the semen motility, spermatheca was separated from its tracheal net, photographed, and stored (Figure 3). To prepare the Kiev medium, 2.43 g of Na citrate·2H₂O, 0.21 g of NaHCO₃, 0.41 g of KCl, and 0.3 g of glucose were dissolved in 50-70 ml of ultrapure or double-distilled water, then adjusted to 100 ml. The pH (8.1) and osmolarity (385 mOsm/kg) were checked. After adding 2 g of BSA, the medium was refrigerated to ensure proper dilution. Both Kiev and Kiev-BSA diluents were sterilized using a 0.22 µm syringe filter, aliquoted into 1 ml vials, and stored at -20°C. During the procedure, the Kiev and Kiev-BSA vials were warmed to 35°C. A 1 µl semen sample was diluted in 100 µl of Kiev medium and assessed within 30 minutes. The sample was carefully mixed, and a 2 µl aliquot was further diluted in 100 µl of Kiev-BSA. A 3 µl portion was placed in a pre-warmed Makler chamber and maintained at 35°C for 5 minutes. At least 100 sperm cells were counted and categorized as motile or static, with results expressed as a percentage.

Equipment Used: Special tweezers FST by DUMONT 4 (11241-30 Dumaxel) and FST by DUMONT 55 (11255-20 Inox-Biology), dissection scissors FST 91501-09, stereo microscope, Petri dishes, entomological pins, Eppendorf tubes (0.5, 1.5 ml, and 2 ml), pipettes and tips, physiological solution, ethanol, 10% formalin, and distilled water.

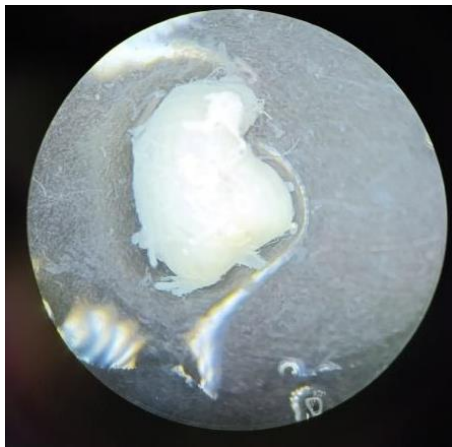


Fig. 1 Ovaries under the stereo microscope

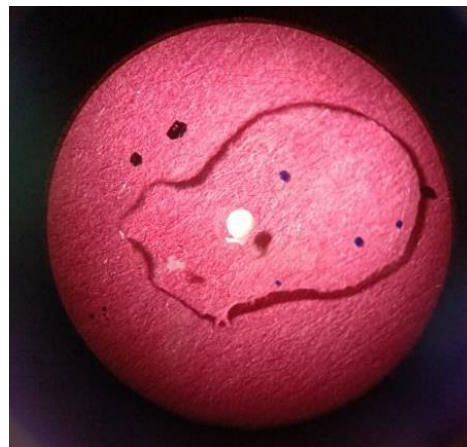


Fig. 2 Spermatheca under the stereo microscope

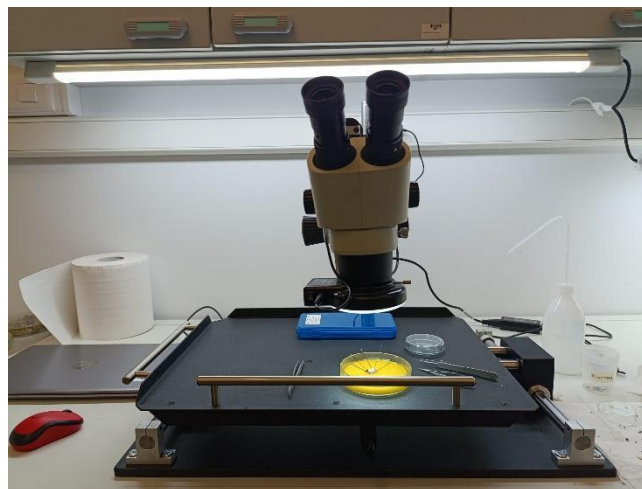


Fig. 3 Stereo microscope and dissection materials

RESULTS AND DISCUSSION

The morphometric analysis revealed substantial variability in key physical and reproductive traits (Table 1), which can have profound implications for colony health and productivity.

Table 1. Key physical and reproductive traits.

	Queen weight (g)	Thorax width (cm)	Number of ovarioles in left ovary	Weight of left ovary (g)	Semen mobility (%)
Mean	0.20	0.52	129.90	0.06	0.66
Minimum	0.16	0.40	107.00	0.01	0.12
Maximum	0.25	0.60	167.00	0.12	0.95
N	26	25	21	26	20
SD	0.02	0.06	14.36	0.02	0.20

The mean queen weight was found to be 0.20 g, ranging from 0.16 g to 0.25 g. This aligns with previous studies that suggest queen weight is a critical factor influencing reproductive potential. Woyke (1989) reported that heavier queens generally possess more developed ovaries with a higher number of ovarioles, leading to increased egg-laying capacity. The relatively low standard deviation (SD = 0.02 g) indicates a degree of uniformity in queen weight within the population studied, which may result from consistent queen rearing practices and similar environmental conditions during larval development.

Thorax width averaged 0.52 cm with a range from 0.40 cm to 0.60 cm. Flight muscles, essential for mating flights, are located in the thorax, and its size can impact the queen's ability to successfully mate with multiple drones (Harbo & Harris, 1999). A moderate standard deviation (SD = 0.06 cm) suggests some variability. Larger thorax width may correlate with stronger flight capability, enhancing the chances of successful mating (Hatch et al., 1999).

The number of ovarioles per ovary, averaging 130 with a range from 107 to 167, is an indicator of a queen's fecundity. Higher ovariole counts brings greater egg-laying potential (Al-Ghamdi et al., 2011). The considerable standard deviation (SD = 14.36) reflects significant variability, which could influence colony population dynamics. Environmental factors, genetics, and larval nutrition are known to affect ovariole development (Kefuss & Taber, 1983). The mean weight of ovarioles was 0.06 g, ranging from 0.01 g to 0.12 g. Ovariole weight can be indicative of reproductive organ development and functionality. Woyke (1971) found that heavier ovarioles correlate with increased reproductive output.

Semen motility averaged 66%, with a wide range from 12% to 95%. High semen motility is crucial for effective fertilization and long-term sperm storage in the queen's spermatheca (Collins, 2000). The high standard deviation (SD = 20%) indicates substantial differences in semen quality among drones, which can be affected by genetics, age, health status, and environmental conditions (Rinderer et al., 2010). Low semen motility may lead to reduced fertilization success and could impact the overall productivity and genetic diversity of the colony.

CONCLUSION

The observed variability in these morphometric and reproductive traits of queen bees underlines the importance of selective breeding and proper management practices. Future studies should investigate the genetic basis of these traits to facilitate marker-assisted selection in breeding programs. Additionally, understanding the environmental factors affecting these morphometric characteristics could lead to improved rearing techniques that maximize queen quality. With global challenges like diseases, parasites, and environmental stressors affecting honey bee populations, improving queen quality is important for sustainable beekeeping.

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DISPERSION OF FRUIT PRODUCTION IN THE ORGANIC PRODUCTION SYSTEM IN VOJVODINA

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Summary: Organic fruit production is based on natural processes and excludes synthetic chemicals. This method supports ecosystem preservation and promotes human and animal health. The aim of the research is to analyze the changes in the area under fruit cultivation within the organic production system in Vojvodina and its distribution across administrative districts. The research also seeks to identify the predominant fruit species in the structure of organic fruit production in Vojvodina. The analysis covers the period from 2018 to 2022, using data sourced from the statistical database of the Ministry of Agriculture, Water Management, and Forestry of the Republic of Serbia. The results of the research can be useful for producers, processors, and policymakers aiming to improve organic fruit production and increase market competitiveness.

Keywords: fruit, organic production, dispersion, administrative district, Vojvodina

INTRODUCTION

"Organic production represents an ecological production management system that promotes and enhances biodiversity, biological cycles, and soil biological activity. It is based on minimal use of external (off-farm) inputs, as well as management practices that establish, ensure, and strengthen ecological harmony" (Liebhardt, 2003). The use of synthetic chemical substances, growth regulators, and additives is excluded, and the use of genetically modified organisms is not permitted. In this way, it aligns with the concept of sustainable development, which our country has adopted as part of its long-term agricultural development strategy. This type of production is desirable in agriculture of any country that favors renewable resources and environmentally acceptable processes that do not harm the environment, human health, or the health and well-being of animals. Additionally, it is increasingly popular as a response to global issues such as climate change, pollution, and loss of biodiversity (Vlahović, Puškarić, Bijelić, 2023).

The most significant international organization that unifies all forms of organic production is IFOAM (International Federation of Organic Agriculture Movements). It was founded in 1972 when five international organizations dealing with organic production issues integrated. Today, it has more than 700 members from 100 countries who have adopted clearly defined principles for the production, processing, and marketing of organic products, which must be adhered to by both member countries and all others wishing to export to those countries (<https://IFOAM.Organics.International>).

The term "organic agriculture" refers to a method of food production on one hand, and a specific quality of products resulting from natural growth conditions, organic incentives from microorganisms, and methods of processing on the other (Cifrić, 2003). The primary goal of organic production is to produce quality food characterized by high nutritional value, ensuring that production is carried out according to principles that preserve ecosystems. According to some authors (Seal, Brandt, 2007; Crinnion, 2010; Blair, 2012), food produced in organic agricultural systems may contain higher levels of certain nutrients, vitamins, and minerals compared to conventionally produced food, such as vitamin C, Mg, P, β -carotene, and phenolic compounds; however, there is still no solid evidence based on studies to support these claims. The benefits of this production method also include maintaining and increasing soil fertility, stability, and biodiversity, as well as preventing soil erosion. The maximum use of renewable energy sources and the maintenance of genetic diversity are practiced.

The aim of the research is to analyze the changes in the area under organic fruit cultivation in the Vojvodina region, both during the conversion period and in the status of true organic products. The research also seeks to determine the most prevalent fruit species in the structure of organic fruit production in Vojvodina and to identify the administrative districts dominant in organic fruit production. The districts included in the analysis are as follows: Northern Banat, Central Banat, Southern Banat, Northern Bačka, Western Bačka, Southern Bačka, and Srem. The results of this study may be important for producers, processors, and exporters of

organically produced fruit. Additionally, the results can provide policymakers in Vojvodina with insights into the current state of organic fruit production, supporting implementation of appropriate measures for producers aimed at increased agricultural fruit production and competitiveness in the market.

MATERIALS AND METHODS

The analysis covers a five-year period from 2018 to 2022, during which adequate statistical documentation is maintained. Data were sourced from the statistical database of the Ministry of Agriculture, Water Management, and Forestry of the Republic of Serbia, available at: <http://www.minpolj.gov.rs/organska/>. Standard statistical-mathematical methods were used for data analysis and interpretation. The intensity of changes was represented by calculating rates of change using trendline functions fitted to the original data. The influence of trends manifests over a longer time period, and trend analysis should indicate the overall tendency of occurrences while disregarding short-term fluctuations. The most common method for observing trends is the graphical method, which was employed in this study. The mathematical function of the trend has the following form: $Y_i = a + bX_i$ (Mutavdžić, Tekić, Novaković, 2023). Tables and cartograms were used to additionally illustrate the most significant phenomena.

RESULTS AND DISCUSSION

The average area under fruit cultivation in the organic production system, including the conversion period in the Vojvodina region, amounted to 261 hectares (2018-2022). During the analyzed period, the area exhibited a significant upward trend with an average annual growth rate of 11.90% (Figure 1). In the initial year, the area was 220 hectares, while in the final analyzed year (2022), it reached 345 hectares. This means that between the boundary years, there was a substantial increase of 125 hectares, or 56.8%. The smallest cultivated area was recorded in 2020, totaling 162 hectares, while the largest cultivated area was in the final analyzed year, reaching to 345 hectares.

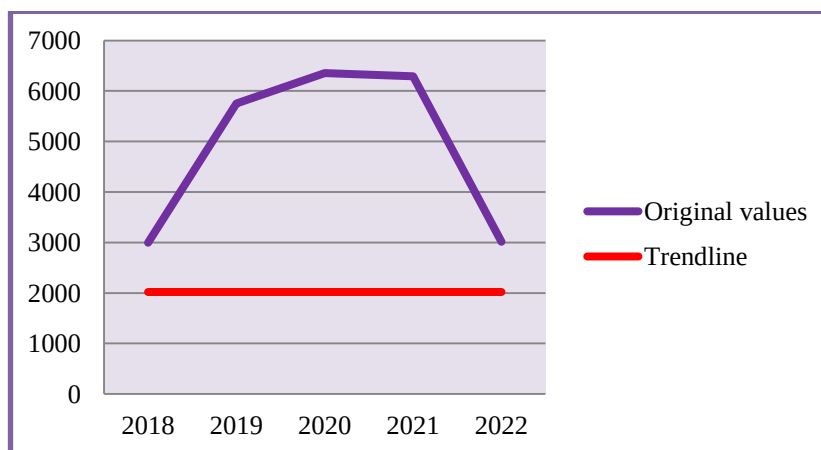


Fig. 1 Area under Fruit Cultivation in the Organic Production System in the Vojvodina Region (ha)

The average area under fruit cultivation in Vojvodina with true organic status from 2018 to 2022 was 91 hectares, accounting for 35% of the total area. During the analyzed period, it exhibited a significant upward trend with an average annual growth rate of 16.9% (Figure 2).

The average area under fruit cultivation during the conversion period was 170 hectares, representing 65% of the total area, and during the analyzed period, it achieved a moderate upward trend with an average annual growth rate of 9.43% (Figure 2), despite a significant decline in the final analyzed year.

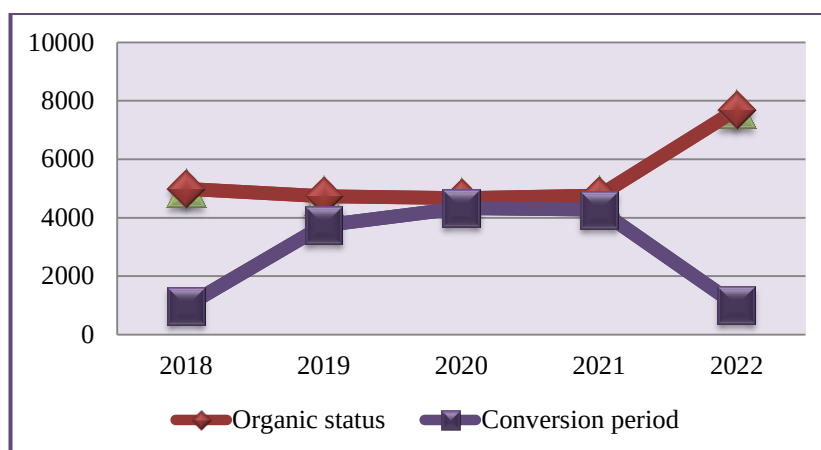


Fig. 2 Changes in the Area under Fruit Cultivation with Organic Status and During the Conversion Period in the Vojvodina Region (ha)

There is a noticeable increase in the area under fruit cultivation with organic status and during the conversion period (Figure 2). It would be highly desirable for this growth trend to continue, as it represents a significant step towards the development of organic fruit production and demonstrates a growing interest among farmers in organic fruit cultivation.

In the structure of average areas under organic fruit cultivation in Vojvodina, grapes are dominant fruit with an average area of 76 hectares, accounting for 33.8% of the total area, and exhibiting a substantial growth trend with an average annual rate of 95.29% (Figure 3). Following grapes, cherries rank second, with an average area of 43 hectares, accounting for 19.1% of the total area, although they demonstrate a significant declining trend with an average annual rate of 14.97%. Hawthorn ranks third, with an area of 36 hectares (16%) and a significant growth trend at a rate of 27.79%. It is followed by hazelnut with 30 hectares (13.3%) and an intensive growth trend at a rate of 22.09%. Raspberry ranks fifth, with an area of 27 hectares, accounting for 12% of the structure, and showing a marked declining trend at a rate of 65.55%.

In addition to these five dominant fruit species, numerous other species are cultivated, including aronia, apple, pear, walnut, plum, and others. However, these are grown in negligible quantities, with average areas not exceeding single-digit figures, collectively accounting for a very small share of 5.8%.

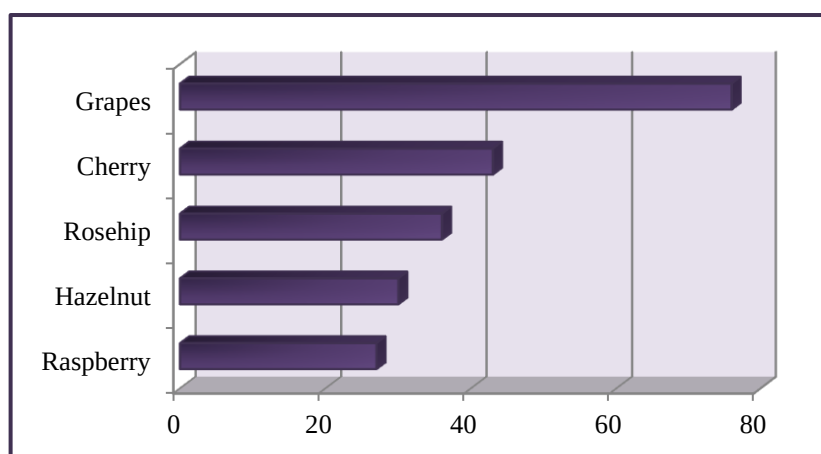


Fig. 3 Area under the Most Important Fruit Species in the Organic Production System in the Vojvodina Region (2018-2022), ha

Based on the average areas of fruit in the organic production system and yield data from the statistical database, the estimated production can be calculated. The estimated production is determined as 70% of conventional production, utilizing yield data from 2023 for this calculation.

Table 1: Calculation of Estimated Production of the Most Important Fruit Species in the Organic Production System in the Vojvodina Region*

Fruit Type	Average Area (ha)	Yield (t/ha)	Conventional Production (t)	Estimated Production in the Organic System (t)
Grapes	76	7.7	585.2	409.6
Cherry	46	5.7	262.2	183.5
Raspberry	27	4.2	113.4	79.4

*No official yield data are available for hazelnut and hawthorn.

Source: Author's calculation based on <https://www.stat.gov.rs/>

Analyzing the areas under organic fruit cultivation by administrative districts in Vojvodina, it can be concluded that the largest area is in the Srem district, totaling 75 hectares (Map 1). This accounts for 29% of the total area, showing a declining trend with an average annual rate of 2.99%. In second place is the South Bačka district with a slightly smaller area of 73 hectares, representing 28% of the total area and demonstrating a substantial growth trend at an average annual rate of 29.41%. These two districts are dominant in organic fruit production in Vojvodina, collectively constituting more than half of the total production, specifically 57%.

The remaining production is distributed across other administrative districts, including the North Bačka district with an average area of 34 hectares (14%), showing a pronounced decline at a rate of 35.32%, and the South Banat district with an area of 30 hectares (11%) and an intense growth trend at a rate of 36.78%. Figure 4 illustrates the average relative contribution of these administrative districts in Vojvodina in the structure of total organic fruit production.

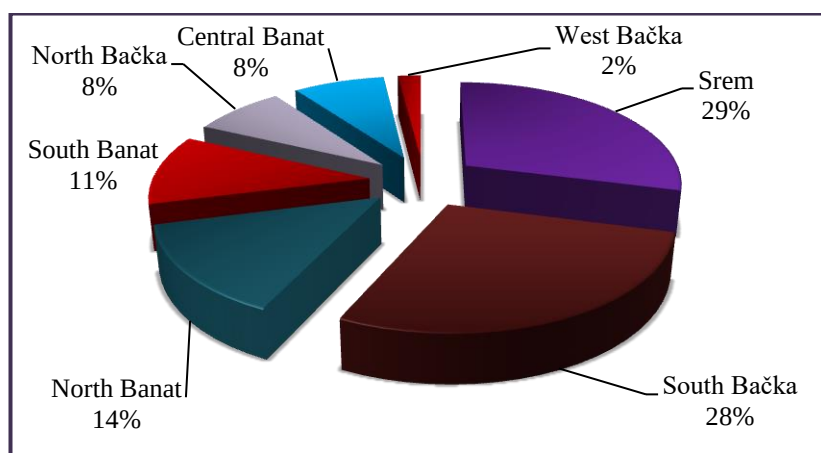


Fig. 4 Structure of Areas under Fruit Cultivation in the Organic Production System in the Vojvodina Region (2018-2022), %

During the analyzed period (2018-2022), four administrative districts showed a positive rate of change: North Bačka, South Bačka, Central Banat, and South Banat. The highest growth rate was observed in the South Bačka district, at 36.78% (Table 2). In other districts, the rate of change was negative, with the West Bačka district experiencing the most intense decline at 36.11%.

Table 2: Area under Fruit Cultivation by Administrative Districts in the Organic Production System in the Vojvodina Region (2018-2022)

Administrative District	Area (%)	Share (%)	Rate of Change (%)
Srem	75	29	-2.99
South Bačka	73	28	29.41
North Banat	34	14	-35.32
South Banat	30	11	36.78
Central Banat	22	8	26.11
North Bačka	21	8	23.15
West Bačka	5	2	-36.11
Vojvodina	261	100.0	11.90

Source: Author's calculation based on <https://www.stat.gov.rs/>

CONCLUSION

Organic fruit production in Vojvodina holds significant potential for regional development, but it is unevenly distributed across districts. The average area under organic fruit cultivation in Vojvodina is 261 hectares, with a significant growth trend at an average annual rate of 11.90%. The most significant districts are Srem and South Bačka, which together account for over half of the total area (57%). The dominant fruit type is grapes, with an average area of 76 hectares, constituting 33.8% of the overall structure.

Factors contributing to the underdevelopment of certain districts may include inadequate climatic and soil conditions, high production costs, labor shortages, insufficient infrastructure, rejection of products by local population and others. For successful organic production and effective marketing of organic products, it is essential to develop a long-term strategy for organic agriculture that provides guidelines for avoiding or mitigating potential investment risks, as well as implementing agricultural, technical, and management practices (Sredojević et al., 2015). Additionally, educating producers on organic methods and standards, along with marketing strategies based on the health and environmental benefits of organic production, are crucial. Further development of organic fruit production will depend on expert knowledge and its application, as well as a shift in public awareness regarding the importance of quality, food safety, and environmental conservation (Vlahović, Užar, 2021).

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SUSCEPTIBILITY OF DIFFERENT POTATO CULTIVARS TO *PECTOBACTERIUM* SPP.

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Pectobacterium spp. are bacterial pathogens responsible for soft rot and blackleg disease in a wide range of crops, particularly potatoes. These infections can occur both in the field and during post-harvest storage, leading to significant economic losses by reducing crop yield, quality, and marketability. Bacterial disease control relies on preventive measures like using healthy seed, selecting resistant cultivars, crop rotation, and copper-based treatments. This study aimed to evaluate the susceptibility of potato tubers from the cultivars Kiebitz, Desiree, and VR808 to three *Pectobacterium* species – *P. carotovorum*, *P. brasiliense*, and *P. punjabense* – which are present in the AP Vojvodina region and known to cause soft rot. The assesment was carried out using the toothpick piercing method. Inoculated tubers were placed in moist containers and incubated at 26°C for 48 hours. After incubation, rot development was assessed, measuring the weight of the rotten tissue. Results showed that all cultivars exhibited susceptibility to soft rot disease and that VR808 cultivar was found highly susceptible to *Pectobacterium* strains, significantly higher compared to Kiebitz and Desiree cultivars. The study results demonstrated that different potato cultivars exhibit different levels of susceptibility to various *Pectobacterium* strains. These results indicate the importance of conducting susceptibility tests on different potato genotypes to pectolytic bacteria, selecting varieties that exhibit the lowest susceptibility to soft rot disease.

Key words: *Pectobacterium*, soft rot, potato, cultivars, susceptibility

INTRODUCTION

Potatoes hold significant importance in human diet, as they are rich in essential nutrients like carbohydrates, vitamins, and minerals, providing affordable source of energy and nourishment. Globally, potatoes rank as the fourth most widely cultivated food, following corn, wheat, and rice (Haverkort, 1990). The global annual production of potatoes is estimated to approximately 388 million tons (FAO, 2019). In Republic of Serbia, potato is grown on 38,472 ha, yielding 589,241 tons of tubers annually (FAO, 2019).

Pectobacterium genus members are ranked among the 10 most important bacterial pathogens of cultivated plants and have a significant economic impact on agricultural productivity and crop quality (Mansfield et al., 2012). *Pectobacterium* spp. cause soft rot and blackleg disease in various crops, including potatoes. Bacterial infections of plants in the field, as well as infections of fruits, roots and tubers in storage lead to complete collapse of infected plants/stored products, with the appearance of foul odor. Potato is one of the major hosts for these pathogens, with average annual yield losses ranging from 10-15% and reaching up to 50% in years with high disease incidence (Statsyuk and Kuznetsova, 2018).

In Serbia, *Pectobacterium atrosepticum* and *P. carotovorum*, the pathogens causing soft rot and blackleg in potatoes, were initially detected in the 1990s. More recently, newly identified *Pectobacterium* species have been reported in the region, including *P. brasiliense* (Grahovac et al., 2019; Loc et al., 2019; Marković et al., 2021) and *P. punjabense* (Loc et al., 2021).

Control of bacterial diseases primarily relies on preventive measures, such as using healthy planting or seed material, selecting less sensitive or resistant cultivars, practicing crop rotation, and applying copper-based treatments. The aim of the study was to determine whether there are differences in the susceptibility of potato tubers of different cultivars to *Pectobacterium* spp., and to test the pathogenicity on the tubers of certain cultivars.

MATERIAL AND METHODS

In order to test the susceptibility of different potato cultivars to species of the genus *Pectobacterium*, three bacterial isolates (*P. carotovorum* - Pc, *P. brasiliense* - Pb, *P. punjabense* - Pp) originating from the tubers and potato plant stems from potato fields in province of Vojvodina (Serbia) were used in the experiment (Table 1).

Two reference isolates of *P. carotovorum* and *P. brasiliense* (DAPP-PG 751 and LMG 21370), originating from Lebanon and Brazil were also included in susceptibility test (Table 1). All isolates belong to the Microbial culture collection of the Laboratory for Biological Research and Pesticides, Department of Plant and Environmental Protection, Faculty of Agriculture, University of Novi Sad, Serbia.

Table 1. Bacterial isolates used in this study

Isolate codes	Species	Location	GenBank Accession Numbers
KC19	<i>Pb</i>	Kula, Serbia	MT240620
DC1 1	<i>Pc</i>	Despotovo, Serbia	MZ427888
ZCVR2	<i>Pp</i>	Zobnatica, Serbia	MZ048662
DAPP-PG 751	<i>Pc</i>	Kfarmayka (Akkar), Liban	KP151644
LMG 21370	<i>Pb</i>	-, Brazil	JF311606

* *Pb* – *P. brasiliense*, *Pc* – *P. carotovorum*, *Pp* – *P. punjabense*

The used strains were identified to the species level based on pathogenic, morphological and biochemical-physiological properties, followed by PCR amplification and sequencing of the 16S rRNA region. The sequencing data was analyzed using NCBI BLASTn algorithm online at <http://blast.ncbi.nlm.nih.gov/>. The sequences of the isolates are deposited in the NCBI GeneBank base (Table 1).

For the trial purposes, the isolates were grown in test tubes on nutrient agar medium (NA) and incubated at 26°C for 24 hours (Figure 1). After incubation, a bacterial suspension of each isolate was prepared and adjusted to a concentration of approximately 1×10^8 CFU/mL. The sterile toothpicks were immersed in the suspension for 10 minutes and used for tuber inoculation.

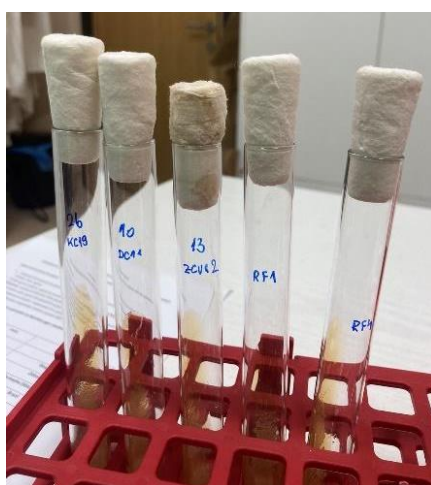


Figure 1. Bacterial isolates in test tubes on nutrient agar (NA) medium

The susceptibility test was carried out on seed potato tubers of the Kiebitz, Desiree and VR808 cultivars using the toothpick piercing method (Duarte et al., 2004; Lebecka, 2017). Prior to the assay, potato tubers were inspected for any sign of disease or injury, washed under tap water and surface sterilized with 70% ethanol, and dried at room temperature over 2 h.

The experiment was conducted in three replicates. After inoculation, tubers were placed in closed plastic containers and sprayed with sterile distilled water to maintain optimal moisture conditions. Potato tubers inoculated with toothpicks dipped in sterile distilled water served as the negative control.

The tubers were incubated at a temperature of 26°C, and the changes were monitored for 48 hours. After incubation, rot development was assessed by measuring the weight of the rotten tissue. The tubers were cut lengthwise and the rotten parts of the tubers were removed. The weight of the tubers was measured before and after the removal of the rotten tissue, and the difference in weight was expressed as a percentage of tissue affected by the pectolytic bacteria.

The obtained data were processed using factorial ANOVA (analysis of variance), and the significance of the differences was calculated by Duncan's multiple range test using the Statistica 14.0.0.15 software package.

RESULTS AND DISCUSSION

In all cultivars where artificial inoculation was performed, symptoms of soft rot appeared, except in the negative control. However, the percentage of tissue affected by soft rot varied, and statistical analysis identified the parameters influencing it.

One-way ANOVA determined that the potato cultivar is a highly significant source of variation for the dependent variable ($p \leq 0.01$), showing that the cultivar has a significant impact on the percentage of tuber tissue affected by soft rot after artificial inoculation with *Pectobacterium* spp. (Table 2).

Table 2. One-way analysis of variance: the effect of potato cultivar on the development of tuber soft rot after artificial inoculation with different *Pectobacterium* isolates

Pectobacterium brasiliense, strain KC19

Source of variation	Sum of squares SS	Degrees of freedom df	Mean square MS	F	p
Cultivar	96.4913	2	48.2456	533.560	0.000000
Error	0.5425	6	0.0904		

Pectobacterium carotovorum, strain DC1 1

Source of variation	Sum of squares SS	Degrees of freedom df	Mean square MS	F	p
Cultivar	128.6042	2	64.3021	253.357	0.000002
Error	1.5228	6	0.2538		

Pectobacterium punjabense, strain ZVCR 2

Source of variation	Sum of squares SS	Degrees of freedom df	Mean square MS	F	p
Cultivar	275.1104	2	137.5552	2713.121	0.000000
Error	0.3042	6	0.0507		

Pectobacterium carotovorum, reference strain DAPP-PG 751

Source of variation	Sum of squares SS	Degrees of freedom df	Mean square MS	F	p
Cultivar	34.0810	2	17.0405	86.0341	0.000038
Error	1.1884	6	0.1981		

Pectobacterium brasiliense, reference strain LMG 21370

Source of variation	Sum of squares SS	Degrees of freedom df	Mean square MS	F	p
Cultivar	278.7321	2	139.3660	403.388	0.000000
Error	2.0729	6	0.3455		

Duncan's multiple range test revealed that all tested isolates exhibited the highest pathogenicity on the VR808 potato cultivar (Table 3), which can be characterized as significantly more susceptible than the other two tested cultivars - Desiree and Kiebitz.

Furthermore, no significant difference in sensitivity between the cultivars Desiree and Kiebitz was found in the case of most isolates, but in the case of artificial inoculation with *P. brasiliense* isolate KC19 and *P. carotovorum* isolate DAPP-PG 751, cultivar Desiree was found significantly more susceptible than cultivar Kiebitz (Table 3).

Table 3. Duncan's multiple range test: The effect of potato cultivar on the development of tuber soft rot after artificial inoculation with different *Pectobacterium* isolates

<i>Pectobacterium brasiliense</i>, strain KC19	
Cultivar	rotten tissue (%)
Kiebitz	3.19 ^a
Desiree	4.73 ^b
VR808	10.78 ^c
<i>Pectobacterium carotovorum</i>, strain DC1 1	
Cultivar	rotten tissue (%)
Kiebitz	2.74 ^a
Desiree	3.59 ^a
VR808	11.15 ^b
<i>Pectobacterium punjabense</i>, strain ZVCR 2	
Cultivar	rotten tissue (%)
Kiebitz	2.45 ^a
Desiree	2.49 ^a
VR808	14.20 ^b
<i>Pectobacterium carotovorum</i>, reference strain DAPP-PG 751	
Cultivar	rotten tissue (%)
Kiebitz	2.43 ^a
Desiree	4.26 ^b
VR808	7.16 ^c
<i>Pectobacterium brasiliense</i>, reference strain LMG 21370	
Cultivar	rotten tissue (%)
Kiebitz	5.43 ^a
Desiree	5.53 ^a
VR808	17.28 ^b

*Values followed with the same letter are at the same level of significance

When *Pectobacterium* isolates are included as a factor in the analysis of the results of conducted tests two-factor analysis of variance showed that potato cultivars, as well as different species of genus *Pectobacterium* with which potato tubers were inoculated, in interaction and individually, significantly affect the development of soft rot on tubers ($p \leq 0.01$) (Table 4).

Table 4. Two-factor analysis of variance: influence of potato cultivar and *Pectobacterium* spp. on the development of tuber soft rot after artificial inoculation

Source of variation	Sum of squares SS	Degrees of freedom df	Mean square MS	F	P
Isolate	429.720	5	85.944	549.47	0.00
Cultivar	596.440	2	298.220	1906.62	0.00
Isolate*Cultivar	216.579	10	21.658	138.47	0.00
Error	5.631	36	0.156		

Duncan's multiple range test revealed that the Kiebitz cultivar exhibited the lowest susceptibility when inoculated with the isolates *Pc* - DC1 1, *Pc* - DAPP-PG 751, and *Pb* - ZVCR 2. Similarly, the Desiree cultivar showed low susceptibility when inoculated with the isolate *Pb* - ZVCR 2. The percentage of rotten tissue in these cases, was in range from 2.43% to 2.75%.

The VR808 cultivar showed the highest susceptibility to all isolates, with the greatest soft rot registered when infected with the *Pb* - LMG 21370 isolate. In this case, an average of 17.28% of tuber tissue was removed 48 hours post-inoculation (Table 5). The reference isolate *Pb* - LMG 21370 from Brazil was identified as the most aggressive.

Among isolates of the genus *Pectobacterium* from the territory of Vojvodina, soft rot development varied across different cultivars. On the cultivar Desiree, the highest pathogenicity was exhibited by the isolate *Pb* - KC19, while the lowest pathogenicity was observed for the isolate *Pp* - ZVCR 2.

The same isolate showed the lowest pathogenicity on the Kiebitz cultivar, while the pathogenicity of the remaining two isolates on this cultivar was at the same level of significance. However, on the most sensitive cultivar, VR808, isolate ZVCR 2, which was the least pathogenic among the isolates originating from Vojvodina on the other two cultivars, exhibited the highest pathogenicity, while the remaining two isolates exhibited pathogenicity at the same level of significance (Table 5).

Table 5. Duncan's multiple range test: The influence of potato cultivar and isolates of *Pectobacterium* species on the development of tuber soft rot after artificial inoculation with different isolates of *Pectobacterium* spp.

Variety	Isolate	Rotten tissue (%)
Kiebitz	<i>Pb</i> KC19	3.19 ^{cd}
	<i>Pc</i> DC1 1	2.74 ^{bc}
	<i>Pp</i> ZVCR 2	2.45 ^b
	<i>Pc</i> DAPP-PG 751	2.43 ^b
	<i>Pb</i> LMG 21370	5.53 ^f
	Uninoculated control	0.00 ^a
Desiree	<i>Pb</i> KC19	4.73 ^e
	<i>Pc</i> DC1 1	3.59 ^d
	<i>Pp</i> ZVCR 2	2.49 ^b
	<i>Pc</i> DAPP-PG 751	4.26 ^e
	<i>Pb</i> LMG 21370	5.43 ^f
	Uninoculated control	0.00 ^a
VR808	<i>Pb</i> KC19	10.78 ^h
	<i>Pc</i> DC1 1	11.15 ^h
	<i>Pp</i> ZVCR 2	14.20 ⁱ
	<i>Pc</i> DAPP-PG 751	7.16 ^g
	<i>Pb</i> LMG 21370	17.28 ^j
	Uninoculated control	0.00 ^a

*Values followed with the same letter are at the same level of significance

According to Charkowski (2015), there is limited information about plant resistance to soft rot agents, and no commercial potato cultivars are resistant to these bacteria. Nevertheless, some cultivars exhibit tolerance, while resistance has been recorded in certain wild potato varieties. *Pectobacterium* is a diverse genus comprising several species capable of infecting potatoes, with multiple species occurring within the same field or even on the same plant. The results of this study align with Charkowski's (2015) assertion that different species within the *Pectobacterium* genus exhibit varying degrees of pathogenicity and genetic diversity regarding virulence factors, while also sharing numerous similarities. This was particularly evident in the case of the VR808 cultivar's sensitivity. Research by Wegener and Jansen (2007) found that purple-fleshed potato cultivars generally show lower susceptibility to soft rot compared to white- or yellow-fleshed cultivars. In this study, all

cultivars had white or yellow flesh however, it is noteworthy that the most sensitive cultivar, VR808, was the only one with red skin.

CONCLUSION

According to the results, it can be concluded that cultivar VR808 exhibits the highest sensitivity to causal agents of tuber soft rot and potato blackleg from *Pectobacterium* genus (*P. carotovorum*, *P. brasiliense* and *P. punjabense*), compared to Kiebitz and Desiree cultivars.

The susceptibility of the Kiebitz and Desiree cultivars to tested *Pectobacterium* strains was uniform, except for KCI9, DCI 1 and DAPP-PG 751 isolates where the Desiree cultivar shows more significant susceptibility than Kiebitz cultivar.

The highest pathogenicity on all tested cultivars was recorded for the reference isolate of *P. brasiliense*.

Among the isolates originating from Vojvodina, *P. punjabense* showed the highest pathogenicity on the most susceptible cultivar VR808, while *P. brasiliense* was recorded as the most pathogenic on Kiebitz and Desiree cultivars.

Conducted research shows that different species of genus *Pectobacterium* show different pathogenicity levels on potato tubers and that the different potato cultivars exhibit different levels of susceptibility to various *Pectobacterium* strains.

These results indicate the importance of conducting susceptibility tests on different potato cultivars to pectolytic bacteria, in order to select varieties that exhibit the lowest susceptibility to soft rot disease.

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